

Numerical solutions of PDEs and applications

Master in Mathematics

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Abstract

This volume gathers a sequence of chapters on the analysis and numerical approximation of partial differential equations. It begins with the Lax–Milgram theorem, energy minimization, and the Stampacchia theorem for constrained variational problems. Sobolev spaces then provide the framework for elliptic boundary-value problems and conforming finite element methods in one and two space dimensions. The Lax–Richtmyer chapter introduces consistency, stability, and convergence immediately before their applications to finite difference approximations of first-order hyperbolic equations. It includes both a fixed-space operator result and an evolution theorem on changing grid spaces. Exercises accompany the theoretical and numerical developments, and a final chapter gathers their worked solutions.

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Chapter 1

The Lax–Milgram theorem and Stampacchia’s extension

1.1 Motivation: the Poisson equation

Consider first the one-dimensional Dirichlet problem

$$-u'' = f \quad \text{in } (0, 1), \quad u(0) = u(1) = 0.$$

If u is a smooth solution and v is a smooth test function vanishing at both endpoints, integration by parts gives

$$\int_0^1 u' v' \, dx = \int_0^1 f v \, dx.$$

This formulation uses only first derivatives. It also has an energy interpretation: the equation is the first-order condition for the functional

$$F(v) = \frac{1}{2} \int_0^1 |v'|^2 \, dx - \int_0^1 f v \, dx.$$

Chapter 2 will provide the complete Hilbert space $H_0^1(0, 1)$ in which these identities make sense, and Chapter 3 will apply the abstract theorem below to this and other elliptic problems. The key questions are already visible: when is the bilinear form continuous, when does it control the chosen norm, and is the right-hand side continuous?

We begin with the equation on a Hilbert space, which is the main tool for the elliptic and finite element chapters. We then extend it to variational inequalities on closed convex constraint sets.

1.2 Setting, notation, and the Lax–Milgram theorem

Throughout, H is a *real* Hilbert space with scalar product $(\cdot, \cdot)_H$ and norm

$$\|u\|_H = (u, u)_H^{1/2}.$$

Its topological dual is denoted by H' , and duality pairings are written as $\langle \cdot, \cdot \rangle_{H', H}$. The point of the indices is to avoid confusing the Hilbert scalar product with a duality pairing.

Definition 1.1. Let $a : H \times H \rightarrow \mathbb{R}$ be a bilinear form.

- (i) It is *continuous* if there exists $C_a > 0$ such that

$$|a(u, v)| \leq C_a \|u\|_H \|v\|_H \quad \forall u, v \in H.$$

(ii) It is *coercive* if there exists $\alpha > 0$ such that

$$a(v, v) \geq \alpha \|v\|_H^2 \quad \forall v \in H.$$

We shall repeatedly use the Riesz representation theorem on H : every continuous linear functional $\ell \in H'$ is represented in a unique way by an element $f \in H$ such that

$$\langle \ell, v \rangle_{H', H} = (f, v)_H \quad \forall v \in H.$$

Theorem 1.2 (Lax–Milgram). *Let $a : H \times H \rightarrow \mathbb{R}$ be a continuous coercive bilinear form. For every $\varphi \in H'$, there exists a unique $u \in H$ such that*

$$a(u, v) = \langle \varphi, v \rangle_{H', H} \quad \forall v \in H. \quad (1.1)$$

If a is symmetric, then u is the unique minimizer on H of

$$F(v) = \frac{1}{2}a(v, v) - \langle \varphi, v \rangle_{H', H}.$$

1.3 An operator proof of the Lax–Milgram theorem

The first part of the theorem follows by representing the bilinear form and the right-hand side with the Hilbert scalar product.

By the Riesz representation theorem, there exists a unique element $f \in H$ such that

$$\langle \varphi, v \rangle_{H', H} = (f, v)_H \quad \forall v \in H. \quad (1.2)$$

Fix $u \in H$. Since a is continuous, the map

$$H \ni v \longmapsto a(u, v) \in \mathbb{R}$$

is a continuous linear functional on H . By Riesz, there exists a unique vector $Au \in H$ such that

$$a(u, v) = (Au, v)_H \quad \forall v \in H. \quad (1.3)$$

In this way we define a linear operator $A : H \rightarrow H$. It enjoys three key properties:

(a) *Boundedness.* For every $u \in H$,

$$\|Au\|_H \leq C_a \|u\|_H. \quad (1.4)$$

Indeed, by (1.3) and continuity of a ,

$$|(Au, v)_H| = |a(u, v)| \leq C_a \|u\|_H \|v\|_H \quad \forall v \in H.$$

Taking the supremum over $\|v\|_H = 1$ yields (1.4).

(b) *Coercivity in operator form.* For every $u \in H$,

$$(Au, u)_H = a(u, u) \geq \alpha \|u\|_H^2. \quad (1.5)$$

(c) *Strong monotonicity.* For every $u, v \in H$,

$$(Au - Av, u - v)_H \geq \alpha \|u - v\|_H^2. \quad (1.6)$$

This is just coercivity applied to $u - v$:

$$(Au - Av, u - v)_H = a(u - v, u - v) \geq \alpha \|u - v\|_H^2.$$

With the operator defined by (1.3), the equation (1.1) is equivalent to

$$Au = f, \quad (1.7)$$

where $f \in H$ is the Riesz representative of φ , that is, $\langle \varphi, v \rangle_{H',H} = (f, v)_H$ for all $v \in H$.

Proposition 1.3. *The operator $A : H \rightarrow H$ is bijective. Consequently, for every $\varphi \in H'$, equation (1.1) has a unique solution.*

Proof. We prove injectivity, closedness of the range, and density of the range.

(a) Injectivity. If $Au = 0$, then by coercivity,

$$\alpha \|u\|_H^2 \leq (Au, u)_H = 0.$$

Hence $u = 0$, so A is injective.

(b) The range $R(A)$ is closed. First observe that coercivity implies the lower bound

$$\alpha \|v\|_H \leq \|Av\|_H \quad \forall v \in H. \quad (1.8)$$

Indeed, from (1.5) and the Cauchy–Schwarz inequality,

$$\alpha \|v\|_H^2 \leq (Av, v)_H \leq \|Av\|_H \|v\|_H.$$

If $v \neq 0$, dividing by $\|v\|_H$ gives (1.8); if $v = 0$, it is trivial.

Now let $(f_n) \subset R(A)$ be a sequence converging in H to some $f \in H$. Write $f_n = Au_n$ with $u_n \in H$. We claim that (u_n) is a Cauchy sequence. Indeed, applying (1.8) to $u_n - u_m$, we find

$$\alpha \|u_n - u_m\|_H \leq \|A(u_n - u_m)\|_H = \|f_n - f_m\|_H.$$

Since (f_n) converges in H , it is Cauchy; therefore (u_n) is also Cauchy. Because H is complete, $u_n \rightarrow u$ for some $u \in H$. By the continuity of A ,

$$Au = \lim_{n \rightarrow \infty} Au_n = \lim_{n \rightarrow \infty} f_n = f.$$

Hence $f \in R(A)$, so $R(A)$ is closed.

(c) The range $R(A)$ is dense. Suppose $v \in H$ is orthogonal to $R(A)$, meaning that

$$(Au, v)_H = 0 \quad \forall u \in H. \quad (1.9)$$

Taking $u = v$ in (1.9) gives

$$(Av, v)_H = 0.$$

By coercivity,

$$\alpha \|v\|_H^2 \leq (Av, v)_H = 0,$$

so $v = 0$. Therefore

$$R(A)^\perp = \{0\}.$$

In a Hilbert space, Section 1.9 shows that for every linear subspace $M \subset H$,

$$\overline{M} = (M^\perp)^\perp.$$

Applying this to $M = R(A)$, we obtain

$$\overline{R(A)} = (R(A)^\perp)^\perp = \{0\}^\perp = H.$$

So $R(A)$ is dense in H . Since it is also closed by part (b), we conclude that $R(A) = H$.

The operator A is therefore bijective. Solving $Au = f$ is the same as solving (1.1), so the Lax–Milgram theorem follows. $\square$

The solution also satisfies the estimate

$$\|u\|_H \leq \frac{1}{\alpha} \|\varphi\|_{H'}.$$

Indeed, taking $v = u$ in (1.1) gives $\alpha \|u\|_H^2 \leq \langle \varphi, u \rangle_{H', H} \leq \|\varphi\|_{H'} \|u\|_H$; division by $\|u\|_H$ proves the estimate when $u \neq 0$, and the case $u = 0$ is immediate.

1.4 The symmetric case: energy minimization

Assume now that a is symmetric. Define a new scalar product on the underlying vector space H by

$$(u, v)_a := a(u, v), \quad \|u\|_a := (u, u)_a^{1/2} = a(u, u)^{1/2}.$$

Because a is symmetric, continuous, and coercive,

$$\sqrt{\alpha} \|u\|_H \leq \|u\|_a \leq \sqrt{C_a} \|u\|_H \quad \forall u \in H. \quad (1.10)$$

Thus $\|\cdot\|_a$ is equivalent to $\|\cdot\|_H$, so the same vector space becomes another Hilbert space, which we denote by H_a .

Since the two norms are equivalent, the functional φ is continuous both on H and on H_a . By the Riesz representation theorem applied to H_a , there exists a unique $g \in H$ such that

$$\langle \varphi, v \rangle_{H'_a, H_a} = (g, v)_a = a(g, v) \quad \forall v \in H. \quad (1.11)$$

The vector g is the unique solution of (1.1). Expanding the square gives

$$\|g - v\|_a^2 = a(v, v) - 2 \langle \varphi, v \rangle_{H', H} + a(g, g) = 2F(v) + a(g, g), \quad F(v) = \frac{1}{2} a(v, v) - \langle \varphi, v \rangle_{H', H}.$$

Since the squared distance from g is uniquely minimized at $v = g$ on H , the Lax–Milgram solution is the unique minimizer of F . Equivalently, if u denotes this solution, then

$$F(v) - F(u) = \frac{1}{2} a(v - u, v - u) \geq \frac{\alpha}{2} \|v - u\|_H^2 \quad \forall v \in H.$$

This proves the minimization assertion of Theorem 1.2. It also shows why symmetry matters: in general the derivative of $\frac{1}{2} a(v, v)$ involves the symmetric part of a , rather than a itself.

1.5 Stampacchia’s extension to convex constraints

On a closed convex set $K \subset H$, not every direction and its negative are admissible. The appropriate first-order condition is a variational inequality. For example, a positivity constraint describes an obstacle that a displacement must stay above. The following theorem extends Lax–Milgram to this setting.

Theorem 1.4. *Let $a : H \times H \rightarrow \mathbb{R}$ be a continuous coercive bilinear form, and let $K \subset H$ be nonempty, closed, and convex. Given $\varphi \in H'$, there exists a unique $u \in K$ such that*

$$a(u, v - u) \geq \langle \varphi, v - u \rangle_{H', H} \quad \forall v \in K. \quad (1.12)$$

If, in addition, a is symmetric, then the same u is characterized by the minimization property

$$F(u) = \min_{v \in K} F(v), \quad F(v) := \frac{1}{2} a(v, v) - \langle \varphi, v \rangle_{H', H}. \quad (1.13)$$

The classical auxiliary results used in the proof are recalled later in this chapter:

- Section 1.9: projection onto a closed convex set in a Hilbert space, together with the fact that the projection is nonexpansive.
- Section 1.10: Banach’s contraction mapping theorem.
- Section 1.11: Gâteaux and Fréchet derivatives, and the derivative of the functional F .

For background on the functional analysis tools used in this chapter, we may also consult [5].

1.6 Proof of Stampacchia in five steps

1.6.1 Step 1: Reformulation with scalar products.

Let f be the Riesz representative already defined in (1.2). Hence the variational inequality (1.12) can already be rewritten as

$$a(u, v - u) \geq (f, v - u)_H \quad \forall v \in K. \quad (1.14)$$

The advantage is that the right-hand side is now expressed using the Hilbert scalar product on H .

1.6.2 Step 2: Operator form.

Use the operator A defined in (1.3). Its boundedness, coercivity, and strong monotonicity were established in (1.4)–(1.6). Using (1.3) and (1.2), the problem (1.12) is therefore equivalent to

$$(Au - f, v - u)_H \geq 0 \quad \forall v \in K, \quad (1.15)$$

or, equivalently,

$$(f - Au, v - u)_H \leq 0 \quad \forall v \in K. \quad (1.16)$$

1.6.3 Step 3: Fixed-point reformulation.

Let $\rho > 0$. Since

$$(u + \rho(f - Au) - u, v - u)_H = \rho (f - Au, v - u)_H,$$

condition (1.16) is equivalent to

$$(u + \rho(f - Au) - u, v - u)_H \leq 0 \quad \forall v \in K. \quad (1.17)$$

By the characterization of the projection onto a closed convex set (Theorem 1.5 in Section 1.9), (1.17) is equivalent to

$$u = P_K^H(u + \rho(f - Au)). \quad (1.18)$$

Thus, if we define

$$S_\rho(v) := P_K^H(v + \rho(f - Av)) \quad (v \in K),$$

then solving (1.12) is exactly the same as finding a fixed point of $S_\rho : K \rightarrow K$.

1.6.4 Step 4: Contracting property.

Take $v_1, v_2 \in K$. Since the projection P_K^H is nonexpansive (Proposition 1.6),

$$\|S_\rho(v_1) - S_\rho(v_2)\|_H \leq \|(v_1 - v_2) - \rho(Av_1 - Av_2)\|_H.$$

Set $w = v_1 - v_2$. Then

$$\begin{aligned} \|w - \rho Aw\|_H^2 &= \|w\|_H^2 - 2\rho (Aw, w)_H + \rho^2 \|Aw\|_H^2 \\ &\leq \|w\|_H^2 - 2\rho\alpha \|w\|_H^2 + \rho^2 C_a^2 \|w\|_H^2 \\ &= (1 - 2\alpha\rho + C_a^2\rho^2) \|w\|_H^2, \end{aligned}$$

where we used (1.5) and (1.4). Consequently,

$$\|S_\rho(v_1) - S_\rho(v_2)\|_H \leq q \|v_1 - v_2\|_H, \quad q := \sqrt{1 - 2\alpha\rho + C_a^2\rho^2}. \quad (1.19)$$

If

$$0 < \rho < \frac{2\alpha}{C_a^2}, \quad (1.20)$$

then $q < 1$, so S_ρ is a strict contraction on the complete metric space K (closed subsets of Hilbert spaces are complete). By Banach’s fixed-point theorem (Section 1.10), S_ρ has a unique fixed point. By (1.18), this fixed point is exactly the unique solution $u \in K$ of (1.12).

For numerical purposes, the upper bound in (1.19) is minimized by the choice $\rho = \alpha/C_a^2$.

1.6.5 Step 5: The symmetric case and the energy scalar product.

Assume now that a is symmetric. Use the equivalent Hilbert structure H_a of Section 1.4 and the vector g characterized by (1.11). The variational inequality (1.12) becomes

$$(u, v - u)_a \geq (g, v - u)_a \quad \forall v \in K,$$

that is,

$$(g - u, v - u)_a \leq 0 \quad \forall v \in K. \quad (1.21)$$

Because K is convex and closed for $\|\cdot\|_H$, it is also convex and closed for $\|\cdot\|_a$ by norm equivalence. Hence, by the projection theorem in the Hilbert space H_a , (1.21) is equivalent to

$$u = P_K^a g,$$

where P_K^a denotes projection onto K with respect to the scalar product $(\cdot, \cdot)_a$. Therefore u is the unique minimizer of

$$v \longmapsto \|g - v\|_a^2 = (g - v, g - v)_a \quad \text{over } K.$$

Expanding this quantity and using (1.11), we get

$$(g - v, g - v)_a = a(v, v) - 2a(g, v) + a(g, g) = a(v, v) - 2\langle \varphi, v \rangle_{H'_a, H_a} + a(g, g).$$

The constant term $a(g, g)$ is irrelevant for minimization, so minimizing $\|g - v\|_a^2$ over K is equivalent to minimizing

$$F(v) = \frac{1}{2}a(v, v) - \langle \varphi, v \rangle_{H', H} \quad \text{over } K.$$

This proves (1.13).

1.7 Recovering Lax–Milgram from Stampacchia

The theorem on the whole Hilbert space can also be recovered as the special case $K = H$ of Theorem 1.4. This gives a second proof of Theorem 1.2 and clarifies the distinction between an equation and a constrained variational inequality.

Proof. Apply Theorem 1.4 with $K = H$. Since H is closed and convex in itself, there exists a unique $u \in H$ such that

$$a(u, v - u) \geq \langle \varphi, v - u \rangle_{H', H} \quad \forall v \in H. \quad (1.22)$$

The key point is that now *every* direction is admissible, with both signs. Take an arbitrary $w \in H$. Choosing $v = u + w$ in (1.22) gives

$$a(u, w) \geq \langle \varphi, w \rangle_{H', H}. \quad (1.23)$$

Choosing instead $v = u - w$ gives

$$a(u, -w) \geq \langle \varphi, -w \rangle_{H', H},$$

that is,

$$a(u, w) \leq \langle \varphi, w \rangle_{H', H}. \quad (1.24)$$

Combining (1.23) and (1.24), we obtain

$$a(u, w) = \langle \varphi, w \rangle_{H', H} \quad \forall w \in H,$$

which is exactly (1.1).

Uniqueness follows either from the uniqueness part of Theorem 1.4 or directly from coercivity: if u_1 and u_2 both satisfy (1.1), then

$$a(u_1 - u_2, v) = 0 \quad \forall v \in H.$$

Taking $v = u_1 - u_2$ yields

$$\alpha \|u_1 - u_2\|_H^2 \leq a(u_1 - u_2, u_1 - u_2) = 0,$$

so $u_1 = u_2$.

When a is symmetric, Theorem 1.4 with $K = H$ gives the minimization property immediately. $\square$

1.8 The functional F and its derivative

Assume in this section that a is symmetric, and define

$$F(v) = \frac{1}{2} a(v, v) - \langle \varphi, v \rangle_{H', H} \quad (v \in H).$$

Section 1.11 recalls the definitions of Gâteaux and Fréchet derivatives and proves that F is actually Fréchet differentiable, with

$$DF(u)[h] = a(u, h) - \langle \varphi, h \rangle_{H', H} \quad \forall u, h \in H. \quad (1.25)$$

The remainder term is especially simple:

$$F(u + h) - F(u) - DF(u)[h] = \frac{1}{2} a(h, h). \quad (1.26)$$

Because $a(h, h) \geq \alpha \|h\|_H^2$, the functional F is strictly convex; in fact it is strongly convex.

Formula (1.25) makes the variational meaning of the previous results completely transparent:

- The unconstrained equation (1.1) is exactly the condition $DF(u) = 0$ in H' .
- On a closed convex set K , the Stampacchia condition (1.12) is equivalent to

$$DF(u)[v - u] \geq 0 \quad \forall v \in K,$$

which is the standard first-order optimality condition for minimizing a differentiable convex functional on a convex set.

Indeed, using (1.26) with $h = v - u$, we obtain the exact identity

$$F(v) - F(u) = DF(u)[v - u] + \frac{1}{2}a(v - u, v - u).$$

Hence, if $DF(u)[v - u] \geq 0$ for all $v \in K$, then automatically $F(v) \geq F(u)$ for all $v \in K$. Conversely, if u minimizes F on K , then the one-variable function $t \mapsto F(u + t(v - u))$ has a minimum at $t = 0$ on $[0, 1]$, so its right derivative at 0 is nonnegative, which is exactly $DF(u)[v - u] \geq 0$.

1.9 Projection onto a closed convex set

In this section $(X, (\cdot, \cdot)_X)$ denotes a real Hilbert space with norm $\|\cdot\|_X$. We keep the notation generic so that the result can be applied both to $(H, (\cdot, \cdot)_H)$ and to $(H_a, (\cdot, \cdot)_a)$.

Theorem 1.5. *Let $C \subset X$ be nonempty, closed, and convex. Then for every $z \in X$ there exists a unique element $u \in C$ such that*

$$\|z - u\|_X = \text{dist}(z, C) := \inf_{v \in C} \|z - v\|_X.$$

Moreover, u is characterized by

$$(z - u, v - u)_X \leq 0 \quad \forall v \in C. \quad (1.27)$$

We denote this point by $u = P_C^X z$.

Proof. Existence. Let $d := \inf_{v \in C} \|z - v\|_X$, and choose a minimizing sequence $(u_n) \subset C$ such that $\|z - u_n\|_X \rightarrow d$. By the parallelogram identity,

$$\left\| z - \frac{u_n + u_m}{2} \right\|_X^2 + \left\| \frac{u_n - u_m}{2} \right\|_X^2 = \frac{1}{2}(\|z - u_n\|_X^2 + \|z - u_m\|_X^2).$$

Since C is convex, $(u_n + u_m)/2 \in C$, hence

$$\left\| z - \frac{u_n + u_m}{2} \right\|_X^2 \geq d^2.$$

Therefore

$$\left\| \frac{u_n - u_m}{2} \right\|_X^2 \leq \frac{1}{2}(\|z - u_n\|_X^2 + \|z - u_m\|_X^2) - d^2 \xrightarrow{n, m \rightarrow \infty} 0.$$

So (u_n) is Cauchy. Because X is complete and C is closed, $u_n \rightarrow u \in C$. Passing to the limit in $\|z - u_n\|_X \rightarrow d$, we obtain $\|z - u\|_X = d$.

Characterization of the minimizer. Assume first that $u \in C$ minimizes the distance to z . Let $v \in C$, and for $t \in [0, 1]$ set $u_t = (1 - t)u + tv \in C$. Since u is a minimizer,

$$\|z - u\|_X^2 \leq \|z - u_t\|_X^2 = \|(z - u) - t(v - u)\|_X^2.$$

Expanding the square gives

$$0 \leq -2t(z - u, v - u)_X + t^2 \|v - u\|_X^2.$$

Dividing by $t > 0$ and letting $t \downarrow 0$ yields (1.27).

Conversely, assume that (1.27) holds. Then for every $v \in C$,

$$\begin{aligned} \|z - v\|_X^2 &= \|(z - u) - (v - u)\|_X^2 \\ &= \|z - u\|_X^2 - 2(z - u, v - u)_X + \|v - u\|_X^2 \\ &\geq \|z - u\|_X^2. \end{aligned}$$

Thus u is indeed a minimizer.

Uniqueness. If $u_1, u_2 \in C$ both satisfy (1.27), then choosing $v = u_2$ in the inequality for u_1 , and $v = u_1$ in the inequality for u_2 , gives

$$(z - u_1, u_2 - u_1)_X \leq 0, \quad (z - u_2, u_1 - u_2)_X \leq 0.$$

Adding these inequalities yields

$$\|u_1 - u_2\|_X^2 \leq 0.$$

Hence $u_1 = u_2$. □

Proposition 1.6. *Let $C \subset X$ be nonempty, closed, and convex. Then*

$$\|P_C^X z_1 - P_C^X z_2\|_X \leq \|z_1 - z_2\|_X \quad \forall z_1, z_2 \in X.$$

Proof. Set $u_i = P_C^X z_i$ for $i = 1, 2$. By (1.27),

$$(z_1 - u_1, u_2 - u_1)_X \leq 0, \quad (z_2 - u_2, u_1 - u_2)_X \leq 0.$$

Adding them yields

$$\|u_1 - u_2\|_X^2 \leq (z_1 - z_2, u_1 - u_2)_X.$$

By Cauchy–Schwarz,

$$\|u_1 - u_2\|_X^2 \leq \|z_1 - z_2\|_X \|u_1 - u_2\|_X,$$

which proves the claim. □

Corollary 1.7. *Let $M \subset X$ be a closed linear subspace. Then $u = P_M^X z$ if and only if*

$$u \in M \quad \text{and} \quad (z - u, v)_X = 0 \quad \forall v \in M. \quad (1.28)$$

Consequently,

$$X = M \oplus M^\perp, \quad M^\perp := \{w \in X : (w, v)_X = 0 \ \forall v \in M\},$$

and for every linear subspace $M \subset X$ we have

$$\overline{M} = (M^\perp)^\perp. \quad (1.29)$$

Proof. If $u = P_M^X z$, then Theorem 1.5 gives

$$(z - u, v)_X \leq 0 \quad \forall v \in M.$$

Since M is a linear space, we may replace v by $u + tw$ with $w \in M$ and $t \in \mathbb{R}$. This gives

$$t(z - u, w)_X \leq 0 \quad \forall w \in M, \ \forall t \in \mathbb{R},$$

which is possible only if $(z - u, w)_X = 0$ for all $w \in M$. The converse is immediate from Theorem 1.5.

The decomposition $X = M \oplus M^\perp$ follows by applying (1.28) to M . Finally, applying this decomposition with $\overline{M}$ in place of M , every $x \in X$ can be written uniquely as $x = m + r$ with $m \in \overline{M}$ and $r \in M^\perp$. If in addition $x \in (M^\perp)^\perp$, then $r = x - m \in (M^\perp)^\perp \cap M^\perp$, so $r = 0$, hence $x \in \overline{M}$. This proves (1.29). The reverse inclusion $\overline{M} \subset (M^\perp)^\perp$ is obvious. □

1.10 Banach’s fixed-point theorem

Theorem 1.8. *Let (X, d) be a nonempty complete metric space, and let $S : X \rightarrow X$ satisfy*

$$d(Sx, Sy) \leq k d(x, y) \quad \forall x, y \in X,$$

for some constant $0 \leq k < 1$. Then S has a unique fixed point in X .

Proof. Choose $x_0 \in X$ and define the Picard iterates by $x_{n+1} = Sx_n$. Then

$$d(x_{n+1}, x_n) = d(Sx_n, Sx_{n-1}) \leq k d(x_n, x_{n-1}) \leq \cdots \leq k^n d(x_1, x_0).$$

If $m > n$, the triangle inequality gives

$$\begin{aligned} d(x_m, x_n) &\leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \\ &\leq \sum_{j=n}^{m-1} k^j d(x_1, x_0) \leq \frac{k^n}{1-k} d(x_1, x_0). \end{aligned}$$

Thus (x_n) is Cauchy, hence converges to some $u \in X$.

Since S is Lipschitz, it is continuous. Therefore

$$Su = S\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} Sx_n = \lim_{n \rightarrow \infty} x_{n+1} = u.$$

So u is a fixed point.

If u and v are fixed points, then

$$d(u, v) = d(Su, Sv) \leq k d(u, v).$$

Because $k < 1$, this forces $d(u, v) = 0$, hence $u = v$. □

1.11 Gâteaux and Fréchet derivatives

Definition 1.9. Let E be a Banach space, $U \subset E$ an open set, and $G : U \rightarrow \mathbb{R}$. We say that G is *Gâteaux differentiable* at $u \in U$ if for every $h \in E$ the limit

$$\delta G(u; h) := \lim_{t \rightarrow 0} \frac{G(u + th) - G(u)}{t}$$

exists, and if the map $h \mapsto \delta G(u; h)$ is linear and continuous. This linear functional is denoted by $DG(u)$.

Definition 1.10. Let E , U , and G be as above. We say that G is *Fréchet differentiable* at $u \in U$ if there exists a bounded linear functional $L \in E'$ such that

$$\lim_{\|h\|_E \rightarrow 0} \frac{|G(u + h) - G(u) - L(h)|}{\|h\|_E} = 0.$$

The functional L is then written $DG(u)$.

Remark 1.11. Fréchet differentiability is stronger than Gâteaux differentiability. If G is Fréchet differentiable at u , then for every direction h the one-variable expansion

$$G(u + th) = G(u) + t DG(u)[h] + o(|t|)$$

holds as $t \rightarrow 0$. In particular, the Gâteaux derivative exists and equals the Fréchet derivative.

Proposition 1.12. *Assume that $a : H \times H \rightarrow \mathbb{R}$ is continuous and symmetric, and let $\varphi \in H'$. Define*

$$F(v) = \frac{1}{2}a(v, v) - \langle \varphi, v \rangle_{H', H}.$$

Then F is Fréchet differentiable on H , and

$$DF(u)[h] = a(u, h) - \langle \varphi, h \rangle_{H', H} \quad \forall u, h \in H. \quad (1.30)$$

Moreover,

$$F(u + h) - F(u) - DF(u)[h] = \frac{1}{2}a(h, h), \quad (1.31)$$

so the remainder is $O(\|h\|_H^2)$.

Proof. Because a is symmetric,

$$\begin{aligned} F(u + h) - F(u) &= \frac{1}{2}(a(u + h, u + h) - a(u, u)) - \langle \varphi, h \rangle_{H', H} \\ &= a(u, h) + \frac{1}{2}a(h, h) - \langle \varphi, h \rangle_{H', H}. \end{aligned}$$

Thus the natural candidate for the derivative is

$$L_u(h) = a(u, h) - \langle \varphi, h \rangle_{H', H}.$$

This is a bounded linear functional of h , and the previous identity becomes

$$F(u + h) - F(u) - L_u(h) = \frac{1}{2}a(h, h).$$

By continuity of a ,

$$\left| \frac{1}{2}a(h, h) \right| \leq \frac{C_a}{2} \|h\|_H^2 = o(\|h\|_H) \quad \text{as } \|h\|_H \rightarrow 0.$$

Therefore $L_u = DF(u)$, proving (1.30) and (1.31). $\square$

Proposition 1.13. *Let E be a Banach space, $K \subset E$ a convex set, and $U \subset E$ an open set containing K . Let $G : U \rightarrow \mathbb{R}$ be Fréchet differentiable, with convex restriction to K . Then, for $u \in K$, the following are equivalent:*

- (i) u minimizes G on K ;
- (ii) $DG(u)[v - u] \geq 0$ for every $v \in K$.

Proof. Assume first that u minimizes G on K , and let $v \in K$. Since K is convex, the segment $u + t(v - u)$ lies in K for every $t \in [0, 1]$. Hence

$$\psi(t) := G(u + t(v - u))$$

has a minimum at $t = 0$ on $[0, 1]$. Therefore $\psi'(0^+) \geq 0$, i.e.

$$DG(u)[v - u] \geq 0.$$

Conversely, assume that $DG(u)[v - u] \geq 0$ for every $v \in K$. By convexity, for $t \in (0, 1)$,

$$G(u + t(v - u)) \leq (1 - t)G(u) + tG(v).$$

Subtract $G(u)$, divide by t , and let $t \downarrow 0$. Using the Fréchet differentiability of G , we obtain

$$DG(u)[v - u] \leq G(v) - G(u).$$

Since the left-hand side is nonnegative by assumption, $G(v) \geq G(u)$ for every $v \in K$. Thus u minimizes G on K . $\square$

1.12 Exercises

Exercise 1.1 (Abstract Galerkin method and Céa’s lemma). Let $\varphi \in H'$, and let $u \in H$ be the Lax–Milgram solution of

$$a(u, v) = \langle \varphi, v \rangle_{H', H} \quad \forall v \in H.$$

Let $V_n \subset H$ be finite-dimensional, and define $u_n \in V_n$ by

$$a(u_n, v_n) = \langle \varphi, v_n \rangle_{H', H} \quad \forall v_n \in V_n.$$

Prove existence and uniqueness of u_n , and then show that

$$\|u - u_n\|_H \leq \frac{M}{\alpha} \inf_{v_n \in V_n} \|u - v_n\|_H,$$

where M is a continuity constant of a .

Exercise 1.2 (Affine constraints). Let $K = u_0 + M$, where $u_0 \in H$ and $M \subset H$ is a closed linear subspace. Show that the Stampacchia variational inequality on K reduces to a Lax–Milgram problem on M . In particular, Stampacchia contains the “affine-constrained” form of Lax–Milgram.

Exercise 1.3 (Stability with respect to the right-hand side). Let $\varphi_1, \varphi_2 \in H'$, and let $u_1, u_2 \in K$ be the corresponding solutions of Stampacchia’s variational inequality. Show that

$$\|u_1 - u_2\|_H \leq \frac{1}{\alpha} \|\varphi_1 - \varphi_2\|_{H'}.$$

Exercise 1.4 (What can fail without coercivity). On $H = \mathbb{R}^2$, define

$$a((x_1, x_2), (y_1, y_2)) = x_1 y_1.$$

Find one continuous linear functional $\varphi \in H'$ for which the Lax–Milgram problem has no solution, and another one for which it has infinitely many solutions.

Exercise 1.5 (A diagonal example in ℓ^2). Let $(\lambda_n)_{n \geq 1}$ satisfy

$$\alpha \leq \lambda_n \leq M \quad \forall n \geq 1,$$

and define on ℓ^2 the bilinear form

$$a(u, v) = \sum_{n \geq 1} \lambda_n u_n v_n.$$

For $b = (b_n) \in \ell^2$, define

$$\varphi(v) = \sum_{n \geq 1} b_n v_n.$$

Solve the Lax–Milgram problem explicitly.

Exercise 1.6 (The same diagonal example with a positivity constraint). In $H = \ell^2$, let

$$K_+ := \{u = (u_n) \in \ell^2 : u_n \geq 0 \ \forall n\}.$$

For the same diagonal bilinear form and the same right-hand side as in Exercise 5, solve Stampacchia’s variational inequality on K_+ .

Exercise 1.7 (A box constraint in ℓ^2). In $H = \ell^2$, let

$$K_{[0,1]} := \{u = (u_n) \in \ell^2 : 0 \leq u_n \leq 1 \ \forall n\}.$$

For the same diagonal bilinear form and the same right-hand side as in Exercise 5, solve Stampacchia’s variational inequality on $K_{[0,1]}$.

Chapter 2

Sobolev spaces

2.1 Introduction

Notation for derivatives. Throughout the notes, partial derivatives are written with operator notation, for example $\partial_x u$, $\partial_t u$, $\partial_{xx} u$, and $D^\alpha u$. When the functions are smooth, these are classical derivatives. In Sobolev spaces, the same notation denotes weak derivatives. Subscript notation such as u_t is not used for derivatives, since subscripts are also used for indexed families and homotopy parameters.

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. In the finite element method, we do not usually work with classical solutions of boundary value problems. Instead, we rewrite the problem in weak form and look for a solution in a space where only a limited number of derivatives are required.

A model example is the Poisson problem

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (2.1)$$

If u and a test function v are smooth and $v = 0$ on the boundary, integration by parts gives

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx. \quad (2.2)$$

This identity only involves first derivatives of u . It is therefore natural to look for u in a space of square-integrable functions whose first derivatives are also square-integrable. This leads to the Sobolev space $H^1(\Omega)$ and, when the boundary condition is homogeneous Dirichlet, to the subspace $H_0^1(\Omega)$.

2.2 Weak derivatives and Sobolev spaces

2.2.1 Weak derivatives

Definition 2.1. Let $u \in L_{\text{loc}}^1(\Omega)$ and let $i \in \{1, \dots, d\}$. A function $g \in L_{\text{loc}}^1(\Omega)$ is called the *weak partial derivative* of u with respect to x_i if

$$\int_{\Omega} u \, \partial_i \varphi \, dx = - \int_{\Omega} g \, \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\Omega). \quad (2.3)$$

In that case we write $g = \partial_i u$.

The definition is the same as saying that the distributional derivative of u is represented by an L_{loc}^1 function.

Remark 2.2. Weak derivatives are unique up to equality almost everywhere. Indeed, if g_1 and g_2 satisfy the defining identity, then

$$\int_{\Omega} (g_1 - g_2) \varphi \, dx = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega), \quad (2.4)$$

so $g_1 = g_2$ almost everywhere.

Proposition 2.3. *Let $u(x) = |x|$ on $(-1, 1)$. Then u is not differentiable at $x = 0$ in the classical sense, but it has the weak derivative*

$$u'(x) = \text{sign}(x) \quad \text{a.e. on } (-1, 1). \quad (2.5)$$

Proof. Let $\varphi \in C_c^\infty((-1, 1))$. Since $u(x) = -x$ on $(-1, 0)$ and $u(x) = x$ on $(0, 1)$, we have

$$\int_{-1}^1 |x| \varphi'(x) \, dx = \int_{-1}^0 (-x) \varphi'(x) \, dx + \int_0^1 x \varphi'(x) \, dx \quad (2.6)$$

$$= \int_{-1}^0 \varphi(x) \, dx - \int_0^1 \varphi(x) \, dx, \quad (2.7)$$

where the boundary terms vanish because φ has compact support in $(-1, 1)$. Therefore

$$\int_{-1}^1 |x| \varphi'(x) \, dx = - \int_{-1}^1 \text{sign}(x) \varphi(x) \, dx, \quad (2.8)$$

which is exactly the defining identity. $\square$

2.2.2 The spaces $W^{m,p}$ and H^m

We use the standard multi-index notation. If $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$, then

$$|\alpha| = \alpha_1 + \dots + \alpha_d, \quad D^\alpha = \partial_1^{\alpha_1} \dots \partial_d^{\alpha_d}. \quad (2.9)$$

Definition 2.4. Let $m \in \mathbb{N}$ and $1 \leq p \leq \infty$. The Sobolev space $W^{m,p}(\Omega)$ is defined by

$$W^{m,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } \alpha \text{ with } |\alpha| \leq m\}, \quad (2.10)$$

where derivatives are understood in the weak sense. When $p = 2$, we write

$$H^m(\Omega) = W^{m,2}(\Omega). \quad (2.11)$$

For $u \in W^{m,p}(\Omega)$, the norm is

$$\|u\|_{W^{m,p}(\Omega)} = \begin{cases} \left(\sum_{|\alpha| \leq m} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max_{|\alpha| \leq m} \|D^\alpha u\|_{L^\infty(\Omega)}, & p = \infty. \end{cases} \quad (2.12)$$

In particular,

$$\|u\|_{H^m(\Omega)} = \left(\sum_{|\alpha| \leq m} \|D^\alpha u\|_{L^2(\Omega)}^2 \right)^{1/2}. \quad (2.13)$$

The corresponding seminorm is

$$|u|_{H^m(\Omega)} = \left(\sum_{|\alpha|=m} \|D^\alpha u\|_{L^2(\Omega)}^2 \right)^{1/2}. \quad (2.14)$$

For finite elements, the most important cases are $H^1(\Omega)$ and $H^2(\Omega)$. The first one is the natural space for weak formulations of second-order elliptic problems; the second one often appears in interpolation estimates for P_1 elements.

Remark 2.5. When $\Omega = I = (a, b)$ is an interval,

$$H^1(I) = \{u \in L^2(I) : u' \in L^2(I)\} \quad (2.15)$$

and

$$\|u\|_{H^1(I)}^2 = \|u\|_{L^2(I)}^2 + \|u'\|_{L^2(I)}^2. \quad (2.16)$$

2.3 Hilbert structure and density

Theorem 2.6. *For every integer $m \geq 0$, the space $H^m(\Omega)$ is a Hilbert space for the scalar product*

$$(u, v)_{H^m(\Omega)} = \sum_{|\alpha| \leq m} \int_{\Omega} D^{\alpha} u D^{\alpha} v \, dx. \quad (2.17)$$

Proof. Only completeness needs justification. Let (u_n) be a Cauchy sequence in $H^m(\Omega)$. Then for every multi-index α with $|\alpha| \leq m$, the sequence $(D^{\alpha} u_n)$ is Cauchy in $L^2(\Omega)$. Since $L^2(\Omega)$ is complete, there exists $g_{\alpha} \in L^2(\Omega)$ such that

$$D^{\alpha} u_n \rightarrow g_{\alpha} \quad \text{in } L^2(\Omega). \quad (2.18)$$

In particular, for $\alpha = 0$, we have $u_n \rightarrow g_0$ in $L^2(\Omega)$. Set $u = g_0$.

We claim that $D^{\alpha} u = g_{\alpha}$ in the weak sense. Let $\varphi \in C_c^{\infty}(\Omega)$. Then

$$\int_{\Omega} u D^{\alpha} \varphi \, dx = \lim_{n \rightarrow \infty} \int_{\Omega} u_n D^{\alpha} \varphi \, dx \quad (2.19)$$

$$= (-1)^{|\alpha|} \lim_{n \rightarrow \infty} \int_{\Omega} D^{\alpha} u_n \varphi \, dx \quad (2.20)$$

$$= (-1)^{|\alpha|} \int_{\Omega} g_{\alpha} \varphi \, dx. \quad (2.21)$$

Thus g_{α} is the weak derivative $D^{\alpha} u$. Hence $u \in H^m(\Omega)$. Finally,

$$\|u_n - u\|_{H^m(\Omega)}^2 = \sum_{|\alpha| \leq m} \|D^{\alpha} u_n - g_{\alpha}\|_{L^2(\Omega)}^2 \rightarrow 0, \quad (2.22)$$

so $u_n \rightarrow u$ in $H^m(\Omega)$. □

Definition 2.7. The space $H_0^1(\Omega)$ is the closure of $C_c^{\infty}(\Omega)$ in $H^1(\Omega)$:

$$H_0^1(\Omega) = \overline{C_c^{\infty}(\Omega)}^{H^1(\Omega)}. \quad (2.23)$$

Proposition 2.8. *The space $H_0^1(\Omega)$ is a closed subspace of $H^1(\Omega)$, hence a Hilbert space.*

Proof. By definition, $H_0^1(\Omega)$ is closed in $H^1(\Omega)$. Since closed subspaces of Hilbert spaces are Hilbert, the conclusion follows. □

Theorem 2.9. *If Ω is a bounded Lipschitz domain, then $C^{\infty}(\overline{\Omega})$ is dense in $H^1(\Omega)$. Moreover, $C_c^{\infty}(\Omega)$ is dense in $H_0^1(\Omega)$ by definition.*

2.4 The one-dimensional case: continuity and endpoint values

In one dimension, Sobolev spaces have a particularly concrete interpretation.

Theorem 2.10. *Let $I = (a, b)$ be a bounded interval and let $u \in H^1(I)$. Then u admits a unique continuous representative on $[a, b]$, still denoted by u , such that*

$$u(x) = u(y) + \int_y^x u'(t) dt \quad \text{for all } x, y \in [a, b]. \quad (2.24)$$

In particular, $H^1(I) \hookrightarrow C^0([a, b])$ continuously.

Remark 2.11. In one dimension, the continuous representative of an H^1 function is absolutely continuous. Its classical derivative exists almost everywhere and agrees almost everywhere with the weak derivative. Actually, the result above can be extended to the Sobolev space $W^{1,1}$.

Corollary 2.12. *There exists a constant C , depending only on a and b , such that*

$$\|u\|_{L^\infty(I)} \leq C \|u\|_{H^1(I)} \quad \text{for all } u \in H^1(I). \quad (2.25)$$

Thus the endpoint values $u(a)$ and $u(b)$ are well-defined and depend continuously on u .

Theorem 2.13. *On a bounded interval $I = (a, b)$,*

$$H_0^1(I) = \{u \in H^1(I) : u(a) = u(b) = 0\}. \quad (2.26)$$

Corollary 2.14. *If $u \in H^2(I)$, then $u' \in H^1(I)$ and therefore $u' \in C^0([a, b])$. Hence*

$$H^2(I) \hookrightarrow C^1([a, b]). \quad (2.27)$$

2.5 Trace theorem and boundary conditions in several dimensions

The one-dimensional picture extends to Lipschitz domains in a weaker but still very useful way.

Theorem 2.15. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. There exists a continuous linear map*

$$\gamma : H^1(\Omega) \rightarrow L^2(\partial\Omega) \quad (2.28)$$

such that $\gamma u = u|_{\partial\Omega}$ whenever u is smooth up to the boundary. Moreover,

$$\|\gamma u\|_{L^2(\partial\Omega)} \leq C_{\text{tr}} \|u\|_{H^1(\Omega)} \quad \text{for all } u \in H^1(\Omega), \quad (2.29)$$

and

$$\ker \gamma = H_0^1(\Omega). \quad (2.30)$$

2.6 Poincaré and Poincaré–Wirtinger inequalities

2.6.1 The global inequalities

Theorem 2.16. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. There exists a constant $C_P > 0$ such that*

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)} \quad \text{for all } u \in H_0^1(\Omega). \quad (2.31)$$

Corollary 2.17. *The formula*

$$(u, v)_V := \int_{\Omega} \nabla u \cdot \nabla v \, dx \quad (2.32)$$

defines a scalar product on $V = H_0^1(\Omega)$, and the associated norm is equivalent to the usual H^1 norm.

Theorem 2.18. *Assume in addition that Ω is connected. There exists a constant $C_{PW} > 0$ such that for every $u \in H^1(\Omega)$,*

$$\|u - u_{\Omega}\|_{L^2(\Omega)} \leq C_{PW} \|\nabla u\|_{L^2(\Omega)}, \quad u_{\Omega} := \frac{1}{|\Omega|} \int_{\Omega} u \, dx. \quad (2.33)$$

This inequality says that on a connected domain, the gradient controls a function up to an additive constant.

2.6.2 Proofs in one dimension

Proposition 2.19. *Let $I = (a, b)$ and let $u \in H_0^1(I)$. Then*

$$\|u\|_{L^2(I)} \leq (b - a) \|u'\|_{L^2(I)}. \quad (2.34)$$

Proof. Since $u(a) = 0$, the fundamental theorem of calculus gives

$$u(x) = \int_a^x u'(t) \, dt. \quad (2.35)$$

By the Cauchy–Schwarz inequality,

$$|u(x)| \leq \sqrt{x - a} \|u'\|_{L^2(I)} \leq \sqrt{b - a} \|u'\|_{L^2(I)}. \quad (2.36)$$

We square and integrate in x over I to obtain

$$\|u\|_{L^2(I)}^2 \leq (b - a)^2 \|u'\|_{L^2(I)}^2. \quad (2.37)$$

□

Proposition 2.20. *Let $I = (a, b)$ and let $u \in H^1(I)$. Denote by*

$$u_I = \frac{1}{b - a} \int_a^b u(x) \, dx \quad (2.38)$$

the average of u on I . Then

$$\|u - u_I\|_{L^2(I)} \leq (b - a) \|u'\|_{L^2(I)}. \quad (2.39)$$

Proof. The continuous function $u - u_I$ has zero average on I , so it must vanish at some point $\xi \in [a, b]$. Therefore,

$$u(x) - u_I = \int_{\xi}^x u'(t) \, dt. \quad (2.40)$$

The same estimate as above then gives the result. □

2.6.3 Local forms used in interpolation estimates

For finite elements, we often need inequalities on a single mesh element.

Corollary 2.21. *Let K be an interval, triangle, or tetrahedron of diameter h_K . Then*

$$\|v - v_K\|_{L^2(K)} \leq Ch_K |v|_{H^1(K)} \quad \text{for all } v \in H^1(K), \quad (2.41)$$

where

$$v_K = \frac{1}{|K|} \int_K v \, dx. \quad (2.42)$$

The constant C depends only on the shape of a reference element; on an interval we may take $C = 1$.

Remark 2.22. This local estimate is one of the basic tools in interpolation theory for finite elements. In one dimension it is exactly the estimate used to prove the H^1 error bound for nodal interpolation.

2.6.4 The zero-mean space and Neumann problems

In this subsection, assume that Ω is connected. The pure Neumann Poisson problem is not coercive on all of $H^1(\Omega)$, because constants have zero gradient. The natural way to remove this indeterminacy is to work in the zero-mean subspace

$$H_\star^1(\Omega) = \{u \in H^1(\Omega) : u_\Omega = 0\}. \quad (2.43)$$

By the Poincaré–Wirtinger inequality, the seminorm $|u|_{H^1(\Omega)} = \|\nabla u\|_{L^2(\Omega)}$ is a norm on $H_\star^1(\Omega)$.

Corollary 2.23. *The formula*

$$(u, v)_\star := \int_\Omega \nabla u \cdot \nabla v \, dx \quad (2.44)$$

defines a scalar product on $H_\star^1(\Omega)$, and the induced norm is equivalent to the full H^1 norm.

Proof. If $(u, u)_\star = 0$, then $\nabla u = 0$ almost everywhere, so u is almost everywhere constant on the connected domain Ω . Since $u_\Omega = 0$, this constant is zero. Thus $(\cdot, \cdot)_\star$ is a scalar product. Norm equivalence follows from Poincaré–Wirtinger. $\square$

For $f \in L^2(\Omega)$, this is the functional framework for the weak formulation of the pure Neumann problem

$$-\Delta u = f \quad \text{in } \Omega, \quad \partial_n u = 0 \quad \text{on } \partial\Omega. \quad (2.45)$$

For a smooth solution, integrating over Ω and using Green’s formula gives the compatibility condition

$$\int_\Omega f \, dx = 0. \quad (2.46)$$

The weak formulation is then: find $u \in H_\star^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \quad \text{for all } v \in H_\star^1(\Omega). \quad (2.47)$$

The right-hand side is continuous on $H_\star^1(\Omega)$ because Poincaré–Wirtinger controls $\|v\|_{L^2(\Omega)}$ by $\|\nabla v\|_{L^2(\Omega)}$.

2.7 The dual space $H^{-1}(\Omega)$

For many elliptic problems, the right-hand side does not need to belong to $L^2(\Omega)$. The natural dual space is

$$H^{-1}(\Omega) := (H_0^1(\Omega))'. \quad (2.48)$$

If $F \in H^{-1}(\Omega)$ and $v \in H_0^1(\Omega)$, the duality pairing is denoted by

$$\langle F, v \rangle_{H^{-1}, H_0^1}. \quad (2.49)$$

The corresponding norm is

$$\|F\|_{H^{-1}(\Omega)} = \sup_{v \in H_0^1(\Omega), v \neq 0} \frac{|\langle F, v \rangle|}{|v|_{H^1(\Omega)}}. \quad (2.50)$$

Proposition 2.24. *Every function $f \in L^2(\Omega)$ defines an element of $H^{-1}(\Omega)$ through*

$$\langle f, v \rangle := \int_{\Omega} f v \, dx. \quad (2.51)$$

Moreover,

$$\|f\|_{H^{-1}(\Omega)} \leq C_P \|f\|_{L^2(\Omega)}. \quad (2.52)$$

Proof. For $v \in H_0^1(\Omega)$, the Cauchy–Schwarz and Poincaré inequalities give

$$\left| \int_{\Omega} f v \, dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C_P \|f\|_{L^2(\Omega)} |v|_{H^1(\Omega)}. \quad (2.53)$$

This is exactly the required continuity estimate. $\square$

Remark 2.25. Thus the weak formulation of

$$-\Delta u = F \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (2.54)$$

may be written for every $F \in H^{-1}(\Omega)$ as

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \langle F, v \rangle \quad \text{for all } v \in H_0^1(\Omega). \quad (2.55)$$

The right-hand side no longer needs to be an integral against an L^2 function.

2.8 Green formulas and weak formulations

The passage from strong to weak formulations relies on integration by parts in Sobolev spaces.

Theorem 2.26. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $u, v \in H^1(\Omega)$. Then, for each $i = 1, \dots, d$,*

$$\int_{\Omega} \partial_i u \, v \, dx = - \int_{\Omega} u \, \partial_i v \, dx + \int_{\partial\Omega} \gamma(u) \gamma(v) n_i \, ds, \quad (2.56)$$

where $n = (n_1, \dots, n_d)$ is the outward unit normal.

Remark 2.27. The proof is by density: we first write the formula for smooth functions and then pass to the limit, using the continuity of the trace operator.

A more familiar form is obtained when u is more regular.

Corollary 2.28. *If $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$, then*

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = - \int_{\Omega} (\Delta u) v \, dx + \int_{\partial\Omega} \partial_n u \, \gamma(v) \, ds, \quad (2.57)$$

where $\partial_n u = \nabla u \cdot n$ is the normal derivative.

Remark 2.29. If $v \in H_0^1(\Omega)$, then $\gamma(v) = 0$, so the boundary term disappears. This is the key point in the weak formulation of homogeneous Dirichlet problems.

Consider, for instance,

$$-\Delta u + cu = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (2.58)$$

where $c \in L^\infty(\Omega)$ with $c \geq 0$ almost everywhere and $f \in L^2(\Omega)$. The weak formulation reads: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in H_0^1(\Omega), \quad (2.59)$$

with

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx + \int_{\Omega} c w v \, dx, \quad (2.60)$$

$$\ell(v) = \int_{\Omega} f v \, dx. \quad (2.61)$$

By the Cauchy–Schwarz and Poincaré inequalities,

$$|a(w, v)| \leq \|\nabla w\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + \|c\|_{L^\infty(\Omega)} \|w\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \quad (2.62)$$

$$\leq \left(1 + \|c\|_{L^\infty(\Omega)} C_P^2\right) |w|_{H^1(\Omega)} |v|_{H^1(\Omega)}, \quad (2.63)$$

and

$$a(v, v) = |v|_{H^1(\Omega)}^2 + \int_{\Omega} c |v|^2 \, dx \geq |v|_{H^1(\Omega)}^2. \quad (2.64)$$

Thus the bilinear form is continuous and coercive on $H_0^1(\Omega)$ for the energy norm

$$\|v\|_V := |v|_{H^1(\Omega)}. \quad (2.65)$$

2.9 A few embeddings to keep in mind

Theorem 2.30. *Let $I = (a, b)$ be a bounded interval. Then*

$$H^1(I) \hookrightarrow C^0([a, b]). \quad (2.66)$$

If moreover $u \in H^2(I)$, then

$$H^2(I) \hookrightarrow C^1([a, b]). \quad (2.67)$$

Theorem 2.31. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain.*

(i) *If $d = 2$ or 3 , then $H^1(\Omega)$ is not in general contained in $C^0(\overline{\Omega})$. Point values are therefore not defined for arbitrary H^1 functions.*

(ii) *If $d \leq 3$, then $H^2(\Omega) \hookrightarrow C^0(\overline{\Omega})$.*

2.10 Exercises

In the following exercises, $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain. Applications to the Dirichlet and Neumann problems are collected in Section 3.6 of the next chapter.

Exercise 2.1 (A basic weak derivative). Let $I = (-1, 1)$ and let $u(x) = |x|$.

1. Show that u in $H^1(I)$ and that its weak derivative is

$$u'(x) = \text{sign}(x) \quad \text{a.e. in } I.$$

2. Show that $u \notin H^2(I)$.

Exercise 2.2 (Poincaré inequality on an interval). Let $I = (a, b)$, and let u in $H^1(I)$ satisfy $u(a) = 0$.

1. Show that, for every x in $[a, b]$,

$$u(x) = \int_a^x u'(t) dt.$$

2. Deduce that

$$|u(x)| \leq \sqrt{b-a} \|u'\|_{L^2(I)} \quad \text{for all } x \in [a, b].$$

3. Deduce the inequalities

$$\|u\|_{L^\infty(I)} \leq \sqrt{b-a} \|u'\|_{L^2(I)}, \quad \|u\|_{L^2(I)} \leq (b-a) \|u'\|_{L^2(I)}.$$

4. Conclude that on $H_0^1(I)$, the seminorm $|u|_{H^1(I)} = \|u'\|_{L^2(I)}$ is a norm.

Exercise 2.3 (Poincaré–Wirtinger and its local form). Let $I = (a, b)$ and let u in $H^1(I)$. Denote by

$$u_I = \frac{1}{b-a} \int_a^b u(x) dx$$

the average of u on I .

1. Show that there exists ξ in $[a, b]$ such that $u(\xi) = u_I$.
2. Deduce that

$$\|u - u_I\|_{L^2(I)} \leq (b-a) \|u'\|_{L^2(I)}.$$

3. Let $K = (\alpha, \beta)$ be any interval, let $h_K = \beta - \alpha$, and let v in $H^1(K)$. Show that

$$\|v - v_K\|_{L^2(K)} \leq h_K |v|_{H^1(K)}.$$

Exercise 2.4 (The energy scalar product). Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, and set $V = H_0^1(\Omega)$.

1. Show that

$$(u, v)_V := \int_{\Omega} \nabla u \cdot \nabla v dx$$

defines a scalar product on V .

2. Show that the associated norm

$$\|u\|_V := \|\nabla u\|_{L^2(\Omega)}$$

is equivalent to the usual $H^1(\Omega)$ norm on V .

3. Let f in $L^2(\Omega)$. Show that

$$\ell(v) := \int_{\Omega} f v \, dx$$

defines a continuous linear form on V when V is equipped with the norm $\|\cdot\|_V$.

Exercise 2.5 (From L^2 and divergence form to H^{-1}). Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain.

1. Let f in $L^2(\Omega)$. Show that

$$\langle F, v \rangle := \int_{\Omega} f v \, dx \quad \text{for } v \in H_0^1(\Omega)$$

defines an element F in $H^{-1}(\Omega)$, and prove

$$\|F\|_{H^{-1}(\Omega)} \leq C_P \|f\|_{L^2(\Omega)},$$

where C_P is a Poincaré constant for Ω .

2. Let g in $L^2(\Omega)^d$. Show that

$$\langle G, v \rangle := - \int_{\Omega} g \cdot \nabla v \, dx \quad \text{for } v \in H_0^1(\Omega)$$

defines an element G in $H^{-1}(\Omega)$, and prove

$$\|G\|_{H^{-1}(\Omega)} \leq \|g\|_{L^2(\Omega)}.$$

3. Interpret G as the distribution $\operatorname{div} g$.

Exercise 2.6 (Point values are not continuous on $H^1(B)$ in two dimensions). Let $B = B(0, 1) \subset \mathbb{R}^2$. For $n \geq 2$, define the radial function

$$u_n(x) = \begin{cases} 1, & |x| \leq n^{-1}, \\ \frac{\log(1/|x|)}{\log n}, & n^{-1} < |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Show that u_n in $H_0^1(B) \cap C^0(\overline{B})$, that $u_n(0) = 1$ for all n , and that

$$|u_n|_{H^1(B)} \rightarrow 0 \quad \text{and} \quad \|u_n\|_{L^2(B)} \rightarrow 0$$

as $n \rightarrow \infty$. Conclude that point evaluation at 0 cannot be a continuous linear form on $H^1(B)$.

Chapter 3

Lax–Milgram in action for a few elliptic problems

This chapter applies the Lax–Milgram theorem proved in Chapter 1 to a few standard elliptic boundary value problems. The functional framework of Chapter 2 is used repeatedly: the Poincaré inequality on $H_0^1(\Omega)$, the Poincaré–Wirtinger inequality on $H_\star^1(\Omega)$, and the trace theorem on bounded Lipschitz domains.

3.1 Homogeneous Dirichlet problems

We work on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$.

3.1.1 The pure Poisson equation

We consider

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (3.1)$$

with $f \in L^2(\Omega)$. The weak formulation is posed on

$$V = H_0^1(\Omega), \quad (3.2)$$

with

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx, \quad (3.3)$$

$$\ell(v) = \int_{\Omega} f v \, dx. \quad (3.4)$$

Proposition 3.1. *The weak formulation*

$$a(u, v) = \ell(v) \quad \text{for all } v \in V \quad (3.5)$$

admits a unique solution $u \in H_0^1(\Omega)$.

Proof. The bilinear form is continuous by the Cauchy–Schwarz inequality:

$$|a(w, v)| \leq \|\nabla w\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}. \quad (3.6)$$

On $H_0^1(\Omega)$, the Poincaré inequality of Chapter 2 shows that $\|\nabla \cdot\|_{L^2(\Omega)}$ is a norm equivalent to the full H^1 norm. Hence

$$a(v, v) = \|\nabla v\|_{L^2(\Omega)}^2 \geq \alpha \|v\|_{H^1(\Omega)}^2 \quad (3.7)$$

for some $\alpha > 0$. Finally,

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C_P \|f\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}. \quad (3.8)$$

We conclude by Theorem 1.2. $\square$

Remark 3.2. This is the basic Dirichlet model. The coercivity does not come from the differential operator alone; it comes from the choice of the space $H_0^1(\Omega)$.

3.1.2 Adding a nonnegative lower-order term

We now consider

$$-\Delta u + cu = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (3.9)$$

where $c \in L^\infty(\Omega)$ and $c(x) \geq 0$ almost everywhere. The weak formulation is again posed on $V = H_0^1(\Omega)$, with

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx + \int_{\Omega} c w v \, dx. \quad (3.10)$$

Proposition 3.3. *The associated weak problem has a unique solution in $H_0^1(\Omega)$.*

Proof. Continuity is immediate:

$$|a(w, v)| \leq \|\nabla w\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + \|c\|_{L^\infty(\Omega)} \|w\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}. \quad (3.11)$$

The Poincaré inequality absorbs the L^2 terms. For coercivity we simply write

$$a(v, v) = \|\nabla v\|_{L^2(\Omega)}^2 + \int_{\Omega} c |v|^2 \, dx \geq \|\nabla v\|_{L^2(\Omega)}^2. \quad (3.12)$$

Thus Theorem 1.2 applies exactly as before. $\square$

Remark 3.4. The case $c \equiv 1$ is especially useful pedagogically. The lower-order term does not change the Dirichlet space, but it completely changes the Neumann and periodic situations.

3.1.3 A negative lower-order term

The sign of the lower-order term matters. Consider

$$-\Delta u - \lambda u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (3.13)$$

with $\lambda \geq 0$. The natural bilinear form is

$$a_\lambda(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx - \lambda \int_{\Omega} w v \, dx. \quad (3.14)$$

Proposition 3.5. *Assume that $\lambda < C_P^{-2}$, where C_P is a Poincaré constant on $H_0^1(\Omega)$:*

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \quad \text{for all } v \in H_0^1(\Omega). \quad (3.15)$$

Then a_λ is coercive on $H_0^1(\Omega)$. In particular, the weak problem has a unique solution.

Proof. For every $v \in H_0^1(\Omega)$,

$$a_\lambda(v, v) = \|\nabla v\|_{L^2(\Omega)}^2 - \lambda \|v\|_{L^2(\Omega)}^2 \geq (1 - \lambda C_P^2) \|\nabla v\|_{L^2(\Omega)}^2. \quad (3.16)$$

The factor in parentheses is positive by assumption, so coercivity follows. $\square$

Remark 3.6. This is the simplest place where the Lax–Milgram framework may fail. If λ is too large, the form is no longer coercive. With the optimal Poincaré constant, the threshold is the first Dirichlet eigenvalue $\lambda_1(\Omega)$. On an interval $I = (a, b)$, we have $\lambda_1(I) = \pi^2/(b - a)^2$.

3.2 Pure Neumann problems

In this section we assume in addition that Ω is connected.

3.2.1 The problem with a coercive lower-order term

We first treat

$$-\Delta u + u = f \quad \text{in } \Omega, \quad \partial_n u = g \quad \text{on } \partial\Omega, \quad (3.17)$$

where $f \in L^2(\Omega)$ and $g \in L^2(\partial\Omega)$. The weak formulation is posed on the full space

$$V = H^1(\Omega), \quad (3.18)$$

with

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx + \int_{\Omega} wv \, dx, \quad (3.19)$$

$$\ell(v) = \int_{\Omega} f v \, dx + \int_{\partial\Omega} g v \, ds. \quad (3.20)$$

Proposition 3.7. *The weak problem has a unique solution $u \in H^1(\Omega)$.*

Proof. The bilinear form is continuous on $H^1(\Omega)$. For the linear form, the trace theorem of Chapter 2 gives

$$\|v\|_{L^2(\partial\Omega)} \leq C_{\text{tr}} \|v\|_{H^1(\Omega)}, \quad (3.21)$$

so

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \|v\|_{L^2(\partial\Omega)} \leq C \|v\|_{H^1(\Omega)}. \quad (3.22)$$

Finally,

$$a(v, v) = \|\nabla v\|_{L^2(\Omega)}^2 + \|v\|_{L^2(\Omega)}^2 = \|v\|_{H^1(\Omega)}^2, \quad (3.23)$$

so a is coercive on the whole space $H^1(\Omega)$. Theorem 1.2 therefore applies directly. $\square$

Remark 3.8. This example is worth isolating. The Neumann boundary condition does not create any difficulty here because the term $+u$ removes the constant kernel.

3.2.2 The pure Neumann Poisson equation

We now remove the lower-order term and consider

$$-\Delta u = f \quad \text{in } \Omega, \quad \partial_n u = g \quad \text{on } \partial\Omega, \quad (3.24)$$

with $f \in L^2(\Omega)$ and $g \in L^2(\partial\Omega)$. The formal weak formulation is

$$\int_{\Omega} \nabla u \cdot \nabla \varphi \, dx = \int_{\Omega} f \varphi \, dx + \int_{\partial\Omega} g \varphi \, ds \quad \text{for all } \varphi \in H^1(\Omega). \quad (3.25)$$

However, on $H^1(\Omega)$ the form

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx \quad (3.26)$$

is not coercive, because every constant belongs to its kernel.

We first read the compatibility condition in the original weak formulation, where the constant function 1 is an admissible test function. If u is a weak solution in $H^1(\Omega)$, then taking $\varphi = 1$ gives

$$\int_{\Omega} f \, dx + \int_{\partial\Omega} g \, ds = 0. \quad (3.27)$$

To recover coercivity we introduce the mean-zero space

$$H_\star^1(\Omega) = \left\{ v \in H^1(\Omega) : \int_\Omega v \, dx = 0 \right\}. \quad (3.28)$$

On this space, the Poincaré–Wirtinger inequality of Chapter 2 yields

$$\|v\|_{L^2(\Omega)} \leq C_{PW} \|\nabla v\|_{L^2(\Omega)}. \quad (3.29)$$

Theorem 3.9. *The pure Neumann problem admits a weak solution if and only if*

$$\int_\Omega f \, dx + \int_{\partial\Omega} g \, ds = 0. \quad (3.30)$$

When this condition holds, there exists a unique solution $u \in H_\star^1(\Omega)$, and every weak solution in $H^1(\Omega)$ differs from u by an additive constant.

Proof. Necessity of the compatibility condition was established by testing with $\varphi = 1$.

Assume now that the compatibility condition holds. We work on $V = H_\star^1(\Omega)$ with the norm

$$\|v\|_V := \|\nabla v\|_{L^2(\Omega)}. \quad (3.31)$$

The bilinear form a is continuous on V , and

$$a(v, v) = \|\nabla v\|_{L^2(\Omega)}^2 = \|v\|_V^2 \quad \text{for all } v \in V, \quad (3.32)$$

so a is coercive on V . For the linear form

$$\ell(v) = \int_\Omega f v \, dx + \int_{\partial\Omega} g v \, ds, \quad (3.33)$$

the Poincaré–Wirtinger inequality and the trace theorem give

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \|v\|_{L^2(\partial\Omega)} \leq C \|\nabla v\|_{L^2(\Omega)} = C \|v\|_V. \quad (3.34)$$

Hence Theorem 1.2 yields a unique $u \in V$ such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in H_\star^1(\Omega). \quad (3.35)$$

It remains to check that this solution satisfies the original weak formulation on the whole space $H^1(\Omega)$. Let $\varphi \in H^1(\Omega)$ and define the average

$$\bar{\varphi} := \frac{1}{|\Omega|} \int_\Omega \varphi \, dx. \quad (3.36)$$

Then $\varphi - \bar{\varphi} \in H_\star^1(\Omega)$ and

$$a(u, \varphi) = a(u, \varphi - \bar{\varphi}) = \ell(\varphi - \bar{\varphi}) = \ell(\varphi) - \bar{\varphi} \ell(1). \quad (3.37)$$

Since the compatibility condition is exactly $\ell(1) = 0$, we obtain

$$a(u, \varphi) = \ell(\varphi) \quad \text{for all } \varphi \in H^1(\Omega). \quad (3.38)$$

Finally, if u_1 and u_2 are two weak solutions in $H^1(\Omega)$, then

$$\int_\Omega \nabla(u_1 - u_2) \cdot \nabla \varphi \, dx = 0 \quad \text{for all } \varphi \in H^1(\Omega). \quad (3.39)$$

Taking $\varphi = u_1 - u_2$ gives $\nabla(u_1 - u_2) = 0$ almost everywhere. Since Ω is connected, $u_1 - u_2$ is constant. $\square$

Remark 3.10. The compatibility condition is not needed to solve the reduced problem on $H_\star^1(\Omega)$. It is needed exactly when we pass from test functions in $H_\star^1(\Omega)$ to arbitrary test functions in $H^1(\Omega)$ through the decomposition

$$\varphi = (\varphi - \bar{\varphi}) + \bar{\varphi}.$$

Remark 3.11. In one dimension, on $I = (a, b)$, the homogeneous Neumann problem

$$-u'' + u = f, \quad u'(a) = u'(b) = 0,$$

is treated in $H^1(I)$ in the same way. Without the term $+u$, constants again form the kernel and the zero-mean normalization becomes necessary.

3.3 Mixed Dirichlet–Neumann problems

In this section, assume that Ω is a bounded connected Lipschitz domain. Let the boundary be decomposed as

$$\partial\Omega = \Gamma_D \cup \Gamma_N \cup \Sigma, \quad \Gamma_D \cap \Gamma_N = \emptyset, \quad (3.40)$$

where Γ_D and Γ_N are relatively open, the interface $\Sigma = \partial\Omega \setminus (\Gamma_D \cup \Gamma_N)$ has zero surface measure, and Γ_D has positive surface measure. This includes the usual mixed boundary conditions on adjacent boundary segments: their meeting points belong to Σ . We consider

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \Gamma_D, \quad \partial_n u = g \quad \text{on } \Gamma_N, \quad (3.41)$$

with $f \in L^2(\Omega)$ and $g \in L^2(\Gamma_N)$.

The natural space is

$$V_{\Gamma_D} = \{v \in H^1(\Omega) : \gamma v = 0 \text{ on } \Gamma_D\}. \quad (3.42)$$

The assumption that Γ_D has positive surface measure is used through the following Poincaré inequality with partial homogeneous trace: there exists a constant $C_{\Gamma_D} > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C_{\Gamma_D} \|\nabla v\|_{L^2(\Omega)} \quad \text{for all } v \in V_{\Gamma_D}. \quad (3.43)$$

Consequently $\|\nabla v\|_{L^2(\Omega)}$ is a norm on V_{Γ_D} , equivalent to the usual $H^1(\Omega)$ norm. The positivity of the measure of Γ_D is essential. If the Dirichlet part is empty, or if it is invisible for the trace in the sense of surface measure, nonzero constants have zero gradient and are not excluded by the boundary condition, so (3.43) cannot hold.

A standard proof of (3.43) is by contradiction. If the inequality failed, there would exist v_n in V_{Γ_D} such that $\|v_n\|_{L^2(\Omega)} = 1$ and $\|\nabla v_n\|_{L^2(\Omega)} \rightarrow 0$. The sequence is bounded in $H^1(\Omega)$; by the Rellich compactness theorem, up to a subsequence, it converges strongly in $L^2(\Omega)$ to some v . Passing to the limit in weak derivatives gives $\nabla v = 0$. Since Ω is connected, v is constant. In fact $v_n \rightarrow v$ in $H^1(\Omega)$, since both the functions and their gradients converge in L^2 . The continuity of the trace operator therefore gives $\gamma v = 0$ on Γ_D . Because Γ_D has positive surface measure, this constant must be zero, contradicting $\|v\|_{L^2(\Omega)} = 1$. References for this form of the Poincaré inequality include [5, Chapter 9] and [2, Chapter 3].

The weak formulation is

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx + \int_{\Gamma_N} g v \, ds \quad \text{for all } v \in V_{\Gamma_D}. \quad (3.44)$$

Proposition 3.12. *Assume that Γ_D has positive measure. Then the mixed weak problem has a unique solution $u \in V_{\Gamma_D}$.*

Proof. We use the Poincaré inequality on V_{Γ_D} : there exists $C_{\Gamma_D} > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C_{\Gamma_D} \|\nabla v\|_{L^2(\Omega)} \quad \text{for all } v \in V_{\Gamma_D}. \quad (3.45)$$

Thus the bilinear form

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx \quad (3.46)$$

is continuous and coercive on V_{Γ_D} . The linear form

$$\ell(v) = \int_{\Omega} f v \, dx + \int_{\Gamma_N} g v \, ds \quad (3.47)$$

is continuous by the trace theorem on Γ_N together with the Poincaré inequality on V_{Γ_D} . Theorem 1.2 then yields existence and uniqueness. $\square$

Remark 3.13. This is the basic model for the distinction between essential and natural boundary conditions. The Dirichlet part is built into the space V_{Γ_D} , while the Neumann part appears in the right-hand side after integration by parts.

Remark 3.14. If nonzero Dirichlet data are prescribed on Γ_D , we first choose a lifting G with the correct trace on Γ_D and then solve for $U = u - G$ in the homogeneous space V_{Γ_D} .

3.4 Periodic problems

We treat the periodic case in one dimension, where the functional setting is especially transparent. Let $I = (0, 1)$ and define

$$H_{\text{per}}^1(I) = \{v \in H^1(I) : v(0) = v(1)\}. \quad (3.48)$$

3.4.1 Periodic reaction–diffusion

Consider

$$-u'' + u = f \quad \text{in } I, \quad u(0) = u(1), \quad u'(0) = u'(1), \quad (3.49)$$

with $f \in L^2(I)$. The weak formulation is: find $u \in H_{\text{per}}^1(I)$ such that

$$\int_0^1 u' v' \, dx + \int_0^1 u v \, dx = \int_0^1 f v \, dx \quad \text{for all } v \in H_{\text{per}}^1(I). \quad (3.50)$$

Proposition 3.15. *The periodic reaction–diffusion problem has a unique weak solution in $H_{\text{per}}^1(I)$.*

Proof. On $H_{\text{per}}^1(I)$ we use the full H^1 norm. The bilinear form is continuous and satisfies

$$a(v, v) = \|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2 = \|v\|_{H^1(I)}^2. \quad (3.51)$$

Thus it is coercive, and Theorem 1.2 applies. $\square$

3.4.2 Pure periodic Poisson

We now consider

$$-u'' = f \quad \text{in } I, \quad u(0) = u(1), \quad u'(0) = u'(1). \quad (3.52)$$

Assume again that $f \in L^2(I)$. Here the bilinear form

$$a(w, v) = \int_0^1 w' v' \, dx \quad (3.53)$$

is not coercive on $H_{\text{per}}^1(I)$ because constants belong to its kernel. We therefore introduce

$$H_{\text{per},\star}^1(I) = \left\{ v \in H_{\text{per}}^1(I) : \int_0^1 v(x) dx = 0 \right\}. \quad (3.54)$$

Proposition 3.16. *Assume that*

$$\int_0^1 f(x) dx = 0. \quad (3.55)$$

Then there exists a unique function $u \in H_{\text{per},\star}^1(I)$ such that

$$\int_0^1 u'v' dx = \int_0^1 f v dx \quad \text{for all } v \in H_{\text{per},\star}^1(I). \quad (3.56)$$

Any periodic weak solution is unique up to an additive constant.

Proof. On $H_{\text{per},\star}^1(I)$ the periodic Poincaré–Wirtinger inequality gives

$$\|v\|_{L^2(I)} \leq C \|v'\|_{L^2(I)}. \quad (3.57)$$

Hence $\|v'\|_{L^2(I)}$ is a norm on this space, a is coercive, and the linear form is continuous. Theorem 1.2 yields existence and uniqueness in $H_{\text{per},\star}^1(I)$.

To recover the original periodic problem, take any $v \in H_{\text{per}}^1(I)$ and write $v = v_\star + \bar{v}$, where $\bar{v} = \int_0^1 v dx$ and $v_\star \in H_{\text{per},\star}^1(I)$. Compatibility gives

$$\int_0^1 u'v' dx = \int_0^1 u'v_\star' dx = \int_0^1 f v_\star dx = \int_0^1 f v dx. \quad (3.58)$$

Thus the identity holds for all periodic H^1 test functions. Compactly supported tests give $-u'' = f$ in distributions, so $u \in H^2(I)$. Integration by parts then gives

$$0 = u'(1)v(1) - u'(0)v(0) = (u'(1) - u'(0))v(0), \quad (3.59)$$

and the common endpoint value of a periodic test function is arbitrary. Hence $u'(0) = u'(1)$; the other periodic condition already follows from $u \in H_{\text{per}}^1(I)$. Testing the difference of two weak solutions against itself shows that the difference is constant. $\square$

Remark 3.17. The periodic case is the exact analogue of the pure Neumann Poisson problem. The kernel is again the space of constants, and the natural normalization is again the zero-mean condition.

3.5 A regularity complement for the Dirichlet model

The variational arguments above provide the weak solution in $H_0^1(\Omega)$. For finite element error estimates, we often need one more derivative.

Proposition 3.18. *Consider the problem*

$$-\Delta u + cu = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (3.60)$$

with f in $L^2(\Omega)$ and c in $L^\infty(\Omega)$ satisfying $c(x) \geq 0$ almost everywhere. Assume that either $\partial\Omega$ is sufficiently smooth (for instance of class $C^{1,1}$), or, in dimension two, that Ω is a convex polygon. Then the weak solution u in $H_0^1(\Omega)$ actually belongs to $H^2(\Omega)$, and there exists a constant $C > 0$ such that

$$\|u\|_{H^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}. \quad (3.61)$$

The constant depends on Ω , on $\|c\|_{L^\infty(\Omega)}$, and on the coercivity constants of the Dirichlet problem.

Proof. We only explain the reduction to the Poisson case. Since u solves the weak problem, we have

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} cuv \, dx = \int_{\Omega} f v \, dx \quad \text{for all } v \in H_0^1(\Omega). \quad (3.62)$$

Taking $v = u$ and using $c \geq 0$ together with the Poincaré inequality, we obtain

$$\|u\|_{H_0^1(\Omega)} \leq C \|f\|_{L^2(\Omega)}. \quad (3.63)$$

In particular,

$$\|cu\|_{L^2(\Omega)} \leq \|c\|_{L^\infty(\Omega)} \|u\|_{L^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}. \quad (3.64)$$

Hence $g := f - cu$ belongs to $L^2(\Omega)$ and satisfies $\|g\|_{L^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}$. The equation rewrites as

$$-\Delta u = g \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (3.65)$$

Applying the standard elliptic regularity result for the homogeneous Dirichlet Poisson problem yields $u \in H^2(\Omega)$ together with the stated estimate. $\square$

Remark 3.19. For the Poisson case $c \equiv 0$, the H^2 estimate quoted in the proof is stated in [4, Lemma 4.6] under the assumption that $\partial\Omega$ is sufficiently regular or that Ω is a convex polygon. For smooth domains, compare also [2, Proposition 4.18].

Why the extra derivative depends on geometry

In one dimension the Poisson regularity step is elementary: on $I = (a, b)$, the distributional identity $u'' = -g \in L^2(I)$, together with $u \in H^1(I)$, is precisely what is needed for $u \in H^2(I)$. Moreover,

$$\|u\|_{H^2(I)} \leq C(\|u\|_{H^1(I)} + \|g\|_{L^2(I)}). \quad (3.66)$$

The energy estimate and $g = f - cu$ therefore prove Proposition 3.18 completely in dimension one.

In higher dimensions, $\Delta u \in L^2$ alone does not control each second derivative up to an arbitrary boundary. On a $C^{1,1}$ domain, the standard proof first uses difference quotients to estimate second derivatives in the interior. Near the boundary it flattens the boundary and estimates tangential derivatives, using the homogeneous Dirichlet trace; the equation then controls the remaining normal derivative. A finite covering and localization give

$$\|u\|_{H^2(\Omega)} \leq C(\|g\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)}). \quad (3.67)$$

This is an outline of the Poisson regularity argument, not a replacement for the boundary estimates in the cited theorem. The reduction $g = f - cu$ above requires no derivatives of c .

For a polygon, the role of convexity can be seen near a corner with interior angle ω . In polar coordinates, the harmonic function

$$s(r, \theta) = r^\lambda \sin(\lambda\theta), \quad \lambda = \frac{\pi}{\omega}, \quad 0 < \theta < \omega, \quad (3.68)$$

vanishes on both sides of the corner. Its second derivatives behave like $r^{\lambda-2}$, so their squared integral near the corner is controlled by

$$\int_0^\varepsilon r^{2\lambda-3} \, dr, \quad (3.69)$$

which is finite when $\lambda > 1$. At every genuine corner of a convex polygon, $\omega < \pi$ and hence $\lambda > 1$. This explains the geometric hypothesis; a full proof additionally decomposes the solution into regular and corner parts. At a reentrant corner, $\omega > \pi$ and $\lambda < 1$: let $w = \chi(r)s$, where χ is a smooth radial cutoff equal to one near the vertex and supported in a sufficiently small corner neighborhood. Then $w \in H_0^1(\Omega)$ and $-\Delta w \in L^2(\Omega)$, but $w \notin H^2(\Omega)$. Thus smooth interior data alone do not imply the estimate on a nonconvex polygon.

Remark 3.20. This regularity improvement is specific to the homogeneous Dirichlet setting under suitable geometric assumptions. For mixed boundary conditions, global H^2 regularity is more delicate; singularities may appear near corners of $\partial\Omega$ or near interface points where the boundary condition changes type, so the full H^2 estimate may fail.

3.6 Exercises

The following exercises use the Sobolev tools of Chapter 2 to establish well-posedness and to distinguish the roles of Dirichlet and Neumann boundary conditions. The first treats a coercive problem directly; the second separates removal of the constant kernel from compatibility of the data.

Exercise 3.1 (The Dirichlet problem: weak formulation and well-posedness). Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, let $c \in L^\infty(\Omega)$ with $c \geq 0$ almost everywhere, and let $f \in L^2(\Omega)$. Set $V = H_0^1(\Omega)$ with $\|v\|_V = \|\nabla v\|_{L^2(\Omega)}$, and let C_P be a Poincaré constant on V .

1. Define

$$a(w, v) = \int_{\Omega} \nabla w \cdot \nabla v \, dx + \int_{\Omega} c w v \, dx, \quad \ell(v) = \int_{\Omega} f v \, dx.$$

Prove continuity and coercivity of a and continuity of ℓ . Apply Lax–Milgram to obtain a unique $u \in V$ satisfying $a(u, v) = \ell(v)$ for every $v \in V$.

2. Prove the energy estimate

$$\|\nabla u\|_{L^2(\Omega)} \leq C_P \|f\|_{L^2(\Omega)}.$$

3. Suppose that $w \in H^2(\Omega) \cap H_0^1(\Omega)$ satisfies $-\Delta w + cw = f$ almost everywhere. Use Green's formula to show that w satisfies the weak formulation, and conclude that $w = u$.
4. In one dimension, with $\Omega = (a, b)$, prove conversely that the weak solution belongs to $H^2(a, b)$ and satisfies $-u'' + cu = f$ almost everywhere, together with $u(a) = u(b) = 0$.

Exercise 3.2 (The Neumann problem: compatibility and well-posedness). Let $\Omega \subset \mathbb{R}^d$ be a bounded connected Lipschitz domain and set

$$V = H_\star^1(\Omega) = \{v \in H^1(\Omega) : v_\Omega = 0\}, \quad v_\Omega = \frac{1}{|\Omega|} \int_{\Omega} v \, dx.$$

Let C_{PW} be a Poincaré–Wirtinger constant on V .

1. Show that V is a closed subspace of $H^1(\Omega)$ and that $(w, v)_\star = \int_{\Omega} \nabla w \cdot \nabla v \, dx$ is a scalar product on V .
2. Prove that $\|v\|_\star = \|\nabla v\|_{L^2(\Omega)}$ is equivalent to the full H^1 norm on V . Conclude that V is complete for $\|\cdot\|_\star$.
3. For $f \in L^2(\Omega)$, prove continuity of $\ell(v) = \int_{\Omega} f v \, dx$ on V . Show that the full weak formulation

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \text{for every } v \in H^1(\Omega)$$

requires the compatibility condition $\int_{\Omega} f \, dx = 0$.

4. Assume this compatibility condition. Use Lax–Milgram to construct the unique solution $u \in V$ of the reduced problem with tests in V , and prove

$$\|\nabla u\|_{L^2(\Omega)} \leq C_{PW} \|f\|_{L^2(\Omega)}.$$

Then extend the identity to every $H^1(\Omega)$ test function. Explain which step needs compatibility and which steps remain valid without it.

5. Prove that all solutions of the full weak problem differ by additive constants and that the zero-mean normalization selects a unique solution.

Chapter 4

One-dimensional P_1 finite elements

4.1 The model problem

We fix an interval $I = (a, b)$ and we consider the boundary value problem

$$-u'' + cu = f \quad \text{in } I, \quad u(a) = u(b) = 0, \quad (4.1)$$

where f in $L^2(I)$ and c in $L^\infty(I)$ with $c(x) \geq 0$ almost everywhere.

Definition 4.1. We set

$$V = H_0^1(I). \quad (4.2)$$

On V , we use the $H_0^1(I)$ norm

$$\|v\|_{H_0^1(I)} := \|v'\|_{L^2(I)} = \left(\int_a^b |v'(x)|^2 dx \right)^{1/2}. \quad (4.3)$$

On $H_0^1(I)$, this is a norm by the Poincaré inequality.

We introduce the bilinear form and the linear form

$$a(w, v) = \int_a^b w'(x)v'(x) dx + \int_a^b c(x)w(x)v(x) dx, \quad (4.4)$$

$$\ell(v) = \int_a^b f(x)v(x) dx. \quad (4.5)$$

The weak formulation is: find u in V such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V. \quad (4.6)$$

Theorem 4.2. *The weak formulation of Definition 4.1 admits a unique solution u in V . Moreover, u belongs to $H^2(I)$.*

Proof. Let C_P be a constant such that

$$\|v\|_{L^2(I)} \leq C_P \|v\|_{H_0^1(I)} \quad \text{for all } v \in V. \quad (4.7)$$

For all w, v in V , we have

$$|a(w, v)| \leq \|w'\|_{L^2(I)} \|v'\|_{L^2(I)} + \|c\|_{L^\infty(I)} \|w\|_{L^2(I)} \|v\|_{L^2(I)} \quad (4.8)$$

$$\leq \left(1 + \|c\|_{L^\infty(I)} C_P^2\right) \|w\|_{H_0^1(I)} \|v\|_{H_0^1(I)}. \quad (4.9)$$

Thus a is continuous on $V \times V$. Moreover,

$$a(v, v) = \|v\|_{H_0^1(I)}^2 + \int_a^b c(x)|v(x)|^2 dx \geq \|v\|_{H_0^1(I)}^2, \quad (4.10)$$

so a is coercive on V with coercivity constant $\alpha = 1$. Finally,

$$|\ell(v)| \leq \|f\|_{L^2(I)} \|v\|_{L^2(I)} \leq C_P \|f\|_{L^2(I)} \|v\|_{H_0^1(I)}. \quad (4.11)$$

We conclude by the Lax–Milgram theorem.

Since u solves the weak problem, we have $-u'' = f - cu$ in the sense of distributions. As $f - cu$ in $L^2(I)$, it follows that u'' in $L^2(I)$, hence u in $H^2(I)$. $\square$

4.2 The mesh, the global finite element space, the hat basis, and the nodal interpolant

Definition 4.3. A mesh of $[a, b]$ is a partition

$$a = x_0 < x_1 < \cdots < x_N = b. \quad (4.12)$$

For $i = 1, \dots, N$, we set

$$K_i = (x_{i-1}, x_i), \quad h_i = x_i - x_{i-1}, \quad h = \max_{1 \leq i \leq N} h_i. \quad (4.13)$$

Definition 4.4. Let V be the energy space of a weak formulation. A finite-dimensional space V_h is called conforming for this weak formulation if

$$V_h \subset V. \quad (4.14)$$

The associated Galerkin method is then called conforming. Thus the discrete trial and test functions satisfy, by construction, all the conditions included in the space V , in particular the essential boundary conditions. The continuous bilinear and linear forms can therefore be restricted directly to V_h .

Remark 4.5. Conformity is relative to the weak formulation. For instance, in the standard homogeneous Dirichlet formulation of the one-dimensional model problem, the energy space is $V = H_0^1(I)$, so a conforming space must be a subspace of $H_0^1(I)$. For a mixed boundary value problem, the space V would instead encode only the essential part of the boundary condition.

Definition 4.6. The global conforming P_1 finite element space is

$$V_h = \left\{ v_h \in C^0([a, b]) : v_h|_{K_i} \text{ is affine for every } i, v_h(a) = v_h(b) = 0 \right\}. \quad (4.15)$$

This space is conforming in the sense of Definition 4.4, because every continuous piecewise affine function belongs to $H^1(I)$, and the endpoint conditions give $V_h \subset H_0^1(I) = V$.

Remark 4.7. The constrained space V_h describes the admissible Galerkin trial and test functions. An implementation may nevertheless store values at every mesh node, including the endpoints. Its full nodal space is

$$W_h = \{w_h \in C^0([a, b]) : w_h|_{K_i} \text{ is affine for every } i\}, \quad \dim W_h = N + 1, \quad (4.16)$$

and $V_h = \{w_h \in W_h : w_h(a) = w_h(b) = 0\}$ has dimension $N - 1$. Keeping endpoint entries in a vector does not make those entries free unknowns. Section 4.6.1 shows explicitly how full assembly followed by Dirichlet enforcement gives the same solution as the Galerkin method on V_h .

Definition 4.8. For $i = 1, \dots, N-1$, the global hat function φ_i is the unique function in V_h such that

$$\varphi_i(x_j) = \delta_{ij} \quad \text{for } j = 1, \dots, N-1. \quad (4.17)$$

Equivalently,

$$\varphi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h_i}, & x \in [x_{i-1}, x_i], \\ \frac{x_{i+1} - x}{h_{i+1}}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases} \quad (4.18)$$

These are global basis functions attached to the interior nodes $x_1, \dots, x_{N-1}$.

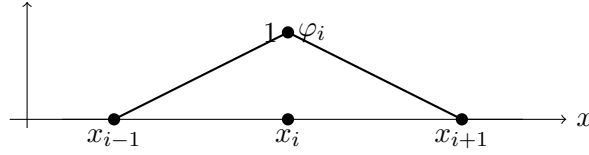


Figure 4.1: A typical hat function.

Proposition 4.9. The family $(\varphi_i)_{i=1, \dots, N-1}$ is a basis of V_h . In particular,

$$\dim V_h = N - 1. \quad (4.19)$$

Moreover, every function v_h in V_h satisfies the nodal representation formula

$$v_h(x) = \sum_{i=1}^{N-1} v_h(x_i) \varphi_i(x) \quad \text{for all } x \in [a, b]. \quad (4.20)$$

Proof. We first prove linear independence. Let $(\lambda_i)_{i=1, \dots, N-1}$ be scalars such that

$$\sum_{i=1}^{N-1} \lambda_i \varphi_i = 0. \quad (4.21)$$

We evaluate at x_j with $j = 1, \dots, N-1$. Since $\varphi_i(x_j) = \delta_{ij}$, we obtain

$$0 = \sum_{i=1}^{N-1} \lambda_i \varphi_i(x_j) = \sum_{i=1}^{N-1} \lambda_i \delta_{ij} = \lambda_j. \quad (4.22)$$

Hence all coefficients vanish.

We next prove that the family spans V_h . Let v_h in V_h and define

$$\tilde{v}_h := \sum_{i=1}^{N-1} v_h(x_i) \varphi_i. \quad (4.23)$$

Then $\tilde{v}_h$ in V_h . For each node x_j , with $j = 0, \dots, N$, we have

$$\tilde{v}_h(x_j) = v_h(x_j). \quad (4.24)$$

Indeed this is clear for $j = 0$ and $j = N$, because both functions vanish there, and for $1 \leq j \leq N-1$ it follows from the Kronecker relation. The difference $v_h - \tilde{v}_h$ is affine on each K_i and vanishes at both endpoints of K_i , hence it vanishes identically on K_i . Therefore $v_h = \tilde{v}_h$ on $[a, b]$. $\square$

Definition 4.10. Because we work in one dimension, every function in $H^1(I)$ has a continuous representative on $[a, b]$. We therefore define the nodal interpolation operator

$$\Pi_h : H_0^1(I) \rightarrow V_h \quad (4.25)$$

by the nodal conditions

$$(\Pi_h v)(x_i) = v(x_i) \quad \text{for } i = 0, \dots, N. \quad (4.26)$$

Equivalently,

$$\Pi_h v = \sum_{i=1}^{N-1} v(x_i) \varphi_i. \quad (4.27)$$

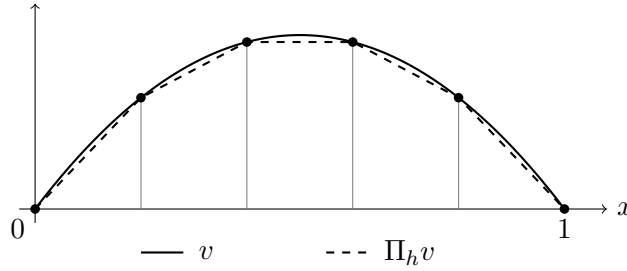


Figure 4.2: The function $v(x) = 4x(1 - x)$ on $[0, 1]$ and its nodal interpolant on five equal elements. Both functions vanish at the endpoints, and the interpolant agrees with v at every mesh node.

Definition 4.11. The discrete Galerkin problem is: find u_h in V_h such that

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (4.28)$$

The inclusion $V_h \subset V$ is the conformity property in Definition 4.4.

4.3 Well-posedness of the discrete problem and the a priori estimate

Theorem 4.12. Let u in V be the solution of Definition 4.1. The discrete problem of Definition 4.11 admits a unique solution u_h in V_h . Moreover, there exists a constant $C > 0$ such that

$$\|u - u_h\|_{H_0^1(I)} \leq Ch \|u''\|_{L^2(I)}. \quad (4.29)$$

More precisely, we may take

$$C = 1 + \|c\|_{L^\infty(I)} C_P^2, \quad (4.30)$$

where C_P is any constant such that

$$\|v\|_{L^2(I)} \leq C_P \|v\|_{H_0^1(I)} \quad \text{for all } v \in H_0^1(I). \quad (4.31)$$

Proof. We argue in three steps.

Step 1. The space V_h is a finite-dimensional subspace of V . The bilinear form a remains continuous and coercive on V_h . By the Lax–Milgram theorem, there exists a unique u_h in V_h such that

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (4.32)$$

Step 2. By Lemma 4.13, with

$$M = 1 + \|c\|_{L^\infty(I)} C_P^2 \quad \text{and} \quad \alpha = 1, \quad (4.33)$$

we obtain

$$\|u - u_h\|_{H_0^1(I)} \leq C \inf_{v_h \in V_h} \|u - v_h\|_{H_0^1(I)}, \quad (4.34)$$

with

$$C = 1 + \|c\|_{L^\infty(I)} C_P^2. \quad (4.35)$$

On an interval, we may choose

$$C_P = \frac{b-a}{\pi}, \quad (4.36)$$

so that

$$C = 1 + \frac{(b-a)^2}{\pi^2} \|c\|_{L^\infty(I)}. \quad (4.37)$$

Step 3. By Theorem 4.2, the solution u belongs to $H^2(I)$. We may therefore choose $v_h = \Pi_h u$ in the previous estimate and use Lemma 4.14:

$$\|u - \Pi_h u\|_{H_0^1(I)} \leq h \|u''\|_{L^2(I)}. \quad (4.38)$$

Combining the last two inequalities gives the announced estimate. $\square$

Lemma 4.13. Assume that a is continuous and coercive on $H_0^1(I)$, with constants M and α , namely

$$|a(w, v)| \leq M \|w\|_{H_0^1(I)} \|v\|_{H_0^1(I)}, \quad (4.39)$$

$$a(v, v) \geq \alpha \|v\|_{H_0^1(I)}^2, \quad (4.40)$$

for all w, v in $H_0^1(I)$. Let u in V and u_h in V_h be the solutions of the continuous and discrete problems. Then

$$\|u - u_h\|_{H_0^1(I)} \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_{H_0^1(I)}. \quad (4.41)$$

Lemma 4.14. For every u in $H^2(I) \cap H_0^1(I)$, the nodal interpolant satisfies

$$\|u - \Pi_h u\|_{H_0^1(I)} \leq h \|u''\|_{L^2(I)}. \quad (4.42)$$

Remark 4.15. In the pure Poisson case $c = 0$, the bilinear form reduces to

$$a(w, v) = \int_a^b w'(x) v'(x) dx. \quad (4.43)$$

For the $H_0^1(I)$ norm, we then have $M = \alpha = 1$, hence we may take $C = 1$ in Theorem 4.12. Therefore

$$\|u - u_h\|_{H_0^1(I)} \leq h \|u''\|_{L^2(I)} = h \|f\|_{L^2(I)}. \quad (4.44)$$

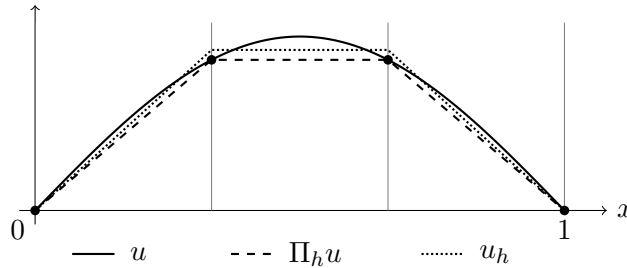


Figure 4.3: For $u(x) = \sin(\pi x)$, $c = 25$, and $f = (\pi^2 + 25) \sin(\pi x)$ on $(0, 1)$, the exact solution, its interpolant, and the Galerkin solution on three equal elements. Exact integration gives $u_h = \beta \Pi_h u$ with $\beta = 54(\pi^2 + 25)/(179\pi^2)$. All three functions satisfy the homogeneous Dirichlet endpoint conditions.

4.4 Proof of Céa's lemma

We now prove Lemma 4.13.

Lemma 4.16. *Let u in V and u_h in V_h be the solutions of the continuous and discrete problems. Then*

$$a(u - u_h, w_h) = 0 \quad \text{for all } w_h \in V_h. \quad (4.45)$$

Proof. For every w_h in V_h , since $V_h \subset V$, we may subtract the discrete problem from the continuous one and obtain

$$a(u - u_h, w_h) = 0. \quad (4.46)$$

□

Proof. Let $e_h = u - u_h$. For any v_h in V_h , coercivity gives

$$\alpha \|e_h\|_{H_0^1(I)}^2 \leq a(e_h, e_h) \quad (4.47)$$

$$= a(e_h, u - v_h) + a(e_h, v_h - u_h). \quad (4.48)$$

Since $v_h - u_h$ in V_h , the second term vanishes by Lemma 4.16. Hence

$$\alpha \|e_h\|_{H_0^1(I)}^2 \leq a(e_h, u - v_h). \quad (4.49)$$

By continuity of a , we obtain

$$\alpha \|e_h\|_{H_0^1(I)}^2 \leq M \|e_h\|_{H_0^1(I)} \|u - v_h\|_{H_0^1(I)}. \quad (4.50)$$

If $e_h \neq 0$, we divide by $\|e_h\|_{H_0^1(I)}$ and get

$$\|u - u_h\|_{H_0^1(I)} \leq \frac{M}{\alpha} \|u - v_h\|_{H_0^1(I)}. \quad (4.51)$$

We then take the infimum over v_h in V_h . If $e_h = 0$, the conclusion is immediate. □

4.5 Proof of the interpolation estimate

We now prove Lemma 4.14.

Definition 4.17. For each element K_i and each function z in $L^1(K_i)$, we define the elementwise average

$$\langle z \rangle_{K_i} := \frac{1}{h_i} \int_{K_i} z(x) dx. \quad (4.52)$$

Lemma 4.18. *Let u in $H^2(I) \cap H_0^1(I)$. On each interval K_i , the derivative of $\Pi_h u$ is constant and satisfies*

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1})}{h_i} = \langle u' \rangle_{K_i}. \quad (4.53)$$

Consequently,

$$(u - \Pi_h u)'|_{K_i} = u' - \langle u' \rangle_{K_i}. \quad (4.54)$$

Proof. The function $\Pi_h u$ is affine on K_i and takes the nodal values $u(x_{i-1})$ and $u(x_i)$. Its derivative is therefore the slope of the secant line. The second identity follows from the fundamental theorem of calculus. □

Lemma 4.19. *Let $K = (\alpha, \beta)$ be an interval of length $h_K = \beta - \alpha$. If z in $H^1(K)$ and*

$$\int_K z(x) dx = 0, \quad (4.55)$$

then

$$\|z\|_{L^2(K)} \leq h_K \|z'\|_{L^2(K)}. \quad (4.56)$$

Proof. Since z in $H^1(K)$, we may identify it with a continuous representative on $\overline{K}$. The condition of zero average implies that z vanishes at some point ξ in K . For every x in K ,

$$z(x) = \int_{\xi}^x z'(t) dt. \quad (4.57)$$

Hence

$$|z(x)|^2 \leq h_K \int_K |z'(t)|^2 dt. \quad (4.58)$$

We integrate with respect to x over K and obtain

$$\int_K |z(x)|^2 dx \leq h_K^2 \int_K |z'(t)|^2 dt. \quad (4.59)$$

This is the desired estimate. $\square$

Proof. On each element K_i , Lemma 4.18 gives

$$(u - \Pi_h u)' = u' - \langle u' \rangle_{K_i}, \quad (4.60)$$

and the right-hand side has zero average on K_i . We apply Lemma 4.19 with

$$z = u' - \langle u' \rangle_{K_i} \quad (4.61)$$

and obtain

$$\|(u - \Pi_h u)'\|_{L^2(K_i)} = \|u' - \langle u' \rangle_{K_i}\|_{L^2(K_i)} \leq h_i \|u''\|_{L^2(K_i)}. \quad (4.62)$$

We square and sum over all elements:

$$\|u - \Pi_h u\|_{H_0^1(I)}^2 = \sum_{i=1}^N \|(u - \Pi_h u)'\|_{L^2(K_i)}^2 \quad (4.63)$$

$$\leq \sum_{i=1}^N h_i^2 \|u''\|_{L^2(K_i)}^2 \quad (4.64)$$

$$\leq h^2 \sum_{i=1}^N \|u''\|_{L^2(K_i)}^2 \quad (4.65)$$

$$= h^2 \|u''\|_{L^2(I)}^2. \quad (4.66)$$

Taking square roots gives

$$\|u - \Pi_h u\|_{H_0^1(I)} \leq h \|u''\|_{L^2(I)}. \quad (4.67)$$

This proves Lemma 4.14. $\square$

4.6 Reference element, local shape functions, local matrices, and assembly

The purpose of this section is to separate clearly the local finite element construction, which is performed on a single element, from the global basis introduced in Section 4.2. Local shape functions live on one element only. The global hat functions are obtained after assembly and are different objects.

We use the following notation. If K is an interval, then

$$P_1(K) := \{p : K \rightarrow \mathbb{R} : p(x) = \alpha + \beta x \text{ for some } \alpha, \beta \in \mathbb{R}\}. \quad (4.68)$$

For a closed interval, the same notation means the restrictions to this closed interval of affine functions on $\mathbb{R}$. Thus $P_1(\widehat{K})$ and $P_1(\overline{K_i})$ are two-dimensional spaces.

Definition 4.20. The reference P_1 finite element is the triple

$$(\widehat{K}, \widehat{P}, \widehat{\Sigma}), \quad (4.69)$$

where

$$\widehat{K} = [0, 1], \quad \widehat{P} = P_1(\widehat{K}), \quad \widehat{\Sigma} = (\widehat{\sigma}_1, \widehat{\sigma}_2), \quad (4.70)$$

with

$$\widehat{\sigma}_1(\widehat{p}) = \widehat{p}(0), \quad \widehat{\sigma}_2(\widehat{p}) = \widehat{p}(1). \quad (4.71)$$

The reference shape functions are the basis $\widehat{\psi}_1, \widehat{\psi}_2$ of $\widehat{P}$ characterized by

$$\widehat{\sigma}_m(\widehat{\psi}_n) = \delta_{mn} \quad m, n = 1, 2. \quad (4.72)$$

Proposition 4.21. The reference shape functions are

$$\widehat{\psi}_1(\widehat{x}) = 1 - \widehat{x}, \quad \widehat{\psi}_2(\widehat{x}) = \widehat{x}, \quad \widehat{x} \in [0, 1]. \quad (4.73)$$

Moreover, every $\widehat{p}$ in $P_1(\widehat{K})$ satisfies the reference nodal decomposition

$$\widehat{p}(\widehat{x}) = \widehat{p}(0)\widehat{\psi}_1(\widehat{x}) + \widehat{p}(1)\widehat{\psi}_2(\widehat{x}) \quad \text{for all } \widehat{x} \in \widehat{K}. \quad (4.74)$$

Proof. The formulas in (4.73) define the unique affine functions on $[0, 1]$ satisfying

$$\widehat{\psi}_1(0) = 1, \quad \widehat{\psi}_1(1) = 0, \quad \widehat{\psi}_2(0) = 0, \quad \widehat{\psi}_2(1) = 1. \quad (4.75)$$

The decomposition formula follows because both sides of (4.74) are affine on $\widehat{K}$ and have the same values at 0 and 1. $\square$

For each element $K_i = (x_{i-1}, x_i)$, we denote by

$$\overline{K_i} = [x_{i-1}, x_i] \quad (4.76)$$

its closure, and we introduce the affine map

$$F_i : \widehat{K} \rightarrow \overline{K_i}, \quad F_i(\widehat{x}) = x_{i-1} + h_i \widehat{x}. \quad (4.77)$$

Definition 4.22. The local finite element attached to $\overline{K_i}$ is the transported triple

$$(\overline{K_i}, P_1(\overline{K_i}), \Sigma_i), \quad (4.78)$$

where

$$\Sigma_i = (\sigma_1^{(i)}, \sigma_2^{(i)}), \quad \sigma_1^{(i)}(p) = p(x_{i-1}), \quad \sigma_2^{(i)}(p) = p(x_i). \quad (4.79)$$

The local shape functions on $\overline{K_i}$ are defined by transport from the reference element:

$$\psi_1^{(i)} = \hat{\psi}_1 \circ F_i^{-1}, \quad \psi_2^{(i)} = \hat{\psi}_2 \circ F_i^{-1}. \quad (4.80)$$

Equivalently, $\psi_1^{(i)}, \psi_2^{(i)}$ are the basis of $P_1(\overline{K_i})$ characterized by

$$\sigma_m^{(i)}(\psi_n^{(i)}) = \delta_{mn} \quad m, n = 1, 2. \quad (4.81)$$

In matrix form, the values of the local degrees of freedom on the local shape functions are

$$\begin{pmatrix} \sigma_1^{(i)}(\psi_1^{(i)}) & \sigma_1^{(i)}(\psi_2^{(i)}) \\ \sigma_2^{(i)}(\psi_1^{(i)}) & \sigma_2^{(i)}(\psi_2^{(i)}) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (4.82)$$

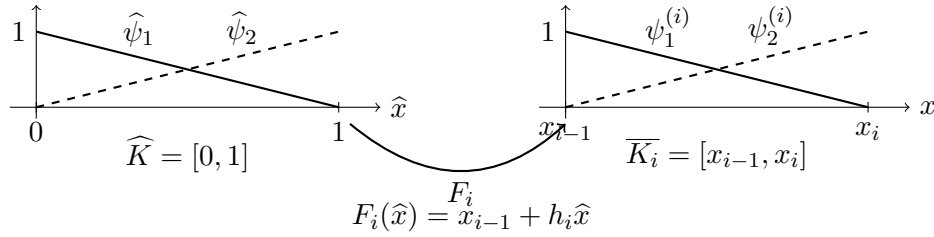


Figure 4.4: Reference and local shape functions related by the affine map F_i . The solid functions correspond to the left endpoint, and the dashed functions correspond to the right endpoint.

Proposition 4.23. *For every $i = 1, \dots, N$, the local shape functions are given by*

$$\psi_1^{(i)}(x) = \frac{x_i - x}{h_i}, \quad \psi_2^{(i)}(x) = \frac{x - x_{i-1}}{h_i}, \quad x \in \overline{K_i}. \quad (4.83)$$

Moreover, every p in $P_1(\overline{K_i})$ satisfies

$$p(x) = p(x_{i-1})\psi_1^{(i)}(x) + p(x_i)\psi_2^{(i)}(x) \quad \text{for all } x \in \overline{K_i}. \quad (4.84)$$

Proof. We combine Proposition 4.21 with the identity $F_i^{-1}(x) = (x - x_{i-1})/h_i$. This gives

$$\psi_1^{(i)}(x) = \hat{\psi}_1\left(\frac{x - x_{i-1}}{h_i}\right) = 1 - \frac{x - x_{i-1}}{h_i} = \frac{x_i - x}{h_i}, \quad (4.85)$$

$$\psi_2^{(i)}(x) = \hat{\psi}_2\left(\frac{x - x_{i-1}}{h_i}\right) = \frac{x - x_{i-1}}{h_i}. \quad (4.86)$$

In particular,

$$\psi_1^{(i)}(x_{i-1}) = 1, \quad \psi_1^{(i)}(x_i) = 0, \quad \psi_2^{(i)}(x_{i-1}) = 0, \quad \psi_2^{(i)}(x_i) = 1, \quad (4.87)$$

which is exactly (4.81). The representation formula follows because both sides are affine on $\overline{K_i}$ and they coincide at the two endpoints. $\square$

Remark 4.24. The functions $\psi_1^{(i)}$ and $\psi_2^{(i)}$ are defined only on $\overline{K_i}$. They are local basis functions attached to the two local degrees of freedom of the element K_i . They are not global hat functions. A global hat function is obtained by gluing together the relevant local shape functions on the neighboring elements that share the same interior node.

Definition 4.25. For each element K_i , we define the local-to-global connectivity map

$$\mathcal{I}_i : \{1, 2\} \rightarrow \{0, \dots, N\}, \quad \mathcal{I}_i(1) = i - 1, \quad \mathcal{I}_i(2) = i. \quad (4.88)$$

The following proposition is the bridge between the global basis and the local bases. The index j labels a global interior node, while the index i labels an element. Since j is in $\{1, \dots, N-1\}$, the node x_j belongs to exactly two elements: K_j , where it is the right endpoint, and K_{j+1} , where it is the left endpoint. Therefore the restriction of the global hat function φ_j is the right local shape function on K_j , the left local shape function on K_{j+1} , and zero on all other elements. This is the elementary index rule behind the assembly formulas below.

Proposition 4.26. *Let j in $\{1, \dots, N-1\}$. For every $i = 1, \dots, N$, we have*

$$\varphi_j|_{\overline{K_i}} = \begin{cases} \psi_2^{(j)} & \text{if } i = j, \\ \psi_1^{(j+1)} & \text{if } i = j+1, \\ 0 & \text{if } i \leq j-1 \text{ or } i \geq j+2. \end{cases} \quad (4.89)$$

Equivalently, with the local-to-global map $\mathcal{I}_i$, this can be written as

$$\varphi_j|_{\overline{K_i}} = \begin{cases} \psi_m^{(i)} & \text{if } \mathcal{I}_i(m) = j \text{ for some } m \in \{1, 2\}, \\ 0 & \text{otherwise.} \end{cases} \quad (4.90)$$

Proof. We fix j in $\{1, \dots, N-1\}$ and i in $\{1, \dots, N\}$. The global function φ_j is the continuous piecewise affine function satisfying

$$\varphi_j(x_k) = \delta_{jk} \quad k = 0, \dots, N. \quad (4.91)$$

On each closed element $\overline{K_i} = [x_{i-1}, x_i]$, the restriction $\varphi_j|_{\overline{K_i}}$ is affine. Hence it is completely determined by its two endpoint values.

We first consider the case $i = j$. Then

$$\overline{K_i} = \overline{K_j} = [x_{j-1}, x_j], \quad (4.92)$$

and x_j is the right endpoint of this element. By (4.91),

$$\varphi_j(x_{j-1}) = 0, \quad \varphi_j(x_j) = 1. \quad (4.93)$$

The local function $\psi_2^{(j)}$ has the same two endpoint values on $\overline{K_j}$, namely

$$\psi_2^{(j)}(x_{j-1}) = 0, \quad \psi_2^{(j)}(x_j) = 1. \quad (4.94)$$

Since both functions are affine on $\overline{K_j}$ and have the same values at x_{j-1} and x_j , they coincide on $\overline{K_j}$. Thus

$$\varphi_j|_{\overline{K_j}} = \psi_2^{(j)}. \quad (4.95)$$

We next consider the case $i = j+1$. Then

$$\overline{K_i} = \overline{K_{j+1}} = [x_j, x_{j+1}], \quad (4.96)$$

and x_j is the left endpoint of this element. By (4.91),

$$\varphi_j(x_j) = 1, \quad \varphi_j(x_{j+1}) = 0. \quad (4.97)$$

The local function $\psi_1^{(j+1)}$ has the same two endpoint values on $\overline{K_{j+1}}$, namely

$$\psi_1^{(j+1)}(x_j) = 1, \quad \psi_1^{(j+1)}(x_{j+1}) = 0. \quad (4.98)$$

Since both functions are affine on $\overline{K_{j+1}}$ and have the same values at x_j and x_{j+1} , they coincide on $\overline{K_{j+1}}$. Thus

$$\varphi_j|_{\overline{K_{j+1}}} = \psi_1^{(j+1)}. \quad (4.99)$$

It remains to consider the elements which do not contain the node x_j . This means

$$i \leq j-1 \quad \text{or} \quad i \geq j+2. \quad (4.100)$$

In both cases, neither endpoint of $\overline{K_i} = [x_{i-1}, x_i]$ is equal to x_j . Therefore (4.91) gives

$$\varphi_j(x_{i-1}) = 0, \quad \varphi_j(x_i) = 0. \quad (4.101)$$

The restriction $\varphi_j|_{\overline{K_i}}$ is affine and vanishes at both endpoints, hence it vanishes identically on $\overline{K_i}$. This proves (4.89).

The connectivity form (4.90) is the same statement in local numbering. Indeed,

$$\mathcal{I}_i(1) = i-1, \quad \mathcal{I}_i(2) = i. \quad (4.102)$$

Thus $\mathcal{I}_i(2) = j$ is exactly the case $i = j$, and $\mathcal{I}_i(1) = j$ is exactly the case $i = j+1$. $\square$

Definition 4.27. The local stiffness matrix, the local mass matrix, the local reaction matrix, the local element matrix, and the local load vector on K_i are defined by

$$S_{mn}^{(i)} := \int_{K_i} (\psi_m^{(i)})'(x) (\psi_n^{(i)})'(x) dx, \quad (4.103)$$

$$M_{mn}^{(i)} := \int_{K_i} \psi_m^{(i)}(x) \psi_n^{(i)}(x) dx, \quad (4.104)$$

$$R_{mn}^{(i)} := \int_{K_i} c(x) \psi_m^{(i)}(x) \psi_n^{(i)}(x) dx, \quad (4.105)$$

$$A^{(i)} := S^{(i)} + R^{(i)}, \quad (4.106)$$

$$b_m^{(i)} := \int_{K_i} f(x) \psi_m^{(i)}(x) dx, \quad (4.107)$$

for $m, n = 1, 2$. The notation $b^{(i)}$ is local, whereas B below denotes the assembled global right-hand side.

Proposition 4.28. For every $i = 1, \dots, N$, we have

$$S^{(i)} = \frac{1}{h_i} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{and} \quad M^{(i)} = \frac{h_i}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (4.108)$$

If c is constant on K_i , equal to a value c_i , then

$$R^{(i)} = c_i M^{(i)} = \frac{c_i h_i}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (4.109)$$

If f is constant on K_i , equal to a value f_i , then

$$b^{(i)} = \frac{f_i h_i}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (4.110)$$

Proof. From Proposition 4.23, we have

$$(\psi_1^{(i)})' = -\frac{1}{h_i} \quad \text{and} \quad (\psi_2^{(i)})' = \frac{1}{h_i}. \quad (4.111)$$

This gives the formula for $S^{(i)}$. The formula for $M^{(i)}$ follows from the direct computation

$$\int_{x_{i-1}}^{x_i} \left(\frac{x_i - x}{h_i} \right)^2 dx = \frac{h_i}{3}, \quad \int_{x_{i-1}}^{x_i} \frac{(x_i - x)(x - x_{i-1})}{h_i^2} dx = \frac{h_i}{6}. \quad (4.112)$$

The formulas for $R^{(i)}$ and $b^{(i)}$ follow immediately when c and f are constant on K_i . $\square$

Definition 4.29. In the global hat basis $(\varphi_1, \dots, \varphi_{N-1})$, we define the global stiffness matrix, the global reaction matrix, the global system matrix, and the global load vector by

$$S_{jk} := \int_a^b \varphi'_k(x) \varphi'_j(x) dx, \quad (4.113)$$

$$R_{jk} := \int_a^b c(x) \varphi_k(x) \varphi_j(x) dx, \quad (4.114)$$

$$A := S + R, \quad (4.115)$$

$$B_j := \ell(\varphi_j), \quad (4.116)$$

for $j, k = 1, \dots, N-1$.

Proposition 4.30. Let u_h, v_h in V_h . We set

$$U_j := u_h(x_j), \quad V_j := v_h(x_j), \quad j = 0, \dots, N, \quad (4.117)$$

so that $U_0 = U_N = V_0 = V_N = 0$. For each element K_i , we also set

$$U^{(i)} := \begin{pmatrix} U_{i-1} \\ U_i \end{pmatrix}, \quad V^{(i)} := \begin{pmatrix} V_{i-1} \\ V_i \end{pmatrix}. \quad (4.118)$$

Then

$$u_h|_{\overline{K_i}} = U_{i-1} \psi_1^{(i)} + U_i \psi_2^{(i)}, \quad (4.119)$$

$$v_h|_{\overline{K_i}} = V_{i-1} \psi_1^{(i)} + V_i \psi_2^{(i)}, \quad (4.120)$$

and consequently

$$\int_{K_i} u'_h(x) v'_h(x) dx = \left(U^{(i)} \right)^T S^{(i)} V^{(i)}, \quad (4.121)$$

$$\int_{K_i} c(x) u_h(x) v_h(x) dx = \left(U^{(i)} \right)^T R^{(i)} V^{(i)}, \quad (4.122)$$

$$\int_{K_i} f(x) v_h(x) dx = \left(b^{(i)} \right)^T V^{(i)}. \quad (4.123)$$

Therefore,

$$a(u_h, v_h) = \sum_{i=1}^N \left(U^{(i)} \right)^T A^{(i)} V^{(i)}, \quad (4.124)$$

$$\ell(v_h) = \sum_{i=1}^N \left(b^{(i)} \right)^T V^{(i)}. \quad (4.125)$$

Proof. The local expansion of $u_h|_{\overline{K_i}}$ and $v_h|_{\overline{K_i}}$ is the representation formula of Proposition 4.23, applied to the affine functions $u_h|_{\overline{K_i}}$ and $v_h|_{\overline{K_i}}$. The matrix identities follow by bilinearity. Summing over all elements gives the global formulas. $\square$

Proposition 4.31. *Let*

$$u_h = \sum_{j=1}^{N-1} U_j \varphi_j. \quad (4.126)$$

The discrete problem of Definition 4.11 is equivalent to the linear system

$$AU = B, \quad (4.127)$$

where $U = (U_1, \dots, U_{N-1})^T$. Moreover, the global matrices and vector are assembled from the local contributions according to

$$S_{jk} = \sum_{i=1}^N \sum_{m,n=1}^2 S_{mn}^{(i)} \delta_{j, \mathcal{I}_i(m)} \delta_{k, \mathcal{I}_i(n)}, \quad (4.128)$$

$$R_{jk} = \sum_{i=1}^N \sum_{m,n=1}^2 R_{mn}^{(i)} \delta_{j, \mathcal{I}_i(m)} \delta_{k, \mathcal{I}_i(n)}, \quad (4.129)$$

$$B_j = \sum_{i=1}^N \sum_{m=1}^2 b_m^{(i)} \delta_{j, \mathcal{I}_i(m)}, \quad (4.130)$$

for $j, k = 1, \dots, N-1$.

Proof. The equivalence with the linear system follows from the definition of the global basis and the identity

$$a(u_h, \varphi_j) = B_j \quad \text{for } j = 1, \dots, N-1. \quad (4.131)$$

The assembly formulas follow from Proposition 4.30 and from Proposition 4.26: on each element, only the two local degrees of freedom indexed by $\mathcal{I}_i(1)$ and $\mathcal{I}_i(2)$ contribute. $\square$

Corollary 4.32. *The matrix A is tridiagonal in the global hat basis.*

Proof. Each element matrix $A^{(i)}$ couples only the two nodes x_{i-1} and x_i . After assembly, the unknown at node x_j can interact only with the unknowns at x_{j-1} , x_j , and x_{j+1} . Therefore $A_{jk} = 0$ whenever $|j - k| \geq 2$. $\square$

4.6.1 Full assembly and enforcement of Dirichlet conditions

The preceding formulas describe the reduced Galerkin matrix directly. For practical assembly, a useful route is to number all nodes first, assemble all local contributions, and then impose the prescribed nodal values. The two routes are algebraically equivalent. Constraints can also be incorporated while local contributions are assembled; full assembly followed by modification is a convenient teaching procedure, not a requirement imposed on every implementation.

Let $(\hat{\varphi}_j)_{j=0}^N$ be the full nodal basis of W_h , characterized by $\hat{\varphi}_j(x_k) = \delta_{jk}$ for $0 \leq j, k \leq N$. For interior indices, $\hat{\varphi}_j = \varphi_j$. Extend the integral formulas for a and ℓ to W_h , and assemble

$$\hat{A}_{jk} = a(\hat{\varphi}_k, \hat{\varphi}_j), \quad \hat{B}_j = \ell(\hat{\varphi}_j), \quad 0 \leq j, k \leq N. \quad (4.132)$$

Thus $\hat{A}$ has size $(N+1) \times (N+1)$ and $\hat{B}$ has $N+1$ entries. Use the same connectivity map $\mathcal{I}_i(1) = i-1$, $\mathcal{I}_i(2) = i$ and the same local matrices as before; now both endpoint indices are included in the assembly sums.

Eliminating prescribed values. Let $F = \{1, \dots, N-1\}$ be the free indices and $D = \{0, N\}$ the Dirichlet indices. Temporarily allow the prescribed values $U_0 = g_0$, $U_N = g_N$, and set

$g_D = (g_0, g_N)^T$. The discrete trial functions then form an affine translate of V_h , while the test functions still belong to V_h . Order the nodal vector as (U_F, U_D) . The free test equations are

$$\hat{A}_{FF}U_F + \hat{A}_{FD}g_D = \hat{B}_F, \quad \hat{A}_{FF}U_F = \hat{B}_F - \hat{A}_{FD}g_D. \quad (4.133)$$

For homogeneous data $g_D = 0$, the reduced matrix and vector already used in this chapter are exactly $A = \hat{A}_{FF}$ and $B = \hat{B}_F$. For nonzero data, the extra term is the algebraic contribution of a Dirichlet lifting.

Keeping the full size and modifying rows and columns. An equivalent symmetric implementation retains the $N + 1$ entries:

1. Using the original boundary columns, set $\tilde{B}_F = \hat{B}_F - \hat{A}_{FD}g_D$.
2. Zero the rows and columns indexed by D , and set their diagonal entries to one.
3. Set $\tilde{B}_D = g_D$.

After the temporary free/boundary reordering, the resulting system is

$$\begin{pmatrix} \hat{A}_{FF} & 0 \\ 0 & I_2 \end{pmatrix} \begin{pmatrix} U_F \\ U_D \end{pmatrix} = \begin{pmatrix} \tilde{B}_F - \hat{A}_{FD}g_D \\ g_D \end{pmatrix}. \quad (4.134)$$

It has size $(N + 1) \times (N + 1)$. Only eliminating the two prescribed endpoint values gives the reduced $(N - 1) \times (N - 1)$ system. For $g_D = 0$, the free right-hand side is unchanged and both endpoint entries of the modified right-hand side must be set to zero. For nonzero data, the correction must be computed before the columns are erased. Zeroing columns without this correction gives a different problem. Zeroing both rows and columns preserves symmetry, and $\hat{A}_{FF}$ is positive definite by coercivity on V_h .

The free equations are precisely the equations obtained by testing with the interior hat functions. The full boundary test equations are replaced by the prescribed nodal values. In particular, the full unconstrained Poisson stiffness matrix is singular because constants are in its kernel; there is no need to solve or invert it before imposing the Dirichlet conditions.

Uniform-mesh check. If $c = c_0$, $f = f_0$, and the mesh is uniform, full assembly gives

$$\hat{A}_{00} = \hat{A}_{NN} = \frac{1}{h} + \frac{c_0 h}{3}, \quad \hat{B}_0 = \hat{B}_N = \frac{f_0 h}{2}, \quad (4.135)$$

$$\hat{A}_{jj} = \frac{2}{h} + \frac{2c_0 h}{3}, \quad \hat{B}_j = f_0 h \quad (1 \leq j \leq N - 1), \quad (4.136)$$

$$\hat{A}_{j-1,j} = \hat{A}_{j,j-1} = -\frac{1}{h} + \frac{c_0 h}{6} \quad (1 \leq j \leq N). \quad (4.137)$$

All other entries vanish. The endpoint diagonal and load entries have one element contribution, whereas each interior diagonal and load entry has two. Selecting the free block gives the matrix in Proposition 4.33. The full matrix, the modified matrix, and the reduced matrix are all tridiagonal in the natural one-dimensional nodal ordering. Their sparsity comes from element connectivity; imposing the boundary values does not create it.

Proposition 4.33. *Assume that the mesh is uniform,*

$$x_i = a + ih \quad \text{for } i = 0, \dots, N, \quad (4.138)$$

with $h = (b - a)/N$, and that $c = c_0$ and $f = f_0$ are constant. Then, for every element K_i ,

$$A^{(i)} = \frac{1}{h} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \frac{c_0 h}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (4.139)$$

$$b^{(i)} = \frac{f_0 h}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (4.140)$$

Here $b^{(i)}$ is the local load vector on K_i . After assembly, the global right-hand side is denoted by B . The coefficients of the global system satisfy

$$A_{ii} = \frac{2}{h} + \frac{2c_0 h}{3}, \quad 1 \leq i \leq N-1, \quad (4.141)$$

$$A_{i,i+1} = A_{i+1,i} = -\frac{1}{h} + \frac{c_0 h}{6}, \quad 1 \leq i \leq N-2, \quad (4.142)$$

$$B_i = h f_0, \quad 1 \leq i \leq N-1. \quad (4.143)$$

Equivalently,

$$A = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 2 \end{pmatrix} + \frac{c_0 h}{6} \begin{pmatrix} 4 & 1 & & & 0 \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ 0 & & & 1 & 4 \end{pmatrix}, \quad B = h f_0 \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{pmatrix}. \quad (4.144)$$

For $c_0 = 0$, the matrix reduces to the classical tridiagonal stiffness matrix for the homogeneous Dirichlet Poisson problem.

Proof. On every element, the local diagonal stiffness contribution is $1/h$, the local off-diagonal stiffness contribution is $-1/h$, the local diagonal reaction contribution is $c_0 h/3$, the local off-diagonal reaction contribution is $c_0 h/6$, and each component of the local load vector is $f_0 h/2$.

Let $1 \leq i \leq N-1$. The node x_i belongs to the two elements $K_i = (x_{i-1}, x_i)$ and $K_{i+1} = (x_i, x_{i+1})$. Therefore the diagonal entry receives two local diagonal contributions:

$$A_{ii} = \left(\frac{1}{h} + \frac{c_0 h}{3} \right) + \left(\frac{1}{h} + \frac{c_0 h}{3} \right) \quad (4.145)$$

$$= \frac{2}{h} + \frac{2c_0 h}{3}. \quad (4.146)$$

If $1 \leq i \leq N-2$, the nodes x_i and x_{i+1} belong together only to K_{i+1} , hence

$$A_{i,i+1} = A_{i+1,i} = -\frac{1}{h} + \frac{c_0 h}{6}. \quad (4.147)$$

If $|i - j| \geq 2$, no element contains both nodes x_i and x_j , hence $A_{ij} = 0$.

For the right-hand side, the global component B_i is the sum of the two local load contributions attached to x_i :

$$B_i = b_2^{(i)} + b_1^{(i+1)} = \frac{f_0 h}{2} + \frac{f_0 h}{2} = h f_0. \quad (4.148)$$

The displayed matrix form is exactly these diagonal, off-diagonal, and right-hand-side entries written at the global level. $\square$

This sparse structure is one of the main reasons why the hat basis is numerically efficient.

4.7 The pure Neumann Poisson problem

We first isolate the genuinely Neumann case, because it is the case in which coercivity fails on the whole space unless constants are removed. In one dimension, the outward normal derivative is

$$\partial_n u(a) = -u'(a) \quad \text{and} \quad \partial_n u(b) = u'(b). \quad (4.149)$$

We consider the pure Neumann Poisson problem

$$-u'' = f \quad \text{in } I, \quad \partial_n u(a) = g_a, \quad \partial_n u(b) = g_b. \quad (4.150)$$

A necessary compatibility condition is obtained by testing the equation against the constant function 1, or equivalently by integrating the equation over I :

$$\int_a^b f(x) dx + g_a + g_b = 0. \quad (4.151)$$

Without this condition, no weak solution exists. Even when it holds, the solution is determined only up to an additive constant, since the derivative does not see constants.

In the continuous and discrete pure Neumann formulations below, we assume (4.151). It is essential for the equivalence between the original Neumann problem and its normalized formulation.

We therefore introduce the zero-mean space

$$H_\star^1(I) = \left\{ v \in H^1(I) : \int_a^b v(x) dx = 0 \right\}, \quad (4.152)$$

endowed with the norm

$$\|v\|_{H_\star^1(I)} := \|v'\|_{L^2(I)}. \quad (4.153)$$

This is a norm by the Poincaré–Wirtinger inequality:

$$\|v\|_{L^2(I)} \leq C_{\text{PW}} \|v'\|_{L^2(I)} \quad \text{for all } v \in H_\star^1(I). \quad (4.154)$$

The weak formulation of (4.150), normalized by zero mean, is: find u in $H_\star^1(I)$ such that

$$a_N(u, v) = \ell_N(v) \quad \text{for all } v \in H_\star^1(I), \quad (4.155)$$

where

$$a_N(w, v) = \int_a^b w'(x) v'(x) dx, \quad (4.156)$$

$$\ell_N(v) = \int_a^b f(x) v(x) dx + g_a v(a) + g_b v(b). \quad (4.157)$$

Proposition 4.34. *Assume that f in $L^2(I)$, g_a, g_b in $\mathbb{R}$, and that the compatibility condition (4.151) holds. Then the weak problem (4.155) admits a unique solution in $H_\star^1(I)$. The set of all weak solutions in $H^1(I)$ is obtained by adding arbitrary constants to this normalized solution.*

Proof. The bilinear form a_N is continuous on $H_\star^1(I)$ and satisfies

$$a_N(v, v) = \|v'\|_{L^2(I)}^2 = \|v\|_{H_\star^1(I)}^2. \quad (4.158)$$

The linear form ℓ_N is continuous on $H_\star^1(I)$ by the Cauchy–Schwarz inequality, the Poincaré–Wirtinger inequality, and the continuity of the endpoint trace maps on $H^1(I)$. The Lax–Milgram theorem gives a unique solution in $H_\star^1(I)$.

To recover the weak formulation with all $H^1(I)$ test functions, let v in $H^1(I)$ and write

$$v = v_\star + \bar{v}, \quad \bar{v} = \frac{1}{b-a} \int_a^b v(x) dx, \quad v_\star \in H_\star^1(I). \quad (4.159)$$

Constants have zero derivative, and compatibility gives

$$\ell_N(1) = \int_a^b f(x) dx + g_a + g_b = 0. \quad (4.160)$$

Hence $a_N(u, v) = a_N(u, v_\star) = \ell_N(v_\star) = \ell_N(v)$. Thus the normalized solution is a weak solution of the original Neumann problem. Conversely, a zero-mean weak solution with all $H^1(I)$ tests satisfies the normalized formulation by restriction. Testing with compactly supported smooth functions gives $-u'' = f$, so u belongs to $H^2(I)$; integration by parts then recovers the two prescribed normal derivatives.

If u is a solution and C is constant, then $u+C$ satisfies the same Neumann equation. Conversely, if u_1 and u_2 are two weak solutions, then $u_1 - u_2$ has zero derivative in $L^2(I)$, hence is constant. $\square$

For the finite element approximation, no endpoint value is prescribed. The conforming P_1 space is therefore

$$W_h = \left\{ v_h \in C^0([a, b]) : v_h|_{K_i} \text{ is affine for every } i \right\}, \quad (4.161)$$

with the full hat basis $(\varphi_0, \dots, \varphi_N)$ characterized by

$$\varphi_j(x_i) = \delta_{ij} \quad \text{for } i, j = 0, \dots, N. \quad (4.162)$$

The normalized discrete space is

$$W_{h,\star} = \left\{ v_h \in W_h : \int_a^b v_h(x) dx = 0 \right\}. \quad (4.163)$$

The discrete pure Neumann problem is: find u_h in $W_{h,\star}$ such that

$$a_N(u_h, v_h) = \ell_N(v_h) \quad \text{for all } v_h \in W_{h,\star}. \quad (4.164)$$

Proposition 4.35. *Assume that the mesh is uniform, with $x_i = a + ih$ and $h = (b - a)/N$, and that $f = f_0$ is constant. In the full nodal basis $(\varphi_0, \dots, \varphi_N)$, the unconstrained pure Neumann stiffness matrix and load vector are*

$$K = \frac{1}{h} \begin{pmatrix} 1 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} \frac{hf_0}{2} + g_a \\ hf_0 \\ \vdots \\ hf_0 \\ \frac{hf_0}{2} + g_b \end{pmatrix}. \quad (4.165)$$

The matrix K is singular and its kernel is generated by the constant vector $(1, \dots, 1)^T$. The compatibility condition is reflected at the discrete level by

$$\sum_{i=0}^N B_i = (b - a)f_0 + g_a + g_b = 0. \quad (4.166)$$

The zero-mean normalization is imposed by the additional constraint

$$h \left(\frac{U_0}{2} + \sum_{i=1}^{N-1} U_i + \frac{U_N}{2} \right) = 0. \quad (4.167)$$

Proof. The endpoint hat functions are supported on only one element, so their diagonal stiffness contribution is $1/h$. Each interior hat function is supported on two elements, so its diagonal stiffness contribution is $2/h$. Two adjacent hat functions overlap on exactly one element and give the off-diagonal contribution $-1/h$. This gives the displayed matrix. The load vector is assembled from one local contribution $hf_0/2$ at each endpoint and two local contributions $hf_0/2$ at each interior node, together with the two Neumann boundary terms. $\square$

Boundary conditions of the form

$$\partial_n u + du = g \quad (4.168)$$

are called *Robin* boundary conditions; in many PDE texts, especially in heat-transfer contexts, they are also called *Fourier* boundary conditions. We use the two names synonymously in this chapter.

Proposition 4.36. *Assume that the mesh is uniform, that $c = c_0$ and $f = f_0$ are constant, and consider Robin, or Fourier, conditions*

$$\partial_n u(a) + d_a u(a) = g_a, \quad \partial_n u(b) + d_b u(b) = g_b, \quad (4.169)$$

with $d_a, d_b \geq 0$. In the full nodal basis $(\varphi_0, \dots, \varphi_N)$, the finite element matrix and load vector are

$$A = \frac{1}{h} \begin{pmatrix} 1 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 1 \end{pmatrix} + \frac{c_0 h}{6} \begin{pmatrix} 2 & 1 & & & 0 \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ 0 & & & 1 & 2 \end{pmatrix} + \begin{pmatrix} d_a & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & d_b \end{pmatrix}, \quad (4.170)$$

and

$$B = \begin{pmatrix} \frac{hf_0}{2} + g_a \\ hf_0 \\ \vdots \\ hf_0 \\ \frac{hf_0}{2} + g_b \end{pmatrix}. \quad (4.171)$$

For $c_0 = 0$ and $d_a = d_b = 0$, this reduces to the pure Neumann stiffness matrix of Proposition 4.35. If the two endpoint values are instead prescribed as homogeneous Dirichlet data and removed from the list of unknowns, the remaining stiffness matrix is the classical Dirichlet tridiagonal matrix displayed in Proposition 4.33.

Proof. The volume stiffness and reaction contributions are the assembled full-basis analogues of Proposition 4.33. The difference is that the endpoint hat functions are supported on one element only. Hence the endpoint diagonal stiffness entries are $1/h$ instead of $2/h$, and the endpoint diagonal reaction entries are $c_0 h/3$ instead of $2c_0 h/3$. The Robin, or Fourier, terms add d_a and d_b to the two endpoint diagonal entries, and the boundary linear form adds g_a and g_b to the two endpoint components of the load vector. $\square$

4.8 Exercises

Use the notation of this chapter. In particular, $I = (a, b)$, $a = x_0 < x_1 < \dots < x_N = b$, $K_i = (x_{i-1}, x_i)$, $h_i = x_i - x_{i-1}$, and $h = \max_i h_i$. The three problems below have the same structure: variational formulation, finite element space, Galerkin approximation, interpolation, a priori estimate, and matrix form.

Problem 4.1 (Dirichlet–Dirichlet boundary condition). Consider

$$-u'' + cu = f \quad \text{in } I, \quad u(a) = u(b) = 0, \quad (4.172)$$

where f in $L^2(I)$ and c in $L^\infty(I)$, with $c(x) \geq 0$ almost everywhere.

Part 1. Variational setting and weak formulation.

Set $V = H_0^1(I)$ and

$$a(w, v) = \int_a^b w'(x)v'(x) dx + \int_a^b c(x)w(x)v(x) dx, \quad (4.173)$$

$$\ell(v) = \int_a^b f(x)v(x) dx. \quad (4.174)$$

Equip V with the norm $\|v\|_V = |v|_{H^1(I)}$.

- (a) Prove that a is continuous on $V \times V$.
- (b) Prove that a is coercive on V .
- (c) Deduce that the weak problem

$$a(u, v) = \ell(v) \quad \text{for all } v \in V \quad (4.175)$$

admits a unique solution.

Part 2. Finite element space, hat basis, and nodal interpolation.

Let

$$V_h = \left\{ v_h \in C^0([a, b]) : v_h|_{K_i} \text{ is affine for every } i, v_h(a) = v_h(b) = 0 \right\}. \quad (4.176)$$

- (a) Prove that $\dim V_h = N - 1$ and that the interior hat functions $(\varphi_1, \dots, \varphi_{N-1})$ form a basis of V_h .
- (b) For v in $H_0^1(I)$, define the nodal interpolant $\Pi_h v$ and write it in the hat basis.

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma.

Let u in V and u_h in V_h be the solutions of the continuous and discrete problems.

- (a) Write the conforming Galerkin problem and prove that it admits a unique solution.
- (b) Establish the Galerkin orthogonality relation

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (4.177)$$

- (c) Assume that there exist $M > 0$ and $\alpha > 0$ such that

$$|a(w, v)| \leq M \|w\|_V \|v\|_V, \quad (4.178)$$

$$a(v, v) \geq \alpha \|v\|_V^2 \quad \text{for all } w, v \in V. \quad (4.179)$$

Prove Céa's lemma,

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (4.180)$$

- (d) In the Poisson case $c = 0$, show that u_h is the orthogonal projection of u onto V_h for the inner product

$$(w, v)_V = \int_a^b w'(x)v'(x) dx, \quad (4.181)$$

and deduce the best-approximation property.

Part 4. Interpolation estimate in the energy norm.

Let u in $H^2(I) \cap H_0^1(I)$.

- (a) Prove that, on each element K_i ,

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1}))}{h_i} = \frac{1}{h_i} \int_{K_i} u'(x) dx. \quad (4.182)$$

Deduce that

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}, \quad (u')_{K_i} := \frac{1}{h_i} \int_{K_i} u'(x) dx. \quad (4.183)$$

- (b) If $K = (\alpha, \beta)$ and z in $H^1(K)$ satisfies $\int_K z(x) dx = 0$, prove

$$\|z\|_{L^2(K)} \leq h_K |z|_{H^1(K)}, \quad h_K = \beta - \alpha. \quad (4.184)$$

- (c) Deduce

$$|u - \Pi_h u|_{H^1(I)} \leq h |u|_{H^2(I)}. \quad (4.185)$$

Part 5. A priori error estimate.

Use Parts 3 and 4 to prove

$$|u - u_h|_{H^1(I)} \leq \frac{M}{\alpha} h |u|_{H^2(I)}. \quad (4.186)$$

In the Poisson case $c = 0$, simplify the constant.

Part 6. Full assembly, Dirichlet enforcement, and explicit Poisson computation.

Assume first that $c = \gamma \geq 0$ is constant. Use the full nodal space W_h and its basis $(\hat{\varphi}_j)_{j=0}^N$, as in Section 4.6.1.

- Compute the local stiffness and reaction matrices on K_i . Write the local load vector in integral form.
- Assemble $\hat{A} \in \mathbb{R}^{(N+1) \times (N+1)}$ and $\hat{B} \in \mathbb{R}^{N+1}$ using all nodes $0, \dots, N$. Give all nonzero matrix entries on a general mesh, including the endpoint entries. If $f = f_0$ is constant, compute all load entries.
- Impose $U_0 = U_N = 0$ by zeroing the corresponding rows and columns, setting their diagonal entries to one, and modifying the right-hand side. State the size of the modified system. Then eliminate the two prescribed entries and identify the reduced system with the Galerkin problem on V_h .
- On a uniform mesh, write the reduced matrix A in tridiagonal form. Explain why both the full and reduced matrices are tridiagonal.
- Temporarily prescribe $U_0 = g_0$ and $U_N = g_N$. Determine the free right-hand side before zeroing the boundary columns. For $N = 3$, $I = (0, 1)$, $\gamma = 0$, $f = 1$, and $(g_0, g_N) = (0, 1)$, compute the corrected free right-hand side and the resulting nodal values. Explain what would go wrong if the column correction were omitted.

(f) For the explicit model problem

$$-u'' = 1 \quad \text{in } (0, 1), \quad u(0) = u(1) = 0, \quad (4.187)$$

with $x_i = ih$ and $h = 1/N$, compute the exact solution, derive the linear system for the nodal values, prove that $U_i = u(x_i)$ for $i = 1, \dots, N-1$, and compute $|u - u_h|_{H^1(0,1)}$ exactly.

Problem 4.2 (Neumann–Neumann boundary condition). Consider the pure Neumann Poisson problem

$$-u'' = f \quad \text{in } I, \quad -u'(a) = g_a, \quad u'(b) = g_b, \quad (4.188)$$

where f in $L^2(I)$ and g_a, g_b in $\mathbb{R}$.

Part 1. Variational setting and weak formulation.

(a) Show that a necessary compatibility condition is

$$\int_a^b f(x) dx + g_a + g_b = 0. \quad (4.189)$$

Assume this compatibility condition throughout the remainder of the problem.

(b) Define

$$V = H_\star^1(I) = \left\{ v \in H^1(I) : \int_a^b v(x) dx = 0 \right\}. \quad (4.190)$$

Use the Poincaré–Wirtinger inequality to show that $\|v\|_V = \|v'\|_{L^2(I)}$ is a norm on V , equivalent to the usual $H^1(I)$ norm.

(c) Derive the weak formulation

$$a_N(u, v) = \ell_N(v) \quad \text{for all } v \in V, \quad (4.191)$$

where

$$a_N(w, v) = \int_a^b w'(x) v'(x) dx, \quad (4.192)$$

$$\ell_N(v) = \int_a^b f(x) v(x) dx + g_a v(a) + g_b v(b). \quad (4.193)$$

(d) Prove existence and uniqueness in V by the Lax–Milgram theorem. Using compatibility, extend the variational identity from zero-mean tests to all tests in $H^1(I)$ by writing $v = v_\star + \bar{v}$. Explain why solutions of the original Neumann problem in $H^1(I)$ are unique only up to additive constants.

Part 2. Finite element space, hat basis, and nodal interpolation.

Define

$$W_h = \left\{ v_h \in C^0([a, b]) : v_h|_{K_i} \text{ is affine for every } i \right\}, \quad (4.194)$$

$$W_{h,\star} = \left\{ v_h \in W_h : \int_a^b v_h(x) dx = 0 \right\}. \quad (4.195)$$

(a) Define the full hat basis $(\varphi_0, \dots, \varphi_N)$ of W_h and prove that $\dim W_h = N + 1$.

(b) Explain why $\dim W_{h,\star} = N$.

- (c) For v in $C^0([a, b])$, define the full nodal interpolant $\Pi_h v$ in W_h . If v has zero mean, explain why $\Pi_h v$ need not have zero mean and how to correct it by subtracting its average.

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma.

- (a) Write the conforming Galerkin problem on $W_{h,\star}$ and prove that it admits a unique solution u_h .

- (b) Prove

$$a_N(u - u_h, v_h) = 0 \quad \text{for all } v_h \in W_{h,\star}. \quad (4.196)$$

- (c) Prove the best-approximation estimate

$$|u - u_h|_{H^1(I)} \leq \inf_{v_h \in W_{h,\star}} |u - v_h|_{H^1(I)}. \quad (4.197)$$

Part 4. Interpolation estimate in the energy norm.

Assume that u in $H^2(I) \cap H_\star^1(I)$.

- (a) Prove the same local derivative identity as in the Dirichlet case:

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}. \quad (4.198)$$

- (b) Set

$$v_h = \Pi_h u - \frac{1}{b-a} \int_a^b \Pi_h u(x) dx. \quad (4.199)$$

Show that v_h in $W_{h,\star}$ and that $(u - v_h)' = (u - \Pi_h u)'$.

- (c) Deduce

$$|u - v_h|_{H^1(I)} \leq h|u|_{H^2(I)}. \quad (4.200)$$

Part 5. A priori error estimate.

Use Parts 3 and 4 to prove

$$|u - u_h|_{H^1(I)} \leq h|u|_{H^2(I)}. \quad (4.201)$$

Part 6. Matrix form on a uniform mesh.

Assume that the mesh is uniform and that $f = f_0$ is constant.

- (a) Write the full stiffness matrix K in the basis $(\varphi_0, \dots, \varphi_N)$.
 (b) Write the load vector B .
 (c) Show that $K(1, \dots, 1)^T = 0$ and explain the role of the compatibility condition.
 (d) Write the discrete zero-mean constraint on the coefficient vector U .

Problem 4.3 (Mixed Dirichlet–Robin boundary condition). In this problem, let $I = (a, b)$ and consider

$$-u'' + cu = f \quad \text{in } I, \quad u(a) = 0, \quad u'(b) + d u(b) = g, \quad (4.202)$$

where $f \in L^2(I)$, $c \in L^\infty(I)$ with $c(x) \geq 0$ almost everywhere, $d \geq 0$, and $g \in \mathbb{R}$. The condition at b is a Robin boundary condition; it is also often called a Fourier boundary condition. Let

$$a = x_0 < x_1 < \dots < x_N = b \quad (4.203)$$

be a mesh of $\bar{I}$, and set

$$K_i = (x_{i-1}, x_i), \quad h_i = x_i - x_{i-1}, \quad h = \max_{1 \leq i \leq N} h_i. \quad (4.204)$$

Part 1. Variational setting and weak formulation.

- (a) Define

$$V = \{v \in H^1(I) : v(a) = 0\}. \quad (4.205)$$

Show that, for every $v \in V$,

$$\|v\|_{L^2(I)} \leq (b-a) \|v'\|_{L^2(I)} \quad \text{and} \quad |v(b)| \leq \sqrt{b-a} \|v'\|_{L^2(I)}. \quad (4.206)$$

Deduce that

$$\|v\|_V := \|v'\|_{L^2(I)} \quad (4.207)$$

is a norm on V , equivalent to the usual $H^1(I)$ norm.

- (b) Derive the weak formulation of the boundary value problem. More precisely, show that if u is a smooth solution, then

$$a(u, v) = \ell(v) \quad \text{for all } v \in V, \quad (4.208)$$

where you must determine the bilinear form $a(\cdot, \cdot)$ and the linear form ℓ .

- (c) Show that a is continuous and coercive on V , and that ℓ is continuous on V for the norm $\|\cdot\|_V$. Deduce that the weak problem admits a unique solution in V .

Part 2. Finite element space, hat basis, and nodal interpolation.

- (a) Define the conforming P_1 space adapted to this problem:

$$V_h = \{v_h \in C^0([a, b]) : v_h|_{K_i} \text{ is affine for every } i, v_h(a) = 0\}. \quad (4.209)$$

Why is $\dim V_h = N$?

- (b) For $i = 1, \dots, N$, define the hat function $\varphi_i \in V_h$ by the nodal conditions

$$\varphi_i(x_j) = \delta_{ij} \quad \text{for } j = 1, \dots, N. \quad (4.210)$$

Write explicitly the endpoint hat function φ_N , and prove that $(\varphi_1, \dots, \varphi_N)$ is a basis of V_h .

- (c) For $v \in C^0([a, b]) \cap V$, define the nodal interpolant $\Pi_h v \in V_h$ and write it in the hat basis. What changes compared with the Dirichlet–Dirichlet case?

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma.

- (a) Write the conforming Galerkin problem associated with the weak formulation, and prove that it admits a unique solution $u_h \in V_h$.
- (b) Prove the Galerkin orthogonality relation

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (4.211)$$

- (c) Let M and α denote continuity and coercivity constants of a on V . Prove Céa's estimate

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (4.212)$$

Part 4. Interpolation estimate in the energy norm.

Assume now that $u \in H^2(I) \cap V$.

- (a) Show that, on each element $K_i = (x_{i-1}, x_i)$,

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1}))}{h_i} = \frac{1}{h_i} \int_{K_i} u'(x) dx. \quad (4.213)$$

Deduce that

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}, \quad (u')_{K_i} := \frac{1}{h_i} \int_{K_i} u'(x) dx. \quad (4.214)$$

- (b) Let $K = (\alpha, \beta)$ be an interval of length $h_K = \beta - \alpha$. Prove that if $z \in H^1(K)$ and

$$\int_K z(x) dx = 0, \quad (4.215)$$

then

$$\|z\|_{L^2(K)} \leq h_K \|z'\|_{L^2(K)}. \quad (4.216)$$

- (c) Deduce the interpolation estimate

$$\|u - \Pi_h u\|_V \leq h \|u''\|_{L^2(I)}. \quad (4.217)$$

Part 5. A priori error estimate.

Use Parts 3 and 4 to prove that

$$\|u - u_h\|_V \leq \frac{M}{\alpha} h \|u''\|_{L^2(I)}. \quad (4.218)$$

Then use the estimates from Part 1 to show that we may take

$$M = 1 + \|c\|_{L^\infty(I)} (b-a)^2 + d(b-a), \quad \alpha = 1. \quad (4.219)$$

Part 6. Matrix form on a uniform mesh.

Assume now that the mesh is uniform,

$$x_i = a + ih, \quad h = \frac{b-a}{N}, \quad (4.220)$$

and that $c = c_0 \geq 0$ and $f = f_0$ are constant. Write

$$u_h = \sum_{j=1}^N U_j \varphi_j. \quad (4.221)$$

- (a) Show that the linear system satisfied by $U = (U_1, \dots, U_N)^T$ has the form

$$AU = B, \quad (4.222)$$

where A is symmetric tridiagonal.

- (b) Compute the non-zero entries of A and B . In particular, determine the interior formulas and the last row corresponding to the endpoint degree of freedom at b .
- (c) Which entries differ from the homogeneous Dirichlet–Dirichlet case, and why?
- (d) Assemble the volume and Robin contributions on the full nodal space W_h before imposing $U_0 = 0$. Impose this Dirichlet condition by a symmetric row-and-column modification, then extract the reduced system. Explain why the Robin node $x_N = b$ remains free and identify the resulting matrix and vector with those computed above.

Chapter 5

Two-dimensional finite elements

5.1 Local finite elements and nodal bases

The finite element method is built from local data. Before introducing a mesh or a boundary value problem, we describe the local compact element, the local polynomial space, the local degrees of freedom, and the corresponding local nodal basis.

Throughout this chapter, the letters have the following roles.

- K denotes a generic compact element of $\mathbb{R}^2$.
- R denotes a rectangle.
- T denotes a triangle.

Throughout this chapter, all rectangles have edges parallel to the coordinate axes, so they are of the form $R = [x_0, x_1] \times [y_0, y_1]$. Their reference maps scale and translate the two coordinates separately. With this convention, the Cartesian definition of $Q_k(R)$ below agrees with transport from the reference square.

Definition 5.1. A finite element is a triple

$$(K, P, \Sigma), \quad (5.1)$$

where K is a compact subset of $\mathbb{R}^2$ with nonempty interior, P is a finite-dimensional space of functions on K , and

$$\Sigma = (\sigma_1, \dots, \sigma_M) \quad (5.2)$$

is a family of linear forms on P with $M = \dim P$. The linear forms σ_i are called the local degrees of freedom.

Definition 5.2. The family $\Sigma = (\sigma_1, \dots, \sigma_M)$ is unisolvent on P if

$$(\sigma_i(p) = 0 \text{ for all } i = 1, \dots, M) \implies p = 0. \quad (5.3)$$

Equivalently, the map

$$P \rightarrow \mathbb{R}^M, \quad p \mapsto (\sigma_1(p), \dots, \sigma_M(p)) \quad (5.4)$$

is an isomorphism.

Definition 5.3. Assume that Σ is unisolvent on P . The local nodal basis associated with (K, P, Σ) is the family $(p_1, \dots, p_M)$ in P characterized by

$$\sigma_i(p_j) = \delta_{ij} \quad \text{for } i, j = 1, \dots, M. \quad (5.5)$$

Proposition 5.4. *If Σ is unisolvent on P , then the local nodal basis exists and is unique. Moreover, every p in P has the expansion*

$$p = \sum_{j=1}^M \sigma_j(p) p_j. \quad (5.6)$$

Proof. The map

$$P \rightarrow \mathbb{R}^M, \quad p \mapsto (\sigma_1(p), \dots, \sigma_M(p)) \quad (5.7)$$

is an isomorphism by unisolvence. For each canonical vector e_j of $\mathbb{R}^M$, there is a unique p_j in P whose image is e_j . This gives $\sigma_i(p_j) = \delta_{ij}$ and proves existence and uniqueness. If

$$q = p - \sum_{j=1}^M \sigma_j(p) p_j, \quad (5.8)$$

then $\sigma_i(q) = 0$ for every i . Unisolvence gives $q = 0$, which proves the expansion. $\square$

Definition 5.5. A Lagrange finite element is a finite element for which the degrees of freedom are point values. Thus we are given $M = \dim P$ distinct nodes

$$a_1, \dots, a_M \in K, \quad (5.9)$$

and

$$\sigma_i(p) = p(a_i), \quad i = 1, \dots, M. \quad (5.10)$$

When these point evaluations are unisolvent on P , the local nodal basis is characterized by

$$p_j(a_i) = \delta_{ij} \quad \text{for } i, j = 1, \dots, M. \quad (5.11)$$

The word nodal has a precise meaning in Definition 5.5: the coefficients of the local expansion are exactly the nodal values.

5.2 Polynomial spaces and reference elements

Definition 5.6. The polynomial spaces used in this chapter are the following.

- On a rectangle R , we define

$$Q_k(R) = \{p(x_1, x_2) \text{ polynomial of degree at most } k \text{ in each variable}\}. \quad (5.12)$$

In particular,

$$Q_1(R) = \{a_0 + a_1 x_1 + a_2 x_2 + a_3 x_1 x_2\}. \quad (5.13)$$

The rectangular Q_1 element is bilinear.

- On a triangle T , we define

$$P_k(T) = \{p(x_1, x_2) \text{ polynomial of total degree at most } k\}. \quad (5.14)$$

In particular,

$$P_1(T) = \{a_0 + a_1 x_1 + a_2 x_2\}. \quad (5.15)$$

The triangular P_1 element is affine.

The dimensions are

$$\dim Q_k(R) = (k+1)^2, \quad \dim P_k(T) = \frac{(k+1)(k+2)}{2}. \quad (5.16)$$

Indeed, $Q_k(R)$ is spanned by the monomials $x_1^r x_2^s$ with $0 \leq r, s \leq k$, whereas $P_k(T)$ is spanned by the monomials $x_1^r x_2^s$ with $r+s \leq k$.

Definition 5.7. Let $\widehat{K}$ be a reference compact element and let

$$F_K : \widehat{K} \rightarrow K, \quad F_K(\widehat{x}) = B_K \widehat{x} + b_K, \quad (5.17)$$

be an affine bijection. If $(\widehat{p}_r)$ is a local basis on $\widehat{K}$, the corresponding local basis on K is defined by

$$p_{K,r}(x) = \widehat{p}_r(F_K^{-1}(x)). \quad (5.18)$$

If $\widehat{a}_r$ is a reference node, the corresponding physical node is

$$a_{K,r} = F_K(\widehat{a}_r). \quad (5.19)$$

This transport from a reference compact element is the two-dimensional analogue of the construction used in dimension one on a reference interval.

5.3 Rectangular Lagrange finite elements

We start with rectangles because their local basis functions are tensor products of one-dimensional Lagrange polynomials.

Definition 5.8. The reference rectangle is

$$\widehat{R} = [0, 1]^2. \quad (5.20)$$

For an integer $k \geq 1$, the canonical rectangular Q_k nodes on $\widehat{R}$ are

$$\widehat{a}_{ij} = \left(\frac{i}{k}, \frac{j}{k} \right), \quad 0 \leq i, j \leq k. \quad (5.21)$$

They form a uniform tensor-product lattice with $k+1$ points in each coordinate direction. The local degrees of freedom are the values at these nodes.

There are $(k+1)^2$ canonical Q_k nodes. This number matches the dimension of $Q_k(\widehat{R})$ because the monomials

$$\xi^r \eta^s, \quad 0 \leq r \leq k, \quad 0 \leq s \leq k, \quad (5.22)$$

form a basis of $Q_k(\widehat{R})$ and are also counted by $(k+1)^2$.

For $k=1$, the four reference nodes are

$$\widehat{S}_1 = (0, 0), \quad \widehat{S}_2 = (1, 0), \quad \widehat{S}_3 = (1, 1), \quad \widehat{S}_4 = (0, 1). \quad (5.23)$$

The Q_1 reference shape functions are

$$\widehat{p}_1(\xi, \eta) = (1 - \xi)(1 - \eta), \quad \widehat{p}_2(\xi, \eta) = \xi(1 - \eta), \quad (5.24)$$

$$\widehat{p}_3(\xi, \eta) = \xi\eta, \quad \widehat{p}_4(\xi, \eta) = (1 - \xi)\eta. \quad (5.25)$$

Proposition 5.9. The four functions $(\widehat{p}_i)_{i=1,\dots,4}$ form the local nodal basis of the rectangular Q_1 element on $\widehat{R}$.

Proof. The identities

$$\hat{p}_i(\hat{S}_j) = \delta_{ij} \quad \text{for } i, j = 1, \dots, 4 \quad (5.26)$$

follow directly from the formulas. Let q in $Q_1(\hat{R})$ vanish at the four vertices. We write

$$q(\xi, \eta) = a_0 + a_1\xi + a_2\eta + a_3\xi\eta. \quad (5.27)$$

The values at $(0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$ give successively $a_0 = 0$, $a_1 = 0$, $a_2 = 0$, and $a_3 = 0$. Hence $q = 0$. The four point values are unisolvent, and the displayed functions are the local nodal basis. $\square$

For general k , let $\ell_0, \dots, \ell_k$ be the one-dimensional Lagrange polynomials on $[0, 1]$ associated with the nodes $0, 1/k, \dots, 1$:

$$\ell_i(s) = \prod_{m=0, m \neq i}^k \frac{s - m/k}{i/k - m/k}, \quad 0 \leq i \leq k. \quad (5.28)$$

Thus $\ell_i(m/k) = \delta_{im}$ for $0 \leq i, m \leq k$.

Proposition 5.10. *The functions*

$$\hat{p}_{ij}(\xi, \eta) = \ell_i(\xi)\ell_j(\eta), \quad 0 \leq i, j \leq k, \quad (5.29)$$

form the local nodal basis of the rectangular Q_k Lagrange element on $\hat{R}$.

On a physical rectangle

$$R = [x_0, x_1] \times [y_0, y_1], \quad (5.30)$$

the canonical Q_k nodes are

$$a_{R,ij} = \left(x_0 + \frac{i}{k}(x_1 - x_0), y_0 + \frac{j}{k}(y_1 - y_0) \right), \quad 0 \leq i, j \leq k, \quad (5.31)$$

and the corresponding local nodal basis is

$$p_{R,ij}(x, y) = \ell_i\left(\frac{x - x_0}{x_1 - x_0}\right) \ell_j\left(\frac{y - y_0}{y_1 - y_0}\right). \quad (5.32)$$

Proof. At the reference node $\hat{a}_{mn}$,

$$\hat{p}_{ij}(\hat{a}_{mn}) = \ell_i(m/k)\ell_j(n/k) = \delta_{im}\delta_{jn}. \quad (5.33)$$

The family contains $(k+1)^2$ functions, which is $\dim Q_k(\hat{R})$. Hence the Kronecker relations give a local nodal basis. The formula on R is obtained by composing the reference functions with the affine coordinate map from R to $\hat{R}$. $\square$

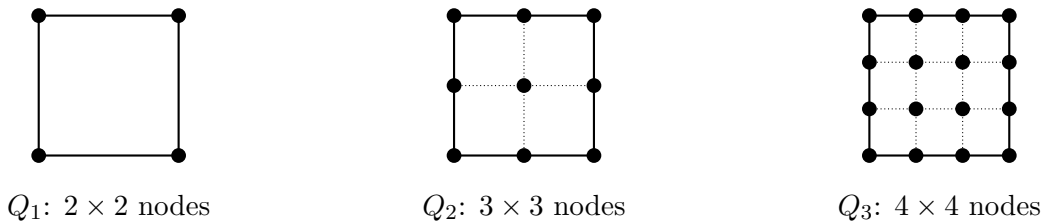


Figure 5.1: Canonical rectangular Q_k Lagrange nodes form a tensor-product lattice. There are $(k+1)^2 = \dim Q_k$ nodes.

5.4 Barycentric coordinates and triangular Lagrange finite elements

We now turn to triangular elements. Barycentric coordinates provide the natural affine coordinates on a triangle and make the formulas for triangular Lagrange bases particularly simple.

Definition 5.11. Let T be a nondegenerate triangle with vertices a_1, a_2, a_3 . The barycentric coordinates on T are the three affine functions $\lambda_1, \lambda_2, \lambda_3$ characterized by

$$\lambda_i(a_j) = \delta_{ij} \quad \text{for } i, j = 1, 2, 3. \quad (5.34)$$

Proposition 5.12. For every x in T , the barycentric coordinates satisfy

$$\lambda_1(x) + \lambda_2(x) + \lambda_3(x) = 1 \quad (5.35)$$

and

$$x = \lambda_1(x)a_1 + \lambda_2(x)a_2 + \lambda_3(x)a_3. \quad (5.36)$$

They are uniquely determined by these two relations.

Proof. The affine map

$$F : \hat{T} \rightarrow T, \quad F(\xi, \eta) = a_1 + (a_2 - a_1)\xi + (a_3 - a_1)\eta, \quad (5.37)$$

transports the reference triangle

$$\hat{T} = \{(\xi, \eta) : \xi \geq 0, \eta \geq 0, \xi + \eta \leq 1\} \quad (5.38)$$

to T . On $\hat{T}$, the barycentric coordinates are

$$\hat{\lambda}_1 = 1 - \xi - \eta, \quad \hat{\lambda}_2 = \xi, \quad \hat{\lambda}_3 = \eta. \quad (5.39)$$

Composing with F^{-1} gives the coordinates on T . The displayed relations follow from this construction. Conversely, if three affine functions satisfy these relations, their values at the three vertices are fixed, hence the affine functions are fixed. $\square$

The determinant formulas are useful in computations. With

$$\Delta_T = \det(a_2 - a_1, a_3 - a_1), \quad (5.40)$$

we have

$$\lambda_1(x) = \frac{\det(a_2 - x, a_3 - x)}{\Delta_T}, \quad (5.41)$$

with the two analogous cyclic formulas for λ_2 and λ_3 . The area of T is $|T| = |\Delta_T|/2$.

Proposition 5.13. For the triangular P_1 Lagrange element with canonical nodes a_1, a_2, a_3 , the local nodal basis is

$$p_1 = \lambda_1, \quad p_2 = \lambda_2, \quad p_3 = \lambda_3. \quad (5.42)$$

Proof. Each λ_i is affine, hence belongs to $P_1(T)$. By definition,

$$\lambda_i(a_j) = \delta_{ij}. \quad (5.43)$$

Since $\dim P_1(T) = 3$, these three functions form the local nodal basis. $\square$

Definition 5.14. For an integer $k \geq 1$, the canonical triangular P_k nodes on T are indexed by

$$\mathcal{A}_k = \{\alpha = (\alpha_1, \alpha_2, \alpha_3) \in \{0, \dots, k\}^3 : \alpha_1 + \alpha_2 + \alpha_3 = k\}. \quad (5.44)$$

The node associated with α is the point a_α with barycentric coordinates

$$\lambda_i(a_\alpha) = \frac{\alpha_i}{k}, \quad i = 1, 2, 3. \quad (5.45)$$

Equivalently,

$$a_\alpha = \frac{\alpha_1}{k}a_1 + \frac{\alpha_2}{k}a_2 + \frac{\alpha_3}{k}a_3. \quad (5.46)$$

Let $N_k(T)$ denote the number of canonical triangular P_k nodes on T . Then

$$N_k(T) = \#\mathcal{A}_k = \frac{(k+1)(k+2)}{2} = \dim P_k(T). \quad (5.47)$$

Indeed, once α_1 is fixed, with $0 \leq \alpha_1 \leq k$, the index α_2 may take $k - \alpha_1 + 1$ values, and then α_3 is fixed by $\alpha_3 = k - \alpha_1 - \alpha_2$. Hence

$$\#\mathcal{A}_k = \sum_{\alpha_1=0}^k (k - \alpha_1 + 1) = 1 + 2 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}. \quad (5.48)$$

This is also the dimension of $P_k(T)$, since the monomials $x_1^r x_2^s$ with $r + s \leq k$ are counted by the same triangular sum.

The position of a canonical triangular node is determined by the zero entries of its index.

- The three vertices correspond to $(k, 0, 0)$, $(0, k, 0)$, and $(0, 0, k)$.
- The edge nodes are the nodes for which one barycentric coordinate is zero.
- The interior nodes are the nodes for which the three barycentric coordinates are positive. They appear for $k \geq 3$.

Proposition 5.15. *The local nodal basis function associated with α in $\mathcal{A}_k$ is*

$$p_\alpha(x) = \prod_{i=1}^3 \prod_{r=0}^{\alpha_i-1} \frac{k\lambda_i(x) - r}{\alpha_i - r}, \quad (5.49)$$

with the convention that an empty product is equal to 1.

Proof. The degree of p_α is $\alpha_1 + \alpha_2 + \alpha_3 = k$, so p_α belongs to $P_k(T)$. Let β in $\mathcal{A}_k$. At the node a_β , we have $k\lambda_i(a_\beta) = \beta_i$, hence

$$p_\alpha(a_\beta) = \prod_{i=1}^3 \prod_{r=0}^{\alpha_i-1} \frac{\beta_i - r}{\alpha_i - r}. \quad (5.50)$$

If $\beta = \alpha$, every factor is equal to 1. If $\beta \neq \alpha$, then, since $\sum_i \alpha_i = \sum_i \beta_i = k$, there exists i such that $\beta_i < \alpha_i$. The factor with $r = \beta_i$ occurs in the product and is zero. Hence

$$p_\alpha(a_\beta) = \delta_{\alpha\beta}. \quad (5.51)$$

The number of functions equals $\dim P_k(T)$, so they form the local nodal basis. $\square$

Remark 5.16. The nodal basis in Proposition 5.15 should be distinguished from the Bernstein basis

$$\frac{k!}{\alpha_1! \alpha_2! \alpha_3!} \lambda_1^{\alpha_1} \lambda_2^{\alpha_2} \lambda_3^{\alpha_3}, \quad \alpha \in \mathcal{A}_k. \quad (5.52)$$

The Bernstein functions form a basis of $P_k(T)$. For $k \geq 2$, this basis does not satisfy the Kronecker nodal relations at the canonical Lagrange nodes. For $k = 1$, the Bernstein and Lagrange bases both consist of the three barycentric coordinates.

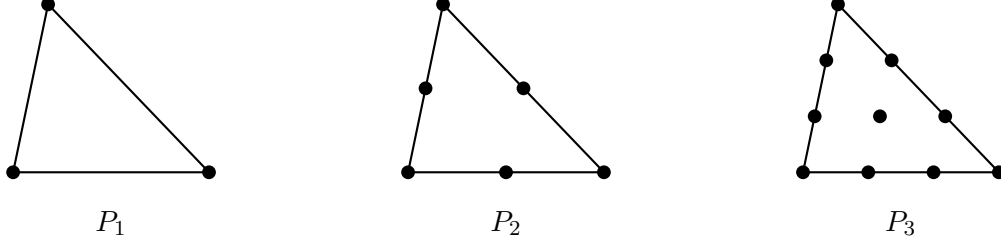


Figure 5.2: Canonical triangular Lagrange nodes for P_1 , P_2 , and P_3 .

Proposition 5.17. *For the triangular P_2 Lagrange element, the six functions*

$$p_1 = \lambda_1(2\lambda_1 - 1), \quad p_{1,2} = 4\lambda_1\lambda_2, \quad p_2 = \lambda_2(2\lambda_2 - 1), \quad (5.53)$$

$$p_3 = \lambda_3(2\lambda_3 - 1), \quad p_{2,3} = 4\lambda_2\lambda_3, \quad p_{3,1} = 4\lambda_3\lambda_1 \quad (5.54)$$

form the local nodal basis of the triangular P_2 element.

Proof. The six P_2 nodes have barycentric coordinates

$$(1, 0, 0), \quad (1/2, 1/2, 0), \quad (0, 1, 0), \quad (0, 0, 1), \quad (0, 1/2, 1/2), \quad (1/2, 0, 1/2). \quad (5.55)$$

At a vertex, one barycentric coordinate is 1 and the other two are 0; at an edge midpoint, two barycentric coordinates are 1/2 and the remaining one is 0. Therefore each displayed function is equal to 1 at its own node and to 0 at the five other nodes. These are exactly the Kronecker relations for the six nodes. Since $\dim P_2(T) = 6$, the six functions form the local nodal basis. $\square$

5.5 The homogeneous Dirichlet model problem

We now introduce the model problem. Let Ω be a bounded polygonal domain in $\mathbb{R}^2$. We consider

$$-\Delta u + cu = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (5.56)$$

where f in $L^2(\Omega)$ and c in $L^\infty(\Omega)$ with $c(x) \geq 0$ almost everywhere.

We set

$$V = H_0^1(\Omega), \quad (5.57)$$

and we use the energy norm

$$\|v\|_{H_0^1(\Omega)} := \|\nabla v\|_{L^2(\Omega)}. \quad (5.58)$$

The weak formulation is: find u in V such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V, \quad (5.59)$$

where

$$a(w, v) = \int_{\Omega} (\nabla w \cdot \nabla v + cwv) \, dx, \quad (5.60)$$

$$\ell(v) = \int_{\Omega} f v \, dx. \quad (5.61)$$

5.6 Meshes, conforming spaces, and global degrees of freedom

A mesh of $\bar{\Omega}$ is a finite family of closed compact elements K such that

$$\bar{\Omega} = \bigcup_{K \in \mathcal{T}_h} K. \quad (5.62)$$

The mesh size is

$$h = \max_{K \in \mathcal{T}_h} \text{diam}(K). \quad (5.63)$$

Definition 5.18. A mesh is admissible if the intersection of two distinct elements is either empty, a whole common edge, or a common vertex.

Definition 5.19. A family $(\mathcal{T}_h)_{h>0}$ of meshes is regular if the elements do not degenerate when $h \rightarrow 0$.

- For rectangles, this means that the aspect ratios stay uniformly bounded.
- For triangles, this means that the angles stay away from 0 uniformly.

For a fixed degree k , the conforming Lagrange spaces are defined as follows.

- On a rectangular mesh,

$$W_h = \{v_h \in C^0(\bar{\Omega}) : v_h|_R \text{ in } Q_k(R) \text{ for every rectangle } R \in \mathcal{T}_h\}, \quad (5.64)$$

$$V_h = \{v_h \in W_h : v_h = 0 \text{ on } \partial\Omega\}. \quad (5.65)$$

- On a triangular mesh,

$$W_h = \{v_h \in C^0(\bar{\Omega}) : v_h|_T \text{ in } P_k(T) \text{ for every triangle } T \in \mathcal{T}_h\}, \quad (5.66)$$

$$V_h = \{v_h \in W_h : v_h = 0 \text{ on } \partial\Omega\}. \quad (5.67)$$

The full space W_h contains the degrees of freedom at all nodes, including boundary nodes. The space V_h contains the admissible trial and test functions for the homogeneous Dirichlet problem. These are compatible descriptions: one may assemble on W_h and impose the Dirichlet constraints algebraically, or incorporate the same constraints during assembly. We will make their equivalence explicit below, following Section 4.6.1.

5.6.1 Shared edges and global conformity

Proposition 5.20. *Let two elements share an edge e . Assume that each element is either a triangular P_k Lagrange element or a rectangular Q_k Lagrange element. If the two local polynomials have the same values at the $k+1$ canonical Lagrange nodes on e , then their traces on e are equal.*

Proof. For a triangular P_k polynomial, the trace on an edge is a one-dimensional polynomial of degree at most k . For a rectangular Q_k polynomial obtained by transport from the reference rectangle, one reference coordinate is constant on each edge, and the trace is again a one-dimensional polynomial of degree at most k . Therefore the difference of the two traces is a one-dimensional polynomial of degree at most k . It vanishes at the $k+1$ common Lagrange nodes on the edge, hence it vanishes identically on e . $\square$

Lemma 5.21. *Let $\mathcal{T}_h$ be an admissible polygonal mesh of Ω . Assume that v belongs to $H^1(K)$ on every compact element K of the mesh and that the traces of v coincide on every common edge. Then v belongs to $H^1(\Omega)$.*

Proof. We prove the statement at the level of distributional derivatives. Let φ in $C_c^\infty(\Omega)$. On each element K , integration by parts gives

$$\int_K v \partial_i \varphi \, dx = - \int_K \partial_i v \varphi \, dx + \int_{\partial K} v \varphi n_i^K \, ds, \quad (5.68)$$

where n^K is the outward normal to K . We sum over all elements. Boundary terms on interior edges cancel because the two neighboring elements have opposite normals and the same trace of v on the common edge. The terms on $\partial\Omega$ vanish because φ has compact support in Ω . Hence

$$\int_\Omega v \partial_i \varphi \, dx = - \sum_{K \in \mathcal{T}_h} \int_K \partial_i v \varphi \, dx. \quad (5.69)$$

The distributional derivative $\partial_i v$ is therefore the piecewise L^2 function obtained by taking the derivative on each element. Thus $\partial_i v$ belongs to $L^2(\Omega)$ for $i = 1, 2$, and v belongs to $H^1(\Omega)$. $\square$

Proposition 5.22. *On an admissible rectangular Q_k mesh or triangular P_k mesh, shared nodal values produce a globally continuous finite element function. Consequently, such a function belongs to $H^1(\Omega)$.*

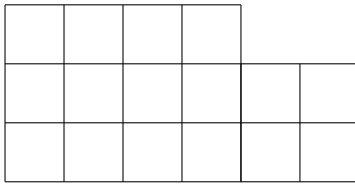
Proof. Let e be a common edge of two elements. The two traces on e have the same values at the $k + 1$ canonical Lagrange nodes of e , so Proposition 5.20 gives equality of the traces on the whole edge. The function is continuous across every interior edge. Since it is polynomial on each element, it is piecewise H^1 . Lemma 5.21 gives membership in $H^1(\Omega)$. $\square$

Proposition 5.23. *Assume that the finite element function is built from rectangular Q_k or triangular P_k Lagrange elements and that all canonical boundary nodal values are zero. Then its trace is zero on $\partial\Omega$. In particular, if the function is globally conforming, then it belongs to $H_0^1(\Omega)$.*

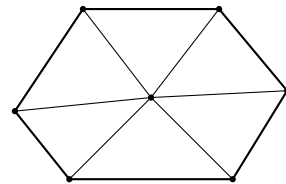
Proof. On each boundary edge, the trace is a one-dimensional polynomial of degree at most k . It has zero values at the $k + 1$ canonical Lagrange nodes of that edge. Hence the trace is identically zero on the whole edge. Since the boundary of the polygonal domain is a union of such edges, the trace is zero on $\partial\Omega$. Together with Proposition 5.22, this gives membership in $H_0^1(\Omega)$. $\square$

Consequently, the space V_h defined above satisfies

$$V_h \subset H_0^1(\Omega). \quad (5.70)$$



A rectangular mesh



A triangular mesh

Figure 5.3: Admissible meshes in dimension two.

5.6.2 Free nodes, global basis functions, and local-to-global maps

Let $Z_1, \dots, Z_{N_{\text{all}}}$ be all global canonical Lagrange nodes of the mesh. The full nodal basis $(\hat{w}_i)_{i=1, \dots, N_{\text{all}}}$ of W_h is characterized by

$$\hat{w}_i(Z_j) = \delta_{ij}, \quad 1 \leq i, j \leq N_{\text{all}}. \quad (5.71)$$

Every $v_h \in W_h$ has the expansion

$$v_h = \sum_{i=1}^{N_{\text{all}}} v_h(Z_i) \hat{w}_i. \quad (5.72)$$

For the homogeneous Dirichlet problem, the free nodes are the nodes that do not lie on $\partial\Omega$:

$$\mathcal{I}_0 = \{i : Z_i \notin \partial\Omega\}. \quad (5.73)$$

Write $\mathcal{F} = \mathcal{I}_0$ and $\mathcal{D} = \{1, \dots, N_{\text{all}}\} \setminus \mathcal{F}$ for the free and Dirichlet node indices. Let $N = \#\mathcal{F}$. The active numbering map is

$$\text{act} : \{1, \dots, N_{\text{all}}\} \rightarrow \{0, 1, \dots, N\}. \quad (5.74)$$

If Z_i is a free node and is the m th free node in the chosen ordering, then $\text{act}(i) = m$. If Z_i is a boundary node, then $\text{act}(i) = 0$.

The global nodal basis of V_h is denoted by $w_1, \dots, w_N$. If Z_i is free and $\text{act}(i) = m$, then

$$w_m(Z_i) = 1, \quad w_m(Z_j) = 0 \text{ for every free node } Z_j \neq Z_i. \quad (5.75)$$

All global basis functions vanish at boundary nodes. Every v_h in V_h has the nodal expansion

$$v_h = \sum_{m=1}^N v_h(Z_{i_m}) w_m, \quad (5.76)$$

where Z_{i_m} is the free node with active number m . In particular, $w_m = \hat{w}_{i_m}$: the reduced basis is the part of the full nodal basis attached to free nodes.

Let $z_{K,1}, \dots, z_{K,n_{\text{loc}}}$ be the local nodes of a compact element K . The local-to-global connectivity map of K is

$$\text{lg}_K : \{1, \dots, n_{\text{loc}}\} \rightarrow \{1, \dots, N_{\text{all}}\} \quad (5.77)$$

defined by

$$z_{K,r} = Z_{\text{lg}_K(r)}. \quad (5.78)$$

Thus $\text{lg}_K(r)$ is the global number of the r th local node of K .

The connection between the global basis and the local shape functions is as follows. If $p_{K,r}$ is the r th local nodal basis function on K , then

$$w_m|_K = \sum_{\substack{1 \leq r \leq n_{\text{loc}} \\ \text{act}(\text{lg}_K(r))=m}} p_{K,r}. \quad (5.79)$$

The sum is either empty or contains a single term. Thus, on a given compact element, a global basis function is either zero or equal to the local basis function attached to the corresponding local node. The same global basis function usually has nonzero restrictions on several neighboring elements.

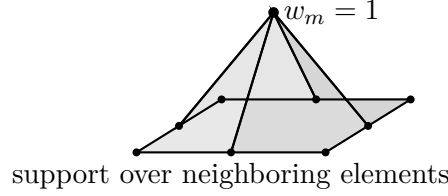


Figure 5.4: A global nodal basis function is assembled from several local shape functions. The picture represents a typical basis function attached to an interior node and nonzero on the neighboring elements.

5.6.3 A two-rectangle local-to-global example

Consider two Q_1 rectangles

$$R_1 = [0, 1] \times [0, 1], \quad R_2 = [1, 2] \times [0, 1], \quad (5.80)$$

with global nodes

$$Z_1 = (0, 0), \quad Z_2 = (1, 0), \quad Z_3 = (2, 0), \quad Z_4 = (0, 1), \quad Z_5 = (1, 1), \quad Z_6 = (2, 1). \quad (5.81)$$

With the local Q_1 ordering bottom-left, bottom-right, top-right, top-left, the local-to-global maps are

$$\text{lg}_{R_1} = (1, 2, 5, 4), \quad \text{lg}_{R_2} = (2, 3, 6, 5). \quad (5.82)$$

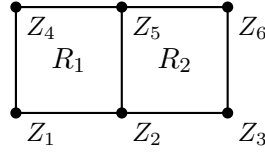


Figure 5.5: Two rectangles sharing one vertical edge. The shared global nodes are Z_2 and Z_5 .

Let $E^{(1)} = (E_{rs}^{(1)})_{1 \leq r, s \leq 4}$ and $E^{(2)} = (E_{rs}^{(2)})_{1 \leq r, s \leq 4}$ be the two local element matrices. If no boundary condition is eliminated, the assembled 6×6 matrix is

$$\begin{pmatrix} E_{11}^{(1)} & E_{12}^{(1)} & 0 & E_{14}^{(1)} & E_{13}^{(1)} & 0 \\ E_{21}^{(1)} & E_{22}^{(1)} + E_{11}^{(2)} & E_{12}^{(2)} & E_{24}^{(1)} & E_{23}^{(1)} + E_{14}^{(2)} & E_{13}^{(2)} \\ 0 & E_{21}^{(2)} & E_{22}^{(2)} & 0 & E_{24}^{(2)} & E_{23}^{(2)} \\ E_{41}^{(1)} & E_{42}^{(1)} & 0 & E_{44}^{(1)} & E_{43}^{(1)} & 0 \\ E_{31}^{(1)} & E_{32}^{(1)} + E_{41}^{(2)} & E_{42}^{(2)} & E_{34}^{(1)} & E_{33}^{(1)} + E_{44}^{(2)} & E_{43}^{(2)} \\ 0 & E_{31}^{(2)} & E_{32}^{(2)} & 0 & E_{34}^{(2)} & E_{33}^{(2)} \end{pmatrix}. \quad (5.83)$$

The load vector is assembled in the same way:

$$\begin{pmatrix} F_1^{(1)} \\ F_2^{(1)} + F_1^{(2)} \\ F_2^{(2)} \\ F_4^{(1)} \\ F_3^{(1)} + F_4^{(2)} \\ F_3^{(2)} \end{pmatrix}. \quad (5.84)$$

The entries attached to the shared nodes receive contributions from both rectangles. This is the local-to-global assembly mechanism in its simplest two-dimensional form.

5.7 Global Lagrange interpolation estimates

The preceding construction of global nodes makes the interpolation operator global. Let u be continuous on $\bar{\Omega}$. For each compact element K , let $z_{K,1}, \dots, z_{K,n_{\text{loc}}}$ be its local nodes and let $p_{K,1}, \dots, p_{K,n_{\text{loc}}}$ be the corresponding local nodal basis. The nodal interpolant $\Pi_h u$ is defined element by element by

$$(\Pi_h u)|_K = \sum_{r=1}^{n_{\text{loc}}} u(z_{K,r}) p_{K,r}. \quad (5.85)$$

On two neighboring elements, the two local interpolants have the same values at the common edge nodes. By Proposition 5.20, their traces coincide on the whole common edge. Thus $\Pi_h u$ is globally continuous.

If $u = 0$ on $\partial\Omega$, then $\Pi_h u$ vanishes at all boundary nodes. Proposition 5.23 gives $\Pi_h u$ in V_h .

In dimension two, nodal interpolation is not defined for arbitrary H^1 functions, since $H^1(\Omega)$ is not embedded in $C^0(\bar{\Omega})$. For the estimates below, u in $H^{k+1}(\Omega)$ is sufficient because $k \geq 1$ and $H^{k+1}(\Omega) \hookrightarrow C^0(\bar{\Omega})$ in dimension two.

The notation $|\cdot|_{H^m(\Omega)}$ denotes the Sobolev seminorm introduced in Definition 2.4.

Theorem 5.24. *Let $(\mathcal{T}_h)_{h>0}$ be a regular family of admissible meshes of Ω .*

- *On triangular meshes, let Π_h be the nodal interpolation operator associated with the P_k Lagrange element.*
- *On rectangular meshes, let Π_h be the nodal interpolation operator associated with the Q_k Lagrange element.*

Then, for $m = 0, 1$, there exists a constant $C > 0$, independent of h and of u , such that

$$|u - \Pi_h u|_{H^m(\Omega)} \leq Ch^{k+1-m} |u|_{H^{k+1}(\Omega)} \quad \text{for all } u \in H^{k+1}(\Omega). \quad (5.86)$$

In particular,

$$\|u - \Pi_h u\|_{H_0^1(\Omega)} \leq Ch^k \|u\|_{H^{k+1}(\Omega)} \quad \text{for all } u \in H^{k+1}(\Omega) \cap H_0^1(\Omega). \quad (5.87)$$

Reference-element proof sketch. Let $\hat{K}$ be the fixed reference triangle or square, and let $\hat{I}$ denote its nodal interpolation operator. The proof has three steps.

1. Bounded interpolation and polynomial reproduction. Since $k+1 \geq 2$ in dimension two, point evaluation is continuous on $H^{k+1}(\hat{K})$. The nodal formula and the fixed, finite collection of basis functions therefore give

$$\|\hat{I}v\|_{H^m(\hat{K})} \leq C \|v\|_{H^{k+1}(\hat{K})}, \quad m = 0, 1. \quad (5.88)$$

Unisolvence implies $\hat{I}p = p$ for every shape polynomial. In particular it reproduces every polynomial of total degree at most k : this is immediate for P_k , and follows from $P_k \subset Q_k$ for the square.

2. Remove the polynomial part. The Bramble–Hilbert polynomial approximation lemma on the fixed reference element states

$$\inf_{p \in P_k} \|v - p\|_{H^{k+1}(\hat{K})} \leq C |v|_{H^{k+1}(\hat{K})}. \quad (5.89)$$

It is a higher-order counterpart of the Poincaré inequality: subtracting a constant is replaced by subtracting a polynomial. One may choose p so that the integrals of all derivatives of $v - p$

of order at most k vanish; these moment conditions determine p by descending degree. On this normalized subspace the full H^{k+1} norm is controlled by the highest-order seminorm. A contradiction proof uses compactness in H^k : a normalized sequence with unit norm and vanishing highest derivatives would converge to a polynomial of degree at most k with all these moments zero, hence to zero. This proves (5.89).

For any $p \in P_k$, polynomial reproduction gives $v - \hat{I}v = (v - p) - \hat{I}(v - p)$. Step 1 and (5.89) yield

$$|v - \hat{I}v|_{H^m(\hat{K})} \leq C|v|_{H^{k+1}(\hat{K})}, \quad m = 0, 1. \quad (5.90)$$

3. Scale to each mesh element. Write the affine map as $F_K(\hat{x}) = B_K\hat{x} + b_K$, set $\hat{u} = u \circ F_K$, and denote the local interpolant by $I_K u$. Matching nodes gives $(I_K u) \circ F_K = \hat{I}\hat{u}$. The change of variables and chain rule give, with $e = u - I_K u$,

$$|e|_{H^m(K)} \leq C|\det B_K|^{1/2}\|B_K^{-1}\|^m|\hat{u} - \hat{I}\hat{u}|_{H^m(\hat{K})}, \quad (5.91)$$

$$|\hat{u}|_{H^{k+1}(\hat{K})} \leq C|\det B_K|^{-1/2}\|B_K\|^{k+1}|u|_{H^{k+1}(K)}. \quad (5.92)$$

Here the matrix norms are Euclidean operator norms. Mesh regularity bounds the largest singular value of B_K by Ch_K and its smallest singular value below by ch_K . The determinant factors cancel, so (5.90) implies

$$|u - I_K u|_{H^m(K)} \leq Ch_K^{k+1-m}|u|_{H^{k+1}(K)}. \quad (5.93)$$

Square, sum over $K \in \mathcal{T}_h$, and use $h_K \leq h$ to obtain the global estimate. Admissibility ensures global continuity; hence for $m = 1$ these elementwise derivatives are the global weak derivatives. The constant is independent of the mesh because the shape bound is uniform over the family; it can depend on k and that bound. No quasi-uniformity assumption is needed. $\square$

Detailed interpolation results are given in [3, Theorems 4.4 and 4.5] and [4, Chapter 3]. The argument above treats P_k and Q_k together: the square element requires the ordinary, isotropic space H^{k+1} , containing derivatives of total order at most $k + 1$, rather than derivatives of order $k + 1$ in each variable simultaneously.

The case $k = 1$ gives, for both triangular P_1 and rectangular Q_1 elements,

$$\|u - \Pi_h u\|_{H_0^1(\Omega)} \leq Ch \|u\|_{H^2(\Omega)}. \quad (5.94)$$

On a triangle, P_1 interpolation reproduces affine functions, so the Hessian controls the remaining error. The same estimate holds for Q_1 because it also reproduces affine functions (and additionally the bilinear term $x_1 x_2$). Thus the second derivatives alone suffice in both cases.

Combined with Céa's lemma, this gives an $O(h)$ error in the energy norm *provided that* the exact solution belongs to $H^2(\Omega)$. Proposition 3.18 supplies this property for the homogeneous Dirichlet model on a convex polygon with $f \in L^2(\Omega)$ and bounded nonnegative c . On a nonconvex polygon, or with mixed boundary conditions, Lax–Milgram still gives a weak solution, but it does not by itself give H^2 regularity or this rate. In particular, the mixed Robin exercise below explicitly assumes $u \in H^2(\Omega)$ when deriving its error estimate.

The case $k = 2$ gives the triangular P_2 estimate

$$\|u - \Pi_h u\|_{H_0^1(\Omega)} \leq Ch^2 \|u\|_{H^3(\Omega)}. \quad (5.95)$$

5.8 Local matrices, explicit formulas, and sparse assembly

For the homogeneous Dirichlet model problem, the local stiffness matrix, local mass matrix, local reaction matrix, local element matrix, and local load vector on a compact element K are

$$S_{rs}^K = \int_K \nabla p_{K,s} \cdot \nabla p_{K,r} \, dx, \quad (5.96)$$

$$M_{rs}^K = \int_K p_{K,s} p_{K,r} dx, \quad (5.97)$$

$$C_{rs}^K = \int_K c p_{K,s} p_{K,r} dx, \quad (5.98)$$

$$A_{rs}^K = S_{rs}^K + C_{rs}^K, \quad (5.99)$$

$$F_r^K = \int_K f p_{K,r} dx, \quad (5.100)$$

for $r, s = 1, \dots, n_{\text{loc}}$. If $c = c_0$ is constant on K , then

$$A^K = S^K + c_0 M^K. \quad (5.101)$$

5.8.1 Full assembly and Dirichlet constraints

First assemble the full matrix and load vector on W_h , before applying Dirichlet constraints:

$$\hat{A}_{ij} = \sum_{K \in \mathcal{T}_h} \sum_{\substack{1 \leq r, s \leq n_{\text{loc}} \\ \text{lg}_K(r)=i, \text{lg}_K(s)=j}} A_{rs}^K, \quad (5.102)$$

$$\hat{B}_i = \sum_{K \in \mathcal{T}_h} \sum_{\substack{1 \leq r \leq n_{\text{loc}} \\ \text{lg}_K(r)=i}} F_r^K, \quad 1 \leq i, j \leq N_{\text{all}}. \quad (5.103)$$

Equivalently, $\hat{A}_{ij} = a(\hat{w}_j, \hat{w}_i)$ and $\hat{B}_i = \ell(\hat{w}_i)$, using the volume forms above, which also make sense on W_h . These full arrays do not yet impose the Dirichlet conditions.

Order the nodes as free nodes $\mathcal{F}$ followed by Dirichlet nodes $\mathcal{D}$. If the prescribed boundary nodal values are $g_{\mathcal{D}}$, write the coefficient vector as $(U_{\mathcal{F}}, g_{\mathcal{D}})^T$. Testing only with functions in V_h gives

$$\hat{A}_{\mathcal{F}\mathcal{F}} U_{\mathcal{F}} = \hat{B}_{\mathcal{F}} - \hat{A}_{\mathcal{F}\mathcal{D}} g_{\mathcal{D}}. \quad (5.104)$$

The boundary equations are replaced by the prescribed values, rather than retained as equations of the Dirichlet problem.

One may instead retain all N_{all} entries in the solution vector. Subtract the contributions of the original Dirichlet columns from the free right-hand side, zero the Dirichlet rows and columns, put 1 on their diagonal entries, and set their right-hand side to the prescribed values. This produces

$$\begin{pmatrix} \hat{A}_{\mathcal{F}\mathcal{F}} & 0 \\ 0 & I_{\#\mathcal{D}} \end{pmatrix} \begin{pmatrix} U_{\mathcal{F}} \\ U_{\mathcal{D}} \end{pmatrix} = \begin{pmatrix} \hat{B}_{\mathcal{F}} - \hat{A}_{\mathcal{F}\mathcal{D}} g_{\mathcal{D}} \\ g_{\mathcal{D}} \end{pmatrix}. \quad (5.105)$$

The modification preserves symmetry and retains the full system size. Eliminating the prescribed components gives (5.104); both systems therefore give the same finite element function. The right-hand-side correction is the algebraic lifting described in Section 4.6.1 and must be computed before the original boundary columns are zeroed. For homogeneous data, $g_{\mathcal{D}} = 0$, so the free right-hand side is unchanged, while the boundary components are set to zero.

5.8.2 Equivalent assembly with active numbering

For homogeneous Dirichlet data, the reduced matrix A and load vector F are exactly $\hat{A}_{\mathcal{F}\mathcal{F}}$ and $\hat{B}_{\mathcal{F}}$ in the free-node ordering. Rather than storing the full arrays first, one may assemble these quantities directly with the active numbering map. For $m, n = 1, \dots, N$,

$$A_{mn} = \sum_{K \in \mathcal{T}_h} \sum_{\substack{1 \leq r, s \leq n_{\text{loc}} \\ \text{act}(\text{lg}_K(r))=m \\ \text{act}(\text{lg}_K(s))=n}} A_{rs}^K, \quad (5.106)$$

$$F_m = \sum_{K \in \mathcal{T}_h} \sum_{\substack{1 \leq r \leq n_{\text{loc}} \\ \text{act}(\lg_K(r))=m}} F_r^K. \quad (5.107)$$

Boundary nodes do not have active numbers, since $\text{act} = 0$ on such nodes. These formulas retain all element contributions to free rows and columns, and perform the same homogeneous Dirichlet elimination as selecting the free block of the full arrays. For nonzero prescribed values, the additional term $-\hat{A}_{\mathcal{FD}} g_{\mathcal{D}}$ must also be assembled into the free right-hand side.

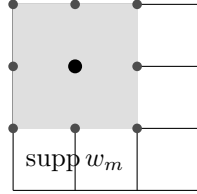
The assembled linear system is

$$AU = F, \quad (5.108)$$

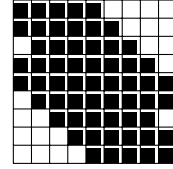
where

$$u_h = \sum_{m=1}^N U_m w_m. \quad (5.109)$$

The matrix is sparse because A_{mn} can be nonzero only when the supports of w_m and w_n overlap, equivalently when the two corresponding nodes belong to at least one common element.



Support of a Q_1 basis function



A typical sparse pattern

Figure 5.6: Local supports produce sparse global matrices.

5.8.3 Explicit local matrices for the lowest-order elements

For a rectangle

$$R = [x_0, x_0 + h_x] \times [y_0, y_0 + h_y], \quad (5.110)$$

with the Q_1 ordering bottom-left, bottom-right, top-right, top-left, set

$$a = \frac{h_y}{h_x}, \quad b = \frac{h_x}{h_y}. \quad (5.111)$$

The local stiffness matrix is

$$S^R = \begin{pmatrix} \frac{a+b}{3} & -\frac{a}{3} + \frac{b}{6} & -\frac{a+b}{6} & \frac{a}{6} - \frac{b}{3} \\ -\frac{a}{3} + \frac{b}{6} & \frac{a+b}{3} & \frac{a}{6} - \frac{b}{3} & -\frac{a+b}{6} \\ -\frac{a+b}{6} & \frac{a}{6} - \frac{b}{3} & \frac{a+b}{3} & -\frac{a}{3} + \frac{b}{6} \\ \frac{a}{6} - \frac{b}{3} & -\frac{a+b}{6} & -\frac{a}{3} + \frac{b}{6} & \frac{a+b}{3} \end{pmatrix}. \quad (5.112)$$

The local mass matrix is

$$M^R = \frac{h_x h_y}{36} \begin{pmatrix} 4 & 2 & 1 & 2 \\ 2 & 4 & 2 & 1 \\ 1 & 2 & 4 & 2 \\ 2 & 1 & 2 & 4 \end{pmatrix}. \quad (5.113)$$

For constant $c = c_0$ on R , the local reaction contribution is $c_0 M^R$.

For a triangle T with vertices $a_i = (x_i, y_i)$, set

$$\Delta_T = \det(a_2 - a_1, a_3 - a_1), \quad |T| = \frac{|\Delta_T|}{2}. \quad (5.114)$$

The P_1 basis is $(\lambda_1, \lambda_2, \lambda_3)$. For each cyclic permutation (i, j, ℓ) of $(1, 2, 3)$, set

$$b_i = y_j - y_\ell, \quad c_i = x_\ell - x_j. \quad (5.115)$$

With the oriented determinant above,

$$\nabla \lambda_i = \frac{1}{\Delta_T} (b_i, c_i). \quad (5.116)$$

Consequently, the local stiffness matrix is

$$S_{ij}^T = |T| \frac{b_i b_j + c_i c_j}{\Delta_T^2} = \frac{b_i b_j + c_i c_j}{4|T|}. \quad (5.117)$$

The local mass matrix is

$$M^T = \frac{|T|}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}. \quad (5.118)$$

For constant $c = c_0$ on T , the local element matrix is $S^T + c_0 M^T$.

5.9 Galerkin approximation and error estimate

The conforming Galerkin approximation of the model problem is: find u_h in V_h such that

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (5.119)$$

The estimate uses the abstract Céa lemma. The lemma is not tied to homogeneous Dirichlet boundary conditions. It only uses a Hilbert space V , a conforming finite-dimensional subspace $V_h \subset V$, and a continuous coercive bilinear form on V .

Lemma 5.25. *Let V be a Hilbert space, let V_h be a finite-dimensional subspace of V , and let a be continuous and coercive on V : there exist $M > 0$ and $\alpha > 0$ such that*

$$|a(w, v)| \leq M \|w\|_V \|v\|_V, \quad (5.120)$$

$$a(v, v) \geq \alpha \|v\|_V^2 \quad \text{for all } w, v \in V. \quad (5.121)$$

Let u in V and u_h in V_h solve

$$a(u, v) = \ell(v) \quad \text{for all } v \in V, \quad (5.122)$$

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (5.123)$$

Then

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (5.124)$$

The proof is the standard Galerkin-orthogonality argument already used in the one-dimensional chapter. It does not use the geometry of the mesh or the nature of the boundary condition; those enter only through the choice of the conforming space $V_h \subset V$.

Theorem 5.26. *Let $(\mathcal{T}_h)_{h>0}$ be a regular family of admissible triangular or rectangular meshes of Ω .*

- *On triangular meshes, let V_h be the conforming Lagrange space of degree k with local space P_k .*

- On rectangular meshes, let V_h be the conforming Lagrange space of degree k with local space Q_k .

Let u be the exact solution of the coercive elliptic model problem, and let u_h be the Galerkin solution in $V_h \subset H_0^1(\Omega)$. If u belongs to $H^{k+1}(\Omega) \cap H_0^1(\Omega)$, then there exists a constant $C > 0$, independent of h , such that

$$\|u - u_h\|_{H_0^1(\Omega)} \leq Ch^k \|u\|_{H^{k+1}(\Omega)}. \quad (5.125)$$

Proof. Céa's lemma, applied with $V = H_0^1(\Omega)$ and $V_h = V_h$, gives

$$\|u - u_h\|_{H_0^1(\Omega)} \leq C \inf_{v_h \in V_h} \|u - v_h\|_{H_0^1(\Omega)}. \quad (5.126)$$

We choose $v_h = \Pi_h u$ and use Theorem 5.24 with $m = 1$. □

The full-assembly route to a two-dimensional computation proceeds as follows.

- We choose a polygonal domain and a triangular or rectangular mesh.
- We compute the local matrices and local load vectors.
- We assemble the full global sparse matrix and load vector through the local-to-global maps, retaining all nodes.
- We impose the Dirichlet data by the symmetric row-and-column modification in (5.105), or by eliminating the prescribed values as in (5.104).
- We solve the resulting linear system and reconstruct the finite element function, including its prescribed boundary values.

The active-numbering route incorporates the same homogeneous constraints during assembly and gives the identical reduced system.

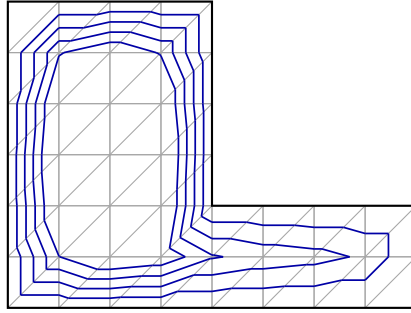


Figure 5.7: A conforming triangular mesh of an L-shaped polygonal domain and level sets of the P_1 Galerkin approximation of $-\Delta u = 1$ with homogeneous Dirichlet data. The contour levels are 0.05, 0.10, 0.15, and 0.20; each contour segment is straight within its triangle.

5.10 Exercises

Let

$$\Omega = (0, 1)^2, \quad \Gamma_R = (0, 1) \times \{1\}, \quad \Gamma_D = \partial\Omega \setminus \Gamma_R. \quad (5.127)$$

Thus the two upper corners belong to Γ_D . Consider

$$-\Delta u + cu = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \Gamma_D, \quad \partial_n u + du = g \quad \text{on } \Gamma_R, \quad (5.128)$$

where f in $L^2(\Omega)$, c in $L^\infty(\Omega)$ with $c(x) \geq 0$ almost everywhere, $d \geq 0$, and g in $L^2(\Gamma_R)$. Let $(\mathcal{T}_h)_{h>0}$ be a regular family of admissible rectangular meshes of $\bar{\Omega}$, and set

$$h = \max_{R \in \mathcal{T}_h} \text{diam}(R). \quad (5.129)$$

This problem applies the rectangular Q_1 construction of Section 5.3 to a mixed boundary condition.

Problem 5.1 (Mixed Dirichlet–Robin boundary condition for rectangular Q_1 finite elements).

Part 1. Variational setting and weak formulation.

(a) Define

$$V = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}. \quad (5.130)$$

Prove that, for every v in V ,

$$\|v\|_{L^2(\Omega)} \leq \|\partial_2 v\|_{L^2(\Omega)} \leq \|\nabla v\|_{L^2(\Omega)}, \quad (5.131)$$

$$\|v\|_{L^2(\Gamma_R)} \leq \|\partial_2 v\|_{L^2(\Omega)} \leq \|\nabla v\|_{L^2(\Omega)}. \quad (5.132)$$

Deduce that

$$\|v\|_V := \|\nabla v\|_{L^2(\Omega)} \quad (5.133)$$

is a norm on V , equivalent to the usual $H^1(\Omega)$ norm.

(b) Derive the weak formulation. More precisely, if u is a smooth solution, show that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V, \quad (5.134)$$

where the bilinear form $a(\cdot, \cdot)$ and the linear form ℓ must be determined.

(c) Prove that a is continuous and coercive on V , and that ℓ is continuous on V for the norm $\|\cdot\|_V$. Deduce that the weak problem admits a unique solution in V .

Part 2. Finite element space, nodal basis, and nodal interpolation.

(a) Define the conforming rectangular spaces

$$W_h = \{v_h \in C^0(\bar{\Omega}) : v_h|_R \text{ in } Q_1(R) \text{ for every rectangle } R \in \mathcal{T}_h\}, \quad (5.135)$$

$$V_h = \{v_h \in W_h : v_h = 0 \text{ on } \Gamma_D\}. \quad (5.136)$$

Justify that $V_h \subset V$.

(b) Let $S_1, \dots, S_N$ be the nodes of the mesh that do not lie on Γ_D . For each i , let w_i in V_h be defined by

$$w_i(S_j) = \delta_{ij} \quad \text{for } i, j = 1, \dots, N. \quad (5.137)$$

Explain why $(w_i)_{i=1, \dots, N}$ is a basis of V_h , and why every function v_h in V_h satisfies the nodal expansion

$$v_h = \sum_{i=1}^N v_h(S_i) w_i. \quad (5.138)$$

Identify the degrees of freedom that are new compared with the homogeneous Dirichlet construction of Section 5.5.

(c) For u in $C^0(\bar{\Omega}) \cap V$, define the nodal interpolant $\Pi_h u$ in V_h and write it in the basis $(w_i)_{i=1, \dots, N}$. Explain what changes compared with the homogeneous Dirichlet case.

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma.

- (a) Write the conforming Galerkin problem associated with the weak formulation, and prove that it admits a unique solution u_h in V_h .
- (b) Prove the Galerkin orthogonality relation

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (5.139)$$

- (c) Let M and α denote continuity and coercivity constants of a on V . Use the abstract Céa lemma to prove

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (5.140)$$

Part 4. Interpolation estimate in the energy norm.

Assume now that u in $H^2(\Omega) \cap V$.

- (a) Since $H^2(\Omega) \subset C^0(\overline{\Omega})$ in dimension two, explain why $\Pi_h u$ is well defined and belongs to V_h .
- (b) Apply Theorem 5.24 with the rectangular Q_1 element, $k = 1$, and $m = 1$, to obtain

$$\|u - \Pi_h u\|_V \leq Ch \|u\|_{H^2(\Omega)}, \quad (5.141)$$

where C is independent of h .

Part 5. A priori error estimate.

Use Parts 3 and 4 to prove that

$$\|u - u_h\|_V \leq Ch \|u\|_{H^2(\Omega)}. \quad (5.142)$$

Then, using the estimates from Part 1, show that we may take

$$M = 1 + \|c\|_{L^\infty(\Omega)} + d, \quad \alpha = 1. \quad (5.143)$$

Part 6. Matrix form, sparsity, and the Robin boundary contribution.

Assume now that $c = c_0 \geq 0$ is constant.

- (a) If $(w_i)_{i=1,\dots,N}$ is the global nodal basis of V_h , write

$$u_h = \sum_{j=1}^N U_j w_j. \quad (5.144)$$

Verify that the discrete problem becomes

$$AU = B, \quad (5.145)$$

with

$$A_{ij} = \int_{\Omega} (\nabla w_j \cdot \nabla w_i + c_0 w_j w_i) dx + d \int_{\Gamma_R} w_j w_i d\Gamma, \quad (5.146)$$

$$B_i = \int_{\Omega} f w_i dx + \int_{\Gamma_R} g w_i d\Gamma. \quad (5.147)$$

- (b) On a structured rectangular mesh, explain why each basis function interacts with at most nine basis functions.

- (c) Let R be a rectangle of the top row, let $e = R \cap \Gamma_R$, and let $p_1^R, \dots, p_4^R$ be the local basis functions obtained from the reference square $\hat{R} = [0, 1]^2$, numbered as in Section 5.3. Compute the local Robin matrix

$$R^{(e)} = \left(\int_e p_j^R p_i^R d\Gamma \right)_{1 \leq i, j \leq 4}. \quad (5.148)$$

- (d) If, in addition, $g = g_0$ is constant on e , compute the local Robin load vector

$$G^{(e)} = \left(\int_e g_0 p_i^R d\Gamma \right)_{1 \leq i \leq 4}. \quad (5.149)$$

- (e) Identify the rows and columns that differ from the homogeneous Dirichlet case, and explain why.
- (f) Assemble the full matrix $\hat{A}$ and load vector $\hat{B}$ on the all-node basis of W_h , including both the volume contributions and the Robin edge contributions. Partition the nodes into those on Γ_D and all remaining nodes. Impose the zero Dirichlet values by symmetric row-and-column modification, and then eliminate the prescribed components. Prove that the resulting free block and load vector are exactly the A and B of part (a). Explain why nodes on Γ_R that do not belong to Γ_D remain unknowns.

Exercise 5.2 (Rectangular Q_1 and triangular P_1 interpolation). *The first three questions are adapted from Question 6 of the June 2026 final examination by F. Sueur and J. S. Hale. The comparison of interpolants extends that question.*

Let $\hat{Q} = [0, 1]^2$ have vertices

$$A_1 = (0, 0), \quad A_2 = (1, 0), \quad A_3 = (1, 1), \quad A_4 = (0, 1),$$

and let $\hat{T} = \{(\xi, \eta) : \xi \geq 0, \eta \geq 0, \xi + \eta \leq 1\}$.

- (a) Recall the definition of a finite element (K, P, Σ) and the meaning of unisolvence of its degrees of freedom.
- (b) Give the four Q_1 Lagrange shape functions on $\hat{Q}$ in the vertex order above. Verify their nodal relations and explain why the four vertex values are unisolvent on $Q_1(\hat{Q})$.
- (c) Give the three barycentric coordinates on $\hat{T}$, in the vertex order $(0, 0), (1, 0), (0, 1)$. Verify that they form the P_1 Lagrange basis on this triangle.
- (d) Let $q(\xi, \eta) = \xi\eta$. Construct three interpolants of its four corner values:
- $I_Q q \in Q_1(\hat{Q})$ on the single square;
 - $I_+ q$, continuous and affine on each of the two triangles obtained by the diagonal $A_1 A_3$;
 - $I_- q$, continuous and affine on each of the two triangles obtained by the diagonal $A_2 A_4$.

Give explicit formulas on each triangle and check continuity across the corresponding diagonal.

- (e) Compare q , $I_Q q$, $I_+ q$, and $I_- q$ at $z = (1/2, 1/2)$. Show that the three interpolants agree on $\partial\hat{Q}$. Explain why their different interior values do not contradict unisolvence.

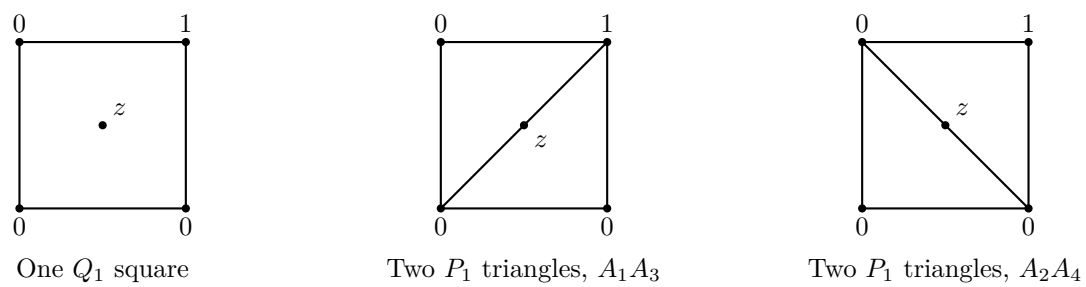


Figure 5.8: Three spaces using the same four corner values of $q(\xi, \eta) = \xi\eta$. The centre z is a comparison point, not an additional degree of freedom.

Chapter 6

The Lax–Richtmyer theorem

The finite element chapters converted approximation estimates into error bounds using coercivity and Céa’s lemma. We now examine consistency, stability, and convergence before applying these ideas to time-dependent finite difference schemes in Chapter 7. We first prove a simple fixed-space operator version. We then give an evolution error estimate and a Lax–Richtmyer equivalence theorem that explicitly connects continuous data with changing grid spaces.

6.1 Operator equations and discretizations

We model a general situation in the numerical solution of PDEs by an operator equation of the form

$$Lx = y, \tag{6.1}$$

where y is given in Y , and the unknown is x in X , where X and Y are two Banach spaces.

Definition 6.1. Let X and Y be Banach spaces and let L be in $\mathcal{L}(X, Y)$, that is a bounded linear operator from X to Y . We say that L is *well posed* if for each y in Y there exists a unique x in X solving (6.1).

Equivalently, L is bijective and, by the open mapping theorem recalled in Section 6.3, L^{-1} belongs to $\mathcal{L}(Y, X)$.

Throughout this section, assume that $L \in \mathcal{L}(X, Y)$ is bijective. Let $H \subset (0, +\infty)$ be a set of admissible discretization parameters with 0 as an accumulation point. All limits $h \rightarrow 0$ are taken within H . We consider a family $(L_h)_{h \in H}$ in $\mathcal{L}(X, Y)$ and, for each h , the discrete problem

$$L_h x_h = y. \tag{6.2}$$

We assume (6.2) is well posed for every h in H , so that L_h is bijective and L_h^{-1} belongs to $\mathcal{L}(Y, X)$.

Definition 6.2. The discretization $(L_h)_{h \in H}$ is:

- *stable* if there exist constants $C > 0$ and $h_0 > 0$ such that

$$\|L_h^{-1}\|_{Y \rightarrow X} \leq C \quad \text{for all } h \in H \text{ with } h < h_0. \tag{6.3}$$

- *consistent* if for every x in X ,

$$\|Lx - L_h x\|_Y \rightarrow 0 \quad \text{as } h \rightarrow 0. \tag{6.4}$$

- *convergent* if for every y in Y , letting $x := L^{-1}y$ solve (6.1) and $x_h := L_h^{-1}y$ solve (6.2), we have

$$\|x - x_h\|_X \rightarrow 0 \quad \text{as } h \rightarrow 0. \tag{6.5}$$

6.2 The fixed-space operator version

Theorem 6.3. *Let X, Y be Banach spaces, and assume that L and every L_h are bounded linear bijections from X to Y . Assume that $(L_h)_{h \in H}$ is consistent, i.e. (6.4) holds for all x in X . Then stability (6.3) is equivalent to convergence (6.5).*

Proof. Let y be fixed in Y , and set $x := L^{-1}y$ and $x_h := L_h^{-1}y$. We use $L_h x_h = y = Lx$ to write

$$\begin{aligned} x - x_h &= L_h^{-1}(L_h x - L_h x_h) \\ &= L_h^{-1}(L_h x - Lx). \end{aligned} \quad (6.6)$$

Stability $\Rightarrow$ convergence. Assume (6.3). Taking norms in (6.6) yields

$$\|x - x_h\|_X \leq \|L_h^{-1}\|_{Y \rightarrow X} \|L_h x - Lx\|_Y \leq C \|L_h x - Lx\|_Y. \quad (6.7)$$

By consistency, the right-hand side tends to 0 as $h \rightarrow 0$, hence (6.5) holds.

Convergence $\Rightarrow$ stability. Assume (6.5) holds for every y in Y . Equivalently, for every y in Y we have

$$L_h^{-1}y \rightarrow L^{-1}y \quad \text{in } X \text{ as } h \rightarrow 0. \quad (6.8)$$

Suppose, for a contradiction, that no uniform inverse bound holds for sufficiently small h . We can then choose $h_m \in H$ with $h_m < 1/m$ and $\|L_{h_m}^{-1}\|_{Y \rightarrow X} > m$. For each fixed $y \in Y$, the sequence $L_{h_m}^{-1}y$ converges, and hence is bounded. The uniform boundedness principle recalled in Section 6.4, applied to this sequence, gives $\sup_m \|L_{h_m}^{-1}\|_{Y \rightarrow X} < +\infty$, a contradiction. Thus (6.3) holds for sufficiently small h . $\square$

Remark 6.4. The theorem is often paraphrased as: for a consistent discretization, the only missing ingredient for convergence is stability (and conversely). A classical reference is [3, Theorem 1.1].

6.3 A reminder on the open mapping theorem

Theorem 6.5. *Let X and Y be Banach spaces and let $L \in \mathcal{L}(X, Y)$ be surjective. Then L is an open map: if $U \subset X$ is open, then $L(U) \subset Y$ is open.*

Corollary 6.6. *Let X and Y be Banach spaces and let $L \in \mathcal{L}(X, Y)$ be bijective. Then $L^{-1} : Y \rightarrow X$ belongs to $\mathcal{L}(Y, X)$. Equivalently, there exists a constant $C > 0$ such that*

$$\|x\|_X \leq C \|Lx\|_Y \quad \text{for all } x \in X.$$

6.4 A reminder on the uniform boundedness principle (Banach–Steinhaus)

Theorem 6.7. *Let X be a Banach space, let I be a nonempty set, let Y be a normed space, and let $(T_i)_{i \in I}$ be a family in $\mathcal{L}(X, Y)$. Assume that for every x in X the set $\{\|T_i x\|_Y : i \in I\}$ is bounded, i.e.*

$$\sup_i \|T_i x\|_Y < +\infty \quad \text{for all } x \text{ in } X. \quad (6.9)$$

Then the operator norms are uniformly bounded:

$$\sup_i \|T_i\|_{X \rightarrow Y} < +\infty. \quad (6.10)$$

Remark 6.8. We recall that $\|T\|_{X \rightarrow Y} := \sup_{\|x\|_X=1} \|Tx\|_Y$ for T in $\mathcal{L}(X, Y)$. The statement is sometimes phrased as *pointwise boundedness implies uniform boundedness*.

The following corollary is tailored to discretization families.

Corollary 6.9. *Let X be a Banach space, let Y be a normed space, and let $(T_h)_{h \in H}$ be a family in $\mathcal{L}(X, Y)$. Assume that for each x in X we have a limit $T_h x \rightarrow Tx$ in Y as $h \rightarrow 0$ for some map $T : X \rightarrow Y$. Then there exists $h_0 > 0$ such that $\sup_{h \in H, h < h_0} \|T_h\|_{X \rightarrow Y} < +\infty$.*

Proof. If the conclusion failed, we could choose $h_m < 1/m$ with $\|T_{h_m}\|_{X \rightarrow Y} > m$. For each $x \in X$, the convergent sequence $T_{h_m}x$ is bounded. The uniform boundedness principle applied to (T_{h_m}) contradicts the choice of its operator norms. $\square$

Remark 6.10. Convergence as $h \rightarrow 0$ says nothing about coarse parameters away from zero. For example, on $H = (0, 1)$ the scalar operators $L = I$ and $L_h = (1 - h)I$ are consistent and convergent, but $\sup_{h \in H} \|L_h^{-1}\| = +\infty$. This is why stability is defined for sufficiently small parameters. For a refinement sequence $H = \{h_m : m \geq 1\}$ with $h_m \rightarrow 0$, adding its finitely many coarse operators does not affect boundedness.

6.5 Stability of time-discrete evolutions

Definition 6.11. Let $(X_h, \|\cdot\|_{X_h})$ be a family of normed spaces, let $\Delta t = \Delta t(h) \rightarrow 0$, and let $G_h \in \mathcal{L}(X_h, X_h)$. Write the linear time-discrete evolution as

$$U^{n+1} = G_h U^n, \quad t^n = n\Delta t. \quad (6.11)$$

We say that this evolution is *stable in X_h on finite time intervals* if, for every $T > 0$, there exists a constant $C_T > 0$, independent of h , Δt , and n , such that, for all sufficiently small admissible h ,

$$\|G_h^n V\|_{X_h} \leq C_T \|V\|_{X_h} \quad \text{for all } V \in X_h \text{ and all } 0 \leq n\Delta t \leq T. \quad (6.12)$$

We say that the evolution is *strongly stable in X_h* if, for all sufficiently small admissible h , its one-step update is non-expansive:

$$\|G_h V\|_{X_h} \leq \|V\|_{X_h} \quad \text{for all } V \in X_h. \quad (6.13)$$

By iteration, strong stability implies (6.12) with $C_T = 1$ on every finite time interval.

Remark 6.12. Definition 6.11 expresses power boundedness for a time evolution. The fixed-space inverse-operator theorem does not itself establish equivalence for this setting; Theorem 6.15 below supplies the connection. The norm is part of the statement. In Chapter 7, for example, $X_h = \ell_h^2$ and stability means boundedness in the topology

$$\ell^\infty(0, T; \ell_h^2), \quad \max_{0 \leq n\Delta t \leq T} \|U^n\|_{\ell_h^2}. \quad (6.14)$$

Strong stability is a non-expansion property in the chosen norm; it is stronger than stability on finite time intervals.

6.6 An error estimate for discrete evolutions

For a time-dependent equation, consistency is measured by inserting an exact trajectory into the update rule. The resulting defect must be divided by Δt : on a finite time interval, there are of order $1/\Delta t$ steps.

Theorem 6.13. Fix $T > 0$ and let $N_T = \lfloor T/\Delta t \rfloor \geq 1$. Assume that $G_h \in \mathcal{L}(X_h, X_h)$ satisfies (6.12). Let $U^{n+1} = G_h U^n$, and let $(W_h^n)_{0 \leq n \leq N_T}$ be reference values in X_h . Define the error and the normalized defect by

$$e^n = U^n - W_h^n, \quad \tau_h^n = \frac{W_h^{n+1} - G_h W_h^n}{\Delta t} \quad (0 \leq n < N_T). \quad (6.15)$$

Then

$$\begin{aligned} \max_{0 \leq n \leq N_T} \|e^n\|_{X_h} &\leq C_T \left(\|e^0\|_{X_h} + \Delta t \sum_{j=0}^{N_T-1} \|\tau_h^j\|_{X_h} \right) \\ &\leq C_T \left(\|e^0\|_{X_h} + T \max_{0 \leq j < N_T} \|\tau_h^j\|_{X_h} \right). \end{aligned} \quad (6.16)$$

In particular, an initial error tending to zero and a normalized defect tending to zero uniformly in time imply convergence to the reference trajectory. If both are $O(h^p)$, the error is $O(h^p)$ on $[0, T]$.

Proof. Subtracting the reference update gives

$$e^{n+1} = G_h e^n - \Delta t \tau_h^n. \quad (6.17)$$

Iteration yields the discrete Duhamel formula

$$e^n = G_h^n e^0 - \Delta t \sum_{j=0}^{n-1} G_h^{n-1-j} \tau_h^j. \quad (6.18)$$

Every power appearing here corresponds to a time at most T . Taking norms and using stability proves (6.16). $\square$

6.7 Lax–Richtmyer equivalence on changing grid spaces

The preceding estimate compares two trajectories on the same grid. To compare numerical solutions with arbitrary continuous data, we must also specify how data are projected onto the grid and how grid values are reconstructed as functions. This avoids using point values of an L^2 equivalence class.

Let X be a Banach space and let $S(t) : X \rightarrow X$, $t \geq 0$, be a strongly continuous semigroup of bounded linear operators: $S(0) = I$, $S(t+s) = S(t)S(s)$, and $t \mapsto S(t)x$ is continuous for every $x \in X$. Assume $\sup_{0 \leq t \leq T} \|S(t)\|_{X \rightarrow X} \leq M_T$ for each $T > 0$. For every admissible h , let X_h be a normed grid space and let

$$P_h : X \rightarrow X_h, \quad E_h : X_h \rightarrow X \quad (6.19)$$

be bounded linear maps satisfying

$$\begin{aligned} \|P_h\|_{X \rightarrow X_h} &\leq 1, & \|E_h V\|_X &= \|V\|_{X_h} \quad \text{for every } V \in X_h, \\ P_h E_h &= I_{X_h}, & E_h P_h x &\rightarrow x \quad \text{for every } x \in X. \end{aligned} \quad (6.20)$$

These assumptions hold for cell averages and piecewise constant reconstruction in L^2 on a periodic interval, as explained in Chapter 7. Let $\Delta t = \Delta t(h) \rightarrow 0$ and $G_h \in \mathcal{L}(X_h, X_h)$.

Definition 6.14. The approximation is *consistent on a dense class* $D \subset X$ if, for every $x \in D$ and every $T > 0$,

$$\max_{0 \leq n < N_T} \left\| \frac{P_h S(t^{n+1})x - G_h P_h S(t^n)x}{\Delta t} \right\|_{X_h} \rightarrow 0, \quad N_T = \lfloor T/\Delta t \rfloor. \quad (6.21)$$

It is *convergent for all data in X* if, for every $x \in X$ and every $T > 0$,

$$\max_{0 \leq n \leq N_T} \|E_h G_h^n P_h x - S(t^n)x\|_X \longrightarrow 0. \quad (6.22)$$

The initial grid vector in this definition is $U^0 = P_h x$.

Theorem 6.15. *Under (6.20) and consistency (6.21) on a dense class $D \subset X$, finite-time stability (6.12) is equivalent to convergence (6.22) for all data in X . Both assertions concern sufficiently small admissible h .*

Proof. Stability implies convergence. For $x \in D$, apply Theorem 6.13 with $W_h^n = P_h S(t^n)x$ and $U^0 = P_h x$. The initial error is zero and consistency makes the defect tend uniformly to zero. Since E_h is isometric, it remains to control $\|E_h P_h S(t^n)x - S(t^n)x\|_X$ uniformly in n . The orbit $\{S(t)x : 0 \leq t \leq T\}$ is compact by strong continuity. The operators $E_h P_h$ have norms at most one and converge strongly to I , so convergence is uniform on this compact orbit: approximate the orbit by a finite ε -net and use the uniform operator bound. This proves (6.22) for $x \in D$. For arbitrary $x \in X$, choose $d \in D$ close to x . Uniform stability and the continuous bound give

$$\max_n \|E_h G_h^n P_h x - S(t^n)x\|_X \leq (C_T + M_T) \|x - d\|_X \quad (6.23)$$

$$+ \max_n \|E_h G_h^n P_h d - S(t^n)d\|_X. \quad (6.24)$$

First let $h \rightarrow 0$, then let $d \rightarrow x$.

Convergence implies stability. Fix $T > 0$. If stability failed for every sufficiently small range of parameters, we could select $h_m \rightarrow 0$ and $0 \leq n_m \Delta t(h_m) \leq T$ such that $\|G_{h_m}^{n_m}\|_{X_{h_m} \rightarrow X_{h_m}} > m$. Set $A_m = E_{h_m} G_{h_m}^{n_m} P_{h_m} \in \mathcal{L}(X, X)$. For each $x \in X$, convergence and the bound M_T imply that $(A_m x)$ is bounded. Banach–Steinhaus therefore yields $\sup_m \|A_m\|_{X \rightarrow X} < +\infty$. But (6.20) implies

$$\|A_m\|_{X \rightarrow X} = \|G_{h_m}^{n_m}\|_{X_{h_m} \rightarrow X_{h_m}} : \quad (6.25)$$

one inequality follows from the bounds for P_h, E_h , and the other by testing A_m on $E_{h_m} V$, since $P_{h_m} E_{h_m} V = V$. This contradicts the choice of n_m, h_m . $\square$

Remark 6.16. This is an evolution form of the Lax–Richtmyer theorem; see [1]. Consistency need only be checked on a dense smooth class, whereas the convergence assertion quantifies over all data in the ambient Banach space. Failure of stability rules out this general convergence assertion. It does not rule out convergence for particular initial data, such as a finite combination of fixed Fourier modes.

6.8 Exercises

Exercise 6.1 (Stability without consistency). Let X be a Banach space and set

$$L = I, \quad L_h = 2I \quad \text{for all } h \in H.$$

Show that:

1. each L_h is bijective;
2. the discretization $(L_h)_{h \in H}$ is stable;
3. it is not consistent;

4. if $x := L^{-1}y$ and $x_h := L_h^{-1}y$, then in general $x_h \not\rightarrow x$ as $h \rightarrow 0$.

Exercise 6.2 (Convergence does not imply consistency). Let $X = Y = \ell^2$, let $(e_n)_{n \geq 1}$ denote the canonical basis of ℓ^2 , and define

$$L = I, \quad P_n x = x_n e_n, \quad L_n = I + n P_n.$$

Show that:

1. each L_n is bijective and

$$L_n^{-1} = I - \frac{n}{n+1} P_n;$$

- 2.

$$\|L_n^{-1}\|_{\ell^2 \rightarrow \ell^2} = 1;$$

3. for every $y \in \ell^2$, we have $L_n^{-1}y \rightarrow y$ in ℓ^2 as $n \rightarrow \infty$;

4. if

$$x = \left(\frac{1}{n} \right)_{n \geq 1} \in \ell^2,$$

then

$$\|L_n x - x\|_{\ell^2} = 1 \quad \text{for all } n \geq 1;$$

5. convergence in the sense of the discrete solutions does not imply consistency.

Exercise 6.3 (A stronger version in operator norm). Let X and Y be Banach spaces, let $L \in \mathcal{L}(X, Y)$ be bijective, and let $(L_h)_{h \in H} \subset \mathcal{L}(X, Y)$ be a family of bijective operators. Assume that

$$\|L_h - L\|_{X \rightarrow Y} \rightarrow 0 \quad \text{and} \quad \sup_{h \in H} \|L_h^{-1}\|_{Y \rightarrow X} < +\infty.$$

Show that:

1. we have the identity

$$L_h^{-1} - L^{-1} = L_h^{-1}(L - L_h)L^{-1};$$

2. consequently,

$$\|L_h^{-1} - L^{-1}\|_{Y \rightarrow X} \leq \|L_h^{-1}\|_{Y \rightarrow X} \|L - L_h\|_{X \rightarrow Y} \|L^{-1}\|_{Y \rightarrow X};$$

3. $L_h^{-1} \rightarrow L^{-1}$ in operator norm from Y to X ;

4. compared with Theorem 6.3, identify precisely what has been strengthened, and what has been gained.

Exercise 6.4 (Consistency only on a dense subspace). Let X be a Banach space, let Y be a normed space, let $L \in \mathcal{L}(X, Y)$, and let $(L_h)_{h \in H} \subset \mathcal{L}(X, Y)$. Let $D \subset X$ be dense. Assume that

$$\sup_{h \in H} \|L_h\|_{X \rightarrow Y} < +\infty \quad \text{and} \quad L_h d \rightarrow Ld \quad \text{in } Y \text{ for all } d \in D.$$

Show that:

1. $L_h x \rightarrow Lx$ in Y for every $x \in X$;

2. under stability, it may be enough to verify consistency only on a dense subspace.

Chapter 7

Finite difference methods for hyperbolic equations

7.1 The scalar transport equation and the method of characteristics

We begin with the constant-coefficient transport problem

$$\partial_t u + a \partial_x u = 0 \quad \text{in } \mathbb{R} \times (0, +\infty), \quad u(x, 0) = u_0(x), \quad (7.1)$$

where a in $\mathbb{R} \setminus \{0\}$ is constant.

Proposition 7.1. *Assume that u_0 in $C^1(\mathbb{R})$. Then the transport problem admits the unique C^1 solution*

$$u(x, t) = u_0(x - at). \quad (7.2)$$

Proof. Let $x(t)$ solve

$$\frac{dx}{dt} = a, \quad x(0) = x_0. \quad (7.3)$$

Then $x(t) = x_0 + at$. Along such a curve,

$$\frac{d}{dt} u(x(t), t) = \partial_t u(x(t), t) + a \partial_x u(x(t), t) = 0. \quad (7.4)$$

Hence u is constant along each characteristic line and

$$u(x_0 + at, t) = u_0(x_0). \quad (7.5)$$

Replacing x_0 with $x - at$ gives the formula. Direct differentiation verifies the equation and the initial condition, and uniqueness follows from the same characteristic argument. $\square$

The initial profile is translated at velocity a without change of shape.

Remark 7.2. On a bounded interval (α, β) , the sign of a determines where a boundary condition is imposed. If $a > 0$, the characteristics enter through $x = \alpha$, so $x = \alpha$ is the inflow boundary and $x = \beta$ is the outflow boundary. If $a < 0$, the rôles are reversed. See [3, Chapter 13, Section 13.1] and [2, Chapter 10, Section 10.2].

7.2 An a priori estimate by the energy method

We next consider the transport-reaction problem

$$\partial_t u + a \partial_x u + a_0 u = f \quad \text{in } (\alpha, \beta) \times (0, +\infty), \quad (7.6)$$

with $a > 0$, $a_0 \geq 0$, initial datum

$$u(x, 0) = u_0(x) \quad \text{for } x \in (\alpha, \beta), \quad (7.7)$$

and inflow boundary condition

$$u(\alpha, t) = \phi(t) \quad \text{for } t > 0. \quad (7.8)$$

Proposition 7.3. *Assume that u is smooth enough. Then, for every $t > 0$,*

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(\alpha, \beta)}^2 + a_0 \|u(t)\|_{L^2(\alpha, \beta)}^2 + \frac{a}{2} |u(\beta, t)|^2 = \int_{\alpha}^{\beta} f(x, t) u(x, t) dx + \frac{a}{2} |\phi(t)|^2. \quad (7.9)$$

Consequently, if $f = 0$ and $\phi = 0$, then

$$\|u(t)\|_{L^2(\alpha, \beta)} \leq \|u_0\|_{L^2(\alpha, \beta)} \quad \text{for all } t \geq 0. \quad (7.10)$$

If $a_0 > 0$, then

$$\|u(t)\|_{L^2(\alpha, \beta)}^2 \leq \|u_0\|_{L^2(\alpha, \beta)}^2 + \frac{1}{a_0} \int_0^t \|f(s)\|_{L^2(\alpha, \beta)}^2 ds + a \int_0^t |\phi(s)|^2 ds. \quad (7.11)$$

Proof. We multiply the equation by u and integrate over (α, β) :

$$\int_{\alpha}^{\beta} \partial_t u u dx + a \int_{\alpha}^{\beta} \partial_x u u dx + a_0 \int_{\alpha}^{\beta} u^2 dx = \int_{\alpha}^{\beta} f u dx. \quad (7.12)$$

The first term is

$$\int_{\alpha}^{\beta} \partial_t u u dx = \frac{1}{2} \frac{d}{dt} \int_{\alpha}^{\beta} u^2 dx. \quad (7.13)$$

The second term gives

$$a \int_{\alpha}^{\beta} \partial_x u u dx = \frac{a}{2} |u(\beta, t)|^2 - \frac{a}{2} |u(\alpha, t)|^2. \quad (7.14)$$

Using $u(\alpha, t) = \phi(t)$ yields the identity. If $f = 0$ and $\phi = 0$, the right-hand side is nonpositive after dropping the outflow term, and the norm is nonincreasing. If $a_0 > 0$, Young's inequality gives

$$\int_{\alpha}^{\beta} f u dx \leq \frac{1}{2a_0} \|f(t)\|_{L^2(\alpha, \beta)}^2 + \frac{a_0}{2} \|u(t)\|_{L^2(\alpha, \beta)}^2, \quad (7.15)$$

and integration in time gives the estimate. $\square$

A corresponding estimate for variable coefficients is given in [3, Chapter 13, Section 13.1.1].

7.3 Linear hyperbolic systems and the wave equation

We consider a first-order linear system

$$\partial_t U + A \partial_x U = 0 \quad \text{in } \mathbb{R} \times (0, +\infty), \quad U(x, 0) = U_0(x), \quad (7.16)$$

where $U(x, t)$ in $\mathbb{R}^p$ and A in $\mathbb{R}^{p \times p}$ is constant.

Definition 7.4. The system is hyperbolic if the matrix A is diagonalizable over $\mathbb{R}$ and has only real eigenvalues.

Proposition 7.5. Assume that $A = T\Lambda T^{-1}$, where

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad \lambda_k \in \mathbb{R}. \quad (7.17)$$

If $W = T^{-1}U$, then the components of W satisfy

$$\partial_t W_k + \lambda_k \partial_x W_k = 0, \quad k = 1, \dots, p. \quad (7.18)$$

In particular,

$$W_k(x, t) = W_k(x - \lambda_k t, 0), \quad U(x, t) = \sum_{k=1}^p W_k(x - \lambda_k t, 0) r_k, \quad (7.19)$$

where r_k is the k th column of T .

Proof. Since $U = TW$, we have

$$T\partial_t W + AT\partial_x W = 0. \quad (7.20)$$

Using $AT = T\Lambda$ and multiplying by T^{-1} gives

$$\partial_t W + \Lambda \partial_x W = 0. \quad (7.21)$$

The diagonal structure gives one scalar transport equation for each component. $\square$

On a bounded interval, the sign of each eigenvalue determines the boundary treatment: each positive eigenvalue requires one incoming condition at the left endpoint, and each negative eigenvalue requires one incoming condition at the right endpoint.

The wave equation as a hyperbolic system

The one-dimensional wave equation, with wave speed $\gamma > 0$,

$$\partial_{tt}u - \gamma^2 \partial_{xx}u = f \quad \text{in } (\alpha, \beta) \times (0, +\infty) \quad (7.22)$$

is a second-order equation in the displacement u . We set

$$v = \partial_t u, \quad q = \gamma \partial_x u, \quad U = \begin{pmatrix} v \\ q \end{pmatrix}. \quad (7.23)$$

Then

$$\partial_t U + \begin{pmatrix} 0 & -\gamma \\ -\gamma & 0 \end{pmatrix} \partial_x U = \begin{pmatrix} f \\ 0 \end{pmatrix}. \quad (7.24)$$

In the homogeneous case $f = 0$, this is a first-order homogeneous hyperbolic system. The characteristic variables

$$z_+ = v - q, \quad z_- = v + q \quad (7.25)$$

satisfy, in the general forced case,

$$\partial_t z_+ + \gamma \partial_x z_+ = f, \quad \partial_t z_- - \gamma \partial_x z_- = f. \quad (7.26)$$

Thus z_+ travels to the right and z_- travels to the left.

For a classical solution on $(0, L)$ with homogeneous Dirichlet data

$$u(0, t) = u(L, t) = 0, \quad (7.27)$$

we have

$$v(0, t) = v(L, t) = 0. \quad (7.28)$$

Since

$$v = \frac{z_+ + z_-}{2}, \quad (7.29)$$

the characteristic variables satisfy the reflection relations

$$z_+(0, t) = -z_-(0, t), \quad z_-(L, t) = -z_+(L, t). \quad (7.30)$$

Thus the incoming characteristic component is prescribed as the opposite of the outgoing component. For non-homogeneous data $u(0, t) = g_0(t)$ and $u(L, t) = g_L(t)$, the same computation gives

$$z_+(0, t) = 2g'_0(t) - z_-(0, t), \quad z_-(L, t) = 2g'_L(t) - z_+(L, t). \quad (7.31)$$

For homogeneous Neumann data

$$\partial_x u(0, t) = \partial_x u(L, t) = 0, \quad (7.32)$$

we have $q(0, t) = q(L, t) = 0$. Since

$$q = \frac{z_- - z_+}{2}, \quad (7.33)$$

the characteristic variables satisfy

$$z_+(0, t) = z_-(0, t), \quad z_-(L, t) = z_+(L, t). \quad (7.34)$$

The Neumann condition is therefore also a reflection relation, but with the opposite sign from the Dirichlet case. For non-homogeneous data $\partial_x u(0, t) = m_0(t)$ and $\partial_x u(L, t) = m_L(t)$, we obtain

$$z_+(0, t) = z_-(0, t) - 2\gamma m_0(t), \quad z_-(L, t) = z_+(L, t) + 2\gamma m_L(t). \quad (7.35)$$

Proposition 7.6. *Assume that u is smooth enough. Then*

$$\frac{d}{dt} \left(\frac{1}{2} \int_{\alpha}^{\beta} \left((\partial_t u)^2 + \gamma^2 (\partial_x u)^2 \right) dx \right) = \int_{\alpha}^{\beta} f \partial_t u \, dx + \gamma^2 [\partial_x u \partial_t u]_{\alpha}^{\beta}. \quad (7.36)$$

In particular, if $f = 0$ and either $u(\alpha, t) = u(\beta, t) = 0$ or $\partial_x u(\alpha, t) = \partial_x u(\beta, t) = 0$, then the wave energy

$$E(t) = \frac{1}{2} \int_{\alpha}^{\beta} \left((\partial_t u)^2 + \gamma^2 (\partial_x u)^2 \right) dx \quad (7.37)$$

is constant in time.

Proof. We multiply the equation by $\partial_t u$ and integrate over (α, β) :

$$\int_{\alpha}^{\beta} \partial_{tt} u \partial_t u \, dx - \gamma^2 \int_{\alpha}^{\beta} \partial_{xx} u \partial_t u \, dx = \int_{\alpha}^{\beta} f \partial_t u \, dx. \quad (7.38)$$

The first term is the derivative of $\frac{1}{2} \int (\partial_t u)^2$. Integration by parts gives

$$-\gamma^2 \int_{\alpha}^{\beta} \partial_{xx} u \partial_t u \, dx = -\gamma^2 [\partial_x u \partial_t u]_{\alpha}^{\beta} + \frac{\gamma^2}{2} \frac{d}{dt} \int_{\alpha}^{\beta} (\partial_x u)^2 \, dx. \quad (7.39)$$

This gives the identity. For homogeneous Dirichlet data, $\partial_t u(\alpha, t) = \partial_t u(\beta, t) = 0$. For homogeneous Neumann data, $\partial_x u(\alpha, t) = \partial_x u(\beta, t) = 0$. In both cases, the boundary term vanishes. $\square$

7.4 Uniform grids and classical finite difference schemes

We return to the scalar transport equation

$$\partial_t u + a \partial_x u = 0. \quad (7.40)$$

On the uniform grid

$$x_j = jh, \quad t^n = n\Delta t, \quad (7.41)$$

we denote the numerical approximation by U_j^n and set

$$\nu = \frac{a\Delta t}{h}, \quad \lambda = \frac{\Delta t}{h}. \quad (7.42)$$

The number $\nu = a\lambda$ is the Courant number. We compare the following four schemes in the same order throughout the chapter.

FTCS

The forward-time centered-space scheme is

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_{j-1}^n}{2h} = 0, \quad (7.43)$$

or equivalently

$$U_j^{n+1} = U_j^n - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n). \quad (7.44)$$

The abbreviation FTCS is used to avoid confusion with finite elements.

Upwind

For $a > 0$, the upwind scheme is

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_j^n - U_{j-1}^n}{h} = 0, \quad (7.45)$$

or

$$U_j^{n+1} = U_j^n - \nu(U_j^n - U_{j-1}^n). \quad (7.46)$$

If $a < 0$, the upwind difference is the forward difference. The subscript U in quantities such as g_U stands for upwind.

Lax–Friedrichs

The Lax–Friedrichs scheme is

$$\frac{U_j^{n+1} - \frac{1}{2}(U_{j+1}^n + U_{j-1}^n)}{\Delta t} + a \frac{U_{j+1}^n - U_{j-1}^n}{2h} = 0, \quad (7.47)$$

or

$$U_j^{n+1} = \frac{1}{2}(U_{j+1}^n + U_{j-1}^n) - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n). \quad (7.48)$$

Lax–Wendroff

The Lax–Wendroff scheme is

$$U_j^{n+1} = U_j^n - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n) + \frac{\nu^2}{2}(U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (7.49)$$

It is obtained by a second-order Taylor expansion in time, together with $\partial_t u = -a \partial_x u$ and $\partial_{tt} u = a^2 \partial_{xx} u$.

On a finite interval with $a > 0$, the inflow value is prescribed at the left boundary and no condition is imposed at the outflow boundary. The upwind scheme simply uses the prescribed left value. Centered three-point schemes such as FTCS, Lax–Friedrichs, and Lax–Wendroff require an adapted boundary closure or a ghost-cell construction near the boundary. For systems, discrete boundary conditions are imposed on incoming characteristic components only. See [3, Chapter 13, Section 13.3.3]. Section 7.7.1 below works out and analyses two concrete outflow closures.

7.5 Consistency and equivalent equations

Definition 7.7. An interior single-step finite difference scheme on the uniform grid is specified by a finite stencil $\mathcal{A} \subset \mathbb{Z}$ and by an update map

$$\mathcal{H}_{h,\Delta t,j}^n : \mathbb{R}^{\mathcal{A}} \rightarrow \mathbb{R}. \quad (7.50)$$

The numerical values satisfy

$$U_j^{n+1} = \mathcal{H}_{h,\Delta t,j}^n \left((U_{j+r}^n)_{r \in \mathcal{A}} \right). \quad (7.51)$$

For an autonomous constant-coefficient scheme, this map does not depend on j or n .

For the four schemes introduced above, the update maps are

FTCS

$$\mathcal{H}_{\text{FTCS}}(V_{-1}, V_0, V_1) = V_0 - \frac{\nu}{2}(V_1 - V_{-1}). \quad (7.52)$$

Upwind

For $a > 0$,

$$\mathcal{H}_U(V_{-1}, V_0) = V_0 - \nu(V_0 - V_{-1}). \quad (7.53)$$

Lax–Friedrichs

$$\mathcal{H}_{\text{LF}}(V_{-1}, V_0, V_1) = \frac{1}{2}(V_1 + V_{-1}) - \frac{\nu}{2}(V_1 - V_{-1}). \quad (7.54)$$

Lax–Wendroff

$$\mathcal{H}_{\text{LW}}(V_{-1}, V_0, V_1) = V_0 - \frac{\nu}{2}(V_1 - V_{-1}) + \frac{\nu^2}{2}(V_1 - 2V_0 + V_{-1}). \quad (7.55)$$

Definition 7.8. Let $\mathcal{H}_{h,\Delta t,j}^n$ be the update map of an interior single-step scheme. For a two-time-level grid function $W = (W_j^n)$, the normalized residual is

$$\text{Res}_{h,\Delta t,j}^n(W) = \frac{W_j^{n+1} - \mathcal{H}_{h,\Delta t,j}^n \left((W_{j+r}^n)_{r \in \mathcal{A}} \right)}{\Delta t}. \quad (7.56)$$

The numerical scheme is equivalently written as

$$\text{Res}_{h,\Delta t,j}^n(U) = 0. \quad (7.57)$$

Definition 7.9. Let u be a smooth exact solution of the differential equation and let

$$(I_h u)_j^n = u(x_j, t^n) \quad (7.58)$$

be its grid restriction. The local truncation error is the normalized residual obtained after inserting the exact grid values into the numerical update:

$$\tau_j^n(u) = \text{Res}_{h,\Delta t,j}^n(I_h u). \quad (7.59)$$

Equivalently,

$$\tau_j^n(u) = \frac{u(x_j, t^{n+1}) - \mathcal{H}_{h,\Delta t,j}^n \left((u(x_{j+r}, t^n))_{r \in \mathcal{A}} \right)}{\Delta t}. \quad (7.60)$$

A scheme is consistent with the differential equation if $\tau_j^n(u) \rightarrow 0$ as $h \rightarrow 0$ and $\Delta t \rightarrow 0$, for every smooth exact solution u , under the chosen relation between h and Δt .

Remark 7.10. This definition concerns the interior stencil. Boundary formulas, when present, have their own truncation errors. The normalization by $1/\Delta t$ gives τ_j^n the same physical dimension as the differential operator $\partial_t u + a \partial_x u$.

Proposition 7.12. *Let u be a smooth solution of $\partial_t u + a\partial_x u = 0$. The local truncation errors of the four classical schemes are as follows.*

FTCS

$$\tau_{j,\text{FTCS}}^n = \frac{\Delta t}{2} \partial_{tt} u + \frac{ah^2}{6} \partial_{xxx} u + O(\Delta t^2 + h^4). \quad (7.61)$$

Upwind

For $a > 0$,

$$\tau_{j,U}^n = \frac{\Delta t}{2} \partial_{tt} u - \frac{ah}{2} \partial_{xx} u + O(\Delta t^2 + h^2). \quad (7.62)$$

Lax–Friedrichs

$$\tau_{j,\text{LF}}^n = \frac{\Delta t}{2} \partial_{tt} u - \frac{h^2}{2\Delta t} \partial_{xx} u + \frac{ah^2}{6} \partial_{xxx} u + O\left(\Delta t^2 + \frac{h^4}{\Delta t} + h^4\right). \quad (7.63)$$

Lax–Wendroff

$$\tau_{j,\text{LW}}^n = \frac{a}{6} (h^2 - a^2 \Delta t^2) \partial_{xxx} u + O(\Delta t^3 + \Delta t h^2 + h^4). \quad (7.64)$$

Under the hyperbolic scaling $\Delta t = \lambda h$, FTCS, upwind, and Lax–Friedrichs are first-order accurate, while Lax–Wendroff is second-order accurate.

Proof. All derivatives below are evaluated at (x_j, t^n) . For FTCS,

$$\frac{u(x_j, t^{n+1}) - u(x_j, t^n)}{\Delta t} = \partial_t u + \frac{\Delta t}{2} \partial_{tt} u + O(\Delta t^2), \quad (7.65)$$

$$\frac{u(x_{j+1}, t^n) - u(x_{j-1}, t^n)}{2h} = \partial_x u + \frac{h^2}{6} \partial_{xxx} u + O(h^4). \quad (7.66)$$

The terms $\partial_t u + a\partial_x u$ cancel by the equation. This gives the FTCS formula.

For the upwind scheme with $a > 0$,

$$\frac{u(x_j, t^n) - u(x_{j-1}, t^n)}{h} = \partial_x u - \frac{h}{2} \partial_{xx} u + O(h^2). \quad (7.67)$$

The announced upwind formula follows after cancellation of $\partial_t u + a\partial_x u$.

For Lax–Friedrichs, the additional average satisfies

$$\frac{u(x_{j+1}, t^n) + u(x_{j-1}, t^n)}{2} = u + \frac{h^2}{2} \partial_{xx} u + O(h^4). \quad (7.68)$$

Substitution in the normalized residual gives

$$\tau_{j,\text{LF}}^n = \partial_t u + a\partial_x u + \frac{\Delta t}{2} \partial_{tt} u - \frac{h^2}{2\Delta t} \partial_{xx} u + \frac{ah^2}{6} \partial_{xxx} u + O\left(\Delta t^2 + \frac{h^4}{\Delta t} + h^4\right), \quad (7.69)$$

and the differential equation cancels the first two terms.

For Lax–Wendroff, the residual is

$$\tau_{j,\text{LW}}^n = \frac{u(x_j, t^{n+1}) - u(x_j, t^n)}{\Delta t} + a \frac{u(x_{j+1}, t^n) - u(x_{j-1}, t^n)}{2h} \quad (7.70)$$

$$- \frac{a^2 \Delta t}{2} \frac{u(x_{j+1}, t^n) - 2u(x_j, t^n) + u(x_{j-1}, t^n)}{h^2}. \quad (7.71)$$

The second difference is

$$\frac{u(x_{j+1}, t^n) - 2u(x_j, t^n) + u(x_{j-1}, t^n)}{h^2} = \partial_{xx} u + \frac{h^2}{12} \partial_{xxxx} u + O(h^4). \quad (7.72)$$

The order-one terms cancel by the equation. The terms

$$\frac{\Delta t}{2} \partial_{tt} u - \frac{a^2 \Delta t}{2} \partial_{xx} u \quad (7.73)$$

also cancel because $\partial_{tt} u = a^2 \partial_{xx} u$. Since $\partial_{ttt} u = -a^3 \partial_{xxx} u$, the leading remaining term is

$$\frac{\Delta t^2}{6} \partial_{ttt} u + \frac{ah^2}{6} \partial_{xxx} u = \frac{a}{6} (h^2 - a^2 \Delta t^2) \partial_{xxx} u. \quad (7.74)$$

□

For smooth periodic solutions, these pointwise estimates imply corresponding finite-time consistency bounds in the discrete norms. For example, if the leading truncation error is $O(h^r)$ uniformly on $[0, T]$, then

$$\max_{0 \leq n \Delta t \leq T} \|\tau^n(u)\|_{\ell_h^2} \leq C_T h^r, \quad (7.75)$$

where C_T depends on a finite number of derivatives of u but not on h .

Equivalent equations, also called modified equations, identify the leading numerical defect of a scheme. We look for a smooth function v that satisfies the difference equation at every point (x, t) , not only at the grid nodes, and then expand the scheme by Taylor formula.

Proposition 7.13. *Assume that $a > 0$ and $\Delta t = \lambda h$, so that $\nu = a\lambda$. The leading equivalent equations of the classical schemes are*

FTCS

$$\partial_t v + a \partial_x v = -\frac{a\nu h}{2} \partial_{xx} v + O(h^2). \quad (7.76)$$

Upwind

$$\partial_t v + a \partial_x v = \frac{ah}{2} (1 - \nu) \partial_{xx} v + O(h^2). \quad (7.77)$$

Lax–Friedrichs

$$\partial_t v + a \partial_x v = \frac{h}{2\lambda} (1 - \nu^2) \partial_{xx} v + O(h^2). \quad (7.78)$$

Lax–Wendroff

$$\partial_t v + a \partial_x v = -\frac{ah^2}{6} (1 - \nu^2) \partial_{xxx} v + O(h^3). \quad (7.79)$$

Proof. The Taylor expansions obtained in the proof of Proposition 7.12 are now interpreted as equations satisfied by v . For example, the upwind scheme gives

$$\partial_t v + a \partial_x v + \frac{\Delta t}{2} \partial_{tt} v - \frac{ah}{2} \partial_{xx} v = O(h^2 + \Delta t^2). \quad (7.80)$$

Using $\partial_{tt} v = a^2 \partial_{xx} v$ to leading order and $\Delta t = \lambda h$ gives

$$\partial_t v + a \partial_x v + \frac{ah}{2} (\nu - 1) \partial_{xx} v = O(h^2), \quad (7.81)$$

which is the announced upwind formula. The other three formulas follow in the same way from the FTCS, Lax–Friedrichs, and Lax–Wendroff expansions in the proof of Proposition 7.12. □

Remark 7.14. The equivalent equation displayed above records only the first non-vanishing modified-equation term. This is useful for identifying the leading artificial viscosity or the leading third-derivative defect, but it does not give the full dispersion picture. The Fourier expansions in Section 7.8 show that all four schemes have a phase error of order θ^3 for generic Courant numbers, although the leading modified equation of upwind and Lax–Friedrichs is diffusive.

7.6 The CFL condition and numerical domains of dependence

The Courant–Friedrichs–Lewy condition expresses that the numerical domain of dependence must contain the exact domain of dependence in the limit. Here we formulate the usual necessity statement under the hyperbolic scaling $\Delta t = \lambda h$ with fixed $\lambda > 0$, hence fixed Courant number $\nu = a\lambda$. For a stencil contained in $\{j-1, j, j+1\}$, the numerical value after n steps depends only on initial nodes in $[x_j - nh, x_j + nh]$. The exact value depends on the initial point $x_j - an\Delta t$. Consequently, convergence for every initial datum requires

$$|a|\Delta t \leq h, \quad \text{that is} \quad |\nu| \leq 1. \quad (7.82)$$

Indeed, if $|\nu| > 1$ is fixed, at a positive fixed time the exact characteristic foot lies outside the numerical domain by a positive distance. Initial data supported there are invisible to the scheme but affect the exact solution. This argument can be made on the whole line, or on a periodic interval for times small enough that the relevant domains do not wrap around the interval.

Under this fixed-Courant scaling, CFL is necessary for convergence for all data, but is not sufficient by itself. FTCS satisfies the same geometric restriction, yet the von Neumann analysis below shows that it is not stable in ℓ_h^2 for fixed nonzero ν .

Remark 7.15 (Courant numbers depending on the mesh). An exact inequality $|\nu_h| \leq 1$ on every mesh is not necessary for finite-time convergence when ν_h varies with h . For example, on the unit periodic interval with $a > 0$, take upwind with $\nu_h = 1 + h$ and $\Delta t_h = h(1 + h)/a$. The one-step operator satisfies

$$\|G_h\|_{\ell_h^2 \rightarrow \ell_h^2} \leq |1 - \nu_h| + \nu_h = 1 + 2h, \quad \|G_h^n\| \leq (1 + 2h)^n \leq e^{2aT} \quad (n\Delta t_h \leq T). \quad (7.83)$$

The scheme is stable on finite time intervals and its normalized residual for smooth transport solutions is $O(h)$. It therefore converges despite the small CFL violation on every grid. Its numerical domain approaches the exact one as $h \rightarrow 0$, since $\nu_h \rightarrow 1$. This does not contradict the fixed-Courant necessity statement above.

7.7 Stability and convergence of the classical schemes

Definition 7.16. Let J_h be the set of grid indices used in the computation. For a grid vector $V = (V_j)_{j \in J_h}$, we define

$$\|V\|_{\ell_h^1} = h \sum_{j \in J_h} |V_j|, \quad (7.84)$$

$$\|V\|_{\ell_h^2} = \left(h \sum_{j \in J_h} |V_j|^2 \right)^{1/2}, \quad (7.85)$$

and

$$\|V\|_{\ell_h^\infty} = \sup_{j \in J_h} |V_j|. \quad (7.86)$$

The factor h in the definitions of ℓ_h^1 and ℓ_h^2 makes these norms discrete analogues of the continuous L^1 and L^2 norms.

Definition 7.17. Consider a one-step finite difference evolution

$$U^{n+1} = G_h U^n. \quad (7.87)$$

Let $1 \leq p \leq \infty$. For a sequence of grid vectors $U = (U^n)$ on a finite time interval $[0, T]$, we use the notation

$$\|U\|_{\ell^\infty(0, T; \ell_h^p)} := \max_{0 \leq n \Delta t \leq T} \|U^n\|_{\ell_h^p}. \quad (7.88)$$

The scheme is *stable in ℓ_h^p on finite time intervals* if, for every $T > 0$, there exists a constant $C_T > 0$, independent of h , Δt , and n , such that

$$\|U\|_{\ell^\infty(0,T;\ell_h^p)} \leq C_T \|U^0\|_{\ell_h^p}. \quad (7.89)$$

Equivalently,

$$\|U^n\|_{\ell_h^p} \leq C_T \|U^0\|_{\ell_h^p} \quad \text{whenever } 0 \leq n\Delta t \leq T. \quad (7.90)$$

The scheme is *strongly stable in ℓ_h^p* if it is non-expansive in this norm, namely

$$\|U^{n+1}\|_{\ell_h^p} \leq \|U^n\|_{\ell_h^p} \quad \text{for every } n \geq 0. \quad (7.91)$$

This implies (7.89) with $C_T = 1$ for every finite time interval.

Thus stability in ℓ_h^2 means boundedness in the finite-time topology $\ell^\infty(0,T;\ell_h^2)$. Strong stability in ℓ_h^2 means that the ℓ_h^2 norm cannot increase from one time level to the next.

A direct estimate for the upwind scheme

Proposition 7.18. *Assume that $a > 0$ and $0 \leq \nu \leq 1$, and work on an infinite or periodic grid. Then the upwind scheme satisfies the discrete maximum principle*

$$\|U^{n+1}\|_{\ell_h^\infty} \leq \|U^n\|_{\ell_h^\infty}. \quad (7.92)$$

In particular, it is strongly stable in ℓ_h^∞ .

Proof. The upwind update is

$$U_j^{n+1} = (1 - \nu)U_j^n + \nu U_{j-1}^n. \quad (7.93)$$

If $0 \leq \nu \leq 1$, this is a convex combination of two values at time level n . Hence

$$|U_j^{n+1}| \leq \max\{|U_j^n|, |U_{j-1}^n|\} \leq \|U^n\|_{\ell_h^\infty}. \quad (7.94)$$

Taking the supremum over j gives the result. $\square$

On an infinite or periodic grid, the upwind and Lax–Friedrichs schemes have nonnegative stencil coefficients under the CFL condition. Summing the pointwise convexity estimate and using invariance of the sum under a grid shift gives strong stability in ℓ_h^1 . On a finite interval with prescribed inflow $U_0^n = \phi(t^n)$, the maximum principle instead bounds the solution by the maximum of its initial values and the inflow values imposed up to the current time. The Lax–Wendroff stencil contains a negative coefficient when $0 < |\nu| < 1$, so its ℓ_h^1 stability is not a consequence of monotonicity. No ℓ_h^1 stability statement for Lax–Wendroff is used here; below we use Fourier analysis to obtain the ℓ_h^2 stability needed for convergence.

Von Neumann analysis on the periodic grid

The Fourier analysis below concerns the interior scheme, on an infinite or periodic grid. Boundary conditions are not part of this local computation.

Definition 7.19. For a linear constant-coefficient one-step scheme on an infinite or periodic grid, we insert a Fourier mode

$$U_j^n = \widehat{U}^n e^{ij\theta}, \quad \theta \in [-\pi, \pi]. \quad (7.95)$$

If the update gives

$$\widehat{U}^{n+1} = g(\theta) \widehat{U}^n, \quad (7.96)$$

then $g(\theta)$ is the amplification factor of the scheme. After n steps, the same mode is multiplied by $g(\theta)^n$.

Theorem 7.20. Consider a linear constant-coefficient one-step scheme on a periodic grid, and let $g(\theta)$ be its amplification factor.

(i) If there exists a constant $\beta \geq 0$ such that

$$|g(\theta)| \leq 1 + \beta \Delta t \quad \text{for all Fourier modes } \theta, \quad (7.97)$$

then the scheme is stable in ℓ_h^2 on every finite time interval. If we can take $\beta = 0$, then the scheme is strongly stable in ℓ_h^2 .

(ii) If the scheme is strongly stable in ℓ_h^2 , then $|g(\theta)| \leq 1$ for every Fourier mode.

The sufficient stability statement in (i) is the scalar one-step case of the von Neumann criterion in Quarteroni [3, Chapter 13, Theorem 13.3]. The necessary condition in (ii) is stated in Quarteroni [3, Chapter 13, Remark 13.4]; it also follows immediately by taking a single Fourier mode as initial data.

Below, the FTCS case is treated first by a direct Fourier-mode counterexample. We exhibit a mode that is multiplied at each step by a factor bounded away from 1, so no finite-time stability estimate with a constant independent of h can hold. Then the sufficient implication in (i) is used to prove stability of the upwind, Lax–Friedrichs, and Lax–Wendroff schemes. This is not presented as a general converse to the von Neumann criterion.

Proposition 7.21. The amplification factors of the four classical schemes are

FTCS

$$g_{\text{FTCS}}(\theta) = 1 - i\nu \sin \theta. \quad (7.98)$$

Upwind

For $a > 0$,

$$g_U(\theta) = 1 - \nu(1 - e^{-i\theta}). \quad (7.99)$$

Lax–Friedrichs

$$g_{\text{LF}}(\theta) = \cos \theta - i\nu \sin \theta. \quad (7.100)$$

Lax–Wendroff

$$g_{\text{LW}}(\theta) = 1 - i\nu \sin \theta + \nu^2(\cos \theta - 1). \quad (7.101)$$

Proof. We insert the Fourier mode

$$U_j^n = g^n e^{ij\theta}. \quad (7.102)$$

Then

$$U_{j+1}^n = e^{i\theta} U_j^n, \quad U_{j-1}^n = e^{-i\theta} U_j^n, \quad U_j^{n+1} = g(\theta) U_j^n. \quad (7.103)$$

For FTCS,

$$g(\theta) U_j^n = U_j^n - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n) \quad (7.104)$$

$$= \left(1 - \frac{\nu}{2}(e^{i\theta} - e^{-i\theta})\right) U_j^n. \quad (7.105)$$

Since $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$, we get $g_{\text{FTCS}}(\theta) = 1 - i\nu \sin \theta$.

For the upwind scheme with $a > 0$,

$$g(\theta)U_j^n = U_j^n - \nu(U_j^n - U_{j-1}^n) \quad (7.106)$$

$$= \left(1 - \nu(1 - e^{-i\theta})\right) U_j^n. \quad (7.107)$$

This gives g_U .

For Lax–Friedrichs,

$$g(\theta)U_j^n = \frac{1}{2}(U_{j+1}^n + U_{j-1}^n) - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n) \quad (7.108)$$

$$= \left(\frac{e^{i\theta} + e^{-i\theta}}{2} - \frac{\nu}{2}(e^{i\theta} - e^{-i\theta})\right) U_j^n. \quad (7.109)$$

Using $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ gives $g_{\text{LF}}(\theta) = \cos \theta - i\nu \sin \theta$.

For Lax–Wendroff,

$$g(\theta)U_j^n = U_j^n - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n) + \frac{\nu^2}{2}(U_{j+1}^n - 2U_j^n + U_{j-1}^n) \quad (7.110)$$

$$= \left(1 - \frac{\nu}{2}(e^{i\theta} - e^{-i\theta}) + \frac{\nu^2}{2}(e^{i\theta} - 2 + e^{-i\theta})\right) U_j^n. \quad (7.111)$$

Since $e^{i\theta} - 2 + e^{-i\theta} = 2(\cos \theta - 1)$, we obtain the stated factor. $\square$

Proposition 7.22. *On periodic grids, under the hyperbolic scaling $\Delta t = \lambda h$ with fixed $\nu \neq 0$, the FTCS scheme is not stable in ℓ_h^2 on finite time intervals. If $0 \leq \nu \leq 1$, the upwind scheme is strongly stable in ℓ_h^2 . If $|\nu| \leq 1$, the Lax–Friedrichs scheme is strongly stable in ℓ_h^2 . If $|\nu| \leq 1$, the Lax–Wendroff scheme is strongly stable in ℓ_h^2 .*

Proof. For FTCS,

$$|g_{\text{FTCS}}(\theta)|^2 = 1 + \nu^2 \sin^2 \theta. \quad (7.112)$$

As soon as $\nu \neq 0$, this is larger than 1 for modes with $\sin \theta \neq 0$, so strong stability is impossible. More precisely, on arbitrarily fine periodic grids, choose a Fourier mode with $\sin^2 \theta \geq 1/2$. Then

$$|g_{\text{FTCS}}(\theta)| \geq \sqrt{1 + \frac{\nu^2}{2}} > 1. \quad (7.113)$$

For $n\Delta t \leq T$, this mode is multiplied by $|g_{\text{FTCS}}(\theta)|^n$. Since n is of order $T/\Delta t$ and $\Delta t \rightarrow 0$ under hyperbolic scaling, the amplification is unbounded as $h \rightarrow 0$. Hence no estimate of the form (7.89) can hold in ℓ_h^2 . This is a direct Fourier-mode counterexample to finite-time stability, not an invocation of a converse to Theorem 7.20.

For upwind,

$$|g_U(\theta)|^2 = |1 - \nu + \nu e^{-i\theta}|^2 \quad (7.114)$$

$$= (1 - \nu)^2 + \nu^2 + 2\nu(1 - \nu) \cos \theta \quad (7.115)$$

$$= 1 - 2\nu(1 - \nu)(1 - \cos \theta). \quad (7.116)$$

If $0 \leq \nu \leq 1$, then $|g_U(\theta)| \leq 1$.

For Lax–Friedrichs,

$$|g_{\text{LF}}(\theta)|^2 = \cos^2 \theta + \nu^2 \sin^2 \theta = 1 - (1 - \nu^2) \sin^2 \theta. \quad (7.117)$$

If $|\nu| \leq 1$, then $|g_{\text{LF}}(\theta)| \leq 1$.

For Lax–Wendroff,

$$|g_{\text{LW}}(\theta)|^2 = 1 - 4\nu^2(1 - \nu^2) \sin^4 \frac{\theta}{2}. \quad (7.118)$$

Therefore $|g_{\text{LW}}(\theta)| \leq 1$ when $|\nu| \leq 1$. The von Neumann criterion gives the announced ℓ_h^2 stability statements. $\square$

FTCS is therefore excluded in the hyperbolic finite-time topology $\ell^\infty(0, T; \ell_h^2)$. It is nevertheless stable in ℓ_h^2 under the parabolic restriction

$$\Delta t \leq \frac{h^2}{a^2}, \quad (7.119)$$

with a stability constant on finite time intervals; see Quarteroni [3, Theorem 13.2]. This parabolic restriction is not the natural scaling for the transport equation.

Corollary 7.23. *Consider a smooth solution of the transport equation with period $L > 0$ on a fixed finite time interval $[0, T]$. Set $h = L/N$ and $\Delta t = \lambda h$, where $\lambda > 0$ is fixed. Define the nodal restriction by*

$$(I_h u(t))_j = u(jh, t), \quad j = 0, \dots, N-1, \quad (7.120)$$

with indices understood modulo N , and initialize the numerical solution by $U^0 = I_h u_0$. Convergence for this smooth solution is measured by

$$\max_{0 \leq n, \Delta t \leq T} \|U^n - I_h u(t^n)\|_{\ell_h^2} \longrightarrow 0 \quad \text{as } h \rightarrow 0. \quad (7.121)$$

The evolution error estimate of Theorem 6.13, together with Proposition 7.12 and the stability estimates above, gives the following conclusions, in the same order as the schemes were introduced.

FTCS *The scheme is consistent, but its finite-time ℓ_h^2 stability fails for fixed $\nu \neq 0$. The stability hypothesis of Theorem 6.13 is therefore unavailable. Some smooth initial data give convergence, whereas others give nonconvergence, as explained in Remark 7.24 below.*

Upwind *If $a > 0$ and $0 \leq \nu \leq 1$, the scheme converges in (7.121), with error $O(h)$.*

Lax–Friedrichs *If $|\nu| \leq 1$, the scheme converges in (7.121), with error $O(h)$.*

Lax–Wendroff *If $|\nu| \leq 1$, the scheme converges in (7.121), with error $O(h^2)$.*

Proof. Set $e^n = U^n - I_h u(t^n)$. The normalized truncation error satisfies

$$e^{n+1} = G_h e^n - \Delta t \tau^n(u). \quad (7.122)$$

Since $e^0 = 0$, Theorem 6.13 gives, for each stable scheme,

$$\max_{0 \leq n, \Delta t \leq T} \|e^n\|_{\ell_h^2} \leq C_T T \max_{(n+1)\Delta t \leq T} \|\tau^n(u)\|_{\ell_h^2}. \quad (7.123)$$

The consistency bounds give the stated rates. More generally, an approximate initialization adds $C_T \|e^0\|_{\ell_h^2}$ to this estimate, so the initial error must tend to zero and must have the appropriate order to preserve the rate. The rate statements are upper bounds; at the exact-translation Courant numbers the corresponding schemes may be exact on the sampled solution. $\square$

Remark 7.24. Failure of stability does not imply failure of convergence for every individual datum. For FTCS, any fixed trigonometric polynomial gives convergence under $\Delta t = \lambda h$: for each fixed physical frequency, the numerical amplification over a fixed time interval approaches the exact transport phase.

Smoothness alone is nevertheless insufficient. For period 2π , the real function

$$u_0(x) = \sum_{k \geq 1} e^{-2^k} \cos(4^k x) \quad (7.124)$$

belongs to C^∞ , since every differentiated series converges absolutely. On the grids with $N_k = 4^{k+1}$ points, its frequency 4^k has grid phase $\theta = \pi/2$ and FTCS amplification modulus $q = \sqrt{1 + \nu^2} > 1$. No other term in the series aliases to this Fourier mode: the lower frequencies are distinct and the higher ones alias to the constant mode. After $n_k = \lfloor T/(\lambda h_k) \rfloor$ steps, the numerical norm is bounded below by a positive constant times

$$e^{-2^k} q^{n_k} \longrightarrow +\infty, \quad h_k = \frac{2\pi}{4^{k+1}}, \quad (7.125)$$

because n_k is of order 4^k , whereas the exact smooth solution remains bounded. Thus FTCS can fail to converge even for real smooth initial data.

Convergence for general L^2 initial data

Nodal sampling is defined for the smooth solutions above, but it is not defined on arbitrary L^2 equivalence classes. To formulate convergence for every initial datum in $X = L^2(\mathbb{T}_L)$, where $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, we use cell averages and piecewise constant reconstruction. On $h = L/N$, let

$$(P_h v)_j = \frac{1}{h} \int_{jh}^{(j+1)h} v(x) dx, \quad (E_h V)(x) = V_j \quad \text{for } jh \leq x < (j+1)h. \quad (7.126)$$

Here $X_h = \mathbb{R}^N$ with the periodic ℓ_h^2 norm. Jensen's inequality and the definitions give

$$\|P_h v\|_{\ell_h^2} \leq \|v\|_{L^2(\mathbb{T}_L)}, \quad \|E_h V\|_{L^2(\mathbb{T}_L)} = \|V\|_{\ell_h^2}, \quad P_h E_h = I_{X_h}. \quad (7.127)$$

Moreover, $E_h P_h v \rightarrow v$ in $L^2(\mathbb{T}_L)$ for every $v \in X$. This follows first for smooth functions and then by density and the uniform bounds above.

The exact evolution $S(t)v = v(\cdot - at)$ is a strongly continuous group of isometries on X . For smooth data, averaging the shifted nodal truncation errors over each cell gives

$$\max_{(n+1)\Delta t \leq T} \left\| \frac{P_h S(t^{n+1})v - G_h P_h S(t^n)v}{\Delta t} \right\|_{\ell_h^2} \longrightarrow 0. \quad (7.128)$$

The smooth periodic functions form a dense subspace of X , so all the hypotheses of Theorem 6.15 are satisfied. Consequently, under $\Delta t = \lambda h$ and with initialization $U^0 = P_h u_0$, finite-time ℓ_h^2 stability is equivalent to

$$\max_{0 \leq n\Delta t \leq T} \|E_h U^n - S(t^n)u_0\|_{L^2(\mathbb{T}_L)} \longrightarrow 0 \quad \text{for every } u_0 \in L^2(\mathbb{T}_L). \quad (7.129)$$

Equivalently, the error can be measured by $\max_{0 \leq n\Delta t \leq T} \|U^n - P_h S(t^n)u_0\|_{\ell_h^2}$; the reconstruction of the cell averages converges uniformly on the compact exact orbit $\{S(t)u_0 : 0 \leq t \leq T\}$.

Under their respective CFL restrictions, upwind, Lax–Friedrichs, and Lax–Wendroff therefore converge for every L^2 initial datum in (7.129). FTCS with fixed $\nu \neq 0$ fails this all-data convergence property because it is unstable. This conclusion uses Theorem 6.15 and does not assert nonconvergence for every individual datum. The smooth counterexample of Remark 7.24 also works with cell-average initialization: the modulus of its selected Fourier coefficient acquires the fixed nonzero factor $\sin(\pi/4)/(\pi/4)$, and all its higher frequencies have zero cell average.

7.7.1 Worked boundary closures on an inflow–outflow interval

Periodic Fourier estimates do not by themselves prove stability of a scheme on a bounded interval: the boundary rows must also be specified and analysed. We work out a simple closure for Lax–Friedrichs and Lax–Wendroff on $(0, L)$ with $a > 0$:

$$\partial_t u + a \partial_x u = 0, \quad u(x, 0) = u_0(x), \quad u(0, t) = \phi(t). \quad (7.130)$$

There is no prescribed value at $x = L$. Characteristics give

$$u(x, t) = \begin{cases} u_0(x - at), & x \geq at, \\ \phi(t - x/a), & x < at. \end{cases} \quad (7.131)$$

The corner compatibility $u_0(0) = \phi(0)$ makes these two formulas agree on $x = at$. For the smooth error estimates below we assume a smooth exact solution on $[0, L] \times [0, T]$. Smooth data with the additional compatibilities $\phi^{(k)}(0) = (-a)^k u_0^{(k)}(0)$ to the required order provide such a solution.

Let $N \geq 2$, $h = L/N$, $x_j = jh$, and fix $0 < \nu = a\Delta t/h \leq 1$. Initialize $U_j^0 = u_0(x_j)$ for $0 \leq j \leq N$. At time level n , set

$$U_0^n = \phi(t^n), \quad U_{N+1}^n = 2U_N^n - U_{N-1}^n. \quad (7.132)$$

Use the chosen three-point scheme at $j = 1, \dots, N$, then set $U_0^{n+1} = \phi(t^{n+1})$. The ghost value is an extrapolation used by the numerical stencil; it is not an additional boundary datum for the differential equation.

For Lax–Friedrichs, substituting (7.132) in the last row gives

$$U_N^{n+1} = \frac{1-\nu}{2}(2U_N^n - U_{N-1}^n) + \frac{1+\nu}{2}U_{N-1}^n \quad (7.133)$$

$$= (1-\nu)U_N^n + \nu U_{N-1}^n. \quad (7.134)$$

For Lax–Wendroff, the ghost rule gives

$$U_{N+1}^n - U_{N-1}^n = 2(U_N^n - U_{N-1}^n), \quad U_{N+1}^n - 2U_N^n + U_{N-1}^n = 0, \quad (7.135)$$

so its last row is exactly the same upwind update:

$$U_N^{n+1} = (1-\nu)U_N^n + \nu U_{N-1}^n. \quad (7.136)$$

Thus the value at the outflow is evolved from computed values inside the interval.

Writing the interior stencil as $cU_{j-1}^n + dU_j^n + bU_{j+1}^n$, the complete update for the N unknowns $U_1, \dots, U_N$ is

$$U_1^{n+1} = c\phi(t^n) + dU_1^n + bU_2^n, \quad (7.137)$$

$$U_j^{n+1} = cU_{j-1}^n + dU_j^n + bU_{j+1}^n, \quad 2 \leq j \leq N-1, \quad (7.138)$$

$$U_N^{n+1} = \nu U_{N-1}^n + (1-\nu)U_N^n, \quad (7.139)$$

where

$$(c, d, b) = \begin{cases} ((1+\nu)/2, 0, (1-\nu)/2), & \text{Lax–Friedrichs,} \\ (\nu(1+\nu)/2, 1-\nu^2, -\nu(1-\nu)/2), & \text{Lax–Wendroff.} \end{cases} \quad (7.140)$$

In particular, the first row uses the inflow value at t^n , while the prescribed boundary entry at the new time is $\phi(t^{n+1})$.

Proposition 7.25. *The Lax–Friedrichs update with this closure satisfies*

$$\max_{0 \leq j \leq N} |U_j^n| \leq \max \left\{ \max_{0 \leq j \leq N} |U_j^0|, \max_{0 \leq m \leq n} |\phi(t^m)| \right\}. \quad (7.141)$$

If the exact solution is smooth, the initialization and imposed inflow are exact, and $\nu \in (0, 1]$ is fixed, then

$$\max_{0 \leq n \Delta t \leq T} \max_{0 \leq j \leq N} |U_j^n - u(x_j, t^n)| \leq C_T h. \quad (7.142)$$

Proof. The interior Lax–Friedrichs coefficients are nonnegative and sum to 1; the last row (7.136) is also a convex combination. Induction proves the maximum bound. For the error $e_j^n = U_j^n - u(x_j, t^n)$, we have $e_0^n = 0$ and

$$\max_j |e_j^{n+1}| \leq \max_j |e_j^n| + \Delta t \max_{1 \leq j \leq N} |\tau_j^n|. \quad (7.143)$$

Here τ_j^n is the normalized residual of the actual closed update: at $j = N$ it is the upwind residual, rather than the interior Lax–Friedrichs residual. Both are $O(h)$ for fixed ν , so summing the estimate gives the result. $\square$

Lax–Wendroff is not a convex-combination scheme, but this particular closure admits a direct energy proof. For real grid vectors, define the weighted norm and the unscaled scalar product used in its energy proof by

$$\|V\|_{h,*}^2 = h \left(\sum_{j=1}^{N-1} |V_j|^2 + \frac{1}{2} |V_N|^2 \right), \quad \langle V, W \rangle_* = \sum_{j=1}^{N-1} V_j W_j + \frac{1}{2} V_N W_N. \quad (7.144)$$

Thus $\|V\|_{h,*}^2 = h \langle V, V \rangle_*$. This norm is uniformly equivalent to the usual ℓ_h^2 norm on $j = 1, \dots, N$: $\|V\|_{h,*} \leq \|V\|_{\ell_h^2} \leq \sqrt{2} \|V\|_{h,*}$.

Proposition 7.26. *Let G_h be the Lax–Wendroff update with (7.136) and homogeneous inflow. For $0 \leq \nu \leq 1$,*

$$\|G_h V\|_{h,*} \leq \|V\|_{h,*}. \quad (7.145)$$

Consequently the closed homogeneous scheme is stable in ℓ_h^2 , with a finite-time constant independent of h .

Proof. Set $V_0 = 0$, $V_{N+1} = 2V_N - V_{N-1}$, and $q_j = V_j - V_{j-1}$ for $1 \leq j \leq N$. Define

$$(D_0 V)_j = \frac{V_{j+1} - V_{j-1}}{2}, \quad (D_2 V)_j = V_{j+1} - 2V_j + V_{j-1}. \quad (7.146)$$

At $j = N$, these are q_N and 0, respectively, and $G_h V = V - \nu D_0 V + (\nu^2/2) D_2 V$. Discrete summation gives

$$\langle V, D_0 V \rangle_* = \frac{1}{2} V_N^2, \quad (7.147)$$

$$\langle V, D_2 V \rangle_* = - \sum_{j=1}^N q_j^2 + V_N q_N, \quad (7.148)$$

$$\|D_0 V\|_*^2 = \sum_{j=1}^N q_j^2 - \frac{1}{2} q_1^2 - \frac{1}{4} \sum_{j=1}^{N-1} (q_{j+1} - q_j)^2, \quad (7.149)$$

$$\langle D_0 V, D_2 V \rangle_* = \frac{1}{2} (q_N^2 - q_1^2), \quad (7.150)$$

where $\|V\|_*^2 = \langle V, V \rangle_*$. Expanding the squared norm of the update and using these identities yields

$$\begin{aligned} \|V\|_*^2 - \|G_h V\|_*^2 &= \nu \left[\left(V_N - \frac{\nu}{2} q_N \right)^2 + \frac{\nu^2}{4} q_N^2 \right] + \frac{\nu^2(1-\nu)}{2} q_1^2 \\ &\quad + \frac{\nu^2(1-\nu^2)}{4} \sum_{j=1}^{N-1} (q_{j+1} - q_j)^2 \geq 0. \end{aligned} \quad (7.151)$$

Multiplication by h proves the weighted contraction. Norm equivalence gives $\|G_h^n V\|_{\ell_h^2} \leq \sqrt{2} \|V\|_{\ell_h^2}$. $\square$

For a C^1 nonzero inflow, set $V_j^n = U_j^n - \phi(t^n)$, so $V_0^n = 0$. Since both closed updates preserve constant vectors when the inflow has that same constant value,

$$V^{n+1} = G_h V^n - (\phi(t^{n+1}) - \phi(t^n)) \mathbf{1}. \quad (7.152)$$

The Lax–Wendroff contraction and $\|\mathbf{1}\|_{h,*} \leq \sqrt{L}$ give

$$\|V^n\|_{h,*} \leq \|V^0\|_{h,*} + \sqrt{L} \int_0^{t^n} |\phi'(s)| ds. \quad (7.153)$$

This is a finite-interval estimate including the imposed inflow; it does not use periodic Fourier modes.

Remark 7.27 (Interior order and boundary order). For smooth solutions, the Lax–Wendroff interior residual is $O(h^2)$, whereas its last, upwind row has

$$\tau_N^n = \frac{ah}{2} (\nu - 1) \partial_{xx} u(L, t^n) + O(h^2). \quad (7.154)$$

Thus for $0 < \nu < 1$ the simple closure above generally has an $O(h)$ normalized boundary residual and does not have uniform second-order local consistency. In the weighted norm,

$$\|\tau^n\|_{h,*}^2 \leq C_T (h^4 + h^3), \quad (7.155)$$

since there are $N - 1$ interior entries and only one outflow entry. With exact initialization and inflow, the homogeneous error recurrence and the energy contraction imply

$$\max_{0 \leq n \Delta t \leq T} \|U^n - I_h u(t^n)\|_{h,*} \leq T \max_{(n+1)\Delta t \leq T} \|\tau^n\|_{h,*} \leq C_T h^{3/2}. \quad (7.156)$$

This is the rate established by this elementary estimate; it is not a claim that the rate is sharp. The periodic $O(h^2)$ result cannot simply be transferred to a bounded interval. To prove a global second-order result, one must analyse the accuracy and stability of the chosen boundary closure together.

7.8 Dissipation and dispersion

The exact transport equation also has an amplification factor. Let $\theta = \xi h$. At time t^n , the Fourier mode with wave number ξ has nodal values

$$e^{i\xi x_j} = e^{ij\theta}. \quad (7.157)$$

The exact solution satisfies

$$u(x_j, t^{n+1}) = u(x_j - a\Delta t, t^n). \quad (7.158)$$

Therefore, for this exact mode,

$$u(x_j, t^{n+1}) = e^{i\xi(x_j - a\Delta t)} = e^{-ia\xi\Delta t} e^{i\xi x_j} = e^{-i\nu\theta} e^{ij\theta}. \quad (7.159)$$

Thus, during one time step, the exact mode $e^{ij\theta}$ is multiplied by

$$g_{\text{ex}}(\theta) = e^{-i\nu\theta}, \quad |g_{\text{ex}}(\theta)| = 1. \quad (7.160)$$

Let $g(\theta)$ be the numerical amplification factor and write

$$g(\theta) = |g(\theta)| e^{-i\varphi(\theta)}. \quad (7.161)$$

The modulus measures the change of amplitude of the mode and therefore numerical dissipation. The phase $\varphi(\theta)$ measures the shift during one time step. If $\varphi(\theta) = \nu\theta$ for all θ , then all frequencies propagate at the correct speed. Deviations from this linear law produce numerical dispersion.

Proposition 7.28. *Assume that $|\nu| < 1$. For small θ , the moduli satisfy*

FTCS

$$|g_{\text{FTCS}}(\theta)| = 1 + \frac{\nu^2}{2}\theta^2 + O(\theta^4). \quad (7.162)$$

Upwind

For $0 < \nu < 1$,

$$|g_U(\theta)| = 1 - \frac{\nu(1-\nu)}{2}\theta^2 + O(\theta^4). \quad (7.163)$$

Lax–Friedrichs

$$|g_{\text{LF}}(\theta)| = 1 - \frac{1-\nu^2}{2}\theta^2 + O(\theta^4). \quad (7.164)$$

Lax–Wendroff

$$|g_{\text{LW}}(\theta)| = 1 - \frac{\nu^2(1-\nu^2)}{8}\theta^4 + O(\theta^6). \quad (7.165)$$

The phases satisfy

$$\varphi_{\text{FTCS}}(\theta) = \nu\theta - \frac{\nu(1+2\nu^2)}{6}\theta^3 + O(\theta^5), \quad (7.166)$$

$$\varphi_U(\theta) = \nu\theta + \frac{\nu(1-\nu)(2\nu-1)}{6}\theta^3 + O(\theta^5), \quad (7.167)$$

$$\varphi_{\text{LF}}(\theta) = \nu\theta + \frac{\nu(1-\nu^2)}{3}\theta^3 + O(\theta^5), \quad (7.168)$$

$$\varphi_{\text{LW}}(\theta) = \nu\theta - \frac{\nu(1-\nu^2)}{6}\theta^3 + O(\theta^5). \quad (7.169)$$

Proof. The modulus formulas follow from

$$|g_{\text{FTCS}}(\theta)|^2 = 1 + \nu^2 \sin^2 \theta, \quad (7.170)$$

$$|g_U(\theta)|^2 = 1 - 2\nu(1-\nu)(1-\cos \theta), \quad (7.171)$$

$$|g_{\text{LF}}(\theta)|^2 = 1 - (1-\nu^2) \sin^2 \theta, \quad (7.172)$$

$$|g_{\text{LW}}(\theta)|^2 = 1 - \nu^2(1-\nu^2)(1-\cos \theta)^2. \quad (7.173)$$

Together with $1 - \cos \theta = \theta^2/2 + O(\theta^4)$ and $\sin \theta = \theta + O(\theta^3)$, these identities give the stated modulus expansions.

We detail the phase computation for the upwind scheme. Since

$$e^{-i\theta} = 1 - i\theta - \frac{\theta^2}{2} + \frac{i\theta^3}{6} + O(\theta^4), \quad (7.174)$$

we have

$$g_U(\theta) = 1 - i\nu\theta - \frac{\nu}{2}\theta^2 + \frac{i\nu}{6}\theta^3 + O(\theta^4). \quad (7.175)$$

Expanding $\log g_U(\theta)$ gives

$$\log g_U(\theta) = -i\nu\theta + \frac{\nu(\nu-1)}{2}\theta^2 + i\frac{\nu(2\nu^2-3\nu+1)}{6}\theta^3 + O(\theta^4). \quad (7.176)$$

Because $g_U(\theta) = |g_U(\theta)|e^{-i\varphi_U(\theta)}$, we have

$$\varphi_U(\theta) = -\Im(\log g_U(\theta)) = \nu\theta + \frac{\nu(1-\nu)(2\nu-1)}{6}\theta^3 + O(\theta^5). \quad (7.177)$$

The remainder is written as $O(\theta^5)$ rather than merely $O(\theta^4)$ because $g_U(-\theta) = \overline{g_U(\theta)}$; with the branch of the logarithm chosen near 1, the phase is an odd function of θ . The other phase expansions are obtained by the same logarithmic expansion and the same oddness argument. $\square$

FTCS is antidissipative when $\nu \neq 0$ and $\sin \theta \neq 0$, since then $|g_{\text{FTCS}}(\theta)| > 1$. This agrees with the instability proved above. The constant and Nyquist modes have amplification modulus 1. The upwind scheme is dissipative, except in the limiting exact-translation case $\nu = 1$. Lax–Friedrichs damps low frequencies at order θ^2 , while Lax–Wendroff damps them only at order θ^4 .

The phase expansions show a dispersive defect for all four schemes at order θ^3 , except at special Courant numbers where the corresponding coefficient vanishes. Thus the leading equivalent equation and the small-frequency amplification factor provide complementary information: the former identifies the first modified-equation term, while the latter separates amplitude error and phase error.

7.9 Numerical comparisons on a periodic grid

We now compare the preceding formulas with numerical solutions of

$$\partial_t u + \partial_x u = 0 \quad \text{on } \mathbb{T}_1 = \mathbb{R}/\mathbb{Z}, \quad u(x, t) = u_0(x - t). \quad (7.178)$$

All schemes use the same periodic grid within each experiment, with

$$h = \frac{1}{N}, \quad \nu = \frac{4}{5}, \quad \Delta t = \frac{4}{5N}, \quad T = \frac{2}{5}. \quad (7.179)$$

For even N , exactly $N/2$ time steps reach T , so no adjustment of the Courant number or final time is needed. The choice $\nu = 0.8$ stays away from the exact-translation case $\nu = 1$. The plotted values are computed by the four stencil updates, without filtering or added artificial smoothing.

7.9.1 A smooth profile

Take

$$u_0(x) = \sin(2\pi x) + \frac{1}{4}\sin(4\pi x), \quad (7.180)$$

initialize by $U_j^0 = u_0(jh)$, and use $N = 100$. Figure 7.1 compares the numerical values after 50 steps with the exact translate. Upwind and Lax–Friedrichs damp the profile, while Lax–Wendroff preserves its amplitude more closely. FTCS increases the amplitudes of its nonconstant Fourier components.

The apparently moderate FTCS error in this test must be interpreted carefully. The datum is a finite trigonometric polynomial, a special case for which exact arithmetic can give convergence despite instability, as explained in Remark 7.24. It does not test the response to small grid-scale perturbations.

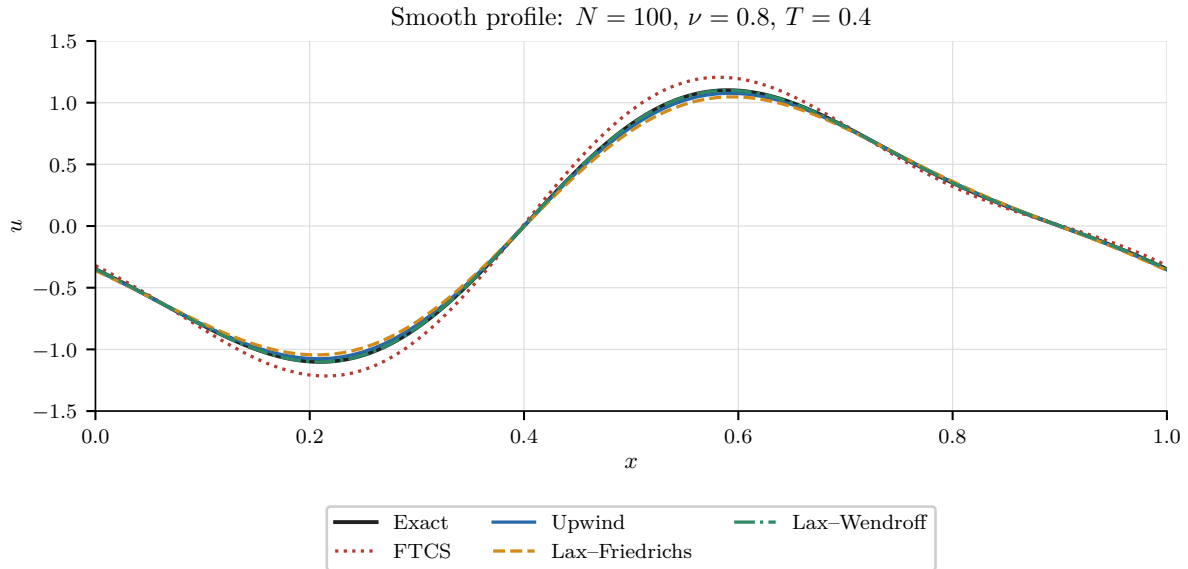


Figure 7.1: The smooth periodic profile (7.180) at $T = 0.4$, with $N = 100$, $\nu = 0.8$, and the same nodal initialization for all four schemes. Lines through numerical nodal values are drawn to aid comparison.

7.9.2 A discontinuous pulse

Next take $u_0 = \mathbf{1}_{[0.15, 0.35]}$ on the periodic interval and again use $N = 100$. We initialize by the cell averages $U^0 = P_h u_0$ and plot the piecewise constant reconstructions $E_h U^{50}$. At $T = 0.4$, the exact pulse occupies $[0.55, 0.75]$; its endpoints lie on cell boundaries. This choice makes the exact comparison independent of any convention for point values at a jump.

Figure 7.2 shows diffusion in the upwind and Lax-Friedrichs approximations. Lax-Wendroff has sharper transitions but creates overshoots and undershoots: its computed minimum and maximum are approximately -0.1554 and 1.1554 . This is consistent with its nonmonotone stencil and dispersive behavior. FTCS produces much larger oscillations, with values of magnitude about 2.23×10^4 after the same 50 steps. It is shown on a separate vertical scale so that the stable-scheme comparison remains readable.

The smooth-data rate estimates of Corollary 7.23 do not apply to this discontinuous datum. The stable schemes still fall within the general L^2 convergence framework; a single fixed-grid picture does not measure their asymptotic convergence rate.

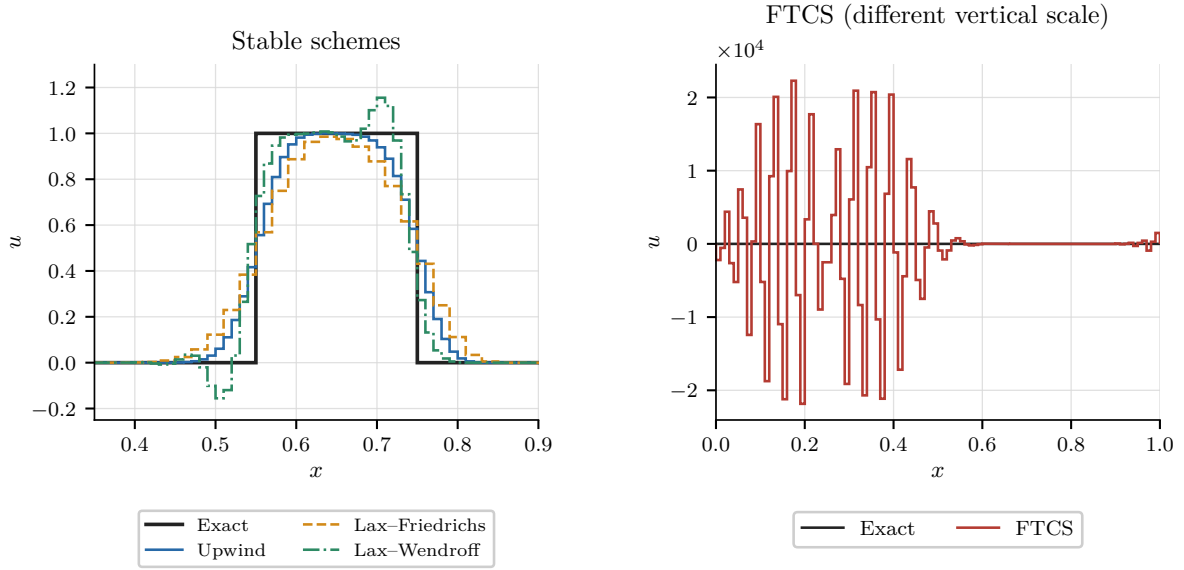


Figure 7.2: Transport of a discontinuous pulse at $T = 0.4$, with $N = 100$, $\nu = 0.8$, and cell-average initialization. The left panel shows the three stable schemes near the transported pulse; the right panel shows FTCS on the full interval. The panels have different vertical scales. All numerical curves are piecewise constant cell reconstructions.

7.9.3 Refinement errors and amplification of a perturbation

For the smooth profile (7.180), define the final-time nodal error

$$E_N = \left(\frac{1}{N} \sum_{j=0}^{N-1} |U_j^{N/2} - u_0(j/N - T)|^2 \right)^{1/2}. \quad (7.181)$$

Keep (7.179) and successively refine $N = 50, 100, 200, 400, 800$. The stable-scheme errors are reported in Table 7.1 and plotted in the left panel of Figure 7.3. The last refinement gives observed orders $p = \log_2(E_{400}/E_{800})$ close to one, one, and two, respectively, consistent with the smooth-solution estimates. FTCS is omitted from this stable-scheme refinement study: on increasingly fine grids, roundoff can seed the unstable high-frequency modes even for this special smooth datum.

N	Upwind	Lax-Friedrichs	Lax-Wendroff
50	3.0407×10^{-2}	6.5323×10^{-2}	3.7484×10^{-3}
100	1.5489×10^{-2}	3.4025×10^{-2}	9.4037×10^{-4}
200	7.8189×10^{-3}	1.7380×10^{-2}	2.3527×10^{-4}
400	3.9285×10^{-3}	8.7851×10^{-3}	5.8827×10^{-5}
800	1.9691×10^{-3}	4.4168×10^{-3}	1.4707×10^{-5}
p	0.9965	0.9921	1.9999

Table 7.1: Final-time ℓ_h^2 errors for the smooth profile, with fixed $\nu = 0.8$ and $T = 0.4$. The last row gives the observed order between $N = 400$ and $N = 800$.

To isolate the stability mechanism, use $N = 200$ and perturb the initial grid vector by

$$\delta_j^0 = \varepsilon \cos(\pi j/2), \quad \varepsilon = 10^{-10}. \quad (7.182)$$

For each linear scheme, the difference between runs with and without this perturbation evolves independently as $\delta^{n+1} = G_h \delta^n$. We therefore compute that difference directly, avoiding subtraction of two nearly equal solutions. The grid mode has $\theta = \pi/2$, so

$$\frac{\|\delta^n\|_{\ell_h^2}}{\|\delta^0\|_{\ell_h^2}} = |g(\pi/2)|^n. \quad (7.183)$$

The gains in the right panel of Figure 7.3 agree with this formula. For FTCS, $|g(\pi/2)| = \sqrt{1 + 0.8^2} > 1$, and after 100 steps the gain is about 5.52×10^{10} . Its initial norm $\varepsilon/\sqrt{2}$ has grown to approximately 3.91. The other three schemes damp this mode.

This perturbation depends on the grid and tests the uniform stability requirement. It is not a counterexample based on one fixed continuous initial datum; Remark 7.24 provides such a counterexample separately. Together, the experiments distinguish accuracy on a smooth profile, behavior near a jump, and sensitivity to small perturbations.

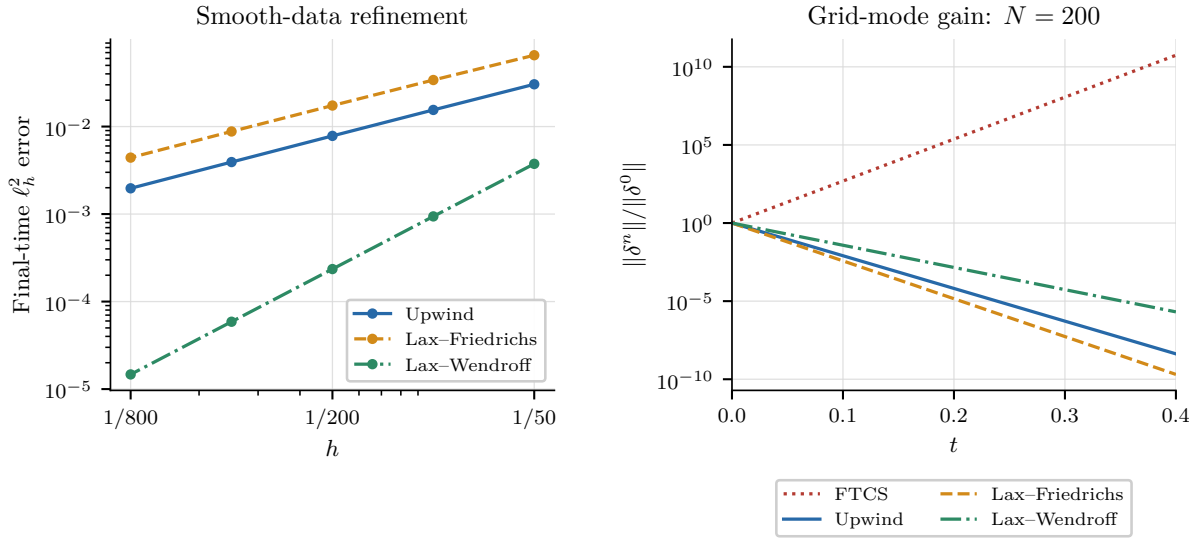


Figure 7.3: Left: final-time smooth-data errors for the three stable schemes under mesh refinement. Right: the ℓ_h^2 gain of the grid mode $\varepsilon \cos(\pi j/2)$ for all four schemes, with $N = 200$ and $\varepsilon = 10^{-10}$. Both panels use $\nu = 0.8$ and $T = 0.4$. The right panel uses a logarithmic vertical axis; its gains are checked against (7.183).

7.10 Characteristic-wise discretization of linear systems

Let

$$\partial_t U + A \partial_x U = 0, \quad A = T \Lambda T^{-1}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p). \quad (7.184)$$

Setting $W = T^{-1}U$ gives p decoupled scalar transport equations. We apply scalar schemes componentwise to W and then transform back to U . For the upwind scheme, set

$$\Lambda^+ = \text{diag}(\max(\lambda_1, 0), \dots, \max(\lambda_p, 0)), \quad \Lambda^- = \text{diag}(\min(\lambda_1, 0), \dots, \min(\lambda_p, 0)), \quad (7.185)$$

and

$$A^+ = T \Lambda^+ T^{-1}, \quad A^- = T \Lambda^- T^{-1}. \quad (7.186)$$

The four characteristic-wise schemes are then written in the same order as in the scalar case:

FTCS

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} A(U_{j+1}^n - U_{j-1}^n). \quad (7.187)$$

Upwind

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{h} \left(A^+(U_j^n - U_{j-1}^n) + A^-(U_{j+1}^n - U_j^n) \right). \quad (7.188)$$

Lax–Friedrichs

$$U_j^{n+1} = \frac{1}{2}(U_{j+1}^n + U_{j-1}^n) - \frac{\Delta t}{2h} A(U_{j+1}^n - U_{j-1}^n). \quad (7.189)$$

Lax–Wendroff

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} A(U_{j+1}^n - U_{j-1}^n) + \frac{\Delta t^2}{2h^2} A^2(U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (7.190)$$

The CFL condition is controlled by the characteristic speeds:

$$\frac{\Delta t}{h} \max_{1 \leq k \leq p} |\lambda_k| \leq 1. \quad (7.191)$$

For further details on finite difference schemes for hyperbolic systems, see [3, Chapter 13, Section 13.4.2].

7.11 Exercises

Use the notation introduced in Chapter 7. Unless otherwise stated, $a > 0$ is a constant, and the transport equation is

$$\partial_t u + a \partial_x u = 0. \quad (7.192)$$

On the uniform grid

$$x_j = jh, \quad t^n = n\Delta t, \quad (7.193)$$

denote by u_j^n an approximation of $u(x_j, t^n)$ and set

$$\lambda = \frac{\Delta t}{h}, \quad \nu = a\lambda = \frac{a\Delta t}{h}. \quad (7.194)$$

When linear systems are considered, use the matrices A , T , Λ and the characteristic variables $W = T^{-1}U$ exactly as in Chapter 7.

Exercise 7.1 (Transport with source term and Duhamel formula). Let $a > 0$ and consider

$$\partial_t u + a \partial_x u = f \quad \text{in } \mathbb{R} \times (0, +\infty), \quad u(x, 0) = u_0(x). \quad (7.195)$$

Assume that u_0 and f are smooth enough.

(a) Using characteristics, prove the Duhamel formula

$$u(x, t) = u_0(x - at) + \int_0^t f(x - a(t - s), s) ds. \quad (7.196)$$

(b) Deduce the estimate

$$\|u(t)\|_{L^\infty(\mathbb{R})} \leq \|u_0\|_{L^\infty(\mathbb{R})} + \int_0^t \|f(s)\|_{L^\infty(\mathbb{R})} ds. \quad (7.197)$$

Exercise 7.2 (Inflow transport on a finite interval). Let $\alpha < \beta$, and consider the inflow problem

$$\partial_t u + a \partial_x u = 0 \quad \text{in } (\alpha, \beta) \times (0, +\infty), \quad (7.198)$$

with initial condition

$$u(x, 0) = u_0(x) \quad \text{for } x \in (\alpha, \beta), \quad (7.199)$$

and boundary condition

$$u(\alpha, t) = \phi(t) \quad \text{for } t > 0. \quad (7.200)$$

Assume that u is smooth enough.

(a) Using characteristics, show that for every $(x, t) \in (\alpha, \beta) \times (0, +\infty)$,

$$u(x, t) = \begin{cases} u_0(x - at), & \text{if } x - at > \alpha, \\ \phi\left(t - \frac{x - \alpha}{a}\right), & \text{if } x - at \leq \alpha. \end{cases} \quad (7.201)$$

(b) Deduce the bound

$$\|u(t)\|_{L^\infty(\alpha, \beta)} \leq \max\left\{\|u_0\|_{L^\infty(\alpha, \beta)}, \|\phi\|_{L^\infty(0, t)}\right\} \quad \text{for all } t \geq 0. \quad (7.202)$$

(c) Show that

$$\|u(t)\|_{L^2(\alpha, \beta)}^2 \leq \|u_0\|_{L^2(\alpha, \beta)}^2 + a \int_0^t |\phi(s)|^2 ds \quad \text{for all } t \geq 0. \quad (7.203)$$

Exercise 7.3 (Fourier analysis of the classical schemes). Consider the four classical schemes on the periodic grid:

FTCS

$$u_j^{n+1} = u_j^n - \frac{\nu}{2}(u_{j+1}^n - u_{j-1}^n). \quad (7.204)$$

Upwind

$$u_j^{n+1} = u_j^n - \nu(u_j^n - u_{j-1}^n). \quad (7.205)$$

Lax–Friedrichs

$$u_j^{n+1} = \frac{1}{2}(u_{j+1}^n + u_{j-1}^n) - \frac{\nu}{2}(u_{j+1}^n - u_{j-1}^n). \quad (7.206)$$

Lax–Wendroff

$$u_j^{n+1} = u_j^n - \frac{\nu}{2}(u_{j+1}^n - u_{j-1}^n) + \frac{\nu^2}{2}(u_{j+1}^n - 2u_j^n + u_{j-1}^n). \quad (7.207)$$

(a) Insert the Fourier ansatz

$$u_j^n = \hat{u}^n e^{ij\theta}, \quad \theta \in [-\pi, \pi], \quad (7.208)$$

and compute the amplification factors $g_{\text{FTCS}}(\theta)$, $g_{\text{U}}(\theta)$, $g_{\text{LF}}(\theta)$ and $g_{\text{LW}}(\theta)$.

(b) Show that the FTCS scheme is never strongly stable in ℓ_h^2 as soon as $\nu \neq 0$. Then, under the hyperbolic scaling $\Delta t = \lambda h$ with fixed nonzero ν , show that FTCS is not stable in ℓ_h^2 on finite time intervals.

(c) Assume that $0 \leq \nu \leq 1$. Prove that

$$|g_{\text{U}}(\theta)| \leq 1, \quad |g_{\text{LF}}(\theta)| \leq 1, \quad |g_{\text{LW}}(\theta)| \leq 1 \quad \text{for all } \theta \in [-\pi, \pi]. \quad (7.209)$$

Deduce that upwind, Lax–Friedrichs, and Lax–Wendroff are strongly stable in ℓ_h^2 on the periodic grid.

(d) Show directly that the upwind scheme satisfies the discrete maximum principle

$$\sup_{j \in \mathbb{Z}} |u_j^{n+1}| \leq \sup_{j \in \mathbb{Z}} |u_j^n| \quad \text{for all } n \geq 0, \quad (7.210)$$

provided that $0 \leq \nu \leq 1$.

Exercise 7.4 (The wave equation as a first-order hyperbolic system). Consider the one-dimensional wave equation with $\gamma > 0$,

$$\partial_{tt}u - \gamma^2 \partial_{xx}u = 0. \quad (7.211)$$

Set

$$v = \partial_t u, \quad q = \gamma \partial_x u, \quad U = \begin{pmatrix} v \\ q \end{pmatrix}, \quad A = \begin{pmatrix} 0 & -\gamma \\ -\gamma & 0 \end{pmatrix}. \quad (7.212)$$

Then U solves

$$\partial_t U + A \partial_x U = 0. \quad (7.213)$$

(a) Compute the eigenvalues and eigenvectors of A , and write a diagonalization

$$A = T \Lambda T^{-1} \quad (7.214)$$

with

$$\Lambda = \text{diag}(\gamma, -\gamma). \quad (7.215)$$

(b) Setting $W = T^{-1}U$, show that the two components of W satisfy decoupled scalar transport equations, and identify their transport velocities.

(c) Apply the Lax–Wendroff scheme componentwise to W . By transforming back to U , prove that

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} A(U_{j+1}^n - U_{j-1}^n) + \frac{\Delta t^2}{2h^2} A^2(U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (7.216)$$

Then simplify this formula by using $A^2 = \gamma^2 I$.

(d) Deduce the CFL condition for this characteristic-wise Lax–Wendroff discretization.

Chapter 8

Solutions of the exercises

The statements are collected at the end of the corresponding chapters. This chapter gathers the solutions in the same chapter-wise order.

8.1 The Lax–Milgram theorem and Stampacchia’s extension

Exercise 1.1 — Abstract Galerkin method and Céa’s lemma

Let $u_n \in V_n$ be defined by

$$a(u_n, v_n) = \langle \varphi, v_n \rangle_{H', H} \quad \forall v_n \in V_n.$$

Since V_n is finite-dimensional, it is a Hilbert space for the scalar product inherited from H . The restriction of a to $V_n \times V_n$ is still continuous and coercive, so Lax–Milgram on V_n gives existence and uniqueness of u_n .

Now let $u \in H$ be the exact solution. Subtracting the equations for u and u_n , we obtain the Galerkin orthogonality

$$a(u - u_n, v_n) = 0 \quad \forall v_n \in V_n.$$

Fix any $v_n \in V_n$. Since $v_n - u_n \in V_n$, orthogonality gives

$$a(u - u_n, v_n - u_n) = 0.$$

Hence

$$a(u - u_n, u - u_n) = a(u - u_n, u - v_n).$$

Using coercivity and continuity,

$$\alpha \|u - u_n\|_H^2 \leq a(u - u_n, u - u_n) \leq M \|u - u_n\|_H \|u - v_n\|_H.$$

If $u \neq u_n$, divide by $\|u - u_n\|_H$; otherwise there is nothing to prove. This yields

$$\|u - u_n\|_H \leq \frac{M}{\alpha} \|u - v_n\|_H \quad \forall v_n \in V_n.$$

Taking the infimum over $v_n \in V_n$ proves Céa’s estimate.

□

Exercise 1.2 — Affine constraints

Let $K = u_0 + M$, with $M \subset H$ a closed linear subspace. Write any $u \in K$ as $u = u_0 + m$ with $m \in M$. Likewise, every $v \in K$ can be written as $v = u_0 + w$ with $w \in M$, so

$$v - u = w - m \in M.$$

The Stampacchia variational inequality

$$a(u, v - u) \geq \langle \varphi, v - u \rangle_{H', H} \quad \forall v \in K$$

is therefore equivalent to

$$a(u, z) \geq \langle \varphi, z \rangle_{H', H} \quad \forall z \in M.$$

Because M is a linear subspace, $z \in M$ implies $-z \in M$. Applying the same inequality to z and to $-z$ forces equality:

$$a(u, z) = \langle \varphi, z \rangle_{H', H} \quad \forall z \in M.$$

Now write $u = u_0 + m$. Then

$$a(m, z) = \langle \varphi, z \rangle_{H', H} - a(u_0, z) \quad \forall z \in M.$$

This is a Lax–Milgram problem on the Hilbert space M , with unknown $m \in M$. Since the restricted form is still continuous and coercive, there is a unique solution m , hence a unique $u = u_0 + m \in K$. □

Exercise 1.3 — Stability with respect to the right-hand side

For $i = 1, 2$, let $u_i \in K$ solve the variational inequality with right-hand side φ_i . Use $v = u_2$ in the inequality for u_1 :

$$a(u_1, u_2 - u_1) \geq \langle \varphi_1, u_2 - u_1 \rangle_{H', H}.$$

Use $v = u_1$ in the inequality for u_2 :

$$a(u_2, u_1 - u_2) \geq \langle \varphi_2, u_1 - u_2 \rangle_{H', H}.$$

Adding these two relations gives

$$a(u_1 - u_2, u_1 - u_2) \leq \langle \varphi_1 - \varphi_2, u_1 - u_2 \rangle_{H', H}.$$

By coercivity and by the definition of the dual norm,

$$\alpha \|u_1 - u_2\|_H^2 \leq \|\varphi_1 - \varphi_2\|_{H'} \|u_1 - u_2\|_H.$$

If $u_1 \neq u_2$, divide by $\|u_1 - u_2\|_H$; otherwise the estimate is trivial. Hence

$$\|u_1 - u_2\|_H \leq \frac{1}{\alpha} \|\varphi_1 - \varphi_2\|_{H'}.$$

□

Exercise 1.4 — What can fail without coercivity

Every continuous linear functional on $\mathbb{R}^2$ has the form

$$\varphi(y_1, y_2) = b_1 y_1 + b_2 y_2.$$

We look for $u = (x_1, x_2) \in \mathbb{R}^2$ such that

$$a(u, v) = \varphi(v) \quad \forall v = (y_1, y_2) \in \mathbb{R}^2.$$

Since

$$a(u, v) = x_1 y_1,$$

this means

$$x_1 y_1 = b_1 y_1 + b_2 y_2 \quad \forall (y_1, y_2) \in \mathbb{R}^2.$$

This can hold only if $b_2 = 0$, because the left-hand side does not depend on y_2 . When $b_2 = 0$, we must have $x_1 = b_1$, while x_2 remains arbitrary.

Therefore:

- if $\varphi(y_1, y_2) = y_2$, then $b_2 = 1$, so there is no solution;
- if $\varphi(y_1, y_2) = y_1$, then $b_1 = 1$ and $b_2 = 0$, so every vector of the form $(1, x_2)$ is a solution.

Thus, without coercivity, existence may fail and uniqueness may fail as well.

□

Exercise 1.5 — A diagonal example in ℓ^2

We seek $u = (u_n) \in \ell^2$ such that

$$\sum_{n \geq 1} \lambda_n u_n v_n = \sum_{n \geq 1} b_n v_n \quad \forall v = (v_n) \in \ell^2.$$

Choose for v the sequence supported at a single coordinate n . Then

$$\lambda_n u_n = b_n \quad \forall n \geq 1.$$

Hence the only possible solution is

$$u_n = \frac{b_n}{\lambda_n} \quad \forall n \geq 1.$$

Because $\lambda_n \geq \alpha > 0$, we have

$$\sum_{n \geq 1} |u_n|^2 \leq \frac{1}{\alpha^2} \sum_{n \geq 1} |b_n|^2 < \infty.$$

So $u \in \ell^2$, and the Lax–Milgram solution is exactly

$$u = \left(\frac{b_n}{\lambda_n} \right)_{n \geq 1}.$$

□

Exercise 1.6 — The same diagonal example with a positivity constraint

We now solve the variational inequality on

$$K_+ = \{u = (u_n) \in \ell^2 : u_n \geq 0 \ \forall n\}.$$

It reads

$$\sum_{n \geq 1} (\lambda_n u_n - b_n)(v_n - u_n) \geq 0 \quad \forall v \in K_+.$$

Fix a coordinate n , and choose $v \in K_+$ equal to u on every coordinate except the n -th one, where we set $v_n = t \geq 0$. Then

$$(\lambda_n u_n - b_n)(t - u_n) \geq 0 \quad \forall t \geq 0.$$

This is a one-dimensional variational inequality on the half-line. It implies

$$u_n = \max\left\{\frac{b_n}{\lambda_n}, 0\right\}.$$

Indeed, if $b_n \leq 0$, the only admissible value is $u_n = 0$; if $b_n > 0$, then the unconstrained minimizer b_n/λ_n is already nonnegative, so it is the solution.

Moreover,

$$0 \leq u_n \leq \left|\frac{b_n}{\lambda_n}\right|,$$

so $u \in \ell^2$. Therefore the constrained solution is the coordinatewise positive part of the unconstrained one. □

Exercise 1.7 — A box constraint in ℓ^2

We now impose the box constraint

$$K_{[0,1]} = \{u = (u_n) \in \ell^2 : 0 \leq u_n \leq 1 \ \forall n\}.$$

The variational inequality becomes

$$\sum_{n \geq 1} (\lambda_n u_n - b_n)(v_n - u_n) \geq 0 \quad \forall v \in K_{[0,1]}.$$

As in Exercise 1.6, fix n and vary only the n -th coordinate. For every $t \in [0, 1]$, we obtain

$$(\lambda_n u_n - b_n)(t - u_n) \geq 0 \quad \forall t \in [0, 1].$$

This scalar variational inequality means exactly that u_n is the projection of b_n/λ_n onto the interval $[0, 1]$:

$$u_n = \Pi_{[0,1]}\left(\frac{b_n}{\lambda_n}\right) := \min\left\{1, \max\left\{0, \frac{b_n}{\lambda_n}\right\}\right\}.$$

As before,

$$0 \leq u_n \leq \left|\frac{b_n}{\lambda_n}\right|,$$

so $u \in \ell^2$. Hence the solution is obtained by coordinatewise truncation of the unconstrained solution. □

8.2 Sobolev spaces

Exercise 2.1 — A basic weak derivative

1. Since u in $L^2(I)$ and $\text{sign}(x)$ in $L^2(I)$, it is enough to check the weak derivative identity. Let φ in $C_c^\infty(I)$. Splitting the integral at 0 and integrating by parts on $(-1, 0)$ and $(0, 1)$, we obtain

$$\begin{aligned} \int_{-1}^1 |x| \varphi'(x) dx &= \int_{-1}^0 (-x) \varphi'(x) dx + \int_0^1 x \varphi'(x) dx \\ &= - \int_{-1}^0 (-1) \varphi(x) dx - \int_0^1 (1) \varphi(x) dx \\ &= - \int_{-1}^1 \text{sign}(x) \varphi(x) dx. \end{aligned}$$

Therefore the weak derivative of u is $u' = \text{sign}(x)$, and u in $H^1(I)$.

2. If u belonged to $H^2(I)$, then u' in $H^1(I)$. By the one-dimensional Sobolev theorem recalled in Chapter 2, every function in $H^1(I)$ has a continuous representative on $[-1, 1]$. But $u' = \text{sign}(x)$ almost everywhere, and this function has a jump discontinuity at 0. This is impossible. Hence $u \notin H^2(I)$.

□

Exercise 2.2 — Poincaré inequality on an interval

1. By the one-dimensional theorem from Chapter 2, u has a continuous representative on $[a, b]$ and satisfies

$$u(x) = u(a) + \int_a^x u'(t) dt.$$

Since $u(a) = 0$, this becomes

$$u(x) = \int_a^x u'(t) dt.$$

2. By the Cauchy–Schwarz inequality,

$$|u(x)| \leq \int_a^x |u'(t)| dt \leq \sqrt{x-a} \|u'\|_{L^2(I)} \leq \sqrt{b-a} \|u'\|_{L^2(I)}.$$

3. Taking the supremum over x gives

$$\|u\|_{L^\infty(I)} \leq \sqrt{b-a} \|u'\|_{L^2(I)}.$$

Then

$$\|u\|_{L^2(I)}^2 \leq (b-a) \|u\|_{L^\infty(I)}^2 \leq (b-a)^2 \|u'\|_{L^2(I)}^2,$$

hence

$$\|u\|_{L^2(I)} \leq (b-a) \|u'\|_{L^2(I)}.$$

4. If u in $H_0^1(I)$ and $|u|_{H^1(I)} = 0$, then $u' = 0$ almost everywhere. By part (i), using $u(a) = 0$, we obtain

$$u(x) = \int_a^x 0 \, dt = 0 \quad \text{for all } x \in [a, b].$$

Thus the seminorm vanishes only at 0, so it is a norm on $H_0^1(I)$.

□

Exercise 2.3 — Poincaré–Wirtinger and its local form

1. Since u in $H^1(I)$, it has a continuous representative on $[a, b]$. A continuous function on a compact interval attains its minimum and its maximum, say m and M . Since

$$m \leq u(x) \leq M \quad \text{for all } x \in [a, b],$$

averaging over the interval gives $m \leq u_I \leq M$. By the intermediate value theorem, there exists ξ in $[a, b]$ such that $u(\xi) = u_I$.

2. The function $u - u_I$ belongs to $H^1(I)$ and vanishes at ξ . By Exercise 2.2 applied on the interval I with the point ξ in place of the left endpoint,

$$|u(x) - u_I| = \left| \int_{\xi}^x u'(t) \, dt \right| \leq \sqrt{b-a} \|u'\|_{L^2(I)}.$$

Integrating the square in x yields

$$\|u - u_I\|_{L^2(I)} \leq (b-a) \|u'\|_{L^2(I)}.$$

3. Apply part (ii) to the interval K and to the function v restricted to K . Since v_K is the average of v on K , we obtain

$$\|v - v_K\|_{L^2(K)} \leq h_K \|v'\|_{L^2(K)} = h_K |v|_{H^1(K)}.$$

□

Exercise 2.4 — The energy scalar product

1. The form $(u, v)_V$ is bilinear, symmetric, and nonnegative. It remains to prove positive definiteness. If $(u, u)_V = 0$, then $\nabla u = 0$ almost everywhere. By the Poincaré inequality on $H_0^1(\Omega)$,

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)} = 0,$$

so $u = 0$ almost everywhere. Hence $(\cdot, \cdot)_V$ is a scalar product.

2. We always have

$$\|u\|_V = \|\nabla u\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)}.$$

Conversely, by Poincaré,

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)} = C_P \|u\|_V.$$

Therefore

$$\|u\|_{H^1(\Omega)}^2 = \|u\|_{L^2(\Omega)}^2 + \|\nabla u\|_{L^2(\Omega)}^2 \leq (C_P^2 + 1) \|u\|_V^2.$$

Thus the two norms are equivalent on V .

3. For v in V , the Cauchy–Schwarz inequality and Poincaré give

$$|\ell(v)| = \left| \int_{\Omega} f v \, dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C_P \|f\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} = C_P \|f\|_{L^2(\Omega)} \|v\|_V.$$

Hence ℓ is continuous on $(V, \|\cdot\|_V)$.

□

Exercise 2.5 — From L^2 and divergence form to H^{-1}

1. For v in $H_0^1(\Omega)$, the Cauchy–Schwarz and Poincaré inequalities give

$$|\langle F, v \rangle| = \left| \int_{\Omega} f v \, dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C_P \|f\|_{L^2(\Omega)} |v|_{H^1(\Omega)}.$$

Hence F is a continuous linear form on $H_0^1(\Omega)$, that is, F in $H^{-1}(\Omega)$, and

$$\|F\|_{H^{-1}(\Omega)} \leq C_P \|f\|_{L^2(\Omega)}.$$

2. Again by Cauchy–Schwarz,

$$|\langle G, v \rangle| = \left| \int_{\Omega} g \cdot \nabla v \, dx \right| \leq \|g\|_{L^2(\Omega)^d} \|\nabla v\|_{L^2(\Omega)^d} = \|g\|_{L^2(\Omega)^d} |v|_{H^1(\Omega)}.$$

Therefore G in $H^{-1}(\Omega)$ and

$$\|G\|_{H^{-1}(\Omega)} \leq \|g\|_{L^2(\Omega)^d}.$$

3. For a smooth compactly supported test function φ , the definition of the distributional divergence is

$$\langle \operatorname{div} g, \varphi \rangle = - \int_{\Omega} g \cdot \nabla \varphi \, dx.$$

This is exactly the same formula as in part (ii). Thus the functional G is the distribution $\operatorname{div} g$, viewed as an element of $H^{-1}(\Omega)$.

□

Exercise 2.6 — Point values are not continuous on $H^1(B)$ in two dimensions

The function u_n is radial, continuous on $\overline{B}$, equal to 0 on ∂B , and piecewise C^1 . Hence u_n in $H_0^1(B) \cap C^0(\overline{B})$ and clearly $u_n(0) = 1$.

For $r = |x|$ in $(n^{-1}, 1)$, we have

$$u_n(r) = \frac{\log(1/r)}{\log n}, \quad u'_n(r) = -\frac{1}{r \log n}.$$

Therefore, in polar coordinates,

$$\begin{aligned} |u_n|_{H^1(B)}^2 &= 2\pi \int_{1/n}^1 |u'_n(r)|^2 r \, dr \\ &= \frac{2\pi}{(\log n)^2} \int_{1/n}^1 \frac{1}{r} \, dr \end{aligned}$$

$$\begin{aligned}
&= \frac{2\pi}{(\log n)^2} \log n \\
&= \frac{2\pi}{\log n} \rightarrow 0.
\end{aligned}$$

For the L^2 norm, we split the disk into $\{|x| \leq n^{-1}\}$ and the annulus $\{n^{-1} < |x| < 1\}$. The contribution of the inner disk is bounded by its area:

$$\int_{|x| \leq n^{-1}} |u_n(x)|^2 dx \leq \pi n^{-2} \rightarrow 0.$$

For the annulus,

$$\int_{n^{-1} < |x| < 1} |u_n(x)|^2 dx = \frac{2\pi}{(\log n)^2} \int_{1/n}^1 (\log(1/r))^2 r dr.$$

With the change of variable $t = \log(1/r)$, so that $r = e^{-t}$ and $r dr = -e^{-2t} dt$, this becomes

$$\frac{2\pi}{(\log n)^2} \int_0^{\log n} t^2 e^{-2t} dt \leq \frac{2\pi}{(\log n)^2} \int_0^\infty t^2 e^{-2t} dt,$$

which tends to 0 as $n \rightarrow \infty$. Hence $\|u_n\|_{L^2(B)} \rightarrow 0$.

Thus $\|u_n\|_{H^1(B)} \rightarrow 0$ while $u_n(0) = 1$ for every n . If point evaluation at 0 were a continuous linear form on $H^1(B)$, there would exist a constant C such that

$$|u(0)| \leq C \|u\|_{H^1(B)} \quad \text{for all } u \in H^1(B) \cap C^0(\overline{B}).$$

Applying this to u_n would give $1 \leq C \|u_n\|_{H^1(B)} \rightarrow 0$, a contradiction. Hence point evaluation at 0 is not continuous on $H^1(B)$ in two dimensions. □

8.3 Lax–Milgram in action for a few elliptic problems

Exercise 3.1 — The Dirichlet problem: weak formulation and well-posedness

1. The Sobolev chapter shows that V is a Hilbert space for the energy scalar product. Cauchy–Schwarz and Poincaré give

$$\begin{aligned}
|a(w, v)| &\leq (1 + \|c\|_{L^\infty(\Omega)} C_P^2) \|w\|_V \|v\|_V, \\
|\ell(v)| &\leq C_P \|f\|_{L^2(\Omega)} \|v\|_V.
\end{aligned}$$

Since $c \geq 0$, $a(v, v) \geq \|v\|_V^2$. Theorem 1.2 yields a unique weak solution $u \in V$.

2. Testing with $v = u$ gives

$$\|u\|_V^2 \leq a(u, u) = \ell(u) \leq C_P \|f\|_{L^2(\Omega)} \|u\|_V.$$

Divide by $\|u\|_V$ when $u \neq 0$; the zero solution satisfies the estimate directly.

3. For $v \in H_0^1(\Omega)$, Green's formula gives

$$\int_\Omega \nabla w \cdot \nabla v dx = - \int_\Omega (\Delta w) v dx + \int_{\partial\Omega} \partial_n w \gamma(v) ds.$$

The boundary term vanishes because $\gamma(v) = 0$. Substitution of $-\Delta w + cw = f$ yields $a(w, v) = \ell(v)$. Uniqueness of the weak solution implies $w = u$.

4. Compactly supported smooth test functions belong to V , so the weak identity implies $-u'' = f - cu$ in the sense of distributions on (a, b) . Both f and cu belong to $L^2(a, b)$, hence $u'' \in L^2(a, b)$. Since $u \in H^1(a, b)$ already, $u \in H^2(a, b)$, and the equation holds almost everywhere. The endpoint values vanish because $u \in H_0^1(a, b)$.

□

Exercise 3.2 — The Neumann problem: compatibility and well-posedness

1. The average map is continuous on $H^1(\Omega)$, since $|v_\Omega| \leq |\Omega|^{-1/2} \|v\|_{L^2(\Omega)}$. Its kernel V is therefore closed. The energy form is bilinear, symmetric, and nonnegative. If $\|\nabla v\|_{L^2} = 0$, then v is constant almost everywhere on the connected domain. Its zero mean forces that constant to be zero, proving positive definiteness on V .
2. Poincaré–Wirtinger gives

$$\|v\|_\star \leq \|v\|_{H^1(\Omega)} \leq \sqrt{1 + C_{PW}^2} \|v\|_\star \quad (v \in V).$$

A closed subspace of the Hilbert space $H^1(\Omega)$ is complete; equivalence of the norms transfers completeness to the energy norm.

3. Cauchy–Schwarz and Poincaré–Wirtinger show that

$$|\ell(v)| \leq C_{PW} \|f\|_{L^2(\Omega)} \|v\|_\star.$$

This continuity does not require compatibility. In the full weak formulation, however, $v = 1$ is an admissible test and gives $0 = \int_\Omega f \, dx$.

4. On V , the bilinear form is the energy scalar product, so it is continuous and coercive with constants one. Lax–Milgram gives a unique $u \in V$ satisfying the reduced problem, and testing with u gives the stated energy estimate exactly as in Exercise 3.1. These steps work for every $f \in L^2(\Omega)$. For an arbitrary $v \in H^1(\Omega)$, write $v = (v - v_\Omega) + v_\Omega$. The first term belongs to V , and the constant has zero gradient. Thus

$$\begin{aligned} \int_\Omega \nabla u \cdot \nabla v \, dx &= \ell(v - v_\Omega) \\ &= \ell(v) - v_\Omega \int_\Omega f \, dx = \ell(v). \end{aligned}$$

The final equality is exactly where compatibility is needed. Without it, the reduced identity does not solve the original Neumann problem on the whole test space.

5. Subtract the full weak equations for two solutions u_1, u_2 and test with $u_1 - u_2$. This gives $\|\nabla(u_1 - u_2)\|_{L^2(\Omega)}^2 = 0$. Connectedness implies that $u_1 - u_2$ is constant. Conversely, adding a constant changes neither side of the weak identity. Exactly one member of this family has zero mean.

□

8.4 One-dimensional P_1 finite elements

Problem 4.1 — Dirichlet–Dirichlet boundary condition

Part 1. Variational setting and weak formulation

- (a) Since $V = H_0^1(I)$, the Poincaré inequality gives a constant $C_P > 0$ such that

$$\|v\|_{L^2(I)} \leq C_P \|v\|_V \quad \text{for all } v \in V. \quad (8.1)$$

Hence, for all w, v in V ,

$$|a(w, v)| \leq \left| \int_a^b w'(x) v'(x) dx \right| + \left| \int_a^b c(x) w(x) v(x) dx \right| \quad (8.2)$$

$$\leq \|w\|_V \|v\|_V + \|c\|_{L^\infty(I)} \|w\|_{L^2(I)} \|v\|_{L^2(I)} \quad (8.3)$$

$$\leq (1 + \|c\|_{L^\infty(I)} C_P^2) \|w\|_V \|v\|_V. \quad (8.4)$$

Thus a is continuous on $V \times V$. The same argument gives

$$|\ell(v)| \leq \|f\|_{L^2(I)} \|v\|_{L^2(I)} \leq C_P \|f\|_{L^2(I)} \|v\|_V, \quad (8.5)$$

so ℓ is continuous on V .

- (b) We write

$$a(v, v) = |v|_{H^1(I)}^2 + \int_a^b c(x) |v(x)|^2 dx \geq |v|_{H^1(I)}^2 = \|v\|_V^2. \quad (8.6)$$

Hence a is coercive on V with coercivity constant $\alpha = 1$ for this norm.

- (c) Parts (a) and (b) show that a is continuous and coercive on V , and part (a) also shows that ℓ is continuous on V . The Lax–Milgram theorem therefore yields a unique u in V such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V. \quad (8.7)$$

□

Part 2. Finite element space, hat basis, and nodal interpolation

- (a) A function v_h in V_h is affine on each element and is therefore determined by its nodal values. The values at $x_0 = a$ and $x_N = b$ are prescribed to be zero, so the free nodal values are exactly

$$v_h(x_1), \dots, v_h(x_{N-1}). \quad (8.8)$$

There are $N - 1$ such values. The hat functions $\varphi_1, \dots, \varphi_{N-1}$ satisfy $\varphi_i(x_j) = \delta_{ij}$ for $1 \leq i, j \leq N - 1$. Hence they are linearly independent. Moreover, for every v_h in V_h ,

$$v_h = \sum_{i=1}^{N-1} v_h(x_i) \varphi_i, \quad (8.9)$$

because both sides are continuous, affine on each element, and have the same nodal values. Thus $(\varphi_1, \dots, \varphi_{N-1})$ is a basis of V_h and $\dim V_h = N - 1$.

- (b) Since $H^1(I)$ embeds into $C^0([a, b])$, the nodal values of v are well defined. The nodal interpolant $\Pi_h v$ is characterized by

$$(\Pi_h v)(x_i) = v(x_i), \quad i = 0, \dots, N. \quad (8.10)$$

Because v in $H_0^1(I)$, we have $v(a) = v(b) = 0$, and therefore $\Pi_h v$ belongs to V_h . Its expansion in the hat basis is

$$\Pi_h v = \sum_{i=1}^{N-1} v(x_i) \varphi_i. \quad (8.11)$$

□

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma

- (a) The conforming Galerkin problem is: find
- u_h
- in
- V_h
- such that

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (8.12)$$

Since $V_h \subset V$, continuity and coercivity of a , and continuity of ℓ , remain valid on V_h . This finite-dimensional space is a Hilbert space for the induced norm, so the Lax–Milgram theorem gives a unique discrete solution.

- (b) Since
- u
- solves the continuous problem and
- u_h
- solves the discrete problem, we have

$$a(u, v_h) = \ell(v_h), \quad (8.13)$$

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (8.14)$$

Subtracting the two equalities gives

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (8.15)$$

This is the Galerkin orthogonality relation.

- (c) Let
- $e_h = u - u_h$
- . For any
- v_h
- in
- V_h
- , coercivity gives

$$\alpha \|e_h\|_V^2 \leq a(e_h, e_h). \quad (8.16)$$

We decompose

$$e_h = (u - v_h) + (v_h - u_h), \quad (8.17)$$

so that

$$a(e_h, e_h) = a(e_h, u - v_h) + a(e_h, v_h - u_h). \quad (8.18)$$

The second term vanishes because $v_h - u_h$ in V_h and part (b) applies. Hence

$$\alpha \|e_h\|_V^2 \leq a(e_h, u - v_h). \quad (8.19)$$

By continuity of a ,

$$\alpha \|e_h\|_V^2 \leq M \|e_h\|_V \|u - v_h\|_V. \quad (8.20)$$

If $e_h \neq 0$, we divide by $\|e_h\|_V$ and obtain

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V. \quad (8.21)$$

Taking the infimum over v_h in V_h yields

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (8.22)$$

If $e_h = 0$, the estimate is immediate.

- (d) In the Poisson case
- $c = 0$
- , the bilinear form becomes

$$a(w, v) = \int_a^b w'(x) v'(x) dx. \quad (8.23)$$

Part (b) then reads

$$\int_a^b (u - u_h)'(x) v_h'(x) dx = 0 \quad \text{for all } v_h \in V_h, \quad (8.24)$$

which is exactly the orthogonality condition that characterizes the orthogonal projection of u onto V_h for the inner product $(\cdot, \cdot)_V$. Moreover, for the norm $\|v\|_V = |v|_{H^1(I)}$, we have $M = \alpha = 1$, so part (c) gives

$$|u - u_h|_{H^1(I)} \leq \inf_{v_h \in V_h} |u - v_h|_{H^1(I)}. \quad (8.25)$$

□

Part 4. Interpolation estimate in the energy norm

(a) On the element $K_i = (x_{i-1}, x_i)$, the interpolant $\Pi_h u$ is affine and satisfies

$$(\Pi_h u)(x_{i-1}) = u(x_{i-1}), \quad (\Pi_h u)(x_i) = u(x_i). \quad (8.26)$$

Its derivative is therefore constant on K_i and equal to the slope of the secant line:

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1})}{h_i}. \quad (8.27)$$

By the fundamental theorem of calculus,

$$u(x_i) - u(x_{i-1}) = \int_{x_{i-1}}^{x_i} u'(x) dx, \quad (8.28)$$

so

$$(\Pi_h u)'|_{K_i} = \frac{1}{h_i} \int_{K_i} u'(x) dx = (u')_{K_i}. \quad (8.29)$$

Subtracting gives

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}. \quad (8.30)$$

(b) Since z in $H^1(K)$ and we are in one dimension, z admits a continuous representative on $\overline{K}$. The condition of zero average implies that z takes both a non-positive and a non-negative value on K , hence there exists ξ in K such that $z(\xi) = 0$. For every x in K ,

$$z(x) = \int_{\xi}^x z'(t) dt. \quad (8.31)$$

By the Cauchy–Schwarz inequality,

$$|z(x)|^2 \leq |x - \xi| \int_K |z'(t)|^2 dt \leq h_K \int_K |z'(t)|^2 dt. \quad (8.32)$$

Integrating with respect to x over K yields

$$\int_K |z(x)|^2 dx \leq h_K^2 \int_K |z'(t)|^2 dt, \quad (8.33)$$

that is,

$$\|z\|_{L^2(K)} \leq h_K \|z\|_{H^1(K)}. \quad (8.34)$$

(c) Apply part (b) on each element to

$$z = u' - (u')_{K_i}. \quad (8.35)$$

If u in $H^2(I)$, then z in $H^1(K_i)$ and $\int_{K_i} z(x) dx = 0$. By part (a),

$$\|(u - \Pi_h u)'\|_{L^2(K_i)} = \|u' - (u')_{K_i}\|_{L^2(K_i)} \leq h_i \|u''\|_{L^2(K_i)}. \quad (8.36)$$

We square and sum over all elements:

$$|u - \Pi_h u|_{H^1(I)}^2 = \sum_{i=1}^N \|(u - \Pi_h u)'\|_{L^2(K_i)}^2 \quad (8.37)$$

$$\leq \sum_{i=1}^N h_i^2 \|u''\|_{L^2(K_i)}^2 \quad (8.38)$$

$$\leq h^2 \sum_{i=1}^N \|u''\|_{L^2(K_i)}^2 \quad (8.39)$$

$$= h^2 |u|_{H^2(I)}^2. \quad (8.40)$$

Taking square roots gives

$$|u - \Pi_h u|_{H^1(I)} \leq h |u|_{H^2(I)}. \quad (8.41)$$

□

Part 5. A priori error estimate

(a) Céa's lemma gives

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (8.42)$$

Since $u(a) = u(b) = 0$, the interpolant $\Pi_h u$ also belongs to V_h . Choosing $v_h = \Pi_h u$ and using the norm $\|v\|_V = |v|_{H^1(I)}$, we obtain

$$|u - u_h|_{H^1(I)} \leq \frac{M}{\alpha} |u - \Pi_h u|_{H^1(I)}. \quad (8.43)$$

(b) Insert the interpolation estimate from Part 4:

$$|u - \Pi_h u|_{H^1(I)} \leq h|u|_{H^2(I)}. \quad (8.44)$$

Combining this inequality with part (a) yields

$$|u - u_h|_{H^1(I)} \leq \frac{M}{\alpha} h|u|_{H^2(I)}. \quad (8.45)$$

(c) If $c = 0$, the bilinear form is

$$a(w, v) = \int_a^b w'(x)v'(x) dx, \quad (8.46)$$

so $M = \alpha = 1$ for the norm $|\cdot|_{H^1(I)}$. The estimate of part (b) simplifies to

$$|u - u_h|_{H^1(I)} \leq h|u|_{H^2(I)}. \quad (8.47)$$

□

Part 6(a)–(e). Full assembly and Dirichlet enforcement

(a) The local shape derivatives are $-1/h_i$ and $1/h_i$. The standard integrals of their products give

$$S^{(i)} = \frac{1}{h_i} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad R^{(i)} = \frac{\gamma h_i}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad b_m^{(i)} = \int_{K_i} f \psi_m^{(i)} dx. \quad (8.48)$$

Here $\int_{K_i} (\psi_m^{(i)})^2 dx = h_i/3$ and $\int_{K_i} \psi_1^{(i)} \psi_2^{(i)} dx = h_i/6$. For constant $f = f_0$, $b^{(i)} = (f_0 h_i/2)(1, 1)^T$.

(b) Assemble with the connectivity $(i-1, i)$, retaining both endpoint indices. The full endpoint entries are

$$\hat{A}_{00} = \frac{1}{h_1} + \frac{\gamma h_1}{3}, \quad \hat{A}_{NN} = \frac{1}{h_N} + \frac{\gamma h_N}{3}.$$

For $1 \leq i \leq N-1$, the interior entries are

$$\hat{A}_{ii} = \frac{1}{h_i} + \frac{1}{h_{i+1}} + \frac{\gamma}{3}(h_i + h_{i+1}).$$

The couplings on each element are

$$\hat{A}_{i-1,i} = \hat{A}_{i,i-1} = -\frac{1}{h_i} + \frac{\gamma h_i}{6}, \quad 1 \leq i \leq N,$$

and all other entries are zero. For $f = f_0$,

$$\hat{B}_0 = \frac{f_0 h_1}{2}, \quad \hat{B}_N = \frac{f_0 h_N}{2}, \quad \hat{B}_i = \frac{f_0(h_i + h_{i+1})}{2} \quad (1 \leq i \leq N-1).$$

For a general load, use $\hat{B}_i = \int_a^b f \hat{\varphi}_i dx$.

- (c) Set the two boundary rows and columns to zero, their diagonal entries to one, and the two boundary load entries to zero. The modified system still has $N+1$ rows and unknown entries. Its boundary equations are $U_0 = U_N = 0$. Eliminating these entries leaves $AU = B$ with $A = \hat{A}_{FF}$ and $B = \hat{B}_F$, $F = \{1, \dots, N-1\}$. The free basis functions are the same interior hat functions used in the constrained Galerkin method, which proves equivalence. The reduced size is $N-1$.
- (d) For $h_i = h$, the reduced matrix has

$$A_{ii} = \frac{2}{h} + \frac{2\gamma h}{3}, \quad A_{i,i+1} = A_{i+1,i} = -\frac{1}{h} + \frac{\gamma h}{6}.$$

It is the tridiagonal stiffness-plus-reaction matrix in Proposition 4.33. Full assembly is already tridiagonal because each element couples only its two endpoint nodes. The symmetric boundary modification and extraction of the interior block preserve this pattern in the natural nodal ordering.

- (e) For general prescribed endpoint values, first set

$$\tilde{B}_F = \hat{B}_F - \hat{A}_{FD} g_D, \quad \tilde{B}_D = g_D, \quad g_D = (g_0, g_N)^T.$$

Use the original boundary columns in the first formula, then zero rows and columns and set the two boundary diagonal entries to one. For the specified three-element example,

$$\hat{A}_{FF} = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}, \quad \tilde{B}_F = \begin{pmatrix} 1/3 \\ 10/3 \end{pmatrix}.$$

The full modified right-hand side is $(0, 1/3, 10/3, 1)^T$, and solving gives

$$(U_0, U_1, U_2, U_3) = (0, 4/9, 7/9, 1).$$

These are the nodal values of $u(x) = x + x(1-x)/2$. If the boundary columns were simply erased without the correction, the free right-hand side would remain $(1/3, 1/3)^T$, giving the wrong interior values $(1/9, 1/9)$.

□

Part 6(f). Explicit Poisson computation

- (a) The exact solution is obtained by integrating twice:

$$u'' = -1. \tag{8.49}$$

Hence

$$u'(x) = -x + C, \quad u(x) = -\frac{x^2}{2} + Cx + D. \tag{8.50}$$

The boundary conditions $u(0) = u(1) = 0$ give $D = 0$ and $C = 1/2$. Therefore

$$u(x) = \frac{x(1-x)}{2}. \tag{8.51}$$

(b) For each $i = 1, \dots, N-1$, the Galerkin equation with test function φ_i reads

$$\int_0^1 u'_h(x) \varphi'_i(x) dx = \int_0^1 \varphi_i(x) dx. \quad (8.52)$$

If we write $U_0 = U_N = 0$, the left-hand side becomes

$$\frac{1}{h} (-U_{i-1} + 2U_i - U_{i+1}), \quad (8.53)$$

while the right-hand side is the area of the hat function,

$$\int_0^1 \varphi_i(x) dx = h. \quad (8.54)$$

Therefore the nodal values satisfy

$$-U_{i-1} + 2U_i - U_{i+1} = h^2, \quad i = 1, \dots, N-1, \quad (8.55)$$

with the boundary values

$$U_0 = U_N = 0. \quad (8.56)$$

(c) We now evaluate the exact solution at the nodes:

$$u(x_i) = \frac{ih(1-ih)}{2}. \quad (8.57)$$

A direct computation gives

$$-u(x_{i-1}) + 2u(x_i) - u(x_{i+1}) = h^2, \quad i = 1, \dots, N-1, \quad (8.58)$$

and clearly $u(x_0) = u(x_N) = 0$. Thus the vector $(u(x_i))_{1 \leq i \leq N-1}$ satisfies the same linear system as $(U_i)_{1 \leq i \leq N-1}$. Since the stiffness matrix is invertible, the solution is unique and

$$U_i = u(x_i) \quad \text{for } i = 1, \dots, N-1. \quad (8.59)$$

Therefore u_h is the nodal interpolant of the exact solution.

(d) On the element $K_i = (x_{i-1}, x_i)$, the derivative of u_h is the average slope of u :

$$u'_h|_{K_i} = \frac{u(x_i) - u(x_{i-1})}{h} = \frac{1}{h} \int_{K_i} u'(x) dx. \quad (8.60)$$

Since $u'(x) = 1/2 - x$, this constant equals $1/2 - m_i$, where

$$m_i = \frac{x_{i-1} + x_i}{2} \quad (8.61)$$

is the midpoint of K_i . Hence, for x in K_i ,

$$(u - u_h)'(x) = u'(x) - u'_h|_{K_i} = \left(\frac{1}{2} - x\right) - \left(\frac{1}{2} - m_i\right) = m_i - x. \quad (8.62)$$

Therefore

$$\int_{K_i} |(u - u_h)'(x)|^2 dx = \int_{x_{i-1}}^{x_i} (x - m_i)^2 dx \quad (8.63)$$

$$= \int_{-h/2}^{h/2} t^2 dt \quad (8.64)$$

$$= \frac{h^3}{12}. \quad (8.65)$$

Summing over all N elements, we obtain

$$|u - u_h|_{H^1(0,1)}^2 = N \frac{h^3}{12} = \frac{h^2}{12}. \quad (8.66)$$

Thus

$$|u - u_h|_{H^1(0,1)} = \frac{h}{\sqrt{12}}. \quad (8.67)$$

□

Problem 4.2 — Neumann–Neumann boundary condition

Part 1. Variational setting and weak formulation

(a) Integrating the equation over I gives

$$\int_a^b f(x) dx = \int_a^b -u''(x) dx = -u'(b) + u'(a). \quad (8.68)$$

Since $-u'(a) = g_a$ and $u'(b) = g_b$, we have $u'(a) = -g_a$, and therefore

$$\int_a^b f(x) dx = -g_b - g_a. \quad (8.69)$$

Thus

$$\int_a^b f(x) dx + g_a + g_b = 0. \quad (8.70)$$

As specified in the problem, we assume this condition in all subsequent parts.

(b) On $V = H_\star^1(I)$, the Poincaré–Wirtinger inequality gives

$$\|v\|_{L^2(I)} \leq C_{\text{PW}} \|v'\|_{L^2(I)}. \quad (8.71)$$

Hence $\|v'\|_{L^2(I)}$ controls the full $H^1(I)$ norm on V . It is a norm because $\|v'\|_{L^2(I)} = 0$ implies that v is constant, and the zero-mean condition then implies $v = 0$.

(c) Let u be smooth and let v in V . Multiplying by v and integrating by parts gives

$$\int_a^b f v dx = \int_a^b -u'' v dx \quad (8.72)$$

$$= \int_a^b u' v' dx - u'(b)v(b) + u'(a)v(a). \quad (8.73)$$

Using $u'(a) = -g_a$ and $u'(b) = g_b$, we obtain

$$\int_a^b u' v' dx = \int_a^b f v dx + g_a v(a) + g_b v(b). \quad (8.74)$$

This is the announced weak formulation.

(d) The bilinear form satisfies

$$a_N(v, v) = \|v'\|_{L^2(I)}^2 = \|v\|_V^2, \quad (8.75)$$

so it is coercive with constant 1. It is continuous by Cauchy–Schwarz. The linear form is continuous because

$$\left| \int_a^b f v \, dx \right| \leq \|f\|_{L^2(I)} \|v\|_{L^2(I)} \leq C_{PW} \|f\|_{L^2(I)} \|v\|_V, \quad (8.76)$$

and the endpoint traces are continuous on $H^1(I)$, hence on V . The Lax–Milgram theorem gives a unique solution in V .

For an arbitrary test v in $H^1(I)$, write

$$v = v_\star + \bar{v}, \quad \bar{v} = \frac{1}{b-a} \int_a^b v(x) \, dx, \quad v_\star \in V. \quad (8.77)$$

The constant part contributes zero to a_N , while compatibility gives $\ell_N(1) = 0$. Therefore

$$a_N(u, v) = a_N(u, v_\star) = \ell_N(v_\star) = \ell_N(v). \quad (8.78)$$

This extension to all $H^1(I)$ tests proves that the normalized solution solves the original Neumann problem weakly. In particular, compactly supported tests give $-u'' = f$, hence $u \in H^2(I)$, and integration by parts recovers the endpoint normal derivatives. Without compatibility, existence on the zero-mean test space alone would not imply existence for the original Neumann problem.

Adding a constant does not change u' or the Neumann conditions. Conversely, subtract the variational identities for two solutions and test with their difference to obtain $\|(u_1 - u_2)'\|_{L^2(I)}^2 = 0$. Thus all solutions in $H^1(I)$ differ by an additive constant.

□

Part 2. Finite element space, hat basis, and nodal interpolation

(a) A function in W_h is determined by all its nodal values

$$v_h(x_0), \dots, v_h(x_N). \quad (8.79)$$

There are $N + 1$ such values. The full hat functions $\varphi_0, \dots, \varphi_N$ are characterized by

$$\varphi_j(x_i) = \delta_{ij}, \quad i, j = 0, \dots, N. \quad (8.80)$$

The same nodal argument as in the Dirichlet case gives

$$v_h = \sum_{j=0}^N v_h(x_j) \varphi_j, \quad (8.81)$$

so they form a basis and $\dim W_h = N + 1$.

(b) The space $W_{h,\star}$ is the kernel of the nonzero linear form

$$v_h \mapsto \int_a^b v_h(x) \, dx \quad (8.82)$$

on the $(N + 1)$ -dimensional space W_h . Hence $\dim W_{h,\star} = N$.

(c) The full nodal interpolant is

$$\Pi_h v = \sum_{j=0}^N v(x_j) \varphi_j. \quad (8.83)$$

Even if $\int_a^b v(x) dx = 0$, the quadrature-like quantity $\int_a^b \Pi_h v(x) dx$ is not necessarily zero. We therefore set

$$v_h = \Pi_h v - \frac{1}{b-a} \int_a^b \Pi_h v(x) dx. \quad (8.84)$$

Then v_h belongs to $W_{h,\star}$, and the subtraction of a constant does not change the derivative.

□

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma

(a) The discrete problem is: find u_h in $W_{h,\star}$ such that

$$a_N(u_h, v_h) = \ell_N(v_h) \quad \text{for all } v_h \in W_{h,\star}. \quad (8.85)$$

Since $W_{h,\star}$ is a subspace of V , the bilinear form remains continuous and coercive on $W_{h,\star}$. The Lax–Milgram theorem gives a unique u_h in $W_{h,\star}$. For every $v_h \in W_h$, its decomposition into a zero-mean function and a constant belongs to $W_{h,\star} + \mathbb{R}$. The same compatibility argument as in Part 1(d) extends the discrete identity to all tests in W_h . This justifies the full-basis matrix equations used in Part 6, together with the zero-mean constraint.

(b) Subtracting the discrete equation from the continuous equation gives

$$a_N(u - u_h, v_h) = 0 \quad \text{for all } v_h \in W_{h,\star}. \quad (8.86)$$

(c) Let $e_h = u - u_h$. For any v_h in $W_{h,\star}$,

$$|e_h|_{H^1(I)}^2 = a_N(e_h, e_h) \quad (8.87)$$

$$= a_N(e_h, u - v_h) + a_N(e_h, v_h - u_h). \quad (8.88)$$

The second term vanishes by Galerkin orthogonality. By Cauchy–Schwarz,

$$|e_h|_{H^1(I)}^2 \leq |e_h|_{H^1(I)} |u - v_h|_{H^1(I)}. \quad (8.89)$$

If $e_h \neq 0$, division by $|e_h|_{H^1(I)}$ gives

$$|u - u_h|_{H^1(I)} \leq |u - v_h|_{H^1(I)}. \quad (8.90)$$

Taking the infimum over v_h in $W_{h,\star}$ gives the result. If $e_h = 0$, the estimate is immediate.

□

Part 4. Interpolation estimate in the energy norm

(a) The proof is the same as in the Dirichlet case. On K_i , the function $\Pi_h u$ is affine and takes the values $u(x_{i-1})$ and $u(x_i)$, hence

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1})}{h_i} = \frac{1}{h_i} \int_{K_i} u'(x) dx. \quad (8.91)$$

Therefore

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}. \quad (8.92)$$

(b) By definition,

$$\int_a^b v_h(x) dx = \int_a^b \Pi_h u(x) dx - \int_a^b \frac{1}{b-a} \int_a^b \Pi_h u(y) dy dx = 0. \quad (8.93)$$

Thus v_h belongs to $W_{h,\star}$. Since u has zero mean and v_h differs from $\Pi_h u$ by a constant, we have

$$(u - v_h)' = (u - \Pi_h u)'. \quad (8.94)$$

(c) Applying the local Poincaré–Wirtinger estimate to $u' - (u')_{K_i}$ on each element gives

$$\|(u - \Pi_h u)'\|_{L^2(K_i)} \leq h_i \|u''\|_{L^2(K_i)}. \quad (8.95)$$

After squaring, summing over i , and using $h_i \leq h$, we obtain

$$|u - v_h|_{H^1(I)} = |u - \Pi_h u|_{H^1(I)} \leq h |u|_{H^2(I)}. \quad (8.96)$$

□

Part 5. A priori error estimate

From Part 3,

$$|u - u_h|_{H^1(I)} \leq \inf_{v_h \in W_{h,\star}} |u - v_h|_{H^1(I)}. \quad (8.97)$$

The function constructed in Part 4 belongs to $W_{h,\star}$, so

$$|u - u_h|_{H^1(I)} \leq |u - v_h|_{H^1(I)} \leq h |u|_{H^2(I)}. \quad (8.98)$$

□

Part 6. Matrix form on a uniform mesh

(a) In the full basis $(\varphi_0, \dots, \varphi_N)$, the stiffness entries are

$$K_{ij} = \int_a^b \varphi_j'(x) \varphi_i'(x) dx. \quad (8.99)$$

On a uniform mesh,

$$K = \frac{1}{h} \begin{pmatrix} 1 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 1 \end{pmatrix}. \quad (8.100)$$

The endpoint diagonal entries are $1/h$ because endpoint hats are supported on one element. Interior diagonal entries are $2/h$ because interior hats are supported on two elements.

(b) For constant $f = f_0$, the volume contribution is $hf_0/2$ at each endpoint and hf_0 at each interior node. The boundary terms add g_a and g_b at the two endpoints. Thus

$$B = \begin{pmatrix} \frac{hf_0}{2} + g_a \\ hf_0 \\ \vdots \\ hf_0 \\ \frac{hf_0}{2} + g_b \end{pmatrix}. \quad (8.101)$$

(c) Every row of K sums to zero, hence

$$K(1, \dots, 1)^T = 0. \quad (8.102)$$

This is the discrete counterpart of the fact that constants belong to the kernel of the Neumann operator. For the linear system to be solvable, the right-hand side must be orthogonal to this kernel, namely

$$\sum_{i=0}^N B_i = 0. \quad (8.103)$$

Here this condition is

$$(b-a)f_0 + g_a + g_b = 0, \quad (8.104)$$

which is the compatibility condition.

(d) If

$$u_h = \sum_{i=0}^N U_i \varphi_i, \quad (8.105)$$

then, on a uniform mesh,

$$\int_a^b u_h(x) dx = h \left(\frac{U_0}{2} + \sum_{i=1}^{N-1} U_i + \frac{U_N}{2} \right). \quad (8.106)$$

The discrete zero-mean normalization is therefore

$$h \left(\frac{U_0}{2} + \sum_{i=1}^{N-1} U_i + \frac{U_N}{2} \right) = 0. \quad (8.107)$$

□

Problem 4.3 — Mixed Dirichlet–Robin boundary condition

Part 1. Variational setting and weak formulation

Question 1(a).

Since $v(a) = 0$, for every $x \in [a, b]$ we have

$$v(x) = \int_a^x v'(t) dt. \quad (8.108)$$

By the Cauchy–Schwarz inequality,

$$|v(x)| \leq \sqrt{x-a} \left(\int_a^x |v'(t)|^2 dt \right)^{1/2} \leq \sqrt{b-a} \|v'\|_{L^2(I)}. \quad (8.109)$$

Taking $x = b$ gives

$$|v(b)| \leq \sqrt{b-a} \|v'\|_{L^2(I)}. \quad (8.110)$$

The same pointwise bound yields

$$\|v\|_{L^2(I)}^2 = \int_a^b |v(x)|^2 dx \leq \int_a^b (b-a) \|v'\|_{L^2(I)}^2 dx = (b-a)^2 \|v'\|_{L^2(I)}^2, \quad (8.111)$$

so

$$\|v\|_{L^2(I)} \leq (b-a) \|v'\|_{L^2(I)}. \quad (8.112)$$

If $\|v\|_V = 0$, then $v' = 0$ almost everywhere on I , hence v is constant almost everywhere. Since $v(a) = 0$, that constant is zero, so $v = 0$. Thus $\|\cdot\|_V$ is a norm on V .

Finally,

$$\|v\|_{H^1(I)}^2 = \|v\|_{L^2(I)}^2 + \|v'\|_{L^2(I)}^2 \leq ((b-a)^2 + 1) \|v\|_V^2, \quad (8.113)$$

while trivially

$$\|v\|_V \leq \|v\|_{H^1(I)}. \quad (8.114)$$

Hence $\|\cdot\|_V$ is equivalent to the usual $H^1(I)$ norm on V . In particular, V endowed with $\|\cdot\|_V$ is a Hilbert space. □

Question 1(b).

Let $v \in V$. Multiplying the equation by v and integrating over I gives

$$\int_a^b (-u''(x) + c(x)u(x))v(x) dx = \int_a^b f(x)v(x) dx. \quad (8.115)$$

Integrating the term in u'' by parts, we obtain

$$-\int_a^b u''(x)v(x) dx = -u'(b)v(b) + u'(a)v(a) + \int_a^b u'(x)v'(x) dx. \quad (8.116)$$

Since $v(a) = 0$, the term at a vanishes. Using the boundary condition

$$u'(b) = g - d u(b), \quad (8.117)$$

we get

$$-u'(b)v(b) = -g v(b) + d u(b)v(b). \quad (8.118)$$

Hence

$$\int_a^b u'(x)v'(x) dx + \int_a^b c(x)u(x)v(x) dx + d u(b)v(b) = \int_a^b f(x)v(x) dx + g v(b). \quad (8.119)$$

Therefore the weak formulation is: find $u \in V$ such that

$$a(u, v) = \ell(v) \quad \text{for all } v \in V, \quad (8.120)$$

with

$$a(w, v) = \int_a^b w'(x)v'(x) dx + \int_a^b c(x)w(x)v(x) dx + d w(b)v(b), \quad (8.121)$$

$$\ell(v) = \int_a^b f(x)v(x) dx + g v(b). \quad (8.122)$$

□

Question 1(c).

For $w, v \in V$, we estimate each term separately:

$$|a(w, v)| \leq \|w'\|_{L^2(I)} \|v'\|_{L^2(I)} + \|c\|_{L^\infty(I)} \|w\|_{L^2(I)} \|v\|_{L^2(I)} + d |w(b)| |v(b)| \quad (8.123)$$

$$\leq \left(1 + \|c\|_{L^\infty(I)} (b-a)^2 + d(b-a)\right) \|w\|_V \|v\|_V. \quad (8.124)$$

Thus a is continuous on $V \times V$ with, for example,

$$M = 1 + \|c\|_{L^\infty(I)} (b-a)^2 + d(b-a). \quad (8.125)$$

Moreover,

$$a(v, v) = \|v'\|_{L^2(I)}^2 + \int_a^b c(x)|v(x)|^2 dx + d|v(b)|^2 \geq \|v\|_V^2, \quad (8.126)$$

so a is coercive on V with coercivity constant

$$\alpha = 1. \quad (8.127)$$

Finally,

$$|\ell(v)| \leq \|f\|_{L^2(I)} \|v\|_{L^2(I)} + |g| |v(b)| \quad (8.128)$$

$$\leq \left((b-a) \|f\|_{L^2(I)} + \sqrt{b-a} |g| \right) \|v\|_V, \quad (8.129)$$

so ℓ is continuous on V .

Since V is a Hilbert space for the norm $\|\cdot\|_V$ and since a is continuous and coercive while ℓ is continuous, the Lax–Milgram theorem yields a unique solution $u \in V$ of the weak problem. $\square$

Part 2. Finite element space, hat basis, and nodal interpolation

Question 2(a).

A function $v_h \in V_h$ is completely determined by its nodal values at $x_1, \dots, x_N$. Indeed, it is continuous, affine on each element, and the value at $x_0 = a$ is fixed by the condition $v_h(a) = 0$.

There are therefore exactly N free nodal values:

$$v_h(x_1), \dots, v_h(x_N). \quad (8.130)$$

Hence

$$\dim V_h = N. \quad (8.131)$$

The difference with the Dirichlet–Dirichlet case is precisely that the value at the right endpoint $x_N = b$ is now a degree of freedom. $\square$

Question 2(b).

For $i = 1, \dots, N-1$, the hat functions are the usual interior ones:

$$\varphi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h_i}, & x \in [x_{i-1}, x_i], \\ \frac{x_{i+1} - x}{h_{i+1}}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases} \quad (8.132)$$

The endpoint hat function at b is

$$\varphi_N(x) = \begin{cases} \frac{x - x_{N-1}}{h_N}, & x \in [x_{N-1}, x_N], \\ 0, & x \in [a, x_{N-1}]. \end{cases} \quad (8.133)$$

To prove that $(\varphi_1, \dots, \varphi_N)$ is a basis of V_h , we first show linear independence. If

$$\sum_{i=1}^N \lambda_i \varphi_i = 0, \quad (8.134)$$

then evaluating at x_j for $j = 1, \dots, N$ gives

$$\lambda_j = \sum_{i=1}^N \lambda_i \varphi_i(x_j) = 0. \quad (8.135)$$

So all coefficients vanish.

Next, let $v_h \in V_h$ and define

$$\tilde{v}_h := \sum_{i=1}^N v_h(x_i) \varphi_i. \quad (8.136)$$

Then $\tilde{v}_h \in V_h$ and $\tilde{v}_h(x_j) = v_h(x_j)$ for $j = 1, \dots, N$. Also, both functions vanish at $x_0 = a$. Hence $v_h - \tilde{v}_h$ is affine on each element and vanishes at both endpoints of each element, so it is identically zero on every element. Therefore

$$v_h = \sum_{i=1}^N v_h(x_i) \varphi_i. \quad (8.137)$$

Thus $(\varphi_1, \dots, \varphi_N)$ is a basis of V_h .

□

Question 2(c).

The nodal interpolant $\Pi_h v \in V_h$ is defined by

$$(\Pi_h v)(x_i) = v(x_i) \quad \text{for } i = 0, \dots, N. \quad (8.138)$$

Since $v(a) = 0$, this is compatible with the condition $(\Pi_h v)(a) = 0$. Using the basis of part (b), we obtain

$$\Pi_h v = \sum_{i=1}^N v(x_i) \varphi_i. \quad (8.139)$$

The difference with the Dirichlet–Dirichlet case is that the coefficient of the endpoint basis function φ_N is now present. In other words, the value at b is interpolated and remains a degree of freedom.

□

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma

Question 3(a).

The conforming Galerkin problem is: find $u_h \in V_h$ such that

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (8.140)$$

Since $V_h \subset V$, the bilinear form a and the linear form ℓ may be restricted to V_h . The continuity and coercivity estimates proved in Part 1 remain valid on the subspace V_h .

Because V_h is finite-dimensional and a is continuous and coercive on V_h , the Lax–Milgram theorem yields a unique discrete solution $u_h \in V_h$.

□

Question 3(b).

Let $v_h \in V_h$. Since u solves the continuous problem and u_h solves the discrete problem, we have

$$a(u, v_h) = \ell(v_h), \quad (8.141)$$

$$a(u_h, v_h) = \ell(v_h). \quad (8.142)$$

Subtracting the two equalities gives

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (8.143)$$

This is the Galerkin orthogonality relation.

□

Question 3(c).

Let

$$e_h := u - u_h. \quad (8.144)$$

For any $v_h \in V_h$, coercivity gives

$$\alpha \|e_h\|_V^2 \leq a(e_h, e_h). \quad (8.145)$$

Now decompose

$$e_h = (u - v_h) + (v_h - u_h), \quad (8.146)$$

so that

$$a(e_h, e_h) = a(e_h, u - v_h) + a(e_h, v_h - u_h). \quad (8.147)$$

Since $v_h - u_h \in V_h$, Galerkin orthogonality implies

$$a(e_h, v_h - u_h) = 0. \quad (8.148)$$

Therefore

$$\alpha \|e_h\|_V^2 \leq a(e_h, u - v_h). \quad (8.149)$$

By continuity of a ,

$$\alpha \|e_h\|_V^2 \leq M \|e_h\|_V \|u - v_h\|_V. \quad (8.150)$$

If $e_h \neq 0$, we divide by $\|e_h\|_V$ and obtain

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V. \quad (8.151)$$

Taking the infimum over all $v_h \in V_h$ yields

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (8.152)$$

If $e_h = 0$, the estimate is immediate.

□

Part 4. Interpolation estimate in the energy norm

Question 4(a).

On the element $K_i = (x_{i-1}, x_i)$, the interpolant $\Pi_h u$ is affine and satisfies

$$(\Pi_h u)(x_{i-1}) = u(x_{i-1}), \quad (\Pi_h u)(x_i) = u(x_i). \quad (8.153)$$

Hence its derivative is constant on K_i and equal to the slope of the secant line:

$$(\Pi_h u)'|_{K_i} = \frac{u(x_i) - u(x_{i-1})}{h_i}. \quad (8.154)$$

By the fundamental theorem of calculus,

$$u(x_i) - u(x_{i-1}) = \int_{x_{i-1}}^{x_i} u'(x) dx, \quad (8.155)$$

therefore

$$(\Pi_h u)'|_{K_i} = \frac{1}{h_i} \int_{K_i} u'(x) dx = (u')_{K_i}. \quad (8.156)$$

Subtracting, we obtain

$$(u - \Pi_h u)'|_{K_i} = u' - (u')_{K_i}. \quad (8.157)$$

□

Question 4(b).

Because $z \in H^1(K)$ and we are in one dimension, z admits a continuous representative on $\overline{K}$. Since its average over K is zero, the function must take both a non-positive and a non-negative value on K , hence there exists $\xi \in K$ such that

$$z(\xi) = 0. \quad (8.158)$$

Then, for every $x \in K$,

$$z(x) = \int_{\xi}^x z'(t) dt. \quad (8.159)$$

By the Cauchy–Schwarz inequality,

$$|z(x)|^2 \leq |x - \xi| \int_K |z'(t)|^2 dt \leq h_K \int_K |z'(t)|^2 dt. \quad (8.160)$$

Integrating over K gives

$$\int_K |z(x)|^2 dx \leq h_K^2 \int_K |z'(t)|^2 dt. \quad (8.161)$$

Therefore

$$\|z\|_{L^2(K)} \leq h_K \|z'\|_{L^2(K)}. \quad (8.162)$$

□

Question 4(c).

On each element K_i , define

$$z_i := u' - (u')_{K_i}. \quad (8.163)$$

Then $z_i \in H^1(K_i)$ and

$$\int_{K_i} z_i(x) dx = 0. \quad (8.164)$$

By part (a),

$$(u - \Pi_h u)'|_{K_i} = z_i. \quad (8.165)$$

Applying part (b) to z_i yields

$$\|(u - \Pi_h u)'\|_{L^2(K_i)} = \|z_i\|_{L^2(K_i)} \leq h_i \|z_i'\|_{L^2(K_i)} = h_i \|u''\|_{L^2(K_i)}. \quad (8.166)$$

We square and sum over all elements:

$$\|u - \Pi_h u\|_V^2 = \|(u - \Pi_h u)'\|_{L^2(I)}^2 \quad (8.167)$$

$$= \sum_{i=1}^N \|(u - \Pi_h u)'\|_{L^2(K_i)}^2 \quad (8.168)$$

$$\leq \sum_{i=1}^N h_i^2 \|u''\|_{L^2(K_i)}^2 \quad (8.169)$$

$$\leq h^2 \sum_{i=1}^N \|u''\|_{L^2(K_i)}^2 \quad (8.170)$$

$$= h^2 \|u''\|_{L^2(I)}^2. \quad (8.171)$$

Taking square roots gives

$$\|u - \Pi_h u\|_V \leq h \|u''\|_{L^2(I)}. \quad (8.172)$$

□

Part 5. A priori error estimate

Question 5.

Céa's estimate from Part 3 gives

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (8.173)$$

Since $u \in H^2(I) \cap V$, the nodal interpolant $\Pi_h u$ belongs to V_h , hence

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - \Pi_h u\|_V. \quad (8.174)$$

Using the interpolation estimate from Part 4, we obtain

$$\|u - u_h\|_V \leq \frac{M}{\alpha} h \|u''\|_{L^2(I)}. \quad (8.175)$$

From Part 1(c), one possible choice is

$$M = 1 + \|c\|_{L^\infty(I)} (b-a)^2 + d(b-a), \quad \alpha = 1, \quad (8.176)$$

so the a priori estimate becomes

$$\|u - u_h\|_V \leq \left(1 + \|c\|_{L^\infty(I)} (b-a)^2 + d(b-a)\right) h \|u''\|_{L^2(I)}. \quad (8.177)$$

Any valid continuity constant M obtained from the estimates of Part 1 is acceptable.

□

Part 6. Matrix form on a uniform mesh

Question 6(a).

The Galerkin equations with the basis functions $(\varphi_i)_{1 \leq i \leq N}$ read

$$a(u_h, \varphi_i) = \ell(\varphi_i) \quad \text{for } i = 1, \dots, N. \quad (8.178)$$

Since

$$u_h = \sum_{j=1}^N U_j \varphi_j, \quad (8.179)$$

this becomes

$$\sum_{j=1}^N a(\varphi_j, \varphi_i) U_j = \ell(\varphi_i), \quad i = 1, \dots, N. \quad (8.180)$$

Hence the coefficient vector $U = (U_1, \dots, U_N)^T$ satisfies

$$AU = B, \quad (8.181)$$

with

$$A_{ij} = a(\varphi_j, \varphi_i), \quad B_i = \ell(\varphi_i). \quad (8.182)$$

The matrix A is symmetric because a is symmetric. It is tridiagonal because φ_i and φ_j have disjoint supports whenever $|i - j| \geq 2$, so then

$$A_{ij} = 0. \quad (8.183)$$

□

Question 6(b).

For the diagonal terms attached to the interior basis functions, we keep the same formula as in the Dirichlet–Dirichlet case:

$$A_{ii} = \frac{2}{h} + \frac{2c_0h}{3}, \quad 1 \leq i \leq N-1. \quad (8.184)$$

Likewise, whenever two adjacent hat functions overlap on one element, their coupling is

$$A_{i,i+1} = A_{i+1,i} = -\frac{1}{h} + \frac{c_0h}{6}, \quad 1 \leq i \leq N-1. \quad (8.185)$$

In particular, this includes the new coupling between the penultimate and endpoint basis functions:

$$A_{N-1,N} = A_{N,N-1} = -\frac{1}{h} + \frac{c_0h}{6}. \quad (8.186)$$

For the load vector, the Robin term does not contribute to B_i when $1 \leq i \leq N-1$ because $\varphi_i(b) = 0$. Hence

$$B_i = \int_a^b f_0 \varphi_i(x) dx + g \varphi_i(b) = hf_0, \quad 1 \leq i \leq N-1, \quad (8.187)$$

since the area of an interior hat function is h .

Now consider the endpoint basis function φ_N , given by

$$\varphi_N(x) = \begin{cases} \frac{x - x_{N-1}}{h}, & x \in [x_{N-1}, x_N], \\ 0, & x \in [a, x_{N-1}]. \end{cases} \quad (8.188)$$

Its derivative is $1/h$ on (x_{N-1}, x_N) and zero elsewhere. Hence

$$A_{NN} = \int_{x_{N-1}}^{x_N} |\varphi'_N(x)|^2 dx + c_0 \int_{x_{N-1}}^{x_N} |\varphi_N(x)|^2 dx + d |\varphi_N(b)|^2 \quad (8.189)$$

$$= \int_{x_{N-1}}^{x_N} \frac{1}{h^2} dx + \frac{c_0}{h^2} \int_{x_{N-1}}^{x_N} (x - x_{N-1})^2 dx + d \quad (8.190)$$

$$= \frac{1}{h} + \frac{c_0h}{3} + d. \quad (8.191)$$

The coupling with the previous basis function is

$$A_{N,N-1} = \int_{x_{N-1}}^{x_N} \varphi'_{N-1}(x) \varphi'_N(x) dx + c_0 \int_{x_{N-1}}^{x_N} \varphi_{N-1}(x) \varphi_N(x) dx \quad (8.192)$$

$$= -\frac{1}{h} + \frac{c_0h}{6}. \quad (8.193)$$

By symmetry,

$$A_{N-1,N} = A_{N,N-1}. \quad (8.194)$$

Finally,

$$B_N = \int_a^b f_0 \varphi_N(x) dx + g \varphi_N(b) \quad (8.195)$$

$$= f_0 \int_{x_{N-1}}^{x_N} \frac{x - x_{N-1}}{h} dx + g \quad (8.196)$$

$$= \frac{hf_0}{2} + g. \quad (8.197)$$

Therefore the full system may be written as

$$A = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ 0 & & & -1 & 1 \end{pmatrix} + \frac{c_0 h}{6} \begin{pmatrix} 4 & 1 & & & 0 \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ 0 & & & 1 & 2 \end{pmatrix} + d \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad (8.198)$$

and

$$B = \begin{pmatrix} hf_0 \\ hf_0 \\ \vdots \\ hf_0 \\ \frac{hf_0}{2} + g \end{pmatrix}. \quad (8.199)$$

□

Question 6(c).

Compared with the homogeneous Dirichlet–Dirichlet system, the rows $1, \dots, N-2$ are unchanged. The differences appear near the right endpoint:

- there is an additional basis function φ_N attached to the node b , so the matrix gains one extra column and one extra row,
- the penultimate row acquires the additional coupling

$$A_{N-1,N} = A_{N,N-1} = -\frac{1}{h} + \frac{c_0 h}{6},$$

which was absent in the Dirichlet–Dirichlet system because the value at b was prescribed,

- the new last diagonal entry contains the Robin contribution d ,
- the stiffness and reaction contributions in A_{NN} are those of an endpoint hat function supported on only one element, so they are half of the usual two-element contributions,
- the load vector gains the new last entry

$$B_N = \frac{hf_0}{2} + g.$$

So the change is not only a new last row: the introduction of the endpoint degree of freedom also modifies the row $N-1$ through its coupling with U_N .

□

Question 6(d).

Use the full hats $(\widehat{\varphi}_j)_{j=0}^N$ and assemble

$$\begin{aligned}\widehat{A}_{ij} &= \int_a^b \left(\widehat{\varphi}_j' \widehat{\varphi}_i' + c_0 \widehat{\varphi}_j \widehat{\varphi}_i \right) dx + d \widehat{\varphi}_j(b) \widehat{\varphi}_i(b), \\ \widehat{B}_i &= \int_a^b f_0 \widehat{\varphi}_i dx + g \widehat{\varphi}_i(b), \quad 0 \leq i, j \leq N.\end{aligned}$$

The Dirichlet set is $D = \{0\}$ and the free set is $F = \{1, \dots, N\}$. Only row and column zero are replaced by a unit diagonal, and only the zeroth right-hand-side entry is set to zero. The full modified system has $N + 1$ entries; extraction gives the N -dimensional system $A = \widehat{A}_{FF}$, $B = \widehat{B}_F$ already found above. The Robin condition contributes d to $\widehat{A}_{NN}$ and g to $\widehat{B}_N$. It does not prescribe U_N , which remains a free degree of freedom. Thus the endpoint row at b must be retained.

□

8.5 Two-dimensional finite elements

Problem 5.1 — Mixed Dirichlet–Robin boundary condition for rectangular Q_1 finite elements**Part 1. Variational setting and weak formulation**

- (a) For almost every x_1 in $(0, 1)$ and every x_2 in $(0, 1)$, the vanishing of the trace on the bottom edge gives

$$v(x_1, x_2) = \int_0^{x_2} \partial_2 v(x_1, t) dt. \quad (8.200)$$

By the Cauchy–Schwarz inequality,

$$|v(x_1, x_2)|^2 \leq x_2 \int_0^{x_2} |\partial_2 v(x_1, t)|^2 dt \leq \int_0^1 |\partial_2 v(x_1, t)|^2 dt. \quad (8.201)$$

After integration in x_2 and then in x_1 , we obtain

$$\|v\|_{L^2(\Omega)} \leq \|\partial_2 v\|_{L^2(\Omega)}. \quad (8.202)$$

Likewise,

$$v(x_1, 1) = \int_0^1 \partial_2 v(x_1, t) dt, \quad (8.203)$$

so that

$$|v(x_1, 1)|^2 \leq \int_0^1 |\partial_2 v(x_1, t)|^2 dt. \quad (8.204)$$

Integrating in x_1 gives

$$\|v\|_{L^2(\Gamma_R)} \leq \|\partial_2 v\|_{L^2(\Omega)}. \quad (8.205)$$

Since $|\partial_2 v| \leq |\nabla v|$ almost everywhere, both second inequalities follow. In particular, if $\|v\|_V = 0$, then $\nabla v = 0$ almost everywhere, hence v is almost everywhere constant, and the vanishing trace on Γ_D forces $v = 0$. Thus $\|\cdot\|_V$ is a norm on V . Finally,

$$\|v\|_V \leq \|v\|_{H^1(\Omega)} \quad (8.206)$$

is immediate, and the first inequality gives

$$\|v\|_{H^1(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2 \leq 2 \|\nabla v\|_{L^2(\Omega)}^2 = 2 \|v\|_V^2. \quad (8.207)$$

So the two norms are equivalent on V .

(b) We multiply the equation by v in V and we integrate over Ω :

$$\int_{\Omega} (-\Delta u + cu)v \, dx = \int_{\Omega} f v \, dx. \quad (8.208)$$

After one integration by parts, we obtain

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} cuv \, dx - \int_{\partial\Omega} \partial_n u v \, d\Gamma = \int_{\Omega} f v \, dx. \quad (8.209)$$

Because $v = 0$ on Γ_D , only the Robin part of the boundary remains. Since $\partial_n u = g - du$ on Γ_R , we get

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} cuv \, dx + d \int_{\Gamma_R} uv \, d\Gamma = \int_{\Omega} f v \, dx + \int_{\Gamma_R} gv \, d\Gamma. \quad (8.210)$$

Therefore

$$a(w, v) = \int_{\Omega} (\nabla w \cdot \nabla v + cuv) \, dx + d \int_{\Gamma_R} uv \, d\Gamma, \quad (8.211)$$

$$\ell(v) = \int_{\Omega} f v \, dx + \int_{\Gamma_R} gv \, d\Gamma. \quad (8.212)$$

(c) For all w, v in V , the estimates from part (a) yield

$$|a(w, v)| \leq \|\nabla w\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + \|c\|_{L^\infty(\Omega)} \|w\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \quad (8.213)$$

$$+ d \|w\|_{L^2(\Gamma_R)} \|v\|_{L^2(\Gamma_R)} \quad (8.214)$$

$$\leq (1 + \|c\|_{L^\infty(\Omega)} + d) \|w\|_V \|v\|_V. \quad (8.215)$$

Thus a is continuous on $V \times V$. Likewise,

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\Gamma_R)} \|v\|_{L^2(\Gamma_R)} \leq (\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\Gamma_R)}) \|v\|_V. \quad (8.216)$$

So ℓ is continuous on V . Finally,

$$a(v, v) = \|v\|_V^2 + \int_{\Omega} c|v|^2 \, dx + d \int_{\Gamma_R} |v|^2 \, d\Gamma \geq \|v\|_V^2, \quad (8.217)$$

so a is coercive on V with coercivity constant $\alpha = 1$ for the norm $\|\cdot\|_V$. The weak problem therefore has a unique solution by the Lax–Milgram theorem.

Part 2. Finite element space, nodal basis, and nodal interpolation

(a) On each rectangle K , the restriction $v_h|_K$ is a polynomial in Q_1 , hence it belongs to $H^1(K)$. Since the mesh is admissible and v_h is globally continuous on Ω , the piecewise polynomial function v_h belongs to $H^1(\Omega)$. By definition, the trace of v_h vanishes on Γ_D . Therefore v_h belongs to V , and we have $V_h \subset V$.

- (b) On each rectangle, a Q_1 polynomial is uniquely determined by its four nodal values. Hence a function v_h in V_h is completely determined by its values at the free nodes $S_1, \dots, S_N$, because the values at the nodes on Γ_D are fixed to be 0. The functions w_i are linearly independent by the Kronecker relations

$$w_i(S_j) = \delta_{ij}. \quad (8.218)$$

They also span V_h because every v_h in V_h has the same nodal values as

$$\sum_{i=1}^N v_h(S_i) w_i. \quad (8.219)$$

Thus $(w_i)_{i=1, \dots, N}$ is a basis of V_h and

$$v_h = \sum_{i=1}^N v_h(S_i) w_i. \quad (8.220)$$

Compared with the homogeneous Dirichlet construction of Section 5.5, the new free degrees of freedom are exactly the non-corner nodes on Γ_R .

- (c) Since u belongs to $C^0(\overline{\Omega}) \cap V$, the nodal values of u are well defined and the values on Γ_D are zero. The function $\Pi_h u$ in V_h is therefore characterized by its nodal values at the free nodes, so the basis expansion gives

$$\Pi_h u = \sum_{i=1}^N (\Pi_h u)(S_i) w_i = \sum_{i=1}^N u(S_i) w_i. \quad (8.221)$$

This is the same interpolation formula as in the homogeneous Dirichlet construction of Section 5.5, except that the set of free nodes now also contains the non-corner nodes on Γ_R .

Part 3. Discrete problem, Galerkin orthogonality, and Céa's lemma

- (a) The conforming Galerkin problem is

$$\text{find } u_h \text{ in } V_h \text{ such that } a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (8.222)$$

Since $V_h \subset V$, the bilinear form a remains continuous and coercive on the subspace V_h , and ℓ remains continuous on V_h . Therefore the discrete problem admits a unique solution u_h in V_h .

- (b) The continuous solution satisfies

$$a(u, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h, \quad (8.223)$$

and the discrete solution satisfies

$$a(u_h, v_h) = \ell(v_h) \quad \text{for all } v_h \in V_h. \quad (8.224)$$

Subtracting the two relations gives

$$a(u - u_h, v_h) = 0 \quad \text{for all } v_h \in V_h. \quad (8.225)$$

This is the Galerkin orthogonality relation.

(c) Let $e_h = u - u_h$. For any v_h in V_h , coercivity gives

$$\alpha \|e_h\|_V^2 \leq a(e_h, e_h). \quad (8.226)$$

We write

$$e_h = (u - v_h) + (v_h - u_h), \quad (8.227)$$

so that

$$a(e_h, e_h) = a(e_h, u - v_h) + a(e_h, v_h - u_h). \quad (8.228)$$

The second term vanishes by Galerkin orthogonality, because $v_h - u_h$ belongs to V_h . Hence

$$\alpha \|e_h\|_V^2 \leq a(e_h, u - v_h) \leq M \|e_h\|_V \|u - v_h\|_V. \quad (8.229)$$

If $e_h \neq 0$, we divide by $\|e_h\|_V$ and obtain

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V. \quad (8.230)$$

Taking the infimum over all v_h in V_h yields C  a's estimate.

Part 4. Interpolation estimate in the energy norm

(a) In dimension two, the Sobolev embedding recalled in Chapter 2 gives

$$H^2(\Omega) \hookrightarrow C^0(\overline{\Omega}). \quad (8.231)$$

Hence the nodal values of u are meaningful. Since u belongs to V , its trace vanishes on Γ_D , and continuity up to the boundary implies that the nodal values of u also vanish at the nodes on Γ_D . Therefore the construction from Part 2(c) defines an element $\Pi_h u$ in V_h .

(b) Theorem 5.24, applied with the rectangular Q_1 element, gives the required estimate. The proof is elementwise after transport from the reference square $\widehat{R}$, and the only effect of the mixed boundary condition is that the set of free boundary nodes is different. Summing the local bounds over the mesh, we get

$$|u - \Pi_h u|_{H^1(\Omega)} \leq Ch \|u\|_{H^2(\Omega)}, \quad (8.232)$$

with C independent of h along the regular family $(\mathcal{T}_h)_{h>0}$. Since

$$\|u - \Pi_h u\|_V = \|\nabla(u - \Pi_h u)\|_{L^2(\Omega)} = |u - \Pi_h u|_{H^1(\Omega)}, \quad (8.233)$$

we obtain the desired estimate

$$\|u - \Pi_h u\|_V \leq Ch \|u\|_{H^2(\Omega)}. \quad (8.234)$$

Part 5. A priori error estimate

Part 3 gives C  a's estimate and Part 4 gives the interpolation bound. Choosing $v_h = \Pi_h u$ in C  a's estimate, we obtain

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - \Pi_h u\|_V \leq C \frac{M}{\alpha} h \|u\|_{H^2(\Omega)}. \quad (8.235)$$

From Part 1(c), we may take

$$M = 1 + \|c\|_{L^\infty(\Omega)} + d, \quad \alpha = 1. \quad (8.236)$$

Therefore

$$\|u - u_h\|_V \leq C(1 + \|c\|_{L^\infty(\Omega)} + d)h \|u\|_{H^2(\Omega)}, \quad (8.237)$$

where the constant C is independent of h .

Part 6. Matrix form, sparsity, and the Robin boundary contribution

(a) If

$$u_h = \sum_{j=1}^N U_j w_j, \quad (8.238)$$

then testing the discrete problem with w_i gives

$$\sum_{j=1}^N U_j a(w_j, w_i) = \ell(w_i), \quad i = 1, \dots, N. \quad (8.239)$$

So the coefficient vector $U = (U_1, \dots, U_N)^T$ satisfies

$$AU = B, \quad A_{ij} = a(w_j, w_i), \quad B_i = \ell(w_i), \quad (8.240)$$

with

$$A_{ij} = \int_{\Omega} (\nabla w_j \cdot \nabla w_i + c_0 w_j w_i) dx + d \int_{\Gamma_R} w_j w_i d\Gamma, \quad (8.241)$$

$$B_i = \int_{\Omega} f w_i dx + \int_{\Gamma_R} g w_i d\Gamma. \quad (8.242)$$

(b) A global Q_1 basis function is supported on the union of the rectangles meeting at its node. On a structured rectangular mesh, another basis function can overlap this support only if its node belongs to the 3×3 block centered at the original node. Hence each basis function interacts with at most nine basis functions, itself included.

(c) Along the top edge of the reference square, where $\hat{x}_2 = 1$, the local basis functions satisfy

$$\hat{p}_1 = 0, \quad \hat{p}_2 = 0, \quad \hat{p}_3 = \hat{x}_1, \quad \hat{p}_4 = 1 - \hat{x}_1. \quad (8.243)$$

If we parametrize the physical edge by $s \mapsto F_K(s, 1)$, with s in $(0, 1)$, then $d\Gamma = |e| ds$. Therefore

$$R^{(e)} = |e| \int_0^1 \begin{pmatrix} 0 \\ 0 \\ s \\ 1-s \end{pmatrix} \begin{pmatrix} 0 & 0 & s & 1-s \end{pmatrix} ds. \quad (8.244)$$

Using

$$\int_0^1 s^2 ds = \frac{1}{3}, \quad \int_0^1 s(1-s) ds = \frac{1}{6}, \quad \int_0^1 (1-s)^2 ds = \frac{1}{3}, \quad (8.245)$$

we obtain

$$R^{(e)} = |e| \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{6} \\ 0 & 0 & \frac{1}{6} & \frac{1}{3} \end{pmatrix}. \quad (8.246)$$

The local Robin contribution to the element matrix is therefore $dR^{(e)}$.

(d) If $g = g_0$ is constant on e , then only the two upper basis functions contribute, and we get

$$G^{(e)} = g_0 |e| \int_0^1 \begin{pmatrix} 0 \\ 0 \\ s \\ 1-s \end{pmatrix} ds = g_0 |e| \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}. \quad (8.247)$$

- (e) Compared with the homogeneous Dirichlet rectangular Q_1 case, the unknown vector now also contains the non-corner nodes on Γ_R . Hence the matrix acquires additional rows and columns corresponding to these boundary degrees of freedom. The volume part of the matrix is assembled exactly as before on this enlarged set of unknowns. In addition, each top boundary edge contributes the Robin block $dR^{(e)}$, so only the equations associated with basis functions whose trace on Γ_R is nonzero receive extra boundary-matrix terms. Likewise, the right-hand side gains the boundary load terms

$$\int_{\Gamma_R} g w_i d\Gamma, \quad (8.248)$$

which are nonzero only for basis functions associated with the free nodes on Γ_R .

- (f) Let $Z_1, \dots, Z_{N_{\text{all}}}$ be all mesh nodes and let $(\hat{w}_i)_{i=1, \dots, N_{\text{all}}}$ be the full nodal basis of W_h . First scatter the local volume matrices $A^R = S^R + c_0 M^R$ and load vectors F^R into the full arrays using the local-to-global maps, retaining every node. Then add the contributions $dR^{(e)}$ from part (c) and the general boundary load

$$G_r^{(e)} = \int_e g p_r^R d\Gamma, \quad r = 1, \dots, 4, \quad (8.249)$$

for every Robin edge, again using the global numbers of all local nodes. The explicit vector from part (d) applies only when g is constant on that edge. The resulting arrays are

$$\hat{A}_{ij} = \int_{\Omega} (\nabla \hat{w}_j \cdot \nabla \hat{w}_i + c_0 \hat{w}_j \hat{w}_i) dx + d \int_{\Gamma_R} \hat{w}_j \hat{w}_i d\Gamma, \quad (8.250)$$

$$\hat{B}_i = \int_{\Omega} f \hat{w}_i dx + \int_{\Gamma_R} g \hat{w}_i d\Gamma, \quad 1 \leq i, j \leq N_{\text{all}}. \quad (8.251)$$

Set $\mathcal{D} = \{i : Z_i \in \Gamma_D\}$ and let $\mathcal{F}$ be its complement. In the ordering with free nodes first, the symmetric row-and-column modification gives

$$\begin{pmatrix} \hat{A}_{\mathcal{F}\mathcal{F}} & 0 \\ 0 & I_{\#\mathcal{D}} \end{pmatrix} \begin{pmatrix} U_{\mathcal{F}} \\ U_{\mathcal{D}} \end{pmatrix} = \begin{pmatrix} \hat{B}_{\mathcal{F}} \\ 0 \end{pmatrix}. \quad (8.252)$$

This system retains N_{all} components. Its Dirichlet equations give $U_{\mathcal{D}} = 0$; eliminating these prescribed values leaves

$$\hat{A}_{\mathcal{F}\mathcal{F}} U_{\mathcal{F}} = \hat{B}_{\mathcal{F}}. \quad (8.253)$$

If $S_m = Z_{i_m}$ is the m th free node, then $w_m = \hat{w}_{i_m}$. Thus $A_{mn} = \hat{A}_{i_m i_n}$ and $B_m = \hat{B}_{i_m}$, proving that this reduced system is precisely the one in part (a). Only the values on Γ_D are prescribed. Nodes on Γ_R outside Γ_D are therefore free: their equations retain the Robin matrix and load contributions. With nonzero Dirichlet nodal values, the corresponding free right-hand side would be $\hat{B}_{\mathcal{F}} - \hat{A}_{\mathcal{F}\mathcal{D}} g_{\mathcal{D}}$, as in Section 4.6.1.

Exercise 5.2 — Rectangular Q_1 and triangular P_1 interpolation

- (a) A finite element consists of a compact element K , a finite-dimensional function space P on K , and a family $\Sigma = \{\sigma_1, \dots, \sigma_m\} \subset P'$ of degrees of freedom, with $m = \dim P$. Unisolvence means that

$$P \ni p \mapsto (\sigma_1(p), \dots, \sigma_m(p)) \in \mathbb{R}^m$$

is bijective. Equivalently, each set of prescribed degree-of-freedom values determines exactly one function in P .

(b) In the required order the four square shape functions are

$$\begin{aligned}\psi_1 &= (1 - \xi)(1 - \eta), & \psi_2 &= \xi(1 - \eta), \\ \psi_3 &= \xi\eta, & \psi_4 &= (1 - \xi)\eta.\end{aligned}$$

They belong to $Q_1(\widehat{Q})$ and satisfy $\psi_i(A_j) = \delta_{ij}$. They are therefore linearly independent. Since $\dim Q_1(\widehat{Q}) = 4$, they form a basis, and $p = \sum_{i=1}^4 p(A_i)\psi_i$ for every $p \in Q_1(\widehat{Q})$. This proves unisolvence of the vertex evaluations.

(c) The barycentric coordinates are

$$\lambda_1 = 1 - \xi - \eta, \quad \lambda_2 = \xi, \quad \lambda_3 = \eta.$$

They are affine, sum to one, and have the Kronecker values at the three vertices. Since $\dim P_1(\widehat{T}) = 3$, they form the triangular Lagrange basis and give unisolvent vertex evaluations.

(d) The corner values of q are $(0, 0, 1, 0)$. Consequently $I_Q q = \psi_3 = \xi\eta = q$. On the split along A_1A_3 , the affine function on the lower triangle $\{0 \leq \eta \leq \xi \leq 1\}$ is η , and on the upper triangle $\{0 \leq \xi \leq \eta \leq 1\}$ it is ξ . Both give the specified values at their three vertices. They agree on $\xi = \eta$, so

$$I_+ q(\xi, \eta) = \min(\xi, \eta).$$

On the split along A_2A_4 , all three values on $\{\xi + \eta \leq 1\}$ are zero. On $\{\xi + \eta \geq 1\}$, the affine function with values zero at A_2, A_4 and one at A_3 is $\xi + \eta - 1$. The two pieces agree on the diagonal, hence

$$I_- q(\xi, \eta) = \max(0, \xi + \eta - 1).$$

(e) At $z = (1/2, 1/2)$,

$$q(z) = I_Q q(z) = \frac{1}{4}, \quad I_+ q(z) = \frac{1}{2}, \quad I_- q(z) = 0.$$

All three have boundary traces 0 on $\xi = 0$ and $\eta = 0$, η on $\xi = 1$, and ξ on $\eta = 1$. Nevertheless, the interior functions differ. The square space contains the bilinear cross term $\xi\eta$, whereas the two triangular spaces consist of functions affine on their respective triangles. Unisolvence is a statement within a specified space and mesh. The same nodal data can therefore select different functions in these three different spaces, without any failure of uniqueness.

□

8.6 The Lax–Richtmyer theorem

Exercise 6.1 — Stability without consistency

1. For every $h \in H$, the operator $L_h = 2I$ is bijective, with inverse

$$L_h^{-1} = \frac{1}{2}I.$$

2. Since

$$\|L_h^{-1}\|_{X \rightarrow X} = \left\| \frac{1}{2}I \right\|_{X \rightarrow X} = \frac{1}{2} \quad \text{for all } h \in H,$$

the family $(L_h)_{h \in H}$ is stable.

3. Consistency would mean that

$$\|Lx - L_h x\|_X \rightarrow 0 \quad \text{for every } x \in X.$$

Here,

$$L_h x - Lx = 2x - x = x,$$

hence

$$\|L_h x - Lx\|_X = \|x\|_X.$$

If $x \neq 0$, this quantity does not tend to 0. Therefore the discretization is not consistent.

4. For $y \in X$, the exact solution is

$$x = L^{-1}y = y,$$

whereas the discrete solution is

$$x_h = L_h^{-1}y = \frac{1}{2}y.$$

Therefore

$$x_h - x = -\frac{1}{2}y, \quad \|x_h - x\|_X = \frac{1}{2}\|y\|_X.$$

If $y \neq 0$, this does not tend to 0 as $h \rightarrow 0$. Thus $x_h \not\rightarrow x$ in general.

This shows that stability alone is not enough: without consistency, we cannot expect convergence. $\square$

Exercise 6.2 — Convergence does not imply consistency

1. For $x = (x_k)_{k \geq 1} \in \ell^2$, we have

$$P_n x = x_n e_n, \quad L_n x = x + n x_n e_n.$$

Therefore L_n leaves every coordinate unchanged except the n th one, which is multiplied by $n + 1$. More precisely,

$$L_n x = (x_1, \dots, x_{n-1}, (n+1)x_n, x_{n+1}, \dots).$$

It follows that L_n is bijective, and its inverse is given by

$$L_n^{-1}y = (y_1, \dots, y_{n-1}, \frac{1}{n+1}y_n, y_{n+1}, \dots).$$

Equivalently,

$$L_n^{-1} = I - \frac{n}{n+1}P_n.$$

2. For $y = (y_k)_{k \geq 1} \in \ell^2$,

$$\|L_n^{-1}y\|_{\ell^2}^2 = \sum_{k \neq n} |y_k|^2 + \frac{1}{(n+1)^2} |y_n|^2 \leq \sum_{k \geq 1} |y_k|^2 = \|y\|_{\ell^2}^2.$$

Hence $\|L_n^{-1}\|_{\ell^2 \rightarrow \ell^2} \leq 1$. On the other hand, if we take $y = e_{n+1}$, then $P_n y = 0$, so $L_n^{-1}y = y$ and

$$\|L_n^{-1}y\|_{\ell^2} = \|y\|_{\ell^2}.$$

Therefore $\|L_n^{-1}\|_{\ell^2 \rightarrow \ell^2} = 1$.

3. For $y \in \ell^2$,

$$L_n^{-1}y - y = -\frac{n}{n+1}y_n e_n.$$

Hence

$$\|L_n^{-1}y - y\|_{\ell^2} = \frac{n}{n+1}|y_n|.$$

Since $y \in \ell^2$, we have $y_n \rightarrow 0$ as $n \rightarrow \infty$. Therefore

$$L_n^{-1}y \rightarrow y \quad \text{in } \ell^2.$$

4. Let $x = (1/k)_{k \geq 1}$. Then $x \in \ell^2$ and

$$P_n x = x_n e_n = \frac{1}{n} e_n.$$

Therefore

$$L_n x - x = n P_n x = e_n.$$

Taking norms gives

$$\|L_n x - x\|_{\ell^2} = \|e_n\|_{\ell^2} = 1 \quad \text{for all } n \geq 1.$$

5. Since $L = I$, the exact solution of $Lx = y$ is simply $x = y$. Part 3 shows that, for every $y \in \ell^2$, the discrete solutions

$$x_n := L_n^{-1}y$$

satisfy $x_n \rightarrow y = L^{-1}y$. Thus the scheme is convergent in the sense of the discrete solutions. However, part 4 shows that there exists $x \in \ell^2$ such that

$$\|L_n x - Lx\|_{\ell^2} = \|L_n x - x\|_{\ell^2} = 1 \quad \text{for all } n \geq 1,$$

so $L_n x \not\rightarrow Lx$. Therefore the family (L_n) is not consistent.

This example proves that convergence of the discrete solutions does not imply consistency. $\square$

Exercise 6.3 — A stronger version in operator norm

1. Since $LL^{-1} = I$ and $L_h L_h^{-1} = I$, we can write

$$L_h^{-1} - L^{-1} = L_h^{-1}LL^{-1} - L_h^{-1}L_h L^{-1} = L_h^{-1}(L - L_h)L^{-1}.$$

2. Taking operator norms and using the submultiplicativity of the operator norm gives

$$\|L_h^{-1} - L^{-1}\|_{Y \rightarrow X} \leq \|L_h^{-1}\|_{Y \rightarrow X} \|L - L_h\|_{X \rightarrow Y} \|L^{-1}\|_{Y \rightarrow X}.$$

3. By assumption,

$$\sup_{h \in H} \|L_h^{-1}\|_{Y \rightarrow X} < +\infty \quad \text{and} \quad \|L_h - L\|_{X \rightarrow Y} \rightarrow 0.$$

Since L is bijective between Banach spaces, the bounded inverse theorem gives $L^{-1} \in \mathcal{L}(Y, X)$. Therefore the right-hand side in the estimate of part 2 tends to 0, and we obtain

$$\|L_h^{-1} - L^{-1}\|_{Y \rightarrow X} \rightarrow 0.$$

Hence $L_h^{-1} \rightarrow L^{-1}$ in operator norm.

4. In Theorem 6.3, the assumption is only the pointwise consistency

$$L_h x \rightarrow Lx \quad \text{in } Y \text{ for every } x \in X,$$

together with stability

$$\sup_{h \in H} \|L_h^{-1}\|_{Y \rightarrow X} < +\infty.$$

Under these assumptions, we obtain only the pointwise convergence

$$L_h^{-1} y \rightarrow L^{-1} y \quad \text{in } X \text{ for every } y \in Y.$$

Here, the assumption has been strengthened from pointwise convergence of $L_h x$ to convergence in operator norm:

$$\|L_h - L\|_{X \rightarrow Y} \rightarrow 0.$$

In return, the conclusion is also stronger: we get convergence of the inverses in operator norm, namely

$$\|L_h^{-1} - L^{-1}\|_{Y \rightarrow X} \rightarrow 0,$$

which is equivalent to uniform convergence of $L_h^{-1} y$ to $L^{-1} y$ on the unit ball of Y .

□

Exercise 6.4 — Consistency only on a dense subspace

Set

$$M := \sup_{h \in H} \|L_h\|_{X \rightarrow Y} < +\infty.$$

1. Let $x \in X$ and let $\varepsilon > 0$. Since D is dense in X , choose $d \in D$ such that

$$(M + \|L\|_{X \rightarrow Y})\|x - d\|_X < \frac{\varepsilon}{2}.$$

Then, for every $h \in H$,

$$\begin{aligned} \|L_h x - Lx\|_Y &\leq \|L_h(x - d)\|_Y + \|L_h d - Ld\|_Y + \|L(d - x)\|_Y \\ &\leq M\|x - d\|_X + \|L_h d - Ld\|_Y + \|L\|_{X \rightarrow Y}\|d - x\|_X. \end{aligned}$$

The first and third terms are together $< \varepsilon/2$. Since $d \in D$, we have $L_h d \rightarrow Ld$ in Y , so for h small enough,

$$\|L_h d - Ld\|_Y < \frac{\varepsilon}{2}.$$

Hence $\|L_h x - Lx\|_Y < \varepsilon$ for h small enough. Therefore

$$L_h x \rightarrow Lx \quad \text{in } Y \text{ for every } x \in X.$$

2. Suppose in addition that the discrete problems are well posed and stable. If we can verify $L_h d \rightarrow Ld$ only for d in a dense subspace D , and if we also know that $\sup_{h \in H} \|L_h\|_{X \rightarrow Y} < +\infty$, then part 1 upgrades this dense-subspace consistency to full consistency on all of X . The Lax–Richtmyer theorem then yields convergence. Thus, in practice, it may be enough to check consistency on a dense class of test elements, provided the family (L_h) is uniformly bounded in $\mathcal{L}(X, Y)$.

□

8.7 Finite difference methods for hyperbolic equations

Exercise 7.1 — Transport with source term and Duhamel formula

- (a) Let (x, t) be fixed and consider the backward characteristic

$$X(s) = x - a(t - s) = x - at + as, \quad 0 \leq s \leq t. \quad (8.254)$$

Then $X(t) = x$ and $X(0) = x - at$. Along this curve,

$$\frac{d}{ds}u(X(s), s) = \partial_t u(X(s), s) + a\partial_x u(X(s), s) = f(X(s), s). \quad (8.255)$$

Integrating from 0 to t gives

$$u(x, t) - u_0(x - at) = \int_0^t f(X(s), s) ds. \quad (8.256)$$

Since $X(s) = x - a(t - s)$, we obtain

$$u(x, t) = u_0(x - at) + \int_0^t f(x - a(t - s), s) ds. \quad (8.257)$$

- (b) Taking absolute values in the formula gives

$$|u(x, t)| \leq \|u_0\|_{L^\infty(\mathbb{R})} + \int_0^t \|f(s)\|_{L^\infty(\mathbb{R})} ds. \quad (8.258)$$

Taking the supremum with respect to x gives the announced estimate.

□

Exercise 7.2 — Inflow transport on a finite interval

- (a) Let (x, t) be fixed and consider the characteristic passing through this point. It is given by

$$X(s) = x - a(t - s) = x - at + as. \quad (8.259)$$

Along this curve we have

$$\frac{d}{ds}u(X(s), s) = \partial_t u(X(s), s) + a\partial_x u(X(s), s) = 0, \quad (8.260)$$

so u is constant along the characteristic.

If $x - at > \alpha$, the backward characteristic reaches the initial line at the point $(x - at, 0)$ before touching the boundary. Hence

$$u(x, t) = u_0(x - at). \quad (8.261)$$

If $x - at \leq \alpha$, the backward characteristic reaches the inflow boundary $x = \alpha$ at the time

$$s_* = t - \frac{x - \alpha}{a} \geq 0. \quad (8.262)$$

Therefore

$$u(x, t) = u(\alpha, s_*) = \phi\left(t - \frac{x - \alpha}{a}\right). \quad (8.263)$$

This proves the announced formula.

- (b) The explicit formula of part (a) shows that every value $u(x, t)$ is obtained either from the initial datum or from the inflow boundary datum. Therefore

$$|u(x, t)| \leq \max \left\{ \|u_0\|_{L^\infty(\alpha, \beta)}, \|\phi\|_{L^\infty(0, t)} \right\} \quad \text{for all } x \in (\alpha, \beta). \quad (8.264)$$

Taking the supremum with respect to x yields

$$\|u(t)\|_{L^\infty(\alpha, \beta)} \leq \max \left\{ \|u_0\|_{L^\infty(\alpha, \beta)}, \|\phi\|_{L^\infty(0, t)} \right\}. \quad (8.265)$$

- (c) We multiply the equation by u and integrate over (α, β) :

$$\int_\alpha^\beta \partial_t u u \, dx + a \int_\alpha^\beta \partial_x u u \, dx = 0. \quad (8.266)$$

As in the energy estimate derived in Chapter 7,

$$\int_\alpha^\beta \partial_t u u \, dx = \frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(\alpha, \beta)}^2, \quad (8.267)$$

and

$$a \int_\alpha^\beta \partial_x u u \, dx = \frac{a}{2} |u(\beta, t)|^2 - \frac{a}{2} |u(\alpha, t)|^2 = \frac{a}{2} |u(\beta, t)|^2 - \frac{a}{2} |\phi(t)|^2. \quad (8.268)$$

Thus

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(\alpha, \beta)}^2 + \frac{a}{2} |u(\beta, t)|^2 = \frac{a}{2} |\phi(t)|^2. \quad (8.269)$$

Integrating from 0 to t gives

$$\|u(t)\|_{L^2(\alpha, \beta)}^2 + a \int_0^t |u(\beta, s)|^2 \, ds = \|u_0\|_{L^2(\alpha, \beta)}^2 + a \int_0^t |\phi(s)|^2 \, ds. \quad (8.270)$$

Dropping the nonnegative boundary term yields

$$\|u(t)\|_{L^2(\alpha, \beta)}^2 \leq \|u_0\|_{L^2(\alpha, \beta)}^2 + a \int_0^t |\phi(s)|^2 \, ds. \quad (8.271)$$

□

Exercise 7.3 — Fourier analysis of the classical schemes

- (a) We insert the Fourier mode

$$u_j^n = \hat{u}^n e^{ij\theta} \quad (8.272)$$

into each scheme.

For FTCS,

$$\hat{u}^{n+1} e^{ij\theta} = \hat{u}^n e^{ij\theta} - \frac{\nu}{2} \hat{u}^n (e^{i(j+1)\theta} - e^{i(j-1)\theta}) \quad (8.273)$$

$$= \left(1 - \frac{\nu}{2} (e^{i\theta} - e^{-i\theta}) \right) \hat{u}^n e^{ij\theta}, \quad (8.274)$$

so

$$g_{\text{FTCS}}(\theta) = 1 - i\nu \sin \theta. \quad (8.275)$$

For the upwind scheme,

$$g_{\text{U}}(\theta) = 1 - \nu(1 - e^{-i\theta}). \quad (8.276)$$

For Lax–Friedrichs,

$$g_{\text{LF}}(\theta) = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) - \frac{\nu}{2}(e^{i\theta} - e^{-i\theta}) = \cos \theta - i\nu \sin \theta. \quad (8.277)$$

For Lax–Wendroff,

$$g_{\text{LW}}(\theta) = 1 - \frac{\nu}{2}(e^{i\theta} - e^{-i\theta}) + \frac{\nu^2}{2}(e^{i\theta} - 2 + e^{-i\theta}) \quad (8.278)$$

$$= 1 - i\nu \sin \theta + \nu^2(\cos \theta - 1). \quad (8.279)$$

(b) For FTCS,

$$|g_{\text{FTCS}}(\theta)|^2 = 1 + \nu^2 \sin^2 \theta. \quad (8.280)$$

If $\nu \neq 0$ and $\sin \theta \neq 0$, then

$$|g_{\text{FTCS}}(\theta)| > 1. \quad (8.281)$$

Therefore the amplification factor is not bounded by 1 on $[-\pi, \pi]$, so the scheme cannot be strongly stable in ℓ_h^2 .

Under the hyperbolic scaling, ν is fixed. On arbitrarily fine periodic grids, choosing a mode with $\sin^2 \theta \geq 1/2$ gives

$$|g_{\text{FTCS}}(\theta)| \geq \sqrt{1 + \frac{\nu^2}{2}} > 1. \quad (8.282)$$

After n steps, this mode is multiplied by $|g_{\text{FTCS}}(\theta)|^n$. Since n is of order $T/\Delta t$ on a fixed time interval, this amplification is unbounded as $\Delta t \rightarrow 0$. Hence FTCS is not stable in ℓ_h^2 on finite time intervals under hyperbolic scaling. This conclusion is obtained by this explicit Fourier-mode construction, not by invoking a converse of Theorem 7.20.

(c) We compute the moduli of the stable schemes. For upwind,

$$|g_{\text{U}}(\theta)|^2 = |1 - \nu + \nu e^{-i\theta}|^2 \quad (8.283)$$

$$= (1 - \nu + \nu \cos \theta)^2 + \nu^2 \sin^2 \theta \quad (8.284)$$

$$= 1 - 2\nu(1 - \nu)(1 - \cos \theta) \leq 1. \quad (8.285)$$

For Lax–Friedrichs,

$$|g_{\text{LF}}(\theta)|^2 = \cos^2 \theta + \nu^2 \sin^2 \theta = 1 - (1 - \nu^2) \sin^2 \theta \leq 1, \quad (8.286)$$

because $0 \leq \nu \leq 1$. For Lax–Wendroff, we write

$$g_{\text{LW}}(\theta) = 1 - \nu^2(1 - \cos \theta) - i\nu \sin \theta. \quad (8.287)$$

Hence

$$|g_{\text{LW}}(\theta)|^2 = (1 - \nu^2(1 - \cos \theta))^2 + \nu^2 \sin^2 \theta \quad (8.288)$$

$$= 1 - \nu^2(1 - \nu^2)(1 - \cos \theta)^2 \leq 1. \quad (8.289)$$

The von Neumann criterion stated in Theorem 7.20 then yields strong ℓ_h^2 stability of upwind, Lax–Friedrichs, and Lax–Wendroff on the periodic grid.

(d) The upwind scheme can be rewritten as

$$u_j^{n+1} = (1 - \nu)u_j^n + \nu u_{j-1}^n. \quad (8.290)$$

When $0 \leq \nu \leq 1$, the coefficients form a convex combination. Hence

$$|u_j^{n+1}| \leq (1 - \nu) \sup_k |u_k^n| + \nu \sup_k |u_k^n| = \sup_k |u_k^n|. \quad (8.291)$$

Taking the supremum with respect to j yields

$$\sup_j |u_j^{n+1}| \leq \sup_j |u_j^n|. \quad (8.292)$$

Thus the upwind scheme satisfies the discrete maximum principle.

□

Exercise 7.4 — The wave equation as a first-order hyperbolic system

(a) The characteristic polynomial of A is

$$\det(A - \mu I) = \mu^2 - \gamma^2, \quad (8.293)$$

so the eigenvalues are $\mu = \gamma$ and $\mu = -\gamma$.

For $\mu = \gamma$, we solve

$$\begin{pmatrix} -\gamma & -\gamma \\ -\gamma & -\gamma \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0, \quad (8.294)$$

which gives $y = -x$. We may choose

$$r_+ = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (8.295)$$

For $\mu = -\gamma$, we similarly obtain

$$r_- = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (8.296)$$

Therefore we may take

$$T = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} \gamma & 0 \\ 0 & -\gamma \end{pmatrix}. \quad (8.297)$$

Since $\det T = 2$, we have

$$T^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \quad (8.298)$$

Hence $A = T\Lambda T^{-1}$.

(b) Since $U = TW$, the system

$$\partial_t U + A \partial_x U = 0 \quad (8.299)$$

becomes

$$\partial_t W + \Lambda \partial_x W = 0. \quad (8.300)$$

Writing $W = (w_+, w_-)^T$, we obtain the decoupled equations

$$\partial_t w_+ + \gamma \partial_x w_+ = 0, \quad (8.301)$$

$$\partial_t w_- - \gamma \partial_x w_- = 0. \quad (8.302)$$

Thus w_+ is transported to the right with velocity γ , whereas w_- is transported to the left with velocity $-\gamma$.

Moreover,

$$W = T^{-1}U = \frac{1}{2} \begin{pmatrix} v - q \\ v + q \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \partial_t u - \gamma \partial_x u \\ \partial_t u + \gamma \partial_x u \end{pmatrix}. \quad (8.303)$$

(c) Applying the scalar Lax–Wendroff scheme componentwise to W gives

$$W_j^{n+1} = W_j^n - \frac{\Delta t}{2h} \Lambda (W_{j+1}^n - W_{j-1}^n) + \frac{\Delta t^2}{2h^2} \Lambda^2 (W_{j+1}^n - 2W_j^n + W_{j-1}^n). \quad (8.304)$$

We multiply by T and use $U_j^n = TW_j^n$:

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} T \Lambda T^{-1} (U_{j+1}^n - U_{j-1}^n) \quad (8.305)$$

$$+ \frac{\Delta t^2}{2h^2} T \Lambda^2 T^{-1} (U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (8.306)$$

Since $T \Lambda T^{-1} = A$ and $T \Lambda^2 T^{-1} = A^2$, we obtain

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} A (U_{j+1}^n - U_{j-1}^n) + \frac{\Delta t^2}{2h^2} A^2 (U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (8.307)$$

Now

$$A^2 = \begin{pmatrix} 0 & -\gamma \\ -\gamma & 0 \end{pmatrix}^2 = \begin{pmatrix} \gamma^2 & 0 \\ 0 & \gamma^2 \end{pmatrix} = \gamma^2 I. \quad (8.308)$$

Therefore the scheme simplifies to

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2h} A (U_{j+1}^n - U_{j-1}^n) + \frac{\gamma^2 \Delta t^2}{2h^2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n). \quad (8.309)$$

(d) The characteristic speeds are $\pm\gamma$. Therefore the CFL condition is controlled by the largest absolute eigenvalue:

$$\frac{\Delta t}{h} \max\{|\gamma|, |-\gamma|\} \leq 1. \quad (8.310)$$

Since $\gamma > 0$, the condition is

$$\frac{\gamma \Delta t}{h} \leq 1. \quad (8.311)$$

□

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